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Fact-checked draft, Oct 23, 2026. Notes are at the end. [UNVERIFIED] means no source was found; DISPUTED means sources disagree.

Part 1: The Ledger

1769–1848

The Tall Stick (1769–1776)

Along the lower San Francisquito Creek, where the Stanford campus now sits, lived a community the Spanish records call the Puichon. They were one of roughly forty small, independent tribal groups that outsiders would later lump together as Ohlone, or, in older books, Costanoan, from the Spanish word for coast people. Each group held a small territory, usually a few hundred people in a cluster of villages, connected to neighbors by marriage, trade, and language.[1]

At Jasper Ridge, the biological preserve that Stanford owns in the hills just west of campus, the university's own chronology records more than five thousand years of human occupation.[2] Spanish, Mexican, and American rule together, from 1769 to 2026, is two hundred fifty-seven years. That is about five percent of the human story on this land. If the whole five thousand years were a single day, Europeans would have walked onto that plain at about a quarter to eleven at night.

Before there was a tall stick, there were people who did not need a name for it, because they knew exactly where they were.

Ohlone communities did not plant fields of wheat. Later American newcomers looked at that and called them lazy, or worse, diggers, for the sticks they used to pry up edible bulbs. The settlers got this completely wrong. Malcolm Harris points out what they missed. The Ohlone moved, he writes, "in response to abundance rather than scarcity." The ripening of acorns, the spawning of fish, the arrival of birds on the bay marshes: short bursts of intense work, followed by long stretches of what Harris calls "social luxury."[3]

That is not the absence of an economy. It is an economy built on knowledge instead of storage, on knowing which oak drops first and which creek runs in which month. California before 1769 held one of the densest concentrations of human linguistic diversity anywhere on earth. Historians' usual estimate is more than five hundred distinct tribal groups, speaking about a hundred languages. The total population of what the Spanish called Alta California was just over three hundred thousand. About forty thousand of them lived in the Bay Area alone.[4]

Harris opens his history of Palo Alto with this sentence: "To speak of the Ohlone is to speak of the broken link between the Bay Area's indigenous people and the Bay Area itself."[5]

Some places change rulers and keep their people: the language, the place names, the memory of where the water is. Other places rupture. The people who knew the land are separated from it, and the new arrivals start the clock over.

On the peninsula of Baja California, below today's border, the Jesuits had built a chain of missions. Then, in 1767, King Charles III of Spain expelled the entire Jesuit order from his empire. The man sent to carry out the expulsion in Baja was the new governor, Gaspar de Portolá. The Jesuits gathered at Loreto and left for exile in February 1768. A month later, the king's inspector general, José de Gálvez, handed their missions to the Franciscans, under a friar named Junípero Serra.[6]

Junípero Serra was born Miguel José Serra on November 24, 1713, in Petra, on Mallorca. He held a chair of philosophy at the university in Palma and gave it up in 1749. He walked the Mexican and California coast on a leg injured years earlier, on the road up from Veracruz.[7] A year after the Jesuit-to-Franciscan handoff in Baja, in 1769, Gálvez sent Portolá and Serra north, by land and by sea, to occupy San Diego and Monterey before the Russians or the British could.[8]

In October 1769, marching north near today's Watsonville, the friar who kept the expedition's most detailed diary, Juan Crespí, described very high trees of a red color, not known to them, and wrote that because none of the expedition recognized them, they were named redwood from their color. That is the first written mention of the coast redwood.[9]

November 1769. Sixty-four men, by the standard count of the party that left San Diego on July 14, and a string of mules came down onto a wide valley at the south end of a bay no European had ever seen from land. The party held Spanish soldiers, two Franciscan friars, an engineer, Christian Indians from the Baja missions, and their commander, Gaspar de Portolá. They had been sent to find Monterey. They walked past it without recognizing it.[10]

On Saturday, November 4, their scouts climbed a ridge and saw a body of water nobody expected. That ridge is called Sweeney Ridge now. It sits between San Bruno and Pacifica, inside the Golden Gate National Recreation Area, with a discovery-site marker on the trail. The expedition diary records that November 4 was the feast day of San Carlos Borromeo, the saint in whose honor they had come north to found a presidio and a mission at Monterey, and also the name-day of the king, Carlos III. They

said Mass on the beach. They ate mussels off the rocks. Then they turned up into the hills, and from the summit they saw what one of them called an immense arm of the sea.[11]

A day later, on November 5, Portolá's own diary recorded the view flatly. Before them "extended a great arm of the sea, sixteen to twenty leagues" in extent.[12]

On November 6 they made camp beside a creek that runs down from the hills to the bay. They stayed five nights while the scouts went out again. On November 11 a council of officers voted to turn around.[13]

The story goes that they camped beneath one enormous redwood, a tree so tall and alone on the plain that the Spanish called it *el palo alto*. The tall stick.

The story goes means legend. The record shows means a document. And the record on this tree is more interesting than the story.

That tree is still standing on San Francisquito Creek, at the border of Palo Alto and Menlo Park, about a hundred and ten feet tall, commuter trains a few yards from its roots. It sprouted around the year 940. It named a city, and it sits in the middle of the Stanford seal, between two Latin words, *semper virens*—ever green—though it stands on the east side of the railroad line, and a land map made for Leland Stanford shows it was never on land he owned.[14] Even Malcolm Harris, who grew up in Palo Alto, has them camped five days under that tree. The diaries support the five nights by the creek. They do not name this tree.[15]

Portolá went south, returned the next year, founded Monterey on June 3, 1770, sailed away on July 9, and never came back. He later governed Puebla, in Mexico. The story goes that he discovered the bay. The record shows a soldier carrying out Gálvez's plan. In 1770 Serra's headquarters, Mission San Carlos Borromeo, was founded at Monterey and then moved to Carmel.[16]

The first time the tall redwood on San Francisquito Creek shows up in writing is five years after Portolá's camp, in 1774, when another friar, Francisco Palóu, put up a wooden cross near a big tree on the creek and marked it as a possible mission site.[17]

Still, no European ship had sailed into the water the scouts had seen from Sweeney Ridge. Ships on the Manila trade had been passing the California coast for two hundred years. They knew Point Reyes. They knew the Farallon Islands, thirty miles out. They had not known there was a bay behind the cliffs.[18]

August 5, 1775. Night. A small Spanish supply ship called the *San Carlos*—carrying the saint's name from the day of the 1769 sighting—a packet built at the naval base of San Blas, felt her way through a strait that is, at its widest, only about three miles across, and often hidden in fog. The *San Carlos* was the first European keel in the water. Her commander, Lieutenant Juan Manuel de Ayala, had been shot in the right foot on April 3, by a loaded pistol his predecessor left in the cabin. He wrote in his report that he could not do the work in person. His pilot, José de Cañizares, took the launch and drew the first working map of the bay, south to what is now Suisun. Ayala arrived the night of August 5, stayed forty-four days, and

called it one of the best harbors on the coast. They anchored by an island he named Isla de los Ángeles. Today the cove is Ayala Cove; the island is Angel Island. On September 18, rudder damaged, they ran down to Monterey and arrived the next day. (California Historical Landmark 236, at Aquatic Park, Beach and Larkin in San Francisco, marks the entrance.) Without that chart, the presidio and the mission of 1776 would have been a guess.[19]

In March 1776, the commander Juan Bautista de Anza and Father Pedro Font rode out to inspect the San Francisquito Creek site Palóu had marked with a cross. Font measured the tall redwood with a surveying instrument: fifty varas high, about a hundred and thirty-seven feet. (Whether the surviving tree at the railroad bridge is the exact one Portolá's party camped beside is DISPUTED: traditional accounts and Malcolm Harris place the five-day camp under El Palo Alto, while the 1769 expedition diaries do not name the tree and some local historians argue the actual 1769 campsite tree stood farther downstream and has since been cut down.)[20] Then Anza and Font looked at the creek, decided its flow was too unreliable to water mission fields through a dry summer, and moved on.[21]

That year, 1776, the Spanish founded the Presidio of San Francisco on its cliff and Mission San Francisco de Asís, named for Saint Francis of Assisi, the founder of the Franciscan order. Almost everyone came to call it Mission Dolores, after a nearby creek and lagoon the Anza expedition named for Our Lady of Sorrows, Nuestra Señora de los Dolores.[22]

Baptismal Register (1777–1832)

January 12, 1777. About fifteen miles southeast of the rejected creek with the tall redwood, on the bank of the Guadalupe River, near an Ohlone village the Spanish wrote down as So-co-is-u-ka—laurel wood—Franciscan friars planted a cross and put up a shelter for worship. They named the place Misión Santa Clara de Thámien, after Saint Clare of Assisi and after the local people, the Thámien. It was the eighth mission in Alta California, and the first one named for a woman.[23]

The very first plan for a big institution on San Francisquito Creek had been a mission. More than a century later, someone would try again, with a railroad fortune instead of a royal charter. Santa Clara itself kept moving; floods, fire, and an earthquake eventually forced six rebuilds of the church.[24]

Every person who came into a mission was written into a book. A baptismal book. A marriage book. A burial book. The line gave a Christian name, the parents, the village, a godparent, and the minister's signature. That was the shape of the form. The books themselves were kept with real care, which is why historians can count what happened, and why Ohlone families today, including the Muwekma Ohlone Tribe, can trace their ancestors through the baptismal registers of the Spanish missions at San Francisco and Santa Clara. For the people who were here first, the paper trail of their own ancestry begins in a book written by the people who came to change them.[25]

A California mission was four institutions in one.

First, it was a church. Its stated purpose was conversion, to bring Native people to Catholic Christianity. A baptized Native person was called a *neófito*, a neophyte, a new believer. An unbaptized one was a *gentil*, a gentile.[26]

Second, it was an economy. A mission was a farm, a ranch, a workshop, and a warehouse. California was months by sea from Mexico, and supply ships were small, so the friars needed the neophytes to grow the wheat, tend the cattle, weave the cloth, tan the leather, and render the tallow.[27]

Third, it was a military partner. Every mission sat in the district of a presidio, a fort with a small garrison of soldiers. The soldiers protected the friars, and they also went out to bring back people who left.[28]

Fourth, it was a labor system. Once baptized, a neophyte was expected to live at the mission, under the friars' rules, and could not simply walk away. People who left without permission were treated as runaways, and soldiers were sent to bring them back. Santa Clara University's Ohlone heritage project puts it plainly: Native people from dozens of communities were brought to live together at the mission, and, "laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain's colony in California."[29] Whether the mission labor regime constituted slavery is DISPUTED among historians: neophytes were not bought and sold as chattel, and some entered the missions, at least initially, for food, trade goods, or safety; yet they could not leave freely, worked without pay, and faced physical punishment, leading eyewitnesses such as Richard Henry Dana Jr. to write that they were "virtually slaves."[30]

A rider leaving Mission Santa Clara heading north with a pouch of letters followed a path that had been a footpath long before he was born, an Ohlone trail along the edge of the hills, above the marshes. He passed the creek with the tall redwood. He reached Mission San Francisco de Asís, about forty-five miles up the peninsula. From there, letters went on by ship, or south by horse to Monterey, the capital, and from Monterey down the coast to San Diego, and eventually to Mexico City. Spanish California's long-distance freight came by sea, on supply ships from San Blas, in Mexico, and later on foreign traders. The road was a corridor for friars, animals, tools, letters, orders, and very slow news. There was no single El Camino Real in Spanish times; *camino real* meant royal road, and there were many *caminos reales*, a general term for roads used under the Crown's authority, most of them following older Native trade routes. As the Autry Museum of the American West notes, El Camino Real as a single road lined with missions, each a day's ride from the next, is an American construct of the twentieth century. The road's one hard fact was its slowness. Mexico City was months away, so whoever controlled the landings and the stamps held everyone else.[31]

Junípero Serra served as president of the California missions from 1769 until his death on August 28, 1784, at Mission San Carlos Borromeo in Carmel, when only nine of the twenty-one missions existed. He was buried there.[32] His successor, Fermín Francisco de Lasuén, founded nine more missions between 1786 and 1798, including La Purísima in 1787. By the time all twenty-one missions existed, they stretched from San Diego, founded in 1769, to Sonoma, founded in 1823, about six hundred miles, roughly the distance from London to Milan.[33]

In 1811, north of Spanish California, John Jacob Astor's Pacific Fur Company built Fort Astoria at the mouth of the Columbia River, the first American settlement on the Pacific. Astor had looked west and tried to own it by trade.[34]

Ten years later, in 1821, Mexico won its independence from Spain, and with it, the far-off province of Alta California became Mexican.[35] The year after that, in 1822, an old artilleryman from the Presidio of San Francisco named José Peña began grazing cattle on the ground south of San Francisquito Creek with the mission's permission.[36] Two years later, in 1824, Ohlone people showed a man named Secundino Robles where they dug a red pigment rock south of San José. It was cinnabar, the ore of mercury—quicksilver.[37]

The new Mexican republic had a problem with the missions. They were run by Spanish-born friars loyal to a church still tied to Spain, and they controlled the best land in California, along with most of its labor. Mexican liberals believed in separating church power from the state. Ambitious soldiers and settlers in California wanted that land. And some officials genuinely believed the neophytes ought to be free citizens.[38]

The word was secularization, from the Latin *saeculum*, the world, as opposed to the cloister. A secular priest serves a parish under a bishop. A regular priest, like a Franciscan, lives under a religious order. To secularize a mission meant, on paper, to turn it into an ordinary parish and to release its lands, herds, and people.[39]

It came in steps. On July 25, 1826, California's first Mexican-born governor, José María de Echeandía, issued a proclamation of emancipation that let qualified neophytes in the southern districts leave mission rule and become Mexican citizens.[40]

At Mission Santa Clara, the Native population peaked in 1827 at about 1,464. (In the fall of 2025, Stanford University enrolled 1,866 new first-years, the Class of 2029; Santa Clara's peak was smaller than one incoming college class living in one compound, working the fields without pay.) Across the mission chain, the death rate for children was especially high. At Mission San Gabriel, one missionary reported that three out of four children died before the age of two. Measles epidemics in 1806 and 1828 swept through the crowded compounds. The missions kept needing new converts because the people already there kept dying.[41]

In 1831, about 18,683 Native people were still under mission control across California, compared with 4,342 soldiers, settlers, and what the count called other classes. Four to one.[42]

By December 31, 1832, at the peak of the system, the missions together held about 151,000 head of cattle and nearly 138,000 sheep, and controlled roughly one-sixth of all the land in Alta California. From 1769 through December 31, 1832, the friars across all the California missions recorded 87,787 baptisms. They also recorded 63,789 deaths. For every ten people baptized, there were about seven burials. Santa Clara was the busiest of all. It recorded more baptisms, more marriages, and more burials than any other mission in the chain: 8,536 baptisms and 6,809 burials. In 1832, 1,125 Native people lived at Mission Santa Clara.[43]

Disease, Harris writes, rather than extermination campaigns, killed about half of the roughly three hundred thousand Native people of Alta California under Spain and Mexico.[44] The mission was also a ledger: the first written record, for thousands of people, of their own names.

Silver and Hides (1833–1847)

On August 17, 1833, the Mexican Congress passed the decree for the secularization of the missions of California. On August 9, 1834, Governor José Figueroa issued the rules for how it would actually work. Each head of a family, and everyone over twenty, would receive a lot of mission land, not more than four hundred varas square and not less than a hundred. A vara is a Spanish yard, about thirty-three inches, so that plot came to somewhere between about seven and twenty-eight acres. They would also receive half the livestock, shared out, and half the tools, the seed, and the movable goods, or less.[45]

On paper, it was one of the most sweeping land redistributions in North American history.

On paper.

During the mission era, the land was supposed to be held in trust by the friars, for the benefit of the Native people, until they were judged ready to manage it themselves. As Santa Clara University's Ohlone heritage project puts it, the lands "were supposed to be held in trust by the Franciscans on behalf of Native people."[46] A trust is a promise. A title is a deed with your name on it. And when the moment came to turn the trust into titles, the titles mostly went to someone else.

That same August of 1834, a Boston boy at Harvard named Richard Henry Dana signed on as a common sailor aboard the brig *Pilgrim*, bound around Cape Horn for California. Measles had damaged his eyes so badly he could not read, and his family could not afford Europe. The *Pilgrim* was going to California to collect cowhides.[47]

The missions and the ranches ran enormous herds. There was no railroad and no bank, and beef could not reach a market, so the skins were the crop. Boston houses sent ships around South America with manufactured goods and swapped them for hides, which went home to become shoes.[48]

Writing about Monterey in 1835, Dana observed: "They have no circulating medium but silver and hides, which the sailors call California bank notes."[49]

The price of a hide was fixed by custom at two dollars, roughly sixty-five dollars today. Josiah Royce, born in a California mining town in 1855 and writing in 1886, noted that the traders took them at that uniform value. A hide was money. Rancheros paid their debts in hides.[50]

Dana noticed the prices going the other way. "Their hides, too, which they value at two dollars in money, they give for something which costs seventy-five cents in Boston; and buy shoes, like as not made of their own hides, and which have been carried twice around Cape Horn, at three or four dollars," he wrote. Things sold "at an advance of nearly three hundred per cent upon the Boston prices."[51]

A cow was skinned in California. The hide became a shoe in Boston. The shoe sailed back, and the person who raised the cow paid several times the Boston price. In Mexican California, there was one official custom-house, at Monterey. Every trading ship had to enter its cargo there and pay duties before it could trade anywhere on the coast. There were very few safe harbors—Dana counted San Francisco, Monterey, and San Diego as the only ports safe from the winter southeasters—and very few ships. If you lived inland, you bought what the ships brought, at the ships' price, or you went without. At Santa Barbara, where the ship anchored three miles offshore and the town was nearly a mile farther inland, Dana wrote that so far as the crew were concerned, the town "might almost as well have been in the middle of the Cordilleras."[52] One way in, one way out, and whoever held the gate set the price.

On the other side of that trade were the Californios. Royce collected the memories of people who knew them, writing that the white population was made up mainly of Spanish and Mexican colonists "whose chief industry was raising cattle for the hides and tallow, and whose private lives were free, careless, and on the whole, as this world goes, moderately charming and innocent." They were, he wrote, "gay and jovial, full of good fellowship and hospitality." They paid the Yankee traders a little slowly, but generally paid in the end, in hides. Missing from that picture were the Native vaqueros and laborers who did most of the work of raising those cattle. As Royce wrote in the same passage: "The mission Indians had in large part disappeared."[53]

Meanwhile, secular officials called *administradores*—administrators—ran the handover of the missions. Dana saw the system as it was ending and had a sailor's contempt for them. The law declared all the Indians free and independent rancheros, he wrote, but the change in their condition was only nominal; the neophytes were "virtually slaves," as much as they ever were. And the missions' great possessions were given over to be preyed upon by "the harpies of the civil power," the administrators, who usually ended, in a few years, by making themselves fortunes.[54]

In 1835, on the east side of San Francisquito Creek, a neighboring ranch called the Rinconada del Arroyo de San Francisquito—about 2,229 acres—was granted to Rafael Soto, whose father had come north with the Anza expedition. That land is now, roughly, downtown Palo Alto, and the tall redwood stood at its northwest corner.[55]

In December 1835, Dana's ship sailed into San Francisco Bay. Past the Presidio on its cliff, there was a crumbling mission, a few ranches, and a newly begun settlement, mostly of Yankee Californians, called Yerba Buena—good herb, for a wild mint that grew on the hills. A Russian brig from Sitka, in Russian Alaska, was anchored there, loading tallow and grain. Looking at the bay, Dana wrote: "If California ever becomes a prosperous country, this bay will be the centre of its prosperity."[56]

At San Diego, Dana's company had a hide-house built to hold forty thousand hides, and the crew would go in and stare at it, calculating how long it would take to fill. On May 8, 1836, the ship he sailed home on had finally loaded, in his words, "our forty thousand hides, thirty thousand horns," along with barrels of otter and beaver skins. Forty thousand hides at two dollars was eighty thousand dollars—roughly $2.6 million today. One

ship. One voyage. Dana reached home in September 1836, finished Harvard, and in 1840 published *Two Years Before the Mast*. For a generation back east, it was the picture of California.[57]

That same year, 1836, secularization began at Mission Santa Clara,[58] and in Monterey a local revolt against a governor sent from Mexico City brought a Californio named Juan Bautista Alvarado to power. Born in Monterey in 1809, Alvarado was about twenty-seven when he rose to the governorship.[59]

Now the mission lands moved into a new kind of property. To get a rancho, a citizen petitioned the governor. If the petitioner could not write, a neighbor wrote it. Attached to the petition was a *diseño*, the drawing, a map with no surveyor behind it: a creek bend, a big oak, a hill. The governor asked the local officials and the mission whether the land was free and whether the applicant owned cattle. If they said yes, he issued a *título*, a title. Then a local official walked the bounds with the grantee and put him in juridical possession. The oak was the corner. The neighbors were the survey.[60]

What was missing was a precise line. A grant might say two leagues, more or less. Under Mexican law a governor could grant as little as a third of an acre or as much as eleven square leagues, about fifty thousand acres. Two leagues was most common. In a country with plenty of land and few people, *more or less* was fine. Your neighbors knew where the oak was.[61]

Up the peninsula, where San Francisco International Airport sits today, a Mexican cattle ranch called Rancho Buri Buri covered 14,639 acres—about fifty-nine square kilometers, almost exactly the land area of the island of Manhattan. Today that ground is San Bruno, Millbrae, South San Francisco, and part of Burlingame, where a Buri Buri Park and San Mateo County's Junipero Serra Park sit on the former ranchland. The name had nothing to do with Spanish; it was traced to an Ohlone village recorded as Urebure, of the Yelamu people, whose families had been brought into Mission Dolores.[62]

In 1839, Alexander Forbes, a Scottish merchant in Tepic, Mexico, published in London what is generally called the first full English book about California, describing a rich, thinly governed province. The outside world was already leaning in.[63]

That same year, 1839, in Monterey, Governor Alvarado granted a former soldier, Antonio Buelna, a ranch on the southwest side of San Francisquito Creek: Rancho San Francisquito, 1,471 acres—almost six square kilometers, nearly three times the country of Monaco. That piece of ground is today the northern part of the Stanford University campus.[64]

By the end of 1839, the Native population at Mission Santa Clara had fallen from 1,125 in 1832 to 291—a drop of about seventy-four percent in seven years. Historian Robert Jackson, working through the baptism and burial registers, found that deaths alone do not explain it. Hundreds left, or were pulled onto the new private ranches. The mission's own census, signed December 31, 1839, by Father Jesús María Vásquez del Mercado, remains in the archives.[65]

Some Ohlone people did get land. Mission Santa Clara issued four land grants to Ohlone men who had entered the mission as children and asked for their own ancestral village sites back, places recorded as Ulistac and Posolmi. Four was the most of any Bay Area mission. But as Santa Clara's

historians note, those grants were only a small fraction of the total acreage the mission system controlled. In the Bay Area, nearly all other lands were granted to colonial elites during the 1830s and 1840s. Trust had become title. Just not their title.[66]

In 1841, José Peña—who had gone away to Mexico, come back, and was now teaching school in the little settlement at Santa Clara—received a grant from Governor Alvarado for the ground south of Buelna's ranch: Rancho Rincón de San Francisquito, the corner of the San Francisquito, 8,418 acres. That second, bigger rancho is the southern part of the Stanford campus today.[67]

By 1845, the red pigment cinnabar deposit that Ohlone people had shown to Secundino Robles in 1824 had become the New Almaden mine, which would become one of the most valuable quicksilver mines in North America. Secundino and his brother Teodoro Robles were paid in cash and a share. Mercury is what miners use to pull gold out of crushed rock. The people who knew the pigment did not get the mine. The men who were shown the place did.[68]

In the south that same year, 1845, Governor Pío Pico sold Mission San Juan Capistrano—the whole thing—to his brother-in-law, John Forster, for $710, roughly $25,000 today. An entire mission.[69]

By 1846, the ground under the future campus was divided among Mexican rancheros, held by grants, *diseños*, and memory. Native people who had worked these fields for the mission were now working them for the rancheros, or gone.[70]

On May 13, 1846, the United States Congress declared war on Mexico. Sonoma had not yet heard the news when, on June 14, 1846, armed American settlers rode into town and took Colonel Mariano Vallejo prisoner in his own house. They painted a bear, in berry juice, on a flour sack—what Malcolm Harris, quoting the period, calls "something that they called a bear." The California Republic lasted twenty-five days. On July 9, Navy lieutenant Joseph Revere raised the Stars and Stripes.[71]

The pioneer John Bidwell, who watched it, called it a "pretense," the "Texas game." General John Frisbie later told a Bancroft interviewer that the real object was San Francisco Bay. Ulysses S. Grant, who fought the Mexican-American War as a young officer, wrote in his memoirs that it was "one of the most unjust ever waged by a stronger against a weaker nation."[72]

On January 30, 1847, the American alcalde of Yerba Buena, Navy lieutenant Washington Bartlett, ordered the name San Francisco used in every official paper, because Yerba Buena was unknown beyond the district. Mariano Vallejo had wanted a rival town named Francisca, for his wife. Bartlett got there first. Vallejo used her second name; the town is Benicia.[73]

That same year, 1847, José Peña sold almost all of Rancho Rincón de San Francisquito to the two brothers, Secundino and Teodoro Robles.[74] And in Mexico City, the war ended with Mexico defeated and its capital occupied. An American diplomat named Nicholas Trist had been sent to negotiate the peace. President James K. Polk recalled him, but Trist ignored the order. On December 4, 1847, Trist wrote to his wife: "Knowing it to be the very last

chance and impressed with the dreadful consequences to our country which cannot fail to attend the loss of that chance, I decided today at noon to attempt to make a treaty; the decision is altogether my own.”[75]

Nine Days (1848)

Monday, January 24, 1848. Coloma, on the South Fork of the American River. James Marshall, a carpenter from New Jersey, was building a sawmill for John Sutter, a Swiss immigrant whose Mexican grant was eleven square leagues—the legal maximum, nearly fifty thousand acres, bigger than Liechtenstein.[76] Marshall had been running water through the mill’s tail-race to scour it deeper. In the morning he walked the channel and saw something shining in the gravel. Josiah Royce, born in a California mining town in 1855 and writing in 1886, later warned that the tale is simple in its main incident and tangled in its details; the speeches later put in Marshall’s mouth do not agree, but the date the histories use is January 24, and the flake is the part that holds.[77]

Wednesday, February 2, 1848. Two thousand miles away, in the town of Guadalupe Hidalgo, just outside Mexico City, the American diplomat Nicholas Trist sat down with three Mexican commissioners—Luis Gonzaga Cuevas, Bernardo Couto, and Miguel Atristain—and signed the peace treaty.[78] Trist was not even supposed to be at the table; President James K. Polk had recalled him months earlier, and Trist had ignored the order, writing privately to his wife on December 4, 1847: “Knowing it to be the very last chance and impressed with the dreadful consequences to our country which cannot fail to attend the loss of that chance, I decided today at noon to attempt to make a treaty; the decision is altogether my own.”[79]

Article V of the treaty drew the new border. Article XII, in the English of the signed text, read: “In consideration of the extension acquired by the boundaries of the United States, as defined in the fifth article of the present treaty, the Government of the United States engages to pay to that of the Mexican Republic the sum of fifteen million of dollars.”[80]

Nine days. The gold was found on January 24. The treaty that handed California to the United States was signed on February 2. Nobody at that table knew.[81]

Mexico sold the land without knowing what was in it. The United States bought it the same way.

The treaty transferred more than 525,000 square miles—fifty-five percent of Mexico, on the order of France, Spain, and Germany put together. It became California, Nevada, Utah, most of Arizona and New Mexico, and parts of Colorado and Wyoming. Half a country, on a paper signed by a man the president had already fired. The price was fifteen million dollars, plus up to three and a quarter million in debts the United States agreed to assume. The Louisiana Purchase, forty-five years earlier, had carried the same fifteen-million-dollar sticker price for a larger piece of the continent.[82] This one came with the harbor Richard Henry Dana had named, and a mill race full of gold.

On March 10, 1848, the United States Senate ratified the treaty—by a vote DISPUTED between 34 to 14 and 38 to 14 in official government accounts—after striking out Article X, the article that had explicitly guaranteed Mexican land grants.[83] The treaty still promised to respect the property of Mexicans who stayed, but the explicit guarantee for the *diseños*, the oak trees, and the *more or less* was gone.[84]

Five days later, on March 15, 1848, the first printed report of the gold at Coloma appeared in a small San Francisco newspaper, *The Californian*.[85]

Nineteen days after the Senate vote, on March 29, 1848, John Jacob Astor died in his house in New York City. A German-born immigrant who had made a fortune in furs and then in Manhattan real estate, he was the richest man in America, leaving an estate commonly estimated at about twenty million dollars—roughly $770 million in consumer prices today, though grocery prices understate a fortune that represented so large a share of a smaller country. His will left $400,000 for a library that later became part of the New York Public Library. Thirty-seven years after his Pacific Fur Company built Fort Astoria at the mouth of the Columbia in 1811, the first American settlement on the Pacific, old Eastern money passed just as the West's own fortune was lying in a mill race.[86]

That same spring of 1848, the first Chinese residents of San Francisco arrived, servants brought from China by an American businessman.[87]

Mexico asked Washington what the deletion of Article X meant. On May 26, 1848, in the city of Querétaro, before the ratifications were exchanged, American commissioners Nathan Clifford and Ambrose Sevier sat down with Mexico's foreign minister, Luis de la Rosa, and signed an explanation: the Protocol of Querétaro. On the deleted article, the American government stated that it "did not in any way intend to annul the grants of lands made by Mexico in the ceded territories." Those grants, the protocol declared, preserved the legal value they possessed, and the grantees could cause their legitimate titles to be acknowledged before the American tribunals. *Legitimate*, it added, meant legitimate under Mexican law in California up to May 13, 1846.[88]

It was a promise that you could prove your paper in an American court.

President James K. Polk did not officially confirm the gold discovery to Congress until December 5, 1848, ten months after Marshall looked into the tail-race.[89]

By the end of 1848, the redwood on San Francisquito Creek stood on the northwest corner of the Soto family's rancho, the *Rinconada del Arroyo de San Francisquito* (roughly 2,229 acres granted to Rafael Soto in 1835, now downtown Palo Alto). Across the creek, Antonio Buelna's widow, María Concepción Valencia, held the 1,471-acre Rancho San Francisquito, granted in 1839. To the south, the brothers Secundino and Teodoro Robles held the 8,418-acre Rancho Rincón de San Francisquito, which they had bought from José Peña in 1847.[90] The mission at Santa Clara was a struggling parish. Its fields, which the neophytes had been promised, belonged to rancheros. The Ohlone families who had built that mission were scattered, working for others, or gone. The Native population of Alta California, just

over three hundred thousand in 1769, had been cut roughly in half under Spanish and Mexican rule, primarily by disease. “But when the United States showed up,” as Malcolm Harris writes, “the rules changed.”[91]

Everyone who held land along the creek held it by paper: a *título* signed by a Mexican governor, a *diseño* with an oak tree for a corner, and the memory of neighbors. Two ways of owning land were about to collide. Mexican ranchos worked on memory, on neighbors, on the oak tree. The American system that had just arrived worked on measured lines, filed documents, and courts.[92]

In 1851, the new Catholic bishop of California would transfer the parish at Mission Santa Clara from the Franciscans to the Jesuits—the order King Carlos III of Spain had expelled in 1767 returning eighty-four years later to open Santa Clara College, today Santa Clara University, the only university campus in California built on the site of one of the twenty-one missions.[93] That same year, 1851, Congress would pass a Land Act requiring every rancho owner in California to prove, in an American court, in English, with American lawyers paid by the hour, that their land was really theirs. Exactly 813 claims would be filed, and though most would be confirmed—whether 514 or 604 is DISPUTED depending on how historians count the appeals—confirmation would not be the same thing as keeping the ranch.[94] Buelna’s widow would file her claim in 1853 and not receive her final patent from the United States until 1868, fifteen years later, after squatters had settled on the land and a wealthy San Franciscan named George Gordon had bought out both the squatters and her family; the Robles brothers, too, would lose much of their land to lawyers’ fees and debts by the late 1850s.[95]

In 1848, two thousand miles east in Port Washington, a lake town in Wisconsin, a young lawyer from New York was trying to build a practice and not getting far. His brothers would leave for the California gold fields ahead of him, and after a fire burned out his office in 1852, he would follow them to the mining country and open a store, where the men digging for gold bought shovels and flour from him—because the shovel is a surer business than the gravel.[96] He did not know that yet.

His name was Leland Stanford.

Part 2: The Toll

1848–1869

To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer. — W. N. Shilling, telegraph operator, May 10, 1869[97]

Panama (1848–1850)

September 1, 1849. Monterey. Forty-eight delegates [UNVERIFIED] met in Colton Hall, a two-story schoolhouse of local stone above the old Spanish presidio, at 570 Pacific Street, to decide what California was.[98]

The United States had owned the country, on paper, since the Treaty of Guadalupe Hidalgo the year before. It had not organized it. There was no territorial governor sent from Washington. There was a military man, a flood of gold miners, and a question.

A territory is the waiting room. Congress governs it. The president appoints the governor. The people do not elect senators, and they do not vote for president. A state writes a constitution, elects its own officers, and gets two seats in the Senate no matter how large or small it is. Most of the West sat in that waiting room for decades: Oregon waited more than twelve years from acquisition to statehood, Nevada nearly seventeen, Utah forty-eight, New Mexico and Arizona sixty-four. California walked past it in two years and seven months, about twenty-five times faster than New Mexico and Arizona.[99]

On September 5, 1849, the delegates voted on whether to ask Congress for a territorial government or to write a state constitution and demand admission. The clerk's count was twenty-eight to eight, for a state. They were not asking permission to wait. They were writing the document and daring Washington to refuse it. The constitution they signed on October 13 banned slavery. On November 13, 1849, the voters ratified it, 12,061 to 811, a landslide in a place that, a year earlier, had almost no American electorate at all.[100]

Three forces arrived together. Gold, which by the end of 1849 had dumped a population into a place that had almost no American government. The refusal, at Monterey, to wait. And the Compromise of 1850, the national bargain that let California in as a free state, with no slavery, and postponed the breakup of the Union for a decade. Millard Fillmore, who became president when Zachary Taylor died, signed the admission bill on September 9, 1850. Congress had spent the year bargaining. The men in Colton Hall had not.[101]

A free state on the Pacific was a prize both sides of that fight thought they understood. They did not yet understand what it would do to a railroad. In a few years the question of where to lay the tracks would be the same question as where slavery was allowed to grow.

In 1849, Collis Potter Huntington was twenty-eight, a hardware merchant from upstate New York, and he wanted California.[102]

There was no railroad across the continent and no practical wagon road for a shopkeeper in a hurry. The sea route around Cape Horn was eighteen thousand miles. One army officer's ship, William Tecumseh Sherman's *Lexington*, took 202 days. Looking back at that isolation, the historian Stephen Ambrose wrote that it must be doubted whether so desirable a place had ever been so cut off.[103]

Huntington took the other way, the Isthmus of Panama. His steerage ticket cost more than $250, roughly $10,000 today. The ship that should have met him on the Pacific side was not there. For two months he was stranded in the mud and the rain. He wrote to his brother Solon: "Death is carrying off the Americans most fearfully. There are eight to ten deaths a day."[104]

Cholera, dysentery, malaria, yellow fever. Huntington did not die. He went into business on the beach, selling medicines and food to men who could not leave. He later said he made $3,000 at it—roughly $120,000 today—earned because other people were stuck. He remembered the walk across the isthmus plainly: “It was only twenty-four miles. I walked it.” (Accounts differ on the distance; the biographer Richard Rayner puts his tramp at thirty-nine miles.) He walked it at least twenty times, buying and selling, while the fever worked through the camps.[105]

When a place is hard to reach, the money is not in the thing at the end of the path. It is in the passage. Huntington did not know the last step yet. He knew the first. Do not dig. Sell to the people who are digging, and charge them for the crossing.

Before any of these merchants met in California, there was a story the Stanford family liked to tell, and it has to be marked as a story. According to a biographical sketch in the Senate’s memorial addresses after Leland Stanford’s death, a “Mr. Whitney” spent a night at Josiah Stanford’s farm, Elm Grove, on the road between Albany and Schenectady, when Leland was thirteen, and talked about a railroad to the Pacific. The boy never forgot it. Leland was born on March 9, 1824, so thirteen was about 1837. The record problem is the name. There were two Asa Whitneys. One (1791–1874) made car wheels and worked on the Mohawk and Hudson Railroad. The other (1797–1872) was a China merchant who came home convinced that the United States should be the road to Asia, went to China only in 1842, and began his public campaign in 1844, years after a thirteen-year-old Leland would have sat at that table. The memorial says only “Mr. Whitney.” Gordon Chang’s history of the Chinese railroad workers identifies the visitor as the China merchant, though the dates argue against it.[106]

What is not legend is the merchant’s own book. In 1849 Asa Whitney, the promoter, published *A Project for a Railroad to the Pacific*. He wrote that the change of the route for the commerce with Asia had, since before the time of Solomon, changed the destinies of empires, and that its march had always been westward: “Through us must be the route to Asia, and the change to our continent will be the last, the final change.”[107]

Senator Thomas Hart Benton, in the same season of prophecy, imagined Chinese workers building that West, and said they must talk together, and trade together, and marry together. The China dream and the China panic belonged to the same decade. The storekeepers would get rich on both.[108]

The Store and the Fort (1850–1861)

June 1850, on the trail west. Charles Crocker was not yet a railroad man. At the Green River crossing, another party moved its animals onto ground Crocker’s men had picketed. Crocker drove the animals off.

“That ground is ours,” Crocker said.

They threatened to pull his stakes. He pulled a pistol.

“You do, if you dare.”

They backed down. Ambrose records this as one of several times Crocker used the pistol, and he quotes Crocker's own rule: a man well assured of his own position, who shows a bold front, need not fear anybody. On that same journey Crocker found his horse-guards playing cards, called for the deck, burned it, and told them it would not do; they had to have their wits about them. The weapon the record gives him, on a riverbank in 1850, is the pistol. An ax handle would show up later, in another man's hand, used to discipline workers, not to hire them.[109]

In the early 1850s, Sacramento was a levee town that flooded, burned, and rebuilt, selling tools to men on their way to the gravel. Four shopkeepers found each other. Huntington and Mark Hopkins, the quiet partner with the ledger, were neighbors at 52 K Street, in the hardware trade. Crocker was a big man, around 250 pounds, in dry goods and a freighting business, who would become the foreman. Leland Stanford, a lawyer who had burned out of a practice in Wisconsin, sold groceries. Shovels and flour. The miner's lottery was the customer's problem. The margin on the shovel was the store's.[110]

Downriver in San Francisco, the merchants were learning a different trade.

On May 14, 1856, at the corner of Montgomery and Washington in San Francisco, James King of William, editor of the *Evening Bulletin*, was shot by James P. Casey, a city supervisor whose criminal record King had been printing. King died on May 20. On May 22 the Committee of Vigilance hanged Casey, and with him a gambler named Charles Cora, from the windows of its headquarters at 41 Sacramento Street. The building had sandbags piled against the walls. People called it Fort Gunnybags.[111]

This was the second committee. The first, in 1851, had been merchants and vigilantes hanging men in a gold-rush city that did not yet have a working court. The 1856 committee was larger, more military, and more sure that it was the government. Its members turned out under arms: a fort on a commercial street. California had been a state for less than six years. When the official state was new, slow, or captured, the people with warehouses decided they would be the police and the court. Sometimes they were punishing a murderer. Sometimes they were deciding who counted as the public.[112]

The state already had a habit of building institutions beside the law and then asking the law to bless them. The vigilance committee was one. The railroad company, chartered by the state and subsidized by Congress, would be the lasting one. The committee hanged two men and disbanded. The company stayed, and it charged rent on the only modern road out.

Through the 1850s, that road could not be chosen, because choosing it chose slavery's direction.

In 1853 Congress called for a survey of possible routes. Jefferson Davis of Mississippi, as secretary of war, sent out four teams of surveyors to explore alternatives from the forty-ninth parallel near the Canadian border to the thirty-second near the Mexican—eleven large volumes of reports, drawings, and maps that never explored the forty-second parallel. Davis wanted the thirty-second-parallel line, from New Orleans through southern Texas and the New Mexico and Arizona Territories to San Diego, arguing that it crossed the fewest mountains and encountered the least snow. No free-state

politician was ready to charter or fund a road that would help slavery expand. Free-soil men wanted Chicago, or St. Louis, or Minneapolis as the eastern terminus. No slave-state politician was willing to give the road to them. In 1854 the United States paid $10 million in the Gadsden Purchase for the strip that made Davis's southern route possible on American soil, and the Army experimented with camels for the desert. The camels were a detail. The blockade was the fact.[113]

Theodore Judah, a young railroad engineer who had built the Niagara Gorge Railroad and designed California's first rail line, the Sacramento Valley Railroad, put the deadlock in a pamphlet in 1857, *A Practical Plan for Building the Pacific Railroad*, reaching for the Bible a year before Abraham Lincoln's "House Divided" speech made the verse famous. The Pacific railroad, Judah wrote, was a house divided against itself: it could not be done until the route was defined, and if the route was defined, the opposing interest was powerful enough to defeat it. In January 1859 he wrote that President James Buchanan favored the extreme southern route and would veto a bill for any other.[114]

Seven months later, on August 13, 1859, it was a hot day in Council Bluffs, Iowa, on the east bank of the Missouri River, looking across at the Nebraska village of Omaha. Abraham Lincoln was in town. He was not yet the nominee. He was a full-time politician, a year past the debates with Stephen Douglas, trying to become the Republican candidate for president. That evening he spoke at Concert Hall. The crowd was large; the local Republican paper, the *Nonpareil*, had welcomed the "distinguished 'Sucker'" and told the town: "Go and hear old Abe."[115]

He had not taken the easy way in. At St. Joseph, Missouri, he could have ridden the Hannibal and St. Joseph—the only railroad across the state—and gone home to Illinois. Instead he boarded a stern-wheel steamboat and came nearly two hundred miles up the Missouri to this bluff. The morning after the speech, he walked up a ravine to the top of the bluff with the Puseys, the family he was staying with, and other citizens of the town. Meriwether Lewis and William Clark had stood on that same bluff in 1804. Lincoln looked out over the vast floodplain stretching twenty miles north and south and four miles west to Omaha, and told his friend W. H. M. Pusey: "Not one, but many railroads will center here."[116]

That day, August 14, a group of citizens gathered on the big porch of the Pacific House hotel to hear him answer questions. When he finished, the crowd thinned. Pusey recognized a young man still standing there. Grenville Mellen Dodge was twenty-eight, a Massachusetts-born railroad engineer trained at Norwich University who had surveyed across Iowa for Peter A. Dey on the Mississippi and Missouri Railroad and up the Platte Valley for Henry Farnam and Dr. Thomas "Doc" Durant of the Rock Island. Pusey pointed him out and said Dodge knew more about railroads than any "two men in the country."[117]

Lincoln's head came around. He studied Dodge intently for a moment, and walked across the porch with Pusey to a bench where Dodge was sitting. He sat beside him, crossed his long legs, swung a foot, put his big hand on the younger man's forearm, and went straight at it.

"Dodge, what's the best route for a Pacific railroad to the West?"

Dodge answered at once: "From this town out the Platte Valley."

Lincoln thought that over for a moment, then asked, “Why do you think so?”

Dodge said the route of the forty-second parallel was the “most practical and economic” for building the railroad, which made Council Bluffs the “logical point of beginning.” Why? Lincoln wanted to know. “Because of the railroads building from Chicago to this point,” Dodge answered, and because of the uniform grade along the Platte Valley all the way to the Rocky Mountains. Lincoln kept asking until he had drawn out everything Dodge had learned in his private surveys for the Rock Island. Years later Dodge described the conversation in a soldier’s phrase: “He shelled my woods completely and got all the information I’d collected.”[118]

There was a money angle under the geography, and it was small, and it was real. Lincoln was about to accept seventeen lots in Council Bluffs as collateral on a loan of $3,500—roughly $130,000 today—to a fellow lawyer and political associate, Norman Judd, attorney for the Mississippi and Missouri Railroad. The lots were in the Riddle Tract, down on the Missouri River floodplain, the same tract where Dodge was buying land for the railroad’s shops and yards (provoking the *Council Bluffs Bugle* to rumor that Dodge was bound to run his surveys to insure his own investments). The future president was also looking at his collateral. He was, in that year, a politician and a railroad lawyer running for president.[119]

The railroad was already his subject. Born February 12, 1809, Lincoln had entered politics at twenty-three in his first campaign, for the Illinois state legislature, telling voters that “no other improvement … can equal in utility the rail road,” though he admitted its “heart-stopping cost.” He finished eighth of thirteen. He lost, and he kept the subject. In 1836, as a legislator, he cast the deciding vote to loan $500,000 to support the Illinois and Michigan Canal, and in 1847 he urged the federal government to grant 2,595,000 acres to Illinois for the Illinois Central. Starting in 1852, he defended the seven-hundred-mile Illinois Central in a landmark case over whether the state legislature could exempt the company from county taxes. When the Illinois Supreme Court accepted his argument in January 1856, saving the railroad millions, Lincoln handed the Illinois Central a bill for $2,000. The railroad rejected it, claiming, “This is as much as Daniel Webster himself would have charged.” Lincoln submitted a revised bill for $5,000, sued, and won. Ambrose calls him, on a scholar’s verdict, “one of the foremost railroad lawyers in the West.” In 1857 he had defended the Rock Island Bridge Company, builder of the first railroad bridge across the Mississippi, after the steamboat *Effie Afton* hit one of its piers and burned. Lincoln went to the river, examined the rebuilt bridge, and measured the currents himself. He argued for “travel from East to West, whose demands are not less important than that of the river.” The jury deadlocked, and the court dismissed the case. A hung jury was a win for the bridge. The boats did not get to own the crossing. Rayner quotes him in that case on the manifest destiny of the people to move westward. The destiny was a legal argument about a bridge.[120]

While Lincoln was questioning Dodge on the porch in Iowa, Theodore Judah was preparing to take California’s case back to the East. On October 20, 1859, Judah and his wife, Anna Ferona Pierce Judah, sailed from San Francisco on the *Sonora* for Panama and New York, bound for Washington with the proceedings of the Sacramento Pacific Railroad Convention. On the steamer Judah befriended California Representative John C. Burch, who

agreed to sponsor his bill, and Oregon Senator Joseph Lane. At dinner one night on the *Sonora*, Anna asked Burch and Lane, “Would there be any advantage in establishing a Pacific Railway Exhibit on Capitol Hill?” She had packed her husband’s convention charts, Sierra ore and fossil samples, and her own sketchbook and paintings of the mountains. On December 6, 1859, Judah, Burch, and California Senator William Gwin met at the White House with President Buchanan, who received them graciously. By January 14, 1860, with Burch’s backing, Judah had turned a room in the Capitol into the Pacific Railroad Museum. Yet when Representative Samuel R. Curtis of Iowa, chairman of the House Select Committee on the Pacific Railroad, introduced a bill for a line from Iowa to Sacramento backed by a $60 million loan in 5 percent thirty-year government bonds, Southerners tacked on a parallel southwestern route, Missouri and Iowa fought over the eastern terminus, and the bill was sent back to committee.[121]

The Iowa railroad men did not forget the porch. At the May 1860 Republican convention in Chicago, Dodge, Norman Judd, Rock Island president John A. Dix, Henry Farnam, Doc Durant, H. M. Hoxie, and J. B. Grinnell helped nominate Lincoln, and they got Des Moines attorney John Kasson to write the railroad plank of the Republican platform: “A Railroad to the Pacific Ocean is imperatively demanded by the interests of the whole country; the Federal Government ought to render immediate and efficient aid in its construction.” Still, when Judah sailed home to California in July 1860—charging the convention just $40 for printing out of $2,500 in expenses—he had to report that “the debate on the slavery question, and other matters of little moment, left us no time for the consideration of the Pacific Railroad Bill.” Visitor after visitor to his Capitol museum had asked the same question: how do you propose to get across the Sierra Nevada? Judah made up his mind never to return to Washington until he had an instrumental survey of the mountains in hand.[122]

In early October 1860, a druggist in Dutch Flat, California, named Daniel W. “Doc” Strong wrote to Judah. Strong knew the old emigrant road over Donner Pass, the one abandoned after the Donner party tragedy of 1846–47, and he thought a railroad could use it. Everywhere else the Sierra crested twice, parallel ridges with a deep valley between them; at Donner Pass, the east-west ridge reached the summit on a single crest, so a railroad climbing the divide between the North Fork of the American River and the Bear River could ascend on a grade no steeper than a hundred feet per mile.[123]

Judah rode up with him on horseback. Anna Judah later remembered that the two men nearly got snowed in on their last night in the mountains and had to break camp in the middle of the night to find the trail in the dark. Back in Strong’s store at Dutch Flat, Judah stretched his paper on the counter and would not sleep or rest until the figures were down. Turning to Strong, he said for the first time, “Dr. I shall make my survey over this, the Donner Pass or the Dutch Flat route, above every other.” The next morning Judah asked for writing materials, drew up articles of association for the Central Pacific Railroad, shoved them across the table, and told Strong, “Sign for what you want.”[124]

The ridge was not gentle. At Cape Horn, two miles above what was then called Illinoistown (soon renamed Colfax), the old unused wagon road hung over a drop of about fifteen hundred feet, roughly the height of the Empire State Building. People who had stood there said a railroad was impossible. Judah, who had built the gorge railroad at Niagara Falls, said it was the line.[125]

Under California law, incorporating a railroad required stock subscriptions of $1,000 per mile—$115,000 for the 115 miles Judah estimated from Sacramento to the Nevada state line—with 10 percent paid in cash. Strong canvassed the mining towns of Dutch Flat, Illinoistown, Grass Valley, and Nevada City and raised $46,500. On November 1, 1860, Judah published a pamphlet titled *Central Pacific Railroad Company of California*—the first public use of the name—and went to San Francisco for the remaining $70,000. On the evening of November 15, leaving Anna at the Russ House, he met with San Francisco capitalists in a leading law office. They scoffed at his $70,000-per-mile estimate, asked how he would clear the snow, and told him the road would take twelve to twenty years. Judah returned to the hotel and told Anna: "Pack your bag, for I am going up to Sacramento on the boat to-morrow. Remember what I say to you to-night so you can tell me sometime. Not two years will go over the head of these gentlemen I have left to-night, but they would give all they have to have what they put away to-night; I shall never talk or labor any more with them—I am going to Sacramento to see what I can do with the wealthy business men of that city."[126]

Starting the day after he arrived, Judah held meetings at the St. Charles Hotel on J Street in Sacramento. Among the merchants who came were Strong, Lucius Booth, jeweler James Bailey, future senator Cornelius Cole, surveyor B. F. Leete, Charles Crocker, Mark Hopkins, Leland Stanford, and Collis Huntington. In Anna Judah's later account, Judah told them: "You are tradesmen here in Sacramento city, your property and your business is here—help me make the survey, I'll make you the Central Pacific railroad company and with the [congressional] bill passed, you have the control of business which will make your fortune in trades, etc. etc. If nothing more, why, you can own a wagon road [from Dutch Flat over the mountains] if not a railroad." Several subscribed for fifty shares each; Huntington refused to subscribe at the meeting, then invited Judah to his office the next evening and agreed to help fund a $35,000 instrumental survey, promising nothing beyond that. Crocker said later: "We none of us knew anything about railroad building, but at the same time were enterprising men, and anxious to have a road built, and have it come to Sacramento."[127]

Meanwhile, on the train east from Springfield to his inauguration in early 1861, Lincoln took a turn at driving the locomotive. Dodge, who had come to Washington with Farnam and Durant to lobby the railroad committees and sat in the audience for the inaugural address, wrote to his wife Anne that Old Abe had delivered "the greatest speech of the age. It is backbone all over." Then Dodge added the practical line: "It looks as though we can get all our measures through and then I'll make tracks for home."[128]

In San Francisco, three street signs hold the memory of that moment. Union Square is a shopping square now, a column in the middle, Geary and Powell at the northeast corner. California Historical Landmark 623 records that the name originated in 1860, from public meetings held on that ground in support of the federal Union in the year before the Civil War broke—the Union that California's gold was helping to pay for, and that a Pacific railroad was supposed to pin to the coast. Sanchez Street runs south through the Mission and Noe Valley; the appendix to *The Beginnings of San Francisco* records that it was named for the family of José Antonio Sánchez, a soldier of Juan Bautista de Anza's expedition of 1776 (Rancho Buri Buri, down on the peninsula, carried a Sánchez name too, whether one soldier or a family of them). And Judah Street, in the same appendix, was named for Theodore D. Judah. The map looks neutral. It is a minute of the meeting.[129]

On April 12, 1861, the Confederates fired on Fort Sumter; the Deep South left the Union, and its members left Congress. While Judah spent the spring and summer of 1861 running his survey line over Donner Pass to the Big Bend of the Truckee River, the Sacramento merchants organized the company. On June 28, 1861, they formally incorporated the Central Pacific Railroad of California, with Stanford as president, Huntington as vice president, Hopkins as treasurer, James Bailey as secretary, Judah as chief engineer, and Crocker, Strong, and Charles Marsh joining them as directors.[130]

Their combined personal wealth, according to Huntington, was $159,000, roughly $6 million today. Neither Congress, the state of California, nor any syndicate of capitalists had put a penny into the corporation. The federal subsidy they were about to chase paid $16,000 a mile on flat land. Ten of those miles cost $160,000. Their whole fortune, pooled, was about one flat mile short of ten miles of easy track. And a simple shopping-basket translation of that fortune understates it: in 1861 those dollars were a larger share of a smaller country than groceries can show. As Hubert Howe Bancroft later wrote, for so “stupendous and hazardous an enterprise it appears an act of madness or of inspiration.” They were not rich enough to build a railroad. They were rich enough to hold the franchise while the United States built it with them.[131]

The Elephant (1861–1865)

On September 4, 1861, Leland Stanford was elected governor of California. The Democrats were split, and a Republican who had been a grocer a few years earlier won with a plurality.[132]

Three weeks later, Judah laid his finished engineering report before the partners: a line up the divide with a maximum grade of 105 feet to the mile, avoiding the Sierra’s second summit range entirely and requiring eighteen tunnels, the longest 1,370 feet. On October 9, 1861, the directors resolved that Judah proceed to Washington on the October 11 steamer “as the accredited agent of the Central Pacific Railroad Company of California, for the purpose of procuring appropriations of land and U.S. Bonds from Government.” On November 27, 1861, back in Sacramento—and apparently without Judah’s knowledge—Stanford, Crocker, Huntington, and Hopkins drew up articles of association for the Dutch Flat and Donner Lake Wagon Road Company, issuing four hundred shares at $1,000 each, with Crocker as president, Hopkins as secretary and treasurer, and three toll gates at Dutch Flat, Polley’s Station, and Donner Lake. Whatever Judah got out of Washington, the four partners were going to collect tolls from Nevada miners and California merchants on the path.[133]

Stanford took the gubernatorial oath on January 10, 1862, in the middle of the Great Flood. Sacramento was a lake. The new governor came to the oath by boat, through the streets, and the capitol waiting for him was only temporary. He served one term, to December 10, 1863. For that whole term he was also president of the Central Pacific, an office he held from 1861 to 1890. The house he worked from was the governor’s office and the railroad’s office. The man who signed state paper for the railroad was the railroad’s president. His partners would call him lazy and vain. He would insist, for the rest of his life, on being called Governor. He was good at being the face.[134]

In the assembly chamber he read his inaugural address from a prepared text, and he told the state what kind of settler he wanted. Gordon Chang records the words: “The settlement among us of an inferior race is to be discouraged.”

He said that Asia, with her numberless millions, sent to their shores “the dregs of her population.” He called the Chinese “a degraded and distinct people,” exercising a deleterious influence upon what he called the superior race. The chamber applauded. These remain, Chang notes, his best-known words about the Chinese. On the campaign stump, separately, his phrases had been shorter: “the dregs of Asia,” “that degraded race.” The inaugural was one occasion; the campaign was another. They agreed with each other.[135]

The same year, in the governor’s house, Jane Stanford fell seriously ill with a lung infection. Western doctors did not help her. The family’s cook, Moy Jin Kee, brought a doctor from Sacramento’s Chinese quarter, Yee Fung Cheung, who treated her with herbs. She recovered. Jane became so attached to Jin Kee’s teenage brother, Jin Mun, that she talked of adopting him. When the boy left their service she gave him a gold ring engraved with his name. He wore it for seventy years.[136]

Public speech, private kitchen. The governor who wanted the Chinese discouraged owed his wife’s recovery to a Chinese physician, and his company was about to decide it could not lay the track without Chinese crews.

In Washington, the votes that had blocked every northern route were no longer in the chamber.

Chang’s history states it in one turn: a divided Congress could not agree, and secession let Washington authorize a northern route the South had opposed. The historian Robert Russell adds the cold institutional point: because the Southerners were gone, “Congress was enabled to enact Pacific railway legislation several years earlier than it otherwise could have done.” California’s gold and Nevada’s silver were helping pay for the war just begun. There was loose talk that the state might follow the South out of the Union. The railroad directors, one of them the governor, would have none of secession. They wanted the coast held, and they wanted the subsidy that holding it would justify. Dodge’s account of Lincoln’s motive was military and political at once: Lincoln told him the road had to be built “not only as a military necessity, but as a means of holding the Pacific Coast to the Union.”[137]

On the steamer voyage east in October 1861, Judah had shared the ship with newly elected California Representative Aaron A. Sargent, spending the days showing Sargent his maps and the evenings explaining the project. In New York Judah printed a thousand copies of his Sierra survey report, sending them to President Lincoln, department heads, congressmen, and the *American Railroad Journal*. Back in Washington, he reopened his Pacific Railroad Museum in the Capitol, with his maps, surveys, and Anna’s sketches of Donner Pass on the walls, while California Senator James A. McDougall helped him draft a bill modeled on the Curtis Bill and Huntington arrived right after Christmas to lobby alongside him.[138]

On January 21, 1862, Sargent got the floor of the House in the midst of debate on another subject and spoke at length for the bill. He called the Pacific railroad a military necessity. He argued that it would save millions in transporting troops, munitions, and mail, quell Western Indians, speed emigration, develop the Great Plains, cause trade with Japan and China to jump, and assure California's loyalty so that no foreign army would dare attack the coast. Judah complained that Sargent spoke before an empty hall, but within a week the House appointed a special subcommittee to work on the bill. Judah himself was everywhere the bill was. He got himself appointed secretary of the Senate Pacific Railroad Committee (chaired by McDougall) and clerk of the House subcommittee (where Sargent sat), giving him charge of all committee documents and the privilege of the floor in both houses. The engineer was also the staff.[139]

Representative Thaddeus Stevens of Pennsylvania, who owned an iron foundry, insisted on an amendment requiring that all rails and ironwork be of American manufacture. Someday, Stevens declared, the Southerners would return to Congress "with the same arrogant, insolent dictation which we have cringed to for twenty years, forbidding the construction of any road that does not run along our southern border." Representative Justin Morrill of Vermont grumbled that the nation could hardly afford both guns and railroads, and charged that the capitalists would grab subsidies at either end without building beyond the Nevada silver mines.[140]

The House passed the bill on May 6, 1862, seventy-nine to forty-nine. On May 23, Judah wrote to the editors of the *Sacramento Union*: "The Pacific Railroad is a fixed fact and you can govern yourselves accordingly." The Senate passed it on June 20, thirty-five to five, and the House concurred in the Senate's amendments a few days later. Lincoln signed the Pacific Railroad Act on July 1, 1862, the same day as Malvern Hill, the last of the Seven Days' Battles on the Virginia Peninsula. Ambrose's account gives George B. McClellan's army a thousand dead and three thousand wounded in that fight, and a country greatly depressed by it. In Washington, a pen. In Virginia, a battle the Union won that did not feel like a victory.[141]

The law that resulted (12 Stat. 489) was, in Ambrose's phrase, complicated to an almost incomprehensible degree. The shape of it was simpler.[142]

Two corporations would build toward each other: a newly chartered corporation, the Union Pacific, capitalized at a hundred thousand shares of $1,000 each ($100 million), building west from the Missouri, and the Central Pacific building east from Sacramento. Each got a right of way two hundred feet on either side of the track over public lands, plus grades no steeper than 116 feet to the mile and curves no sharper than ten degrees. Each got land, in a checkerboard. A section is one square mile, 640 acres. The grant was five alternate sections per mile on each side of the line—every other square, ten sections in all, or 6,400 acres for each mile of track. The squares the government kept in between became valuable because the railroad was there.[143]

And the government loaned its credit, in bonds, released after each forty miles of track was approved by government commissioners: $16,000 a mile on flat ground, $32,000 in the foothills, $48,000 in the mountains. It was a thirty-year loan at 6 percent, constituting a first mortgage on the railroads—a loan, not a gift. The bonds were a debt the companies were supposed to repay in thirty years, principal and interest, minus the value of carrying government mail, troops, and stores. In practice the debt would outlive the

builders and land on their heirs, while the land grant was the part that behaved like a gift. A mountain mile paid triple a flat mile; in today's prices, that mountain mile was roughly $1.6 million. The Central Pacific had to complete fifty miles within two years and fifty miles each year thereafter, and finish its line by July 1, 1876, a few days before the nation's hundredth birthday, under pain of forfeiture. So it mattered, enormously, where someone said the mountains began.[144]

From 1861 to 1864, prices in the East jumped, because the Union was printing paper money, the greenbacks. California's economy still ran on gold coin. Sacramento's dollar and Washington's dollar were not the same dollar in those years.[145]

Before leaving Washington with a written testimonial of thanks signed by forty-four House members, seventeen senators, and the secretary of the Senate, Judah wired his partners in Sacramento over the newly completed transcontinental telegraph: "We have drawn the elephant. Now let us see if we can harness him."[146]

An elephant, in the American phrase of that century, was a prize you had won and might not be able to feed. They had the franchise. The Sierra was still there.

The next day, July 2, 1862, Lincoln signed a second land law: Justin Morrill's Land Grant Act, the one Morrill wanted even while he doubted the country could afford guns and railroads both. It granted each loyal state 30,000 acres of federal land for every senator and representative it had under the census of 1860, to endow at least one state college of agriculture and the mechanic arts. New York had thirty-one representatives and two senators, thirty-three seats: 990,000 acres, on the order of 120 times the size of Stanford University's 8,353-acre campus today. Cornell University, chartered in 1865, was the school New York built with that grant. Stanford is not a land-grant college. It would be founded later, privately, by a railroad man, on a fortune that week's other law helped create. Same week, same president: public land turned into a college by a statute, and public land turned into a railroad subsidy.[147]

The triple rate for mountain miles was an invitation, and Charles Crocker accepted it.

He drove the state geologist, Josiah D. Whitney, out to Arcade Creek, about seven miles east of Sacramento. This was not Asa Whitney the promoter, and not the car-wheel Whitney; Josiah Whitney mapped mountains for a living. The ground at the creek was flat. What Crocker showed him was a fan of reddish earth washed down from the hills. Whitney agreed that the Sierra Nevada, for the purposes of the subsidy, began there. On September 29, 1862, Governor Stanford telegraphed Washington about evidence to fix the "western base of the Sierra Nevada." Aaron Sargent, still in the capital, carried the geologist's report to Lincoln. Lincoln said it seemed about right.[148]

The mountains moved about fifteen miles west, on paper. Foothill miles paid $32,000. Mountain miles paid $48,000. The difference was $16,000 a mile. Across fifteen miles, that was $240,000 in extra bonds—in 1863 prices, roughly $6.5 million today. The United States had been talked into paying mountain rates for a fan of red dirt on a valley floor.[149]

Sargent told it as a joke at Lincoln's expense, and at his own: "Here you see, my pertinacity and Abraham's faith moved mountains." (Another report puts a variant of the joke—"Here is a case in which Abraham's faith has moved mountains"—in Lincoln's own mouth.) Faith, in the joke, meant a signature. The mountains did not move. The category did.[150]

The track still had to be laid, but the extra $240,000 was not payment for a steeper grade. It was payment for a definition. Judah refused to sign the affidavit. He told Strong he could not, because "the foothills do not begin here." He wrote to Anna that he had brought his partners a franchise and laid it at their door: "Rightly used it gives them unlimited credit throughout the world, and they would beggar it."[151]

He also told her he could not make these men appreciate the elephant on their shoulders. They would not do what he said they must; there would be trouble in Congress if they went on this way. He could not find the money to stop them. He was out. On November 2, 1863, Theodore Judah died. He was thirty-seven.[152]

In the spring of 1863, while Judah was still fighting his partners, the war and the railroad sat in the same room again. General Ulysses S. Grant ordered Grenville Dodge to Washington. Dodge was alarmed; he thought he was about to be called to account. Instead, Lincoln wanted to talk about the Missouri River, about towns fifty miles up and down the bank that all wanted to be the place the Union Pacific started, even as Robert E. Lee was preparing to march into Pennsylvania. Lincoln remembered the porch in 1859. Dodge's verdict was short: he found Mr. Lincoln "well posted."[153]

A year later the companies came back for more. Lincoln signed the amending Pacific Railroad Act on July 2, 1864. The land grant doubled, to ten alternate sections on each side, about 12,800 acres a mile—land equal to about one and a half Stanford campuses for every mile of finished track. The companies were permitted to issue their own first-mortgage bonds, equal to the government bonds, which pushed the United States into second place in line if the railroad failed. And the government would advance bonds after twenty miles of track instead of forty. Representative Elihu B. Washburne described the galleries, when it passed, as a tempest of the wildest disorder, lobbyists male and female. The Union Pacific's man, Thomas Durant, had a lobbyist, Joseph P. Stewart, spreading $250,000 in Union Pacific bonds, $20,000 of it to Charles T. Sherman, the general's brother. (Years later, in his epilogue, Ambrose would sum up the partners' Washington method even more bluntly: Huntington lied, handed out bribes, and got the president to accept a fictitious map.)[154]

The men who held the statute also decided they should hold the contract to build. The Sacramento partners awarded the construction to Charles Crocker and Company (and later the Contract and Finance Company), a firm they owned. Ambrose calls it an almost identical device to the Crédit Mobilier. In the East, Thomas Durant and a promoter named George Francis Train bought a sleepy Pennsylvania corporation, the Pennsylvania Fiscal Agency, and renamed it the Crédit Mobilier of America. The Union Pacific paid that company, which the Union Pacific's own directors owned, to build the Union Pacific. The profit was in the building, whether or not a ticket was ever sold. Among the insiders was Congressman Oakes Ames of Massachusetts, who with his brother Oliver Ames—later president of the Union Pacific—belonged to the family that made the Ames shovels that

Huntington, Hopkins, and Crocker had sold to miners and that Huntington had visited in 1862 for credit. The store and the subsidy had the same supplier.[155]

On March 18, 1865, while the war was still not over and Lee had not surrendered, Abraham Lincoln signed patents in Washington that conveyed Mission San Juan Capistrano and Mission San Luis Rey to Bishop Joseph Sadoc Alemany of California. The patents confirmed a decree the Land Commission had issued in 1855, on a claim Alemany had filed in 1853; a later Supreme Court case, *Foster v. Mora*, measured the San Juan Capistrano parcel—the buildings and the gardens that a Mexican governor had once sold for $710—at 46.74 acres (though a state-lands note records the patent acreage as 44.40). Lincoln did not return all the missions on that day (San Gabriel's patent was November 19, 1859; San Diego's was May 23, 1862; La Purísima waited until 1874). What he did was narrower. The American patent was how the church got its parcel back, through the same federal title office that was confirming Mexican rancho grants, and that, in the same decade, was giving railroads millions of acres. One office, three kinds of land: a mission church, a Mexican ranch, a checkerboard along a track. The railroad was simply the applicant that arrived with a statute written for it.[156]

The Summit (1864–1867)

James Harvey Strobridge was the Central Pacific's construction boss. Irish, over six feet tall, thirty-seven when he took the work, he was a man who said that laborers generally earned their money when they worked for him.[157] At Bloomer Cut, east of Sacramento, a black-powder blast cost him his right eye.[158] The Chinese workers who learned English called him the "one-eyed bossy man," and white foremen would clap a hand over one eye to make a crew think he was coming.[159]

Strobridge had not wanted Chinese workers hired. His line, as Stephen Ambrose records it, was plain: "I will not boss Chinese."[160] He insisted they could not do the work. Then he did boss them, because the white crews were walking off the grade to head for the Nevada silver mines, and the mountain was not going to wait.[161] The legend goes that Charles Crocker answered Strobridge's objection by pointing to the Great Wall of China, though the company payrolls tell a plainer, earlier, and DISPUTED story.[162] Traditional accounts, including Ambrose's, place the decision to hire Chinese labor in 1865, after white labor shortages stalled the climb; yet Central Pacific payroll records studied by later researchers show Chinese crews working under labor contractors including Hung Wah and Ah Toy as early as January and February of 1864.[163]

By Gordon H. Chang's account, the company soon needed a workforce on a scale it could barely imagine. It printed thousands of copies of a woodcut flyer to circulate in California and in China.[164] No copy survives. Crocker admitted to Collis Huntington that no one on the company side had ever translated the flyer back into English; the men who believed they were buying labor could not read their own advertisement.[165] At peak, somewhere between ten and twelve thousand Chinese men worked on the grade, and across the life of the build the total reached fifteen to twenty

thousand.[166] Chinese workers became roughly 90 percent of the Central Pacific's labor force—in Chang's judgment, by far the largest single workforce in American industry to that date.[167]

When there was trouble among the Chinese crews, Ambrose writes, Strobridge could pick out the leaders with a glance. He confronted them, usually with an ax handle, and they gave way.[168] The handle was Strobridge's; Crocker was the man who hired him. Crocker, for his part, later claimed he quarreled with Strobridge about the way he spoke to the men. "They are human creatures," Crocker said. Strobridge answered that "you have got to do it," and that they were "about as near brutes as they can get." Crocker said he found out that it was true—the same bold front he had shown with a pistol at Green River in 1850, now turned on his own workforce.[169] Meanwhile, Strobridge's wife, Hanna Maria, lived in a passenger car fitted out as a house at the end of track, with an awning and a caged canary by the door.[170] The men called her the "Heroine of the Central Pacific," and Ambrose notes she was the only white woman who saw the work through from beginning to end.[171] The canary was real. It did not soften the ax handle.

While the crews cut the grade toward the pass, the partners' Dutch Flat and Donner Lake Wagon Road—organized on November 27, 1861, with four hundred shares at $1,000 each, Crocker as president, Mark Hopkins as secretary and treasurer, and three toll gates at Dutch Flat, Polley's Station, and Donner Lake—was already charging tolls before the rails arrived.[172] In February 1866 Hopkins wrote Huntington [UNVERIFIED] that snow stopped the mountain work five or six months a year, and that getting freight over the pass meant paying teamsters a dollar and a quarter a ton.[173] Even then, Hopkins wrote, the teamsters would rather stay on the old Pacific Road—more taverns, more waiting girls and barmaids, a familiar road.[174] The partners already owned a toll, and they were learning that people pay for the path they know. Hopkins's same letter put the railroad construction in a builder's arithmetic: reach the desert this season and it was a six-month job; miss it, and it was eighteen months.[175] On the flat, beyond the snow, a day of track was a day of government bonds.

The winter of 1866 into 1867 brought snow so deep, and so bright, that men went snow-blind. E. B. Crocker, Charles's brother and the company's lawyer, wrote to Huntington that they had had to buy up all the cheap goggles in the market.[176] Think of that, he wrote: what an item in railroad building.[177]

Inside the 1,370-foot Summit Tunnel, engineered by Lewis M. Clement through the granite above Donner Lake, the work moved a few inches at a time.[178] Crews of three stood at the rock face, one man holding the drill, two swinging sledges of fourteen to eighteen pounds.[179] They drilled, they packed the hole with black powder, they ran, they waited for the blast, and they walked back into the smoke and did it again. Ambrose's account of that facing is six to twelve inches in a day, with shifts organized by chief engineer Samuel Skerry Montague going around the clock.[180] A month of that work moved the mountain less than the length of a room.

On Christmas Day, 1866, the *Dutch Flat Enquirer* reported that a gang of Chinese workers had been covered by a snow slide and that four or five died before they could be dug out.[181] Near the Summit Tunnel, an avalanche carried away some twenty Chinese men, whose bodies were found only after the spring thaw.[182] How many died on the line, the company did not publish. The historian Thomas W. Chinn's judgment is that the loss of life was heavy.[183] (In June 1870, a train would pass through Sacramento carrying, by a single newspaper report whose masthead remains [UNVERIFIED], the bones of perhaps 1,200 Chinese workers, boxed to be sent home to China.)[184]

On Monday, June 24, 1867, near the summer solstice, in the middle of the building season, three thousand Chinese workers put down their tools at once, from Cisco to Truckee, across nearly thirty miles of mountain.[185]

The demand, as the company understood it, was parity with white workers: $40 a month instead of $35 (in today's prices, $35 a month was roughly $840 a month, or about $10,000 a year), and a workday of ten hours instead of eleven.[186] Five dollars, and an hour. The strike was not a demand to get rich. It was a demand to be paid like the other men on the same rock. A contemporary newspaper [UNVERIFIED] called it the greatest strike the country had known.[187]

Mark Hopkins wrote to Huntington with the fear of a bookkeeper: if they succeeded in this demand, then they controlled.[188] E. B. Crocker wrote the most honest line: "The truth is they are getting smart."[189] His remedy was to inundate California and Nevada with laborers—freedmen, Chinese, Japanese, all kinds.[190]

Charles Crocker went up the mountain. What the strikers demanded and what the strike achieved remain DISPUTED between the company's public memory and the partners' private correspondence.[191] In Charles Crocker's later telling, the strike leaders told him, "Eight hours a day is good for a white man, all the same good for a Chinaman"—asking for eight hours rather than the ten the company wire had named.[192] The hours shifted with the teller; the pay did not. Crocker answered in a pidgin of his own making: "John Chinaman no make laws for me. I make laws for Chinaman. You sell for thirty-five dollars a month, me buy. You sell for forty dollars and eight hours a day, me no buy."[193]

Then the food stopped. On the mountain, the Central Pacific was a monopsony—the single big buyer of labor—and when the workers tried to act as a seller with a price, the buyer answered with a lockout. Charles Crocker claimed he cut off the provisions, and the labor agents stopped supplying the camps.[194] E. B. Crocker wrote that none of the partners went near the strikers for a week; they did not want to exhibit anxiety.[195] After eight days most of the men went back to work.[196]

Histories that follow Charles Crocker's version call the strike a flat defeat. Chang reads the partners' own letters and sees men who were badly shaken, noting that pay for skilled Chinese workers quietly rose over the following months.[197] What the principals wanted next was not a negotiation. It was a flood.

“We want a hundred thousand chinamen here,” E. B. Crocker wrote, “so as to bring the price of labor down.”[198]

Huntington went further: it would be all the better for them and the state, he wrote, if half a million came over in 1868.[199] The governor who had called the Chinese the “dregs” of Asia in his 1862 inaugural address now loved the supply.[200]

On August 3, 1867, Chinese crews holed through the Summit Tunnel and let the air in.[201] Beyond the Sierra lay the Truckee and Humboldt valleys of Nevada, the Great Desert, and the race across Utah against Grenville Dodge and the Casement brothers’ Union Pacific crews—a race in which both companies graded parallel roadbeds past each other to claim the federal bonds and the checkerboard land.[202]

Done (1869)

April 28, 1869. Seven in the morning, a whistle in the Utah desert.[203]

The Central Pacific’s crews set out to lay more track in a single day than anyone had ever laid. Charles Crocker had bet the Union Pacific’s vice president, Thomas “Doc” Durant, $10,000—roughly $250,000 today—that his men could lay ten miles in one day.[204] Whether Durant formally accepted the wager is DISPUTED—Ambrose writes that Durant “reportedly” was sure they could not do it and took the bet.[205]

By seven that evening the crews had laid ten miles and fifty-six feet of track up the slope toward Promontory, bending rails around curves by hand.[206] They moved at almost a mile an hour for twelve hours. The celebratory accounts named the eight Irish rail handlers—Michael Shay, Michael Kennedy, Michael Sullivan, Patrick Joyce, Thomas Daley, George Elliott, Edward Killeen, and Fred McNamara—under track boss George Coley.[207] They did not name a single Chinese worker. Several thousand men made the ten miles; the official story kept eight names.[208] (At the end of construction, Strobridge did seat his Chinese foreman, most likely the labor contractor Hung Wah, at the head of a dinner table, and a newspaper ran the scene under the heading “Chinese Laborers at Table.”[209] The honor was private. The pay packets were the public fact.)

A few days after the ten-mile day, some of those Chinese workers, perhaps including Hung Wah, climbed a bluff near Monument Rock (Monument Point) in Utah with a brown-glazed jug of rice wine that had come from the Pearl River delta.[210] From the rock they could see the newly laid line, and beyond it the Great Salt Lake. They drank, left the jug on the outcrop, and went back to camp. Archaeologists found it a century and a half later. Chang writes that the bottle, though mute, had something to say.[211] It was the celebration that was not on the program.

May 10, 1869. A dry flat in the Utah Territory, north of the Great Salt Lake, a place called Promontory Summit.[212] Water had to be hauled in. There was no band and no stage. Two locomotives faced each other across a gap of a

couple of rail lengths: the Central Pacific's *Jupiter* and the Union Pacific's *No. 119*.[213]

The ceremony was two days late, for a reason that was not smoothed over at the track: Union Pacific workers, owed back wages, had detached Thomas Durant's private car on his way out, set a guard on it, and held him hostage until he wired Oliver Ames for the money and paid them.[214] The builders, as ever, were paid last, when they were paid at all.

Leland Stanford, governor of California turned railroad president, was the only one of his four Sacramento partners who came; chief engineer Samuel Montague and construction boss James Strobridge stood with him, while Grenville Dodge, Jack and Dan Casement, Sidney Dillon, and Samuel B. Reed stood for the Union Pacific.[215] In Stanford's hand was a silver-headed maul. In the tie beneath him, polished California laurel, a golden spike sat in a hole already drilled so the precious metal would not be scarred.[216] Reporters had been telling the country that the blow would be heard farther than any sound a mortal man had ever made. A telegraph operator named W. N. Shilling beat his warning down the wire to every office that was asking when: "To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer."[217]

The locomotive engineers, George Booth and Sam Bradford, pulled the whistles.[218] A Civil War veteran in the crowd said later they all yelled like to bust.[219] Stanford's speech ran less than three minutes. No one at the center of the flat spoke the name of a worker. A squad of Chinese men in clean blue frocks had already carried one rail into the gap, set it, and stepped back. An Irish squad had carried the other.[220] The laurel tie was in place. The spike was seated in its pre-drilled hole, so the gold would not have to take a real blow.

Stanford swung. He missed, and the maul hit the rail.[221]

It did not matter. The operator closed the circuit anyway. The wire went out across the continent: *done*.[222]

The officials announced that the two railroads together had laid 1,776 miles of track, rhyming the iron road with the year of independence.[223] The ceremony was a machine for making a myth—that a golden spike finished a country—and when the dignitaries left, ordinary workers pulled the gold spike and the laurel tie out of the roadbed, put iron and common wood back in, and drove the spikes that actually held the rails.[224] (The laurel tie would burn in the San Francisco fire of 1906; the golden spike sits today in the Cantor Arts Center at Stanford.)[225]

Andrew J. Russell's photograph of the ceremony, *East and West Shaking Hands*, two engines nose to nose, became the enduring image of the railroad.[226] Chang is careful about what the glass plate shows: there is no evidence that Russell arranged the crowd to keep Chinese workers out of the frame.[227] Look closer, and a ghostly figure with his back to the camera, blurred because he was moving, stands in the middle of the picture. From his clothes, his hat, and his stature, he is Chinese.[228] He is in the most famous photograph of the road he helped build, and he is a smear. A century later, on May 10, 1969, at the Promontory centennial, United States

Secretary of Transportation John Volpe would stand on the same flat and ask the crowd, “Who else but Americans could drill tunnels in mountains thirty feet deep in snow?”[229]

Four years after Promontory, in February 1873, when Representative Jeremiah Wilson’s congressional committee investigating railroad construction finances put Collis Huntington on the stand and asked for the books of Charles Crocker and Company and the Contract and Finance Company, Mark Hopkins carried fifteen leather-bound volumes of accounts down to the basement furnace and burned them.[230] Huntington came back to the committee and told the congressmen the records were lost. “My partner Hopkins is a peculiar man,” Huntington said. “He considered the paper no longer worth saving.”[231] Hopkins died five years later, in March 1878, and the furnace kept its secret.[232]

Doctors had already been selling California’s climate before the rails met. In 1852 James Blake wrote that there was not a day in the year when the mind or the body was weakened by heat or numbed by cold, and J. Praslow, a physician practicing in San Francisco from 1849, wrote that the climate was so healthy that illness was rarely found.[233] Across the decades from 1840 to 1900, one historical estimate puts illness behind roughly a quarter of California’s new arrivals, and the great health rush would begin in earnest after 1870, once the trains were running and a state report called California the “sanatorium of the world.”[234] Soon enough the railroad would carry a different kind of passenger west: the coughing rich, a sick young man whose son would matter to a valley of companies, and the Stanfords’ own boy.[235]

In May 1869, the road was finished. The United States had loaned its credit and granted the land; the Chinese crews had drilled the granite and laid the iron; the furnace would eat the construction ledgers. Now the four Sacramento shopkeepers owned the chokepoint. Everyone who wanted to move a bushel of wheat had to pay.[236]

Part 3: The Octopus

1874–1886

Friends (1874–1876)

November 30, 1874. New York. Collis Huntington sat down to write a letter to San Francisco.[237]

The four railroad partners—Leland Stanford, Collis Huntington, Mark Hopkins, and Charles Crocker—did not particularly like each other, and they liked Leland Stanford least.[238] Friendship, in the Gilded Age, was not a feeling. In the history of the transcontinental railroads, it was an arrangement.[239] Measures in California lived or died on who would stand with whom when the vote was called.[240]

Some of the arrangements were with Congress, and they survived on paper because a man named David Colton saved his mail.[241] Colton was the partners' man in San Francisco, a junior associate who wrote constantly to Huntington in New York.[242] Huntington wrote back in his own voice: dry, busy, sure of the price of things. Between the middle of the 1870s and Colton's death in 1878, the letters piled up.[243] Five years after Colton died, during an 1883 lawsuit brought by his widow, Ellen, the letters would be read into the public record and reprinted in a pamphlet whose title did not bother to whisper: *How Members of Congress Are Bribed*.[244] On November 30, 1874, in Letter No. 389 of that collection, looking ahead to the Congress about to sit, Huntington wrote to Colton: "I think this coming session of Congress will be composed of the hungriest set of men that ever got together."[245]

Friendship with a newspaper worked the same way as friendship with a senator. You did not always buy the printing press. You bought the pipe the news came through.[246]

In the 1870s the Associated Press was that pipe.[247] A paper that was not in the association did not get the national wire, which meant it did not get the country, not in time to print it. The wire was a chokepoint, the same shape as a port or a mountain pass. Whoever stood at the gate decided which stories moved.

Huntington understood the gate. He wrote to Mark Hopkins that if the Associated Press agent in Washington were seen to, a great deal of hostile copy could be kept back.[248] His lobbyist, Richard Franchot, once got that Washington reporter onto the Central Pacific payroll.[249] It worked until the Associated Press editor, James Simonton, noticed that the tone of the stories had changed, and suspected why.[250]

In 1875 the partners tried something larger.[251] The *San Francisco Chronicle* had begun ten years earlier, in January of 1865, as a theatrical handbill called the *Daily Dramatic Chronicle*, started by two brothers, Charles and Michael Henry de Young.[252] Charles was the editor who treated scandal as a product. By the 1870s the handbill had become a daily, and for a time the railroad had a hand on it.[253] In 1875 the partners wanted an exclusive Associated Press membership for the *Chronicle*, and they offered double the going rate.[254] Double. The point of paying double was not generosity. It was to cut two other papers, the *Bulletin* and the *Call*, off the wire—papers the railroad hated.[255] The attempt failed.[256] The instinct is the lesson. A news monopoly was worth more than a fair price, because a news monopoly decided what a fair price sounded like in print.

On January 17, 1876, Huntington was back on the rails.[257] A rival, Tom Scott of the Pennsylvania Railroad, was pushing a bill in Congress to help his own Pacific line, the Texas and Pacific, a southern route to the coast.[258] Huntington took the night train to Washington.[259]

"It cost money to fix things so that I would know his bill would not pass," Huntington wrote to Colton in Letter No. 107. "I believe that with $200,000 I can pass our bill."[260]

Two hundred thousand dollars in 1876 is roughly $6.3 million now, as the cost of one bill.[261]

Twelve days later, on January 29, 1876, in Letter No. 110, and again on March 23 in Letter No. 138, Huntington used the sentence he reached for whenever he wanted to sound clean: “I keep on high ground.”[262] The second time he finished the thought. He kept on high ground so that no investigation could hurt them.[263]

Lobbying is arguing for a bill in public, with your name on the argument. Bribery is paying an official to use the office as a lever. Huntington’s letters live in the gap. He did not write that he handed a senator an envelope. He wrote that two senators were switched this week and could be switched back with the proper arrangements.[264] The high ground was not innocence. It was a method: pay in ways that look like friendship, so that if a committee asks, the paper smells like business.

Nineteenth-century Americans were not naive about this.[265] As Richard White observes, they were not shocked by the corruption of the press or of individual lawmakers; they were frightened by what it added up to: a republic whose decisions were for sale.[266] A railroad that could raise a rate and then rent the description of the rate was not only a carrier. It was an editor. The word nineteenth-century citizens used was *corruption*, and they meant it the old way, as rot in the structure, not as one man’s bad manners.[267]

And it was not a California invention. While Huntington priced a bill to fight Scott, farther north Jay Gould was gathering lines.[268] Charles Perkins of the Chicago, Burlington and Quincy said the Union Pacific’s life depended on high rates more than any other company’s.[269] John Murray Forbes, whose Boston family had made its first fortune in the China trade, said Gould had as many aliases as a London professional thief.[270] Stanford’s face was Western. The technique was national.

Nob Hill (1876–1878)

In 1876, up a steep street in San Francisco, the hill did not yet have its fame. It had a view, wind, and sand. The men who had just finished a railroad decided the view was the point.

Leland Stanford built first, at California and Powell.[271] The house was finished in 1876, with interiors commissioned from the New York firm Pottier & Stymus.[272] In the European fashion, the mansion rose through a ground floor, a first floor on the second story, a second floor, and a third floor—more than fifty rooms on about two acres, inside a wall thirty feet high.[273] The front doors were mahogany and rosewood, seven inches thick.[274] A glass dome over the hall sat seventy feet up.[275] The carved framework of the conservatory cost $30,000.[276] A custom library ladder, three rungs, cost $150.[277] The whole production came to $2 million in the money of the 1870s—roughly $63 million now.[278]

In 1899 an economist named Thorstein Veblen would give this a name in *The Theory of the Leisure Class*: conspicuous consumption, spending whose job is to be seen.[279] The mansions came first. The theory arrived to describe them. Stanford's house was not a machine for living. It was a machine for being looked at, from the bay and from the offices of men who paid the railroad's rates. That same year, 1876, the Stanford family also bought a farm south of the city in Santa Clara County, about 650 acres, a place called Mayfield Grange.[280]

California wrote its history on signposts and then forgot the ink. Nob Hill is one of those names. *Nob* is not a saint. The Spanish named towns for Francis, Clare, Joseph, and the rest, and those names are still the mission chain. *Nob* is short for *nabob*.[281] And *nabob* is an English mangling of a word from India, *nawab*, a deputy of the Mughal emperor, a man who governed in someone else's name and got rich doing it.[282] The British brought the insult home for merchants who made a fortune in the East and spent it loudly in London. San Francisco picked it up for a different east: Sacramento grocers and Nevada silver men who hauled their money up one steep ridge and built houses the rest of the city had to crane its neck to see.

So the hill is a joke that froze. Catholic California named places for heaven. Railroad California named a hill for showing off.

Even before the hill was finished, the crowd found the address. In July 1877, the country was four years into the long depression that had started in 1873. In San Francisco, unemployed men gathered on a vacant lot near the city hall, a sandlot, and the meeting that was called against the railroads turned, fast, against the Chinese.[283]

They marched up the hill on Leland Stanford's mansion.[284] The man at the front of the Workingmen's Party of California was Denis Kearney, an Irish-born drayman who hauled freight for a living, and a speaker who had found the sentence that would move a crowd.[285] (In the following year, 1878, when Kearney accused Senator William Sharon of trying to sell his "rotten water works"—the Spring Valley Water Company—to the city for $15 million, Kearney's sandlot audience would chant "Hemp, hemp, hemp!" to wish Sharon hanged.[286])

"I will give the Central Pacific just three months to discharge their Chinamen," Kearney told the sandlot in July 1877, "and if that is not done, Stanford and his crowd will have to take the consequences."[287]

Three months. That was the number in the threat. Kearney's slogan, the one painted on banners, was four words: "The Chinese Must Go."[288] As the historian Malcolm Harris judges him, Kearney was better at being a racist than at being a labor leader.[289] It was easier to make Chinese workers pay for the depression than to make Stanford pay. Stanford had Kearney arrested.[290] Then Stanford did what the very rich have often done when a crowd learns the address. He left town for Mayfield Grange, the 650-acre farm south of the city.[291] The crowd had marched on a mansion. The owner was becoming a country gentleman.

Scapegoating is not just meanness. It is a political technology. The anger in the sandlot was not imaginary. Wages were down. The railroad really was a monopoly. A man who hauled goods for a living had reason to hate a

company that set the price of moving goods. Kearney took that anger and nailed it to a people who could not vote, who had built the railroad, and who were still in the company's workforce because the company wanted them there. In his 1862 gubernatorial inaugural address, read from a text, Stanford had said Asia sent "the dregs of her population," and he had called the Chinese "a degraded and distinct people" [UNVERIFIED].[292] On the stump the line had been shorter: "the dregs of Asia" [UNVERIFIED].[293] By the time the tracks were finished he wanted more of them, because they kept the wage down. In 1877 his neighbors wanted them gone, and he was happy to let the neighbors do the shouting.

The workers and the owner were not on the same side of the door. The crowd was outside the mansion. The men the crowd wanted fired were the railroad's labor, paid by the month, living in camps the mansion never had to see. A monopoly can survive a riot if the riot aims at its workforce instead of its rates. That is what the sandlot gave the Big Four: a villain who was not them.

Up on the hill, the building went on. Mark Hopkins let his wife talk him into a house a block above Stanford's, at California and Mason.[294] He was the quiet partner, the ledger. Even after he was a multimillionaire he kept a small cottage, kept no carriage, and took no vacations.[295] The palace went up anyway.

He never lived in it. On the night of March 28, 1878—some notices date the death the twenty-ninth—a private railroad car stood on a siding at Yuma, on the Colorado River, where California meets Arizona.[296] Hopkins was sick. He had not wanted the trip. Huntington wanted eyes on the ground the Southern Pacific was reaching for, out past the river, and Hopkins knew that Huntington's schemes usually had something behind them. He got off his sickbed, into his own company's car, and went to look. He went to sleep in the car. He did not wake up.[297]

He was the partner the others could stand: the bookkeeper, the man who had answered panic, in the building years, with as few words as possible. He died in the one house he actually used, a car on a company track, at the edge of the map.

"My old friend Mr. Hopkins has gone over the river," Huntington wrote after the death. "His death has made me very sad. I was very fond of Mr. Hopkins. We had been friends from the day of our first meeting, and partners in business, as you know, for many years."[298]

Huntington did not come west for the funeral. He was too busy.[299] He added that there might be other men as good as Hopkins, but that he would not live long enough to form such a friendship again.[300] The affection was real. So was the arithmetic. A partnership is a machine that expects a man to be replaced, and the man who ran the machine from New York was not going to miss a vote in order to stand by a grave.

Two weeks after Hopkins died on the siding, on April 10, 1878, a new cable car line opened up California Street.[301] Getting up the hill was its own advertisement. Cable cars already climbed Clay Street; Andrew Hallidie's line had started in 1873, and it stopped two blocks short of Stanford's door.[302] So Stanford and his railroad friends built another one, up California

Street itself, past the new palaces.[303] A railroad president built a vehicle so the public could reach his front steps, and so he would not have to walk the last two blocks.

A block away, Charles Crocker wanted a whole square: California, Sacramento, Taylor, and Jones.[304] One lot on it, at 1100 California, belonged to a man named Nicholas Yung, an undertaker, comfortable but not rich, who liked the view and would not sell at Crocker's price.[305] The offers climbed—$6,000, then more, and at some point Yung is said to have named $12,000.[306] Crocker stopped bargaining and built a fence.

The height and the ornaments of that fence are disputed in the city's memory. Popular retellings shrink the fence to thirty feet; contemporary press accounts cited by Richard Rayner and preserved in San Francisco archives describe a wooden fence forty feet high around three sides of Yung's house at 1100 California, so the sun barely got in.[307] It was four stories of boards, on three sides of a family's windows. The *San Francisco Examiner* was writing about it by August 17, 1878, under the name the city gave it: a spite fence.[308]

The cost, in the newspaper accounts that survive in later write-ups, was about $3,000—roughly $100,000 now.[309] Crocker could lose that sum the way another man loses a cufflink. What he bought was a lesson. The price of refusing the railroad was darkness at noon.

The story goes that Yung answered by putting a coffin on his roof, skull and crossbones turned toward Crocker's windows; what a *San Francisco Chronicle* report actually preserves is Yung's threat to mount a coffin above the fence, as an advertisement for his undertaking business and a reminder of mortality.[310] Threat or object, the picture stuck. The man who buried the dead, boxed in by the railroad. Yung eventually moved the house to another lot he owned, on Broderick Street, and refused to sell the empty ground.[311]

Two months after the *Examiner* named the spite fence, on the evening of October 8, 1878, a carriage stopped outside David Colton's house on Nob Hill.[312] He was unconscious inside it. What killed him is disputed: the company said a horse had thrown him, while the rumor in the street was a stabbing.[313] He died two days later, on October 10, 1878, and the truth of that night stayed dark.[314]

This was succession, the unromantic kind. Four men had owned the Central Pacific in equal shares and let a construction company they also owned pay them for building it. By the fall of 1878 one of the four had died in a car on a siding, and Colton, the fifth man they had let partway in, was dead on Nob Hill in a story nobody could finish. The survivors inherited the dead men's leverage, and they used it on the widows. Colton's widow, Ellen, was walked through a story in which her husband had been an embezzler, and she was pressed to give the shares back—the pressure that would drive her to court in 1883 and spill Huntington's letters into print.[315] Grief was not a pause in the business. It was an opening in the cap table.

(The palaces on the hill would last thirty years. The fire after the earthquake of April 18, 1906, took the hill in one night, and what replaced the houses is the city's memory pretending to be an address: the Stanford Court Hotel at 905 California on the footprint of Stanford's mansion; the Mark Hopkins

Hotel on Hopkins's corner, after his house first outlived him as an art school; Huntington Park on the lot of the house that had been David Colton's and later Collis Huntington's, given to the city by Huntington's widow, Arabella, in 1915; and Grace Cathedral rising on ground the Crocker family, through William H. Crocker, turned over to the Episcopal church after the fire.[316])

The Wheat Field (1879–1880)

On March 1, 1879, in Washington, President Rutherford B. Hayes vetoed a bill passed by Congress to cut Chinese immigration hard, down toward fifteen passengers on a ship.[317] His reason was a treaty. The United States had promised China, in the Burlingame Treaty, a freer movement of people.[318] Hayes did not veto because California's workers had changed his heart. He vetoed because he did not want to tear up a diplomatic bargain in public. Then he sent a minister to renegotiate the bargain.[319]

In California that same year, the party Denis Kearney fronted did not stay in the street. Voters adopted a fresh state constitution in 1879, over the railroad's opposition.[320] Kearney's Workingmen helped write a streak of anti-Chinese clauses into it, and the same constitution created a state railroad commission—three elected men whose job was to watch the rates.[321]

Also in 1879, a San Francisco newspaperman published the most famous economics book of the century to explain the rise in the land. Henry George's *Progress and Poverty* argued that a railroad, a ditch, a town—all of that improvement—raises the value of the land around it, and the rise flows to whoever holds the title, not to the people who did the digging.[322] He did not need a wheat field to draw the picture. He already had the railroad.

"A railroad company approaches a small town as a highwayman approaches his victim," George wrote. "The threat, 'If you do not accede to our terms we will leave your town two or three miles to one side!' is as efficacious as the 'Stand and deliver,' when backed by a cocked pistol."[323]

Build through the town, or build past it and let it die. After the road is built, charge what the town cannot refuse.

In San Francisco, the Workingmen's Party ran a Baptist minister named Isaac Kalloch for mayor in 1879.[324] The *San Francisco Chronicle*—where the railroad had held influence—went after him personally. That grip had not made the paper gentle; it had made the paper a faction. Kalloch answered from the pulpit and, by the paper's own later account, said things about Charles de Young's mother that a man of that temper would not let stand.[325] In August of 1879, Charles de Young shot Kalloch.[326] The minister lived. The sympathy of the shooting helped elect him.[327]

On the evening of April 23, 1880, at about half past seven, Mayor Kalloch's son, Milton Kalloch, walked into the *Chronicle* office and shot Charles de Young dead at his desk.[328] The *New York Times* ran the wire the next day,

April 24, 1880, plain as a bill of lading, printing the son's initials as J. M. Kalloch: Charles de Young killed; Mayor Kalloch's son; the *Chronicle* office; half past seven.[329]

The son was tried and acquitted.[330] One juror's reported view was that he would have done the same [UNVERIFIED].[331] Michael Henry de Young—M. H. de Young, the surviving brother—took the paper.[332] (He would run the *Chronicle* for decades, serve for twenty-three years as a director of the Associated Press, and in 1894 organize the California Midwinter International Exposition in Golden Gate Park—a world's fair staged in a California winter whose art building grew into the museum that officially took the name M. H. de Young Memorial Museum in 1924.[333])

A newspaper can be a public record. It can also be a political weapon owned by a side. De Young had not shot Kalloch because of a disagreement about grain rates. He shot him because the paper and the candidate had become personal enemies, and in that San Francisco the personal was how politics was conducted. The railroad did not order the shot. It did not have to. It had spent years teaching the city that a paper was a piece of property, to be captured, starved of wire service, or aimed. Once that is normal, editors start carrying the same tools as the men they cover.

Eighteen days after Charles de Young died at his desk, the fight reached the valley.

May 11, 1880. A Tuesday morning in a wheat field, two and a half miles north of a new town called Hanford, in the San Joaquin Valley of California.[334] The valley is the long flat trough that runs down the middle of the state, between the coast mountains and the Sierra Nevada. In spring the wheat stands high enough to hide a man on a horse.

In Hanford that morning, families had come in for a picnic.[335] The children, at least, expected to have fun. The adults had heard who got off the train a few hours after midnight. His name was Alonzo W. Poole. He was a United States marshal, and he was carrying eviction notices.[336]

This was not empty land, and it was not a picnic that went wrong for no reason. In 1877 the Southern Pacific had finally received its federal patents to land in this part of the valley, along a branch line toward the Mussel Slough of the Kings River.[337] Settlers were already on it. Many of them had brought water to ground the county histories call barren, and had made a crop. Railroad agents then announced they would sell at prices from $17 an acre to $40.[338]

Hold the price up to the company's own advertisement. The company itself had advertised the land at "$2.50 an acre upwards."[339] Seventeen dollars is almost seven times that advertised floor. Forty dollars is sixteen times it. The difference was not the dirt. The difference was who got paid for the work. And the Pacific Railway Act of July 1, 1862, the law that gave the railroad its federal land grant in the first place, had named a different ceiling in Section 3: grant land still unsold three years after the entire road was finished was to be open to settlement and preemption at a price not exceeding $1.25 an acre, paid to the company.[340]

Poole rode out of Hanford in a wagon with a Southern Pacific land man beside him.[341] Two neighbors who were buying the disputed ground from the railroad met them in a buggy. Their names were Mills Hartt and Walter Crow. In the buggy they had two shotguns, a Spencer rifle, and three pistols.[342] Hartt had been bragging in town about what he would do to settlers who would not move. Crow was known as one of the best shots in California. He had knocked wild geese out of the air with that rifle.[343]

They stopped at a homestead where nobody was home, put the family's belongings in the road, and lined shotgun shells up on the porch.[344] Then they went on to a place held by Henry Brewer and John Storer, part of which the railroad had sold to Walter Crow. It was about half past ten.[345]

At the picnic, word arrived. Riders came out through the wheat and the alfalfa. How many rode out is disputed: Poole later testified he saw forty or fifty men on horseback, while other witnesses said about fifteen.[346] Testimony cannot settle the number of men on horseback. A legend will.

Poole knew the neighborhood, and the neighborhood liked him. A settler named James Patterson asked him to wait until the court appeals were heard. Poole said he was doing his job. Someone told him to give up his pistol. He would not. Then a nervous horse struck him and knocked him down.[347] He was the most reliable witness, and he was in the dust when the shooting started.

Nobody agrees who fired first. In less than a minute, twenty or thirty shots were fired.[348] Walter Crow did most of the killing. James Harris. Iver Knutson, forty-nine, a father of nine, a man Crow knew. Daniel Kelly. Archibald McGregor, armed with a penknife. John Henderson. Hartt was hit too, and took more than a day to die.[349]

There was a pause. Smoke hung over the wheat. Crow backed the others off with a pistol and said, "One at a time."[350] The marshal told him to take a horse and go to the timber. Crow would not. He walked into the wheat he had bought from the railroad, the field he had told his friends he would harvest that year. A mile and a half on, armed settlers met him. He raised his pistol. He was shot in the back—with his own rifle, according to one report—and he fell dead in a ditch.[351]

"As has long been anticipated, the conflict."

That was how the *Visalia Times* opened its account four days later, on May 15, 1880, as preserved in the *History of Tulare County*.[352] The death toll itself is counted two ways in the historical record: the *Visalia Times* named seven dead—five settlers, and two men who had come to take the land—and listed an eighth man, Edwin Haymaker, among the wounded.[353] Haymaker had only a scalp wound that morning. He cleaned it at the well. He developed pneumonia, and he was dead within the month; Richard Rayner counts eight dead if you count Haymaker.[354] The next legend of the American West, the gunfight at the O.K. Corral in Tombstone on October 26, 1881, a year and a half after this, killed three.[355] The movies kept Tombstone. The wheat field is the one that tells you how California actually worked.

The story goes that this was honest farmers against a monster. The record is meaner. Some of the settlers had fought for the Confederacy.[356] Their league drilled like a militia. Masked men had burned a homestead, and each side blamed the other.[357] Crow knew the men he killed. Rayner concludes that nobody rode out with a settled plan to start a slaughter, and that drawn guns made one likely.[358] Both things are true. The railroad wrote the price. The men in the field pulled the triggers.

The settlers at Mussel Slough had improved the ground. The railroad held the patent. The patent won. In the end the courts sided with the company. Some families left. Some paid the higher price. The Southern Pacific kept the land.[359]

Put the spring of 1880 on one calendar. April 23, a son kills an editor in a newspaper office on the bay. May 11, eighteen days later, seven men die in a wheat field, and an eighth is dying of a scalp wound he does not yet know will kill him. Same state. Same railroad sitting under both stories, not as the finger on every trigger, but as the condition both fights were fought inside. A city and a valley that had come to expect the conflict.

That same year, 1880, Nicholas Yung died, while Charles Crocker's forty-foot fence still stood on Nob Hill, where Yung's heirs kept the feud for years before the lot was finally sold and the boards came down.[360] A miner named George Hearst—who had bought the coast ranch at San Simeon in 1865 and made his fortune across the Comstock, the Homestake, and the Anaconda—picked up the *San Francisco Examiner*, a weak Democratic daily, third in the city.[361] And the federal census takers counted the people Denis Kearney had pointed at from the sandlot: 75,132 Chinese people in California, in a state of 864,894—8.7 percent, or nearly one Californian in twelve.[362] Of all the Chinese people in the United States—105,465 in the 1880 census—seven in ten lived in that one state.[363]

Exclusion and the Octopus (1881–1884)

On February 19, 1881, the *San Francisco Wasp* printed a cartoon titled "In the Hollow of His Hand."[364] In it, Charles Crocker and Collis Huntington stand on the railroads and hold Leland Stanford up, and Stanford holds the state railroad commission in his hand, a commissioner small enough to close his fingers on.[365] Voters had adopted a fresh state constitution in 1879, over the railroad's opposition, and that constitution had created a railroad commission: three elected watchdogs whose job was to watch the rates.[366] Two years later, a humor magazine drew the watchdogs sitting in the palm of the man they were supposed to watch.[367]

Lawyers now call this regulatory capture, though the motion is simpler than the jargon. A public creates a guard to stand between a monopoly and its customers. The monopoly then hires the guard, or elects him, or tires him out, until the guard's reports describe the monopoly's wishes as the public's safety. You do not have to repeal the law. You inhabit it.

Six weeks later, on April 2, 1881, the *Wasp* printed a poem by its editor, Ambrose Bierce.[368] He had the Mussel Slough dead in mind, and he aimed the last stanza at a man who had not been in the field:

> These mounds are green where lie the brave who fought and fell at Hanford. O, point me out, I pray, the grave of Leland Stanford.[369]

Bierce had a nickname ready: “Stealand Landford.”[370]

In Washington, the diplomatic bargain with China had already been pried open. In 1879, Congress had passed a bill to cut Chinese immigration hard, down toward fifteen passengers on a ship, and President Rutherford B. Hayes had vetoed it on March 1, 1879, because the Burlingame Treaty promised a freer movement of people.[371] Hayes did not veto the bill because California’s workers had changed his heart; he vetoed it because he did not want to tear up a diplomatic bargain in public, and he sent a minister to renegotiate the treaty.[372] By 1881, the renegotiated Angell Treaty had come back.[373] It let the United States restrict Chinese immigration, so long as the restriction was “reasonable.”[374] *Reasonable* is a lawyer’s word. It means whatever the next Congress can pass and the next president will sign.

In 1882 Congress tried for a twenty-year ban on Chinese laborers.[375] President Chester A. Arthur vetoed that bill too, as more than the treaty allowed.[376] Congress came back with ten years. On May 6, 1882, Arthur signed the Chinese Exclusion Act.[377]

It took five years from Denis Kearney’s three-month threat on the San Francisco sandlot, and it arrived wearing the language of a treaty.[378] National Archives historians call it the first significant federal law restricting immigration, and the first time federal law barred an ethnic working group because Congress claimed that group endangered the good order of certain places.[379] Precision matters here: it was not the first unfriendly statute. The Page Act, in 1875, had already made it much harder for Chinese women to enter the country, under the stated target of prostitution and forced labor.[380] The 1882 law was wider. It shut out Chinese laborers, skilled and unskilled, for ten years.[381] It made Chinese people ineligible for citizenship.[382] It required papers of the Chinese who were allowed to come and go: merchants, teachers, students, travelers.[383] A workforce that had been recruited, by this same railroad, with handbills the railroad’s own officers could not read, was now a people the federal government would not admit and would not naturalize.[384]

The sequel shows how a panic becomes a policy. In 1892 the Geary Act would extend the ban and order Chinese residents to carry a certificate of residence or face deportation; in 1902 Congress would stretch it again, including to Hawaii and the Philippines; and the exclusion laws would stay on the books until 1943, when China was a wartime ally and Congress repealed them, leaving a quota of 105 people a year.[385] Sixty-one years.[386] A man who had laid rail in the Sierra in 1867 was old, or buried, before the statute that answered him came off the books.

Put the 1880 census back beside the statute: 75,132 Chinese people in California, in a state of 864,894—nearly one Californian in twelve—and about seven in ten of the 105,465 Chinese people in the United States.[387] The men who spoke for the state described an invasion. The count describes a minority, concentrated because the jobs were concentrated: on the railroad,

in the fields, in the laundries and the factories of one coast.[388] Exclusion did not break the railroad monopoly. It fed it a story. The monopoly remained. The people who had built it were barred at the door.

Stanford's own record is the cleanest exhibit. In his 1862 inaugural address as governor, he had said Asia sent "the dregs of her population" and called the Chinese "a degraded and distinct people"; on the campaign stump, the line had been shorter: "the dregs of Asia" [UNVERIFIED].[389] As railroad president, by the time the tracks were finished, he had wanted more of them, because they kept the wage down.[390] Later, as a senator, he would sit in a body that had made the ban normal.[391] The inconsistency is not a mystery and it is not a private paradox. It is a sequence. Use a people when you need their hands. Describe them as a danger when their hands make your neighbors angry. Sign nothing yourself if other men will sign it for you.

That August of 1882—dated August 19, 1882, in standard accounts—a drawing in the *Wasp* called *The Curse of California* put the railroad on the page as an octopus, tentacles wrapped around the farmers, the miners, the whole state.[392] And that fall, California voters tried to break it. They elected an antimonopoly governor, George Stoneman, a Democrat, and a legislature that looked, on paper, like the railroad's worst day.[393] The 1879 constitution already promised a commission and a harder hand on rates.[394] Stoneman wanted taxes, including back taxes, and a bill to outlaw the special contracts by which the railroad charged one shipper a quiet price and another shipper a punishment.[395]

Then, in 1883, Ellen Colton's lawsuit against the railroad partners—filed after the associates had walked David Colton's widow through a story that her late husband was an embezzler and pressed her to give his shares back—brought Colton's saved mail into open court.[396] Huntington's private letters from the 1870s—calling the coming session of Congress "the hungriest set of men that ever got together" (November 30, 1874), pricing a bill against rival Tom Scott at "$200,000" while noting that "[i]t cost money to fix things so that I would know his bill would not pass" (January 17, 1876), and boasting "I keep on high ground" so that no investigation could hurt them (January 29 and March 23, 1876)—were read aloud and reprinted in a pamphlet titled *How Members of Congress Are Bribed*.[397] What the letters did, five years after Colton died on Nob Hill, was turn the private voice of the railroad into a public document.[398] Huntington's defense, for the rest of his life, was sarcasm and a bad memory.[399]

In 1884 Governor Stoneman called a special session of the California legislature to do railroad business and nothing else.[400]

Count the votes he thought he had: a Democratic majority of sixty-one to nineteen in the assembly, and thirty-two to eight in the senate.[401] The bills sailed through the assembly.[402] In the senate, bill after bill ended in a tie, and the lieutenant governor, a Democrat, broke the tie against them.[403] Nine of ten Democratic senators from San Francisco voted the railroad's way.[404] They were the lambs of a man the city called the Blind Boss.[405]

His name was Christopher Buckley.[406] Buckley had publicly supported the antimonopoly bills.[407] Then his senators killed them.[408] The reason was not that Buckley loved Stanford; it was that a fight between a railroad and a

reform wave was an opportunity only a fool would waste.[409] The railroad could taste defeat, or it could sweeten Buckley's pot.[410] Nineteenth-century slang for the dollar was *simoleons*. Buckley liked them.[411]

The Pacific Railway Commission, looking back, estimated that the Southern Pacific spent about $700,000 on politics that year—roughly $26 million now.[412] It did not all go to one envelope. It went into a stew: railroad money, kickbacks, payments from anyone who needed a franchise or a contract, protection, the ordinary boodle of a city boss, mixed until a senator could not have told you which dollar had started in a freight tariff.[413] Once the money is in the stew, every dollar looks like every other dollar.

"The political marriage of Leland Stanford and Christopher Buckley gave birth to the Octopus," Richard White writes.[414]

The public had the image in the *Wasp* before White had the marriage certificate. What the 1884 session added was a date and a method: not a metaphor floating free, but a boss, a senate, a special session, and a pile of money large enough to turn a sure majority into a tie.[415] The voters had built a commission and elected a hostile governor. The railroad did not repeal the commission. It bought the senate votes that kept the commission's teeth in a drawer. The law stayed on the books, where it could be pointed at. The practice stayed in Buckley's pocket, where it could be used. A government that looks like it is governing the railroad, and a railroad that is governing the government: that is the octopus. Not the animal. The arrangement.[416]

It would not last in every market—within a year, rival transcontinental lines were reaching the coast, and a second track is the one thing an octopus in a rate table cannot survive—but between the middle 1880s and the early 1890s, as White notes, the political animal was quite real.[417] When Frank Norris published his novel *The Octopus* in 1901, watching "tentacles of steel" clutching the soil while the railroad raised its tariff to "all the traffic would bear" on the excuse that the road "could not be operated at a legitimate profit," he was giving fiction's name to an arrangement that had been sealed in Sacramento in 1884.[418]

While the legislature sat in California, the Stanfords were in Europe.

March, 1884. Florence, Italy. Leland Stanford was a railroad president and a former governor; the Senate seat was still a year away.[419] The house on Nob Hill had a dome seventy feet high.[420] He and his wife Jane had one child, Leland Stanford Jr., born on May 14, 1868.[421] He was fifteen. In a hotel in Florence, the boy had a fever. What to call the illness is DISPUTED in the Stanford record: David Starr Jordan would later call it "Roman fever," while university histories call it typhoid [UNVERIFIED].[422]

On March 13, 1884, the fever killed him, two months short of his sixteenth birthday.[423] His mother would keep his room as he left it.[424] His father had spent twenty years turning a franchise into a government.

The Caption (1885–1886)

In 1885, a year after his son died in Florence, Leland Stanford—with Christopher Buckley's help—bought his way into the United States Senate, taking his seat on March 4, 1885.[425] Senators were still chosen by state legislatures, not by the voters.[426] A boss who owned a city's delegation owned a piece of the Senate.

A year later, Stanford helped put a second man there. George Hearst had been lucky in mining on a scale that looks like fiction: the Comstock, the Homestake, the Anaconda, claims that became the backbone of American metal.[427] He wanted a Senate seat, and in those years a rich man campaigned by feeding legislators.[428] Hearst got there in 1886 with the support of Leland Stanford, who chipped in because Hearst's opponent was Aaron Sargent, a man in Huntington's orbit.[429] Hearst was appointed to a short vacancy on March 23, 1886 (serving through August 4, 1886), and then chosen by the legislature—under a January 19, 1887 *New York Times* headline that read "Hearst's Boodle Wins"—for a full term that began on March 4, 1887 and lasted until his death in office on February 28, 1891.[430]

Hearst owned a newspaper, the *San Francisco Examiner*, a weak Democratic daily, third in the city, that he had picked up in 1880.[431] On the day his full Senate term began, March 4, 1887, the masthead changed: "W. R. Hearst, proprietor."[432] William Randolph Hearst was in his mid-twenties; he had looked at Joseph Pulitzer's *New York World* and decided the *Examiner* should shout—bigger headlines, fires, adultery, violent crime, and preferably all three in one story.[433] He hired, at high salaries, a staff of famous writers, and Ambrose Bierce was the first of them.[434] In the *Examiner* in 1896, Bierce would call Huntington's head a dromedary (February 11, 1896), give him one leg in the grave, one arm in the treasury, and one eye on the police (April 10, 1896), and write, when Huntington told a committee he could not recall, that he took his hands out of his pockets long enough to be sworn (April 12, 1896).[435] Stanford's feud with his partner financed, at one remove, the most widely read attack on Huntington: a weapon you help load does not ask which enemy you had in mind.

That same year, 1886, a memo prepared for the president of the Union Pacific ran down the New York papers one at a time, the way a quartermaster runs down boots.[436] One sheet existed to blackmail. One would print what it was handed. And the *Star*, the memo said, did "as Huntington directs."[437] As White wrote: "Nineteenth-century Americans were not shocked by the corruption of the press."[438] It was how the trade worked. What frightened people was the use. A railroad that can raise a rate and then rent the description of the rate is not only a carrier. It is an editor.

And in January of 1886, the Southern Pacific's long tax fight reached the Supreme Court of the United States.[439]

California had tried to tax the railroad's property without letting the company deduct its mortgages the way an ordinary landowner could.[440] The Southern Pacific had refused to pay, turning the refusal into some sixty-three railroad tax test cases—for the collection of taxes of 1880, 1881, and 1882, beginning with *San Mateo County v. Southern Pacific Railroad*—that ran

for years while counties spent time and money they did not have.[441] In the *San Mateo* argument before the Supreme Court in 1882, former New York senator Roscoe Conkling, appearing for the railroad, had produced the manuscript journal of the Joint Committee on Reconstruction to suggest that the drafters of the Fourteenth Amendment had intended the word *person* to shield corporations too—what historian Howard Jay Graham later exposed as a deceptive account.[442]

The judge who had heard the California tax cases on the federal circuit was Stephen J. Field, a justice of the United States Supreme Court riding circuit, which justices still did.[443] Leland Stanford had a close relationship with Justice Stephen Field.[444] When the railroad decided to bring the test cases, Field recommended a lawyer, a law professor named John Norton Pomeroy, who admired the way Field wrote opinions and would later publish a celebratory account of Field's work.[445] Recommending a lawyer to a friend is not, by itself, a crime. Field then did the thing modern judicial ethics would forbid: he presided.[446] He did not step aside. And during the litigation, Adam Winkler writes, he shared with the railroad's lawyers confidential memoranda about the case written by other justices.[447]

In his circuit opinions—including the 1883 circuit ruling in *Santa Clara County v. Southern Pacific Railroad*—Field had said corporations were persons protected by the Fourteenth Amendment.[448] The amendment had been ratified in 1868 for a different person: the formerly enslaved, and anyone a state might try to crush.[449] Its promise was that no state shall "deny to any person within its jurisdiction the equal protection of the laws."[450] Field's move was to slide the railroad into the word *person*. Treating the company's property differently from a widow's property was, he wrote, "the very essence of tyranny."[451] He offered a comparison meant to shock: a tax law that changed its rules according to whether the owner was a white man or a Black man would be an outrage; so, he said, was a tax law that changed its rules because the owner was a corporation.[452]

The amendment had been written because states were denying rights to Black Americans. Field used that sentence to protect a railroad's balance sheet.

Now, in January 1886, *Santa Clara County v. Southern Pacific Railroad* was before the full Supreme Court.[453] The question on the table was Field's question: does the Fourteenth Amendment cover corporations? Before the argument, Chief Justice Morrison Waite told the lawyers the Court did not want to hear about it.[454]

When the decision came down on May 10, 1886, the Court ruled for the railroad on the narrowest ground available.[455] The county had included the value of the fences along the tracks in the tax bill.[456] California law did not allow that. Because the case could be decided on that ground, the opinion said it was "not necessary to consider any other questions," including the grave constitutional questions decided below.[457]

The Supreme Court did not hold, in 1886, that corporations are persons under the Fourteenth Amendment. It said it did not need to reach the question. It decided a fence.[458]

Then the court reporter went to work. His name was J. C. Bancroft Davis.[459] His job was to write the syllabus and the headnotes, the summary at the front of a published opinion in the *United States Reports*, so a busy lawyer could see what was decided without reading the decision.[460] Headnotes are not the law. They are a caption. Davis's caption said this:

> The defendant Corporations are persons within the intent of the clause in section one of the Fourteenth Amendment to the Constitution of the United States, which forbids a state to deny to any person within its jurisdiction the equal protection of the laws.[461]

Above the opinion, Davis printed a remark he attributed to the Chief Justice, from before the argument:

> The Court does not wish to hear argument on the question whether the provision in the Fourteenth Amendment which forbids a state to deny to any person within its jurisdiction the equal protection of the laws applies to these corporations. We are all of the opinion that it does.[462]

Davis had checked with Waite before he published. Waite's terse private reply survived:

> I leave it with you to determine whether anything need be said about it in the report inasmuch as we avoided the constitutional question in the decision.[463]

Davis printed it anyway. He had once been president of a railroad himself, the Newburgh and New York Railway.[464] A reader who stopped at the headnote would walk away believing the Court had held that corporations were people. The Court had not.[465]

Three words govern what happened next. A *holding* is the rule a court must use to reach its result. *Dicta* are remarks a judge makes on the way by, which do not control the case. A *headnote* is a reporter's summary, not a vote.[466] *Santa Clara*'s holding was a fence. Three years later, the caption became a citation. In 1889, in *Minneapolis & St. Louis Railway Co. v. Beckwith*, a railroad that had killed three hogs in Iowa and been ordered to pay $24 in damages—roughly $900 today—spent tens of thousands of dollars taking the case to the Supreme Court.[467] The money was cheap if the prize was a stronger shield against every future fence law, livestock law, and tax. Stephen Field wrote the opinion. He upheld the Iowa statute, wrote along the way that corporations are persons under the Fourteenth Amendment, and then added one sentence: "It was so held in Santa Clara County v. Southern Pacific Railroad."[468] It had not been so held. Field had written, three years earlier in a concurring opinion in *San Bernardino County v. Southern Pacific Railroad*, an angry complaint that the Court had dodged the constitutional question.[469] He had not forgotten. He cited the dodge as if it were a decision.[470]

A strange idea becomes a normal one by repetition, in the book every lawyer keeps on the desk. By 1912, when a former congressional and Supreme Court librarian named Charles Wallace Collins counted every Fourteenth Amendment case since 1868 in *The Fourteenth Amendment and the States*—604 in all—only 28 involved African Americans, and those

plaintiffs usually lost.[471] Three hundred and twelve, more than half, involved corporations, which used the same sentences against laws on wages, zoning, and child labor.[472] If those cases were a calendar year, the cases about Black Americans would run out before the end of January; the corporate cases would last until November.[473] Collins called the rebuilt amendment the “Magna Charta of accumulated and organized capital.”[474]

Mussel Slough made the papers for a week. The headnote stayed in the bound volume. The railroad did not need to win every case; it needed the word *person* to be available the next time a state tried to tax a fence. And back in California, grief had already given the fortune a new shape. Grief does not launder a fortune, but it can change what the fortune is for. The octopus had an heir, and the heir would be made of stone, and scholarships, and a clause that said forever.

Part 4: The Boy and the Farm

1884–1905

The Room Across the Hall (1884–1889)

March 13, 1884. Florence. A boy of fifteen died two months short of his sixteenth birthday, and his parents decided the fortune would outlive them.[475]

Leland Stanford Jr. had been born on May 14, 1868 (David Starr Jordan’s memoir calls the illness that killed him “Roman fever”; university histories call it typhoid, and the diagnosis remains disputed).[476] Back in the house on Nob Hill in San Francisco, his bedroom sat across the hall from his mother’s. Jane Lathrop Stanford turned the tutor’s room next door into a study where she worked with her secretary, Bertha Berner. She kept the boy’s bedroom as it was when he died. She later described what she did there: going “into dear Leland’s room and kneel down and tell all my sorrows to the blessed Jesus. I feel my darling boy and my dear husband are pleading with me.”[477]

The sentence that came out of the night in Florence has two wordings, and they are not the same sentence. David Starr Jordan wrote in his memoir, *The Days of a Man*, that after a long and dreary night the father woke with these words on his lips: “The children of California shall be my children.” Jordan said the father told him the story in 1891. Malcolm Harris, in *Palo Alto*, prints a different pronoun, having Leland tell Jane, “The children of California shall be our children,” when they decided to build on the farm.[478] It is a remembered sentence, not a diary line. *My children* makes the university a father’s replacement son. *Our children* makes it a couple’s.

Herbert C. Nash, the boy’s tutor, wrote a biographical sketch for the memorial volume *In Memoriam: Leland Stanford, Jr.* that put Jane at the center: young ladies and the conventions of society had no attractions for the boy, but toward his mother he would display the most attentive gallantry. Jane made Nash the university librarian. The sketch became the life that was not to be questioned.[479]

Six weeks after the boy died, on April 24, 1884, in Paris, Leland Stanford made his will, looking to the endowment of the university. Jane made her will the same way, and copies were sent to America.[480]

In the fall of 1891, after the university opened, a story went around that the school had been founded under spiritualist influence, and that a medium named Maud Lord Drake had been the guiding hand. In 1892 Leland and Jane sat down with Jordan and dictated a statement they wanted kept. Maud Lord Drake, the statement said, had been unknown to them until they met her at a séance with Ulysses S. Grant and his wife, Julia Grant, in October 1884. About that time, the statement added, Mrs. Drake was detected in fraud.[481]

Jane added a sentence of her own: “No spiritualistic influence affected the decision. Mrs. Drake had no more to do with it than a babe unborn.”[482]

No record survives of what was said in that parlor. Jordan added in his printed statement that both Stanfords were, for a time, deeply interested in a spiritualism that seemed to offer a demonstrable life after death. With the Grants they attended several séances in Washington. Jordan wrote that they never got, through mediums, any evidence they found convincing. The wills came first, in Paris. The Drake claim came after, and they spent a page denying it.[483]

The séances the documents describe at the Nob Hill mansion were conducted with the Reverend John Philip Newman and his wife, Julia Newman, friends of the Grants, trying to reach the boy.[484] Mrs. Newman was the chief medium. Amid luminous clouds, the boy did not appear. A different child did, a spirit the Newmans called Ensign, and the medium reported that heaven rejoiced in the decision to dedicate the fortune to God and the church. That was a séance claim, not a deed. Bertha Berner later wrote in her 1935 memoir, *Mrs. Leland Stanford: An Intimate Account*, that after a trip to Australia, where the dead did not answer, Jane never again attended a séance.[485]

The Grants were in the story for a more ordinary reason: patronage, the trade in favors inside a circle that does not have to call them favors. A railroad pass. A pulpit. Later, an embassy.

On June 2, 1884, from New York, Ulysses S. Grant wrote to Leland Stanford. Before the failure of the Wall Street firm Grant and Ward, Grant’s youngest son, Jesse Root Grant (J. R. Grant), had sent his wife, child, and nurse to California and had expected to follow. His ready money had been in the firm.

“He deposited with the firm on the 2nd of May $86,000,” Grant wrote.[486]

Grant asked whether it was too much, under those circumstances, to request a pass from New York out and back for the son and the family. He told Stanford to refuse if it was inconvenient. Eighty-six thousand dollars in 1884—roughly $3.2 million today—had been wiped out in a Wall Street partnership that carried a former president’s name. On July 2, an invitation to Long Branch followed. The circle held. Five years later, in 1889, Frederick Grant, the general’s older son, wrote to Stanford about his appointment as American minister to Austria.[487] A pass, then a post. The office was public. The ask was private.

The pulpit was louder. The Reverend John Philip Newman had been the Grants' pastor at the Metropolitan Memorial Methodist Episcopal Church in Washington and then at the Madison Avenue Congregational Church in New York. In 1884 he was between jobs. The Stanfords brought him west to preach the boy's memorial. Richard White writes that they reportedly promised him $10,000, equal to his yearly salary (roughly $372,000 today), though the payment is not in an account anyone has produced.[488]

The boy traveled west in a funeral car draped in black. While they were bringing the body home, the Stanfords looked at Cornell, Harvard, the Massachusetts Institute of Technology, and Yale. What Jane said about Yale was an insult, and it is on the page. She said graduates of Harvard and Yale were of benefit to no one, including themselves. Her husband, she said, employed "six of them as car-conductors on the Market Street Line now."[489]

Back in California, at the Palo Alto estate, publicity claimed the memorial featured the greatest floral displays on the West Coast, supposedly costing $20,000 (roughly $744,000 today), a figure White marks as a publicity claim rather than a bill. White writes that if it were not for the parents' grief, the praise of a rich dead boy the preacher had never known would have been risible. Newman's sermon drew charges of blasphemy for comparing the boy to Christ, and set off a church fight over whether a minister who was not an Episcopalian should have been in an Episcopal service.[490]

A dead child, a borrowed Christ, a fee the size of a preacher's year. The university had been decided in April, in Paris. The performance was in the winter. Grief was real. So was the stage.

The Dead Hand (1885–1891)

On March 9, 1885, the California legislature passed an enabling act so a private deed of trust could found a university. The title began: "An act to advance learning, the arts and sciences, and to promote the public welfare." It was written as a general law. It had one institution in mind.[491]

On November 11, 1885, Leland Stanford and Jane Lathrop Stanford signed the Founding Grant. By then Leland was a United States senator; senators were still chosen by state legislatures, and he had taken the seat that March. The deed stated their desire to found, on their estate known as the Palo Alto Farm, in San Mateo and Santa Clara counties, "a University for both sexes," with colleges, schools, museums, galleries, and everything necessary to a university of high degree. They conveyed the land to a board of trustees and to their successors forever.[492]

Two clauses in the instrument formed the lock:

"the principal thereof shall forever remain intact, and that the rents, issues, and profits thereof shall be devoted to the foundation and maintenance of the University hereby founded and endowed."[493]

And, inside the rights the founders reserved for themselves while they lived:

"but this reservation does not include the right or power to sell or encumber any of the real property granted."[494]

A trust is property held by people who do not own it for themselves. The trustees spend the income on the purpose named in the deed. The principal stays. Lawyers call a clause that keeps ruling after the giver is dead a dead hand. The same lock can freeze a mistake. The clause that stopped a sale of the land was the clause that would later stop Jane from turning the school into a boys' college.

One name on the trustee list carried over from the railroad's legal wars: Stephen J. Field, a justice of the Supreme Court of the United States. The same man in whose Ninth Circuit courtroom a railroad tax case (*Santa Clara County v. Southern Pacific Railroad*, 18 F. 385, 1883; 118 U.S. 394, 1886) had grown a Supreme Court reporter's caption treating corporations as persons under the Fourteenth Amendment now sat on the board of the railroad president's university.[495] The deed did not blush.

The 1885 grant did not forbid tuition; it left the question to the trustees. What the founders told David Starr Jordan—after Leland had first offered the presidency to Columbia's Nicholas Murray Butler—was a promise of a fortune. The ultimate endowment, they said, would be about $30 million.[496] Jordan had fought to get Indiana University's budget to $50,000 [UNVERIFIED]. He was offered thirty times that, someday.

Someday was not a deposit. Before Leland died, he gave the university three ranches totaling 85,000 acres. The Vina ranch never turned a profit, and its vines died of phylloxera. Gridley's grain was in decline. The Palo Alto stables usually lost $7,000 a year, most of Jordan's $10,000 salary. Trustee George Crothers said the ranches brought "nothing but taxes." White notes that taxes ($6,983 in one estate accounting) did exceed the revenue ($6,606), and the founding papers prohibited a sale.[497]

Frederick Law Olmsted, the landscape architect who had made Central Park, was hired to site the campus. He wanted the buildings in the foothills. The Stanfords wanted flat land. They were building a monument you could march through. Olmsted suggested, and Charles A. Coolidge of Shepley, Rutan and Coolidge—the successor firm of the architect H. H. Richardson—designed, the low buildings of the inner quadrangle, Encina Hall for the men, and a long drive from the railroad station planted in Mexican fan palms and Canary date palms: Palm Drive. Flat land. A quad. An arch. A church at the center.[498]

In *Imperial San Francisco*, Gray Brechin describes the result: a symmetrical composition of sandstone, continuous arcades, and a Romanesque chapel loosely modeled on Trinity Church in Boston, the whole roofed in red tile because the Stanfords wanted the buildings to resemble California's missions. Brechin's word for the place is a "supermission."[499]

At the entrance stood a triumphal arch of San Jose sandstone, more than a hundred feet tall, second in size only to the Arc de Triomphe—a "Babylonian eruption," White writes, in the middle of the mission-style quiet.[500] The frieze on the arch was called *The Progress of Civilization*. The donors put themselves in it, on horseback, ahead of a locomotive entering the Sierra. Jane suggested the outline of that story and gave herself and her husband featured roles. Augustus Saint-Gaudens, the most famous sculptor in the country, turned her suggestions into the program.[501]

Leland wrote to Jordan a sentence Brechin quotes: our graduates are to be "[missionaries] to spread correct ideas of civilization."[502]

The old Franciscan missions had sat on crown land the friars held, in the official story, for the Native people they baptized, before secularization in the 1830s scattered that claim. The families who had passed through the baptismal books at Mission Dolores and Mission Santa Clara were the Muwekma Ohlone; the campus sat in their watershed. Under the missions lay Native labor. Under the quad lay a railroad fortune graded in large part by Chinese workers paid less than white crews, as the Central Pacific's own managers had admitted under oath.[503] Adobe and red tile had become sandstone and red tile.

Jordan explained in his memoir the names chosen for the roads in front of the quad. The road the quadrangle faced they called Serra Avenue, for Father Junípero Serra, who had founded the first of the California missions. Lasuen Street was named for Fermín Lasuén, Serra's successor, the contained man after the impulsive one, who had carried the mission chain farther, including La Purísima. Other campus roads took the names Cabrillo, Portolá, and Crespí. The founders wanted the mission era in the directory, and they got it.[504]

A few miles from the farm, in the same Santa Clara Valley, another widow was building.

Sarah Lockwood Pardee Winchester signed her name Sallie. She was the heir to a repeating rifle. Her husband, William Wirt Winchester—the son of Oliver Winchester, the New Haven manufacturer who had turned a shirt business into the Winchester Repeating Arms Company and had given Yale the land for an observatory in 1871 (built in 1882)—had died of tuberculosis in New Haven on March 7, 1881. In 1886, as historian Mary Jo Ignoffo established from the house's papers, Sarah bought an eight-room farmhouse near San Jose and named it Llanada Villa (later 525 South Winchester Boulevard).[505]

The Stanfords laid their cornerstone on May 14, 1887, the day their son would have turned nineteen. Two grieving households, both rich, started monuments about fifteen miles apart, within roughly a year of each other.[506]

The tourist story has the two widows backward. The story goes that a Boston medium told Sarah Winchester that the rifles had stacked up so many dead that the spirits would haunt her unless the hammers never stopped—producing stairs to nowhere and a window in a floor. The record does not show that medium. The name later books give him, "Adam Coons," has no trace in her lifetime; it first appeared in 1967, in Susy Smith's *Prominent American Ghosts*. Sarah's companion, Henrietta Servers, said she was not superstitious. She had been raised Baptist, and after 1908 she attended St. Paul's Episcopal Church in Burlingame. Damage from the 1906 earthquake explains the broken stairs, after which she largely stopped building; a carpenter added the number thirteens to the house after she was dead, and the first printed mention of them was in 1929. The documented spiritualist in the valley was the railroad widow on the farm. And there is no evidence Jane Stanford and Sarah Winchester ever met or wrote to each other. The person who connected them was a lawyer in San Jose named Samuel Franklin Leib, who was a Stanford trustee and also Sarah

Winchester's legal and financial adviser—handling her property, investments, taxes, will, and estate, as preserved in the Samuel Leib Papers (1865–1922) at History San Jose.[507]

In 1893, nearly a decade after Leland Jr. died, curators from the university museum measured every piece of furniture in the boy's Nob Hill bedroom, noted where it stood, and moved his things into the campus museum. They told Jane they had arranged the room "just as it was in your city home."[508]

Two Suits and a Bicycle (1891–1896)

October 1, 1891. The inner quadrangle was still being finished on the morning of the opening ceremony.

The university's public history counts 555 students in that first year, about 130 of them women, roughly one in four. Richard White's history counts 559, mainly from California, which made it the largest college in the far West (the opening enrollment figure is disputed between the two counts).[509] Four students separate the two counts. David Starr Jordan stood before the pioneer class.

"It is hallowed by no traditions; it is hampered by none," Jordan said. "Its finger posts all point forward."[510]

A school with no traditions had not yet had time to lie about itself.

In the crowd was Herbert Hoover. He had been orphaned at nine in Iowa—his father, a Quaker blacksmith, died first, and then his mother—and relatives had passed him along until he ended up in Oregon with an uncle who did his duty without much tenderness.[511] A visiting engineer had told the boy that mining engineers were scarce and well paid. The East Coast schools were beyond his means, and the new college in California was not going to charge tuition.

In the summer of 1891 Hoover went to Portland and sat the entrance exams. He came up short. Joseph Swain, a Quaker who would teach mathematics at Stanford, gave the exams and watched him attack plane geometry.

"I observed that he put his teeth together with great decision and his whole face and posture showed his determination to pass the examination at any cost," Swain recalled.[512]

Swain admitted him on the condition that he bone up. On August 29, 1891, a few weeks after he turned seventeen, Hoover stood at the Salem station with a ticket for San Francisco, two suits in a satchel, and a bicycle. He boarded the train. At the edge of the campus he studied, failed English, and was waved through again on the promise of effort. He was given the first room assignment under the red tile of Encina Hall, and for the rest of his life he liked to say he had been Stanford's first student. There was no electricity and no hot water for weeks after classes began.[513]

Free, in the grant, meant the college would not send a bill for instruction. It did not pay for the train, the room, or the year of geometry a boy had missed.

Less than two years later, on June 21, 1893, Leland Stanford died.

The Panic of 1893 was already on, the start of the deepest depression the country had been through. The Southern Pacific, the basis of the fortune, was missing dividends and could not pay full interest on its bonds. Within two years of opening, White writes, the new college was in danger of collapsing.[514] Leland's will left the university $2.5 million in cash (roughly $93 million today). It left about $7 million (roughly $260 million today) largely to his relatives, not to Jane's. The bulk of the estate, whatever it was actually worth, went to Jane, and the university lived on her largesse. The three ranches still could not be sold.[515]

Then the federal government sued. In 1894 the Justice Department went after the Stanford estate for $15 million—roughly $580 million today—and froze the assets, including the bequest to the university.[516]

The money was the old government construction loans to the Central Pacific, come due. The Central Pacific was a California corporation, and California law made stockholders personally liable for debts the company did not pay. Stanford and his partners had drained money out of the Central Pacific into their own pockets, sold the hollowed-out railroad to British investors, and had the Southern Pacific, which they still controlled, lease it back. Because the British investors were afraid of California's stockholder-liability law, the Associates had kept the Central Pacific stock in their own names. In 1894 the government came to the names.[517]

"People think that Governor Stanford left me a very rich woman," Jane told a visitor. "I thought so myself, but it now seems that I was left a legacy of debt, trouble and worry."[518]

She kept the university's doors open on the household allowance the probate court allowed her, on an insurance policy, and on the sale of personal belongings. She cut faculty salaries ten percent. The faculty, White writes, would have left if there had been anywhere to go in a depression. Jordan stayed and defended the Stanfords, because the alternative was a dead college. He called it the cause of higher education in America. That year, the cause was a widow selling furniture.[519]

Jane's lawyer, Russell J. Wilson, hired Joseph Choate to argue the case at the Supreme Court. In 1896 the Court ruled that the stockholders were not personally liable, even if California law had said so when the railroad was incorporated; the government would have to collect from the Central Pacific itself. Official university histories treat that ruling as the end of the money trouble. White writes that it solved nothing. Just before the decision, on February 9, 1896, Choate wrote to Jane that a victory might put "perhaps $40 million" in her hands—a guess White calls a "considerable exaggeration." Choate's advice was to rewrite her will and fix a maximum amount the university should be allowed to hold. Her public promise had been to pour the fortune in. Her lawyer was telling her to pour less.[520]

The boy was twelve years dead. The university was five years old. The fortune was a lawsuit with a palm drive.

The Expanding Peoples (1898–1900)

February 25, 1898. Washington. The Secretary of the Navy, John D. Long, had gone home for the afternoon. Theodore Roosevelt, the assistant secretary, was acting secretary for those hours. He cabled Commodore George Dewey in Hong Kong, marking the message secret and confidential: “Keep full of coal. In the event of declaration of war Spain, your duty will be to see that the Spanish squadron does not leave the Asiatic coast, and then offensive operations in Philippine Islands.”[521]

The next day Long wrote in his diary that “the very devil seemed to possess him yesterday afternoon.” Long left the order standing.[522]

Ten days earlier, on February 15, 1898, the battleship *Maine* had exploded in Havana harbor, killing 266 men (Navy inquiries in 1898 and 1911 blamed a mine; a 1976 study led by Admiral Hyman Rickover concluded the likeliest cause was a coal-bunker fire next to a magazine, and the question remains disputed). President William McKinley asked Congress for war, and Congress declared it in April. As Navy historians note, the story that Roosevelt engineered the war single-handedly has largely been refuted; an 1896 Navy war plan already pointed at Manila. Gray Brechin writes that the Navy Department had planned the seizure of the Philippines “two years before Gridley was ready to fire.”[523] What Roosevelt did on one February afternoon was aim the fleet.

On May 1, 1898, Dewey’s squadron slid past the old fort at Corregidor into Manila Bay. “You may fire when you are ready, Gridley,” Dewey told his captain. Six hours of shelling sank the Spanish fleet.[524]

Hawaii came into the American empire the same summer, annexed by joint resolution—the Newlands Resolution—on July 7, 1898, while the war was still on. The resolution carried the name of Representative Francis G. Newlands, an heir by marriage to the William Sharon silver fortune who had laid out Burlingame and Hillsborough, the suburbs just up the peninsula from the Stanford farm. In a private letter dated October 6, 1898, his brother-in-law and law partner, Hall McAllister Jr., wrote to Frederick Sharon that keeping “the entire Philippine group” would “prove of incalculable benefit to San Francisco and would vastly improve the prospects of all business and properties here.”[525]

On December 10, 1898, in the Treaty of Paris, Spain ceded Puerto Rico and Guam, gave up Cuba, and sold the Philippines to the United States for $20 million (roughly $800 million today). Fighting between American and Filipino forces around Manila began on February 4, 1899. Two days later, on February 6, the United States Senate ratified the treaty by a vote of 57 to 27, barely the two-thirds required. The $20 million went to Spain.[526]

Americans called the war that followed an insurrection. At the peak, 75,000 U.S. troops were in the islands. Both sides committed atrocities: according to the State Department’s Office of the Historian, U.S. forces at times burned villages, reconcentrated civilians, and used torture, while Filipino fighters tortured captured soldiers and terrorized civilians who cooperated with the Americans. General William Shafter told the *Chicago News* that it might be necessary to kill half the Filipinos so the remaining half could be advanced to a higher plane of life than their present semi-barbarous state.[527]

David Starr Jordan campaigned in public against annexing the Philippines. His argument was not a sermon on consent. As Harris records, Jordan argued that the colonies were “incapable of civilized self-government” and “not habitable by Anglo-Saxon people,” adding, “The home cannot endure in the climate of the tropics.”[528]

Across the China Sea, the Empress Dowager Cixi had held the real power of the Qing throne since 1861. In 1898 she confined the reforming Guangxu Emperor under house arrest and moved to stop foreign businessmen from treating her realm as a buffet. Western newspaper reports said she had “the soul of a tiger in the body of a woman.” In the summer of 1900 she backed the Boxers, the martial, magical movement that meant to drive the foreigners out.[529]

In June 1900, Herbert Hoover—twenty-five years old, the orphan who had come to Encina Hall with two suits and a bicycle—was a mining engineer in the foreign settlement at Tianjin. On June 14 a Boxer force, described as 25,000 strong, hit the Chinese city and burned the French cathedral. About 2,000 foreign soldiers held the settlement, protecting 700 foreign civilians—300 of them women and children—and thousands of Chinese Christians. Imperial troops sent to scatter the Boxers joined them and turned their guns on the compound. Hoover helped build barricades at the gates out of sacks of sugar and grain, fought fires, and carried food on his bicycle, chewing nuts without shucking them. His wife, Lou Henry Hoover, refused to wait in the basement and ran her own errands with a .38 Mauser on her hip.[530]

Rail and telegraph lines were cut. Back at Stanford, faculty and students read a newspaper report that some of their best alumni were probably dead: “It is almost certain that they have fallen victims to the savage ferocity of the Mongolian hosts, unless they managed to make their escape previous to the bloody massacre, which is not likely.”[531]

They had not fallen; casualties inside the foreign compound were light. And the coal was the point.

Hoover considered the Kaiping mines, northeast of Tianjin, the only successful mining venture in China by Western standards. Nine thousand men worked three collieries directed by Hoover’s employer, Chang Yen-mao. Chinese law kept the mines with the emperor, and Chang could not sell them to foreigners. During the siege the Boxers stopped the coal, the share price fell, and Russian and Japanese troops were said to be planting flags on the pits. The foreigners in the settlement arrested Chang on suspicion that he was negotiating with the Boxers; Whyte notes there was no firm evidence, but they locked him up anyway. From confinement Chang signed a power of attorney telling Gustav Detring to put the company under the British flag on the best terms he could get.[532]

Hoover and Detring wrote the deal and backdated the papers so the reorganization would look as if it had been agreed to before the uprising. The new British firm was capitalized at one million shares of one pound each, of which existing Chinese shareholders were to receive 375,000; what the London promoters would take for themselves was not written down. Hoover refused to let the deed go straight to his London boss, Charles Algernon Moreing. Having already told Moreing that without absolute control the game was not worth the candle, Hoover had the deed made out to himself as trustee.[533]

Whyte calls the takeover “the biggest mining transaction in the history of China.” In *Palo Alto*, Harris quotes the historian Ian Phimister: the previous Chinese owners “were defrauded of their property.”[534]

The Dowager Queen (1899–1904)

In California, Jane Stanford was rearranging her own empire.

In 1899, afraid that a campaign to amend the state constitution in Stanford’s favor might fail and pull her gifts down with it, she deeded the entire university endowment—$25 million, roughly $1 billion today—with no reservations and no trust clauses, to a single man: the San Jose lawyer and Stanford trustee Samuel Franklin Leib.[535] The idea was that if a court ruled her gifts to the university illegal, Leib could deed them back to her. George Crothers later called it a strange procedure; had Leib died or acted on the deed, the university might never have recovered the money.[536] It was the same Samuel Leib who handled the property, investments, taxes, and will of Sarah Winchester down the road at 525 South Winchester in San Jose, and who settled her estate after she died; History San Jose holds the Samuel Leib Papers, including a Sarah Winchester series running from 1865 to 1922.[537] One lawyer’s filing cabinet held the paperwork of the railroad fortune and the paperwork of the rifle fortune.

On May 31, 1899, on the advice of Russell J. Wilson, Jane sold her Southern Pacific stock for a reported price of $11 million (roughly $440 million today).[538] Crothers later called the sale one of the great disasters to befall the university: he calculated at the time that it was $7 million (roughly $280 million today) less than the banking house of Speyer & Co., led by James and Edgar Speyer, got when it sold the same stock the following year to E. H. Harriman, the man who already ran the Union Pacific (by the 1940s, Crothers’s estimates of the loss to the endowment had grown to $2 million a year in income and $100 million in value).[539]

On the same day she sold the stock, May 31, 1899, Jane signed what Crothers called the Great Deed, conveying most of her estate back to the university.[540] Crothers called it a great but defective grant. A clause inserted by Mountford Wilson gave Jane the power “to grant, bargain, sell, convey, or lease” all the property she had just given.[541] She kept so much control that Crothers thought the gift had, in effect, not happened; she treated it lightly enough, he said, to pull millions back out.[542] The dead hand of 1885 and the key to unlock it were in the same folder.

On November 6, 1900, California voters passed the constitutional amendment to Article IX, Section 10 of the 1879 state constitution by a vote of 137,667 to 67,737, about two to one.[543] Drafted with the involvement of George Crothers and Mountford Wilson, it confirmed the founding trusts and let the legislature exempt Stanford’s lands and trust funds from taxation while they were used for education.[544] In exchange, the amendment stated that California students were not to be charged tuition unless the legislature later authorized it (a later education code did).[545] The yes votes subsidized a private university by not collecting property tax from its land.

Five days after the polls closed, on November 11, 1900, David Starr Jordan—whose own daughter, Barbara Jordan, had died suddenly that September—accepted a letter of resignation that the economist and sociologist Edward Alsworth Ross had written back in June.[546] Jordan had held it until after the election, hoping the situation would change. He accepted it, he said, "with great reluctance," praising Ross as a teacher and a man.[547]

Ross and others pointed out that if the dismissal had become news before November 6, the tax-exemption amendment might have failed.[548] With the vote safely counted, Ross—who also corresponded that day with San Francisco Mayor James Duval Phelan—called in the San Francisco reporters and blamed Jane Stanford.[549] She had long disliked his politics, dating back to his 1896 support for William Jennings Bryan and his pamphlet *Honest Dollars* during the battle between bimetallism and the gold standard.[550] More recently, in May 1900, he had spoken publicly at a labor rally against "coolie" labor—the era's slur for Chinese workers—and Japanese immigration, and had advocated municipal ownership of utilities rather than private street-railway and water franchises.[551]

On the morning of November 14, 1900, the *San Francisco Chronicle* ran the front-page headline: "Dr. Edward Ross Forced Out of Stanford University."[552]

Jordan had not yet read Ross's statement when reporters reached him, and his on-the-record praise of Ross ran next to Ross's accusation.[553] Ross publicly thanked his "great chief"—what Richard White calls a Judas kiss that placed the firing squarely on Jane.[554] Telling the story of the Ross affair, White writes that "Jane Stanford played the dowager queen."[555] It was her university. The dismissal was her due. The firing triggered a wave of faculty resignations and dismissals—most notably the head of the history department, George Elliott Howard, in January 1901, along with professors Morton Aldrich, Frank Fetter, Spencer, and Little—and prompted a February 20, 1901 investigation by a Committee of Economists led by Columbia's Edwin R. A. Seligman that helped catalyze the national movement for academic freedom and tenure.[556] Ross lost his job and won the headline; the tax exemption was already in the constitution.

During the same years, Jane moved to close a different door. By 1899, according to the university's own magazine, women made up about two in five Stanford students.[557] The 1885 grant had stipulated "a University for both sexes."[558] On May 31, 1899, the same day she signed the Great Deed and sold the railroad stock, Jane moved to cap female enrollment; Crothers stopped her from issuing an instruction that would have told the trustees to break the grant outright, warning her that breaking it risked the entire trust, and got her to settle for a maximum of 500 women.[559] In 1904 she still called coeducation an unsettled experiment.[560]

In the center of the quad she built Memorial Church, dedicated on January 25, 1903, with an address by the chaplain, the Reverend D. Charles Gardner, and turbulent early tenure by the spiritualist-leaning preacher R. Heber Newton of All Souls Church in New York.[561] Jane called the church her "Koohinoor."[562] The facade read: *Erected to the Glory of God and the Loving Memory of My Husband Leland Stanford*.[563] She chose the wall inscriptions herself, painting a line she attributed to her husband: that the

best form of religion is trust in God and a firm belief in the immortality of the soul.[564] She wanted nondenominational Christianity with the dogma stripped off; White's verdict is that the church "spoke mainly to a sect of one."[565] Meanwhile, in Melbourne, Australia—where Jane and Bertha Berner traveled in 1903—her brother-in-law Thomas Welton Stanford funded a Stanford fellowship in psychic research (later held by John Edgar Coover) and donated his collection of "apports," objects he believed spirits had delivered into séances conducted with the medium Charles Bailey.[566]

From 1899 to 1905, George Crothers—who worked for Jane without a salary so she could not treat him as an employee to be dismissed at will, and who became a trustee in 1902 in a seat some had expected to go to Herbert Hoover—feared she might hand the university away altogether.[567] Catholic Archbishop Patrick Riordan had her ear and had tried, and failed, to stop her from signing the Great Deed in 1899.[568] Jane toyed with turning Stanford into a boys' school, giving it to the Catholic Church, dividing the fortune among her heirs, or giving the endowment to Columbia—a cruel dig at Jordan, since Leland had first offered the Stanford presidency to Columbia's Nicholas Murray Butler.[569] By 1902 she was a catechist, taking instruction to enter the Roman Catholic Church.[570] Crothers argued that a hierarchical church grown rich on the sacraments would betray her hope of a purer Protestantism.[571] She postponed the conversion, though she still summoned Crothers in the middle of the night with rumors of students "queening" in the campus's shady lanes—stories that came back false—and worried that women students endangered the men's celibacy.[572] In 1903, Crothers engineered her formal resignation as surviving founder and her assumption of the presidency of the Board of Trustees on July 6, 1903, followed by a Santa Clara County Superior Court special proceeding to quiet title and lock the university's trusts in place.[573] Even then, by the summer of 1904, aided by German professor Julius Goebel, Jane was privately polling trustee Horace Davis and Columbia's Butler on ways to replace Jordan or strip departments, preparing a July 6, 1904 memorandum before reading a softened address to the trustees on August 1, 1904.[574] Crothers answered the midnight summonses and kept the deed intact.

Outside the farm, the Philippine war ground on. On November 27, 1900, Major Edwin F. Glenn subjected Tobeniano Ealdama, the town *presidente* of Igbaras on Panay, to the "water cure"; court-martialed and convicted, Glenn received a one-month suspension and a $50 fine.[575] On September 28, 1901, at Balangiga on Samar, townspeople attacked a company of the Ninth U.S. Infantry, killing dozens of soldiers (published counts of the American dead disagree) while about 28 villagers died in the attack.[576] Afterward, Brigadier General Jacob H. Smith told Major Littleton Waller he wanted no prisoners: "Kill and burn. The more you kill and burn, the more you please me."[577] Asked the age limit, Smith said, "Everything over ten," ordering Samar turned into a "howling wilderness" where "even the birds could not live."[578] Waller partly countermanded him—"we are not making war on women and children"—and Smith was court-martialed for conduct to the prejudice of good order, admonished, and retired by Roosevelt.[579]

Mark Twain, a vice president of the Anti-Imperialist League from 1901, asked whether the United States would go on conferring its civilization upon "the peoples that sit in darkness."[580] Roosevelt declared the war over on July 4, 1902, though fighting continued—with 857 recorded engagements

following his announcement; in all, according to the State Department, more than 4,200 U.S. combatants, more than 20,000 Filipino combatants, and as many as 200,000 Filipino civilians died of violence, famine, and disease.[581]

On May 13, 1903, in San Francisco's Union Square, President Roosevelt dedicated the Dewey Monument, a Roman victory column with the names of Dewey's ships cut into the base, *Olympia* first.[582] He told the crowd that the proper place of Americans was "with the great expanding peoples, with the people that dare to be great."[583]

Room 120 (1905–1906)

January 14, 1905. At the Nob Hill mansion in San Francisco, seventy-six-year-old Jane Stanford drank from her bedtime bottle of Poland Spring water, tasted something bitter, and forced herself to vomit.[584] Servants including her personal maid Elizabeth Richmond and footman Joe Yeng were in the house; the water was sent to Wakelee's Pharmacy and the Falkenau Assaying Company, where chemists found a massive, lethal dose of strychnine, the alkaloid sold as rat poison.[585] Instead of turning the matter over cleanly to the San Francisco Police Department under Chief George Wittman, Jane's brother Charles G. Lathrop and attorney Mountford Wilson hired Harry Morse's private detective agency, whose operative Jules Callundan investigated the household staff, casting suspicion on Richmond, former butler Albert Beverly, and Chinese servants Ah Wing and Ah Young.[586]

Nine days later, on January 23, Jane went down the peninsula to the Hotel Vendome in San Jose—the same town as Samuel Leib's law office and Sarah Winchester's Llanada Villa—to rest, see old friends, meet with trustees, and fight with Jordan over the university budget.[587] Then, after stopping at the St. Francis Hotel in San Francisco and writing farewell letters to Crothers and trustee May Hopkins on February 15, she decided to sail for Hawaii.[588]

On February 15, 1905, she boarded the steamship *Korea* in San Francisco, bound for Honolulu and Yokohama (the passenger list spelled Bertha Berner's name as "Benner").[589] The dock was crowded and military bands played for a fellow passenger: Major General Arthur MacArthur, commander of the Department of the Pacific, who had served as military governor of the Philippines from May 1900 to July 1901 before clashing with William Howard Taft.[590] Jane had packed $75,000 in jewels—roughly $2.8 million today.[591] On the voyage she spoke freely about the poisoned Poland Spring bottle to General MacArthur, his wife, and a Chicago grain merchant named Charles Sidney Dole.[592] Nothing in the record shows that anyone later asked the general what she said.[593] Meanwhile, on February 18, three days after she sailed, the San Francisco newspapers broke the story of the January poisoning at the Nob Hill mansion.[594]

That same afternoon, February 18, 1905, in the Blue Room of the White House, the Chinese minister Liang Cheng presented President Roosevelt with a portrait of the Empress Dowager Cixi, painted by the American artist

Katharine Carl for the 1904 St. Louis World's Fair.[595] Roosevelt accepted it for the United States.[596] The other dowager's face was a diplomatic gift; the Palo Alto dowager was on a ship in the Pacific, talking about a bottle.

Jane docked in Honolulu looking, in the phrase of the record, "well and hearty."[597] The governor of the Territory of Hawaii was George R. Carter, whom Roosevelt had appointed in 1903 after consulting, among others, David Starr Jordan.[598] On February 21, Jane checked into the Moana Hotel at Waikiki, which had opened on the beach in 1901.[599] She took Room 120. Bertha Berner and Jane's maid, May Hunt, shared Room 122.[600]

Tuesday, February 28, 1905. Jane spent the day being happy. A carriage took her slowly along the windward coast past the Nuuanu Pali and Koolau Cliffs because she wanted the ride to last; May Hunt remembered that she chattered and sang.[601] Around one o'clock they picnicked in a grove. As Berner later told the *Hawaiian Star*, the hotel had packed sandwiches of graham bread, Swiss cheese, tongue and lettuce, hard-boiled eggs, gingerbread, oranges, chocolate candy, and cold coffee.[602] Jane ate heartily of everything except the eggs. They were back at the Moana by four.[603]

At half past six the three women went down to dinner. On the verandah Jane told a guest named Mrs. Grinbaum about the pleasant day.[604] Dinner was a bowl of soup; it took ten minutes. She sat outside until eight or a quarter past, then went upstairs with Berner and asked her to set out her medicine.[605]

Berner put a cascara sagrada capsule on the bed stand; cascara was an over-the-counter laxative whose capsules in that era often contained a trace of strychnine on the theory that it hurried the herb.[606] Then Berner took a small bottle of bicarbonate of soda, two to two and a half inches tall with a mouth wide enough for a teaspoon, from a medicine basket (a bottle later traced to W. E. Jackson at the Stanford Pharmacy in Palo Alto, where Berner had been courted by pharmacist Paul J. Schwab).[607] She asked May Hunt for a spoon, measured the half teaspoon Jane wanted, and left it beside the capsule.[608]

"Give it time to get out of my stomach," Jane said, referring to the glass of water she would drink before taking the medicine.[609] She urged Berner to take some of the soda too. Berner declined.[610]

Berner and Hunt left around nine. Jane called Hunt back because she was having trouble with the door lock; Hunt showed her how it worked and went to bed.[611] Berner thought her employer fell asleep without taking the dose, then woke later and swallowed it.[612]

Hours later, a voice filled the hallway. A guest across the hall, Adam Heunisch, opened his door and found Jane Stanford in her doorway, her arms thrown up.[613]

"Oh I am so ill," she cried. "Get me a doctor."[614]

Heunisch ran for the elevator. Dr. Francis Howard Humphris lived in the hotel; Berner told the elevator boy the doctor should not even stop to dress.[615] May Hunt tried to get Jane back to bed, but she would not go,

gripping the door casing. She said a spasm had thrown her out of bed. When Humphris arrived, she told him she had been poisoned again and asked for a stomach pump.[616] When he asked how she knew, she said she had been poisoned before and turned to Berner. Berner said the earlier Poland Spring bottle had held arsenic, then corrected herself: strychnine.[617] Humphris administered a solution of bromide and chloral hydrate, and Dr. H. V. Murray arrived to assist, followed by Dr. Francis Root Day.[618]

The stomach pump did not arrive in time. Unlike the massive dose in January, the Honolulu dose was a small, precise amount of pure strychnine alkaloid.[619] Her jaws set. Humphris felt her muscles lock and took the glass from her hand. Berner watched the last spasm.[620]

“Oh God, forgive me my sins,” Jane said, her jaw stiffening. “Is my soul prepared to meet my dear ones?”[621]

Then came the words that Richard White notes she would rarely if ever have spoken in life, having spent twenty years calling death a gentle passage into a higher realm:

“This a horrible death to die.”[622]

Her body went rigid. The soles of her feet turned inward toward each other, insteps arched, toes pointing forward, knees wide. Her eyeballs protruded and her pupils dilated. Her jaws fixed, her fingers contracted with the thumbs dug into the palms, and her breathing stopped. From the moment Dr. Humphris entered Room 120 to the final spasm, ten minutes had passed.[623]

The cause of Jane Stanford’s death—and who, if anyone, poisoned her—remains DISPUTED.[624] In Honolulu, Drs. Clifford B. Wood and Francis Root Day performed the autopsy on March 1, and territorial chemists R. A. Duncan and Edmund Shorey analyzed the stomach contents and the medicine bottle.[625] Using the “fading purple” potassium bichromate color test, the chemists found strychnine in the bicarbonate of soda and in her body; the physicians testified that a trace from a cascara capsule could not explain the death.[626] On the evening of March 9, the coroner’s jury, presided over by High Sheriff William Henry, returned its verdict, printed in the newspapers on March 10:

“Jane Lathrop Stanford came to her death at Honolulu, Island of Oahu, Territory of Hawaii, on the twenty-eighth day of February, A.D. 1905 from Strychnine poisoning, said strychnine having been introduced into a bottle of bicarbonate of soda with felonious intent by some person or persons to this Jury unknown and of the contents of which bottle Jane Lathrop Stanford had partaken.”[627]

During the inquest, Judge W. L. Stanley, representing the estate, sat beside the deputy attorney general and whispered questions.[628] The day before Bertha Berner testified on March 6, it became public that Jane’s will left her $15,000 (roughly $560,000 today).[629] Jane’s brother Ariel Lathrop called Berner a medium.[630] No complete official transcript of Berner’s inquest testimony survives in the university’s files; White describes her as bold, flirtatious, charming, quick, and calculating, while noting that none of those adjectives is a conviction.[631] Newspapers and detectives also chased

Chinese servants and acquaintances, including Ah Wing, a mentally ill former servant named Wong Toy Wong (committed to Mendocino State Hospital), and a Chinatown informant using the alias Tay Wang (Charles Eyre).[632] Hawaiian officials concluded the poison had been put into the soda in San Francisco and declined to pay for a prosecution.[633] High Sheriff William Henry told the press that David Starr Jordan, trustee Timothy Hopkins, and their private detectives (Jules Callundan of the Morse agency and San Francisco police detective Harry Reynolds, who sailed to Honolulu with Jordan and Hopkins on the *S.S. Alameda*) had come to Honolulu to cover up a crime, not to solve one.[634]

On the other side of the dispute, the night before the Honolulu funeral service at Central Union Church on March 15, Jordan and Hopkins drew up a statement attributing Jane's death to natural causes: advanced age, unaccustomed exertion, a surfeit of unsuitable food, and the unusual exposure of the picnic, perhaps aggravated by strychnine in the laxative capsule and a small amount in the bicarbonate through a "pharmacist's error."[635] Jordan paid Honolulu physician Dr. Ernest C. Waterhouse to write a medical critique challenging the autopsy and the chemists' color tests, and after returning to San Francisco on the *Alameda* on March 21, recruited Cooper Medical College pathologist Dr. William Ophüls (along with Drs. Charles L. Morgan and H. A. L. Ryfkogel) to argue that she had died of heart disease—coronary sclerosis or angina pectoris, a spasm of the heart.[636] No bicarbonate containing strychnine was ever found on the market, Jordan never released the full report he claimed held his evidence, and Dr. Clifford Wood in Hawaii called the heart theory "simply rot" (a full century later, Stanford neurologist Robert W. P. Cutler's 2003 study *The Mysterious Death of Jane Stanford* and Richard White's 2022 investigation *Who Killed Jane Stanford?* both concluded from the clinical symptoms and chemical tests that she died of strychnine poisoning, while leaving the killer's identity unproven).[637]

The structural conflict of interest is the whole case. Jordan's university inherited the fortune if the death was ordinary and the will stood; a murder investigation threatened to embarrass the estate, invite a contest over the will by disgruntled Stanford relatives (such as Leland Sr.'s brother Asa P. Stanford, whose widow sued the estate for a million dollars in June 1906), and reopen the mishandled January poisoning—and Jordan himself had been the target of Jane's quiet 1904 campaign to replace him.[638] A century later, Malcolm Harris writes that it is no longer particularly controversial to suggest Jordan was part of a conspiracy, most likely with Bertha Berner, though no one has produced definitive proof.[639] The chemists found strychnine. The president found a picnic.

The Southern Pacific—now controlled by E. H. Harriman—found a newspaper. W. H. Mills, an old political operative for the Associates, paid off a wide stretch of the California press, including the *San Francisco Post* and William Dargie's *Oakland Tribune*, on the railroad's behalf, while Fremont Older's *San Francisco Bulletin* accused the authorities of a cover-up.[640] Presumably at the request of Mountford and Russell Wilson and Charles and Ariel Lathrop, the *Post* denounced the murder stories and praised Jordan and Wilson for rejecting them—before spoiling the favor slightly by observing that private detectives are more often hired to hush a scandal than to solve one.[641]

At the funeral in Memorial Church on the farm on Friday, March 24, 1905—following sermons by the Reverend John W. Dinsmore and the Reverend Charles R. Brown, and a eulogy by Rabbi Jacob Voorsanger—the congregation sang “Nearer, My God, to Thee” and the choir sang the Stanford alma mater.[642] The casket was placed inside a steel chest. The next day, March 25, in the family mausoleum, the chest was slid into the marble sarcophagus between her husband and her son, and sealed.[643] For twelve years the inscription had waited with a blank space: *Jane L. Stanford. Born in Mortality, August 2, 1828. Passed to Immortality—*. On March 16, 1905, before the ship carrying her body had even docked in San Francisco, a workman had already chiseled the date: February 28, 1905.[644]

Fourteen months after the workman cut the marble, on the morning of April 18, 1906, the San Andreas Fault broke.[645] Visiting Harvard philosopher William James—who had come west to teach at Stanford for the semester and had written home marveling at the sunny quadrangle—was thrown from his bed on campus; the spire of Jane’s Memorial Church, the new library and gymnasium, the museum, and the 100-foot triumphal arch came crashing down.[646] Five days after the earthquake, in Beijing, the Empress Dowager Cixi issued an edict sending 100,000 taels of silver “as a mark of friendship” for the sufferers in San Francisco, plus 40,000 taels to the Chinese minister in Washington for the city’s Chinese residents; because the Roosevelt administration refused relief gifts from foreign governments, the American minister William Woodville Rockhill returned the 100,000-tael note with assurances that no slight was intended, while the 40,000 taels for Chinatown stayed on its errand.[647] Six months after that, on October 11, 1906, the San Francisco school board ordered the city’s Japanese schoolchildren into the segregated Oriental School in Chinatown where Chinese children had been sent since 1885.[648]

In 1906, the year after Jane died, David Starr Jordan became the inaugural chair of the Committee on Eugenics of the American Breeders’ Association, publishing *The Human Harvest: A Study of the Decay of Races Through the Survival of the Unfit* the following year.[649] In 1909, Sarah Winchester sent $300,000—thirty years of Jordan’s salary—through funds entrusted to Yale to endow the William Wirt Winchester Hospital for tuberculosis in New Haven.[650] More than a century later, on September 13, 2018, Stanford’s trustees voted to rename Serra Mall, the Serra dormitory, and Serra House while keeping Serra Street; on October 7, 2019, the university’s main address changed from 450 Serra Mall to 450 Jane Stanford Way.[651] In October 2020, because of Jordan’s eugenics work, the trustees stripped his name from Jordan Hall (Building 420) and removed the statue of Louis Agassiz from its facade.[652]

The dead hand in the Founding Grant said the land could not be sold. It did not say the sandstone could not crack.

Part 5: The Engineer’s Century

1891–1945

The Orphan and the Mine (1891–1905)

In the summer of 1891, before opening day, Herbert Hoover arrived at a campus that was not finished. The Iowa orphan—born in 1874, raised a Quaker, and sent to Oregon after his parents died—had been let into Stanford on a conditional admission after he passed mathematics and failed the rest of the entrance examination.[653] He lived on the farm while the sandstone was still going up. He ran a laundry agency and a lecture bureau. He wrote a friend that he was working awful hard and had schemes for making more.[654] The hunger was not a joke.

Earlier on that same ground, in 1878, Eadweard Muybridge had set a row of cameras on Leland Stanford's stock farm to see whether a horse ever had all four hooves off the ground.[655] It did. Those frames were the ancestor of the motion picture, paid for by the railroad president. Four years earlier, down the coast, on January 31, 1874, Thomas Balch Elliott and about 150 of his Indianapolis neighbors had formed the Indiana Colony on Rancho San Pascual—the settlement that became Pasadena.[656] Many of them had come for a milder climate, sick with asthma or tuberculosis. In 1882 Emma Bangs kept a Pasadena boarding house the town called the "busted lung brigade"; a parrot there had learned to cough "exactly like a lunger."[657] By the 1890s, when the university opened on the stock farm, the state was already drawing people who needed a new climate or a new start.

At the start of Hoover's second semester, a geologist named John Casper Branner arrived to chair the department. Hoover could type. Branner hired him as an office assistant at $25 a month—roughly $1,000 a month in modern money—and Hoover kept the job until he graduated.[658] It was not a living. It was a door. Hoover's own rule, told to a classmate, was simple: do your work so they notice it, and always be on the job.[659]

A guest lecture by Branner in Monterey captured another student, Lou Henry.[660] She talked her parents into letting her study geology, the only woman in the program. When she first met Hoover in Branner's laboratory, she expected someone brawny. What walked in was speechless, hair uncombed, tanned from summer work. She did not look away.[661]

Hoover graduated in the pioneer class of 1895, in geology, with a degree and almost no money. His first job was pushing ore carts in a Nevada mine.[662] By 1896 he was publishing in the *Mining and Scientific Press* ("Some Notes on 'Crossings'") and working on the litigation between the Carson City Gold and Silver Mining Company and the North Star Mining Company.[663] The degree had not delivered a career. It had delivered a person a London banker could be talked into trusting.

In the 1890s, investors in the City of London—the financial district, not the city of churches—were putting money into holes in Western Australia, South Africa, and China. They could not see the holes. Somebody had to go and look, price the risk, buy the machinery, hire the men, keep books an auditor in London could read, and talk to whatever government claimed the ground. The scarce thing was not dynamite. It was a report a stranger would believe. A degree from a new university on a horse farm was, for a few years, a strange but usable proof.

The recommendation that moved Hoover was not an application he filled out. Louis Janin, a San Francisco mining engineer, passed him to the London firm of Bewick, Moreing and Company. In the spring of 1897, at twenty-three, they sent him to Western Australia.[664]

Coolgardie and Kalgoorlie were inland, hot, short of water, and full of money. Hoover wrote home to his brother Theodore and to his Stanford friend Lester Hinsdale from the goldfields as he scouted properties for the firm.[665] He told Bewick, Moreing which holes to buy. He recommended the Sons of Gwalia. The name was Welsh. The mine made his reputation.[666]

Lou Henry took her geology degree in 1898. On May 1 of that year, Commodore George Dewey's squadron destroyed the Spanish fleet in Manila Bay in a morning. The war was short. The empire was not. By 1899, Bewick, Moreing had sent Hoover to China, where he examined the Kaiping coal mines and coal field in Chihli Province.[667] In February 1899 he married Lou Henry in Monterey, and they sailed the next day—two people who could read the same rock.[668]

On May 12, 1901, President William McKinley broke ground in San Francisco's Union Square for a monument to Dewey's victory at Manila Bay.[669] Union Square was already named, for pro-Union rallies on the eve of the Civil War. The granite shaft rising there was not about that war. It was about an ocean.

Two years later, in May 1903—sources differ on whether the ceremony fell on May 13 or May 14 [DISPUTED]—President Theodore Roosevelt dedicated the finished monument, a granite shaft with a bronze Victory on top, wreath in one hand, trident in the other.[670] Five years had passed from the battle to the column; a morning in Manila had become a stone in a shopping square before the uniforms were out of fashion. Former mayor James D. Phelan presented the shaft to the city, and Mayor Eugene Schmitz accepted it. Roosevelt said Manila had opened the Pacific to American commerce. The shots that count, he said, are the shots that hit.[671]

While Roosevelt was in the city, on May 14, 1903, the Union League Club gave him a farewell banquet with covers for eight hundred. The *Morning Oregonian* printed the toast list the next morning: "The Army, General MacArthur. The Navy, Secretary Moody. California, Governor Pardee. The Citizens' Committee, M. H. de Young."[672] General Arthur MacArthur—Douglas MacArthur's father—had commanded in the Philippines. He toasted the Army at the banquet for a naval victory in the islands he had helped to hold, while M. H. de Young of the *San Francisco Chronicle* toasted the citizens. A lot of people standing under the column in Union Square think it is for General MacArthur. It is for Dewey; Arthur MacArthur was inside at dinner.

Earlier that year, on January 25, 1903, Jane Stanford had dedicated Memorial Church at the center of the Stanford Quadrangle, with Reverend D. Charles Gardner delivering the dedication address.[673] The church was topped by an eighty-foot, twelve-sided spire on a crossing that was not tied hard enough to the roofs around it.[674]

On October 17, 1904, in a remodeled saloon in San Francisco's North Beach, Amadeo Pietro Giannini opened the Bank of Italy.[675] Born in San Jose in 1870, Giannini had left school at thirteen; he did not go to Stanford, and he did not go to Berkeley.[676] The Bank of Italy employed Italian-speaking tellers, took small deposits, and made loans to tradesmen. By the end of 1905, its assets stood above $1 million.[677]

Across the Pacific in 1905, eight years after Hoover first went down the Sons of Gwalia, the parliament of Western Australia appointed a Royal Commission on the Ventilation and Sanitation of Mines.[678] A royal commission is not a royal visit; it is a government inquiry with the power to compel testimony and write a judgment. This one was hard on most of the operators in the goldfields. Workers, the record of the inquiry said, were unduly exposed to dust and fumes and falling rock. At the Sons of Gwalia, the commission found the air "notably bad."[679]

The same parliament was launching six or eight royal commissions a year, some of them on questions as small as stone or stucco for the new law courts.[680] A government that inquires into everything will inquire into you. That took some of the sting out. It did not take the finding out. Hoover was thirty-one when the commission spoke, and that same year he co-authored *The Economics of Mining* with T. A. Rickard, Walter Benton Ingalls, and R. Gilman Brown.[681] The question of how a young engineer got his results followed him.

In France in 1905, Alfred Binet built a test to find children who were falling behind in school, so they could be helped. Binet warned against using it as a measure of inborn worth.[682] In Palo Alto that same year, a house was built at 367 Addison Avenue for Dr. John Spencer, a physician who would become mayor, with a twelve-by-eighteen-foot garage in the back.[683] And on February 23, 1905, the *San Jose Mercury* announced that Stanford had appointed William James—the Harvard philosopher, the most famous philosopher in America, and a leading figure in the American branch of the Society for Psychical Research, who went to séances looking for one honest medium, what he called a "white crow"—to teach at the university, after an invitation initiated the previous September.[684] Less than a week later, on February 28, 1905, Jane Stanford was dead at the Moana Hotel in Honolulu.[685] Whether she died of strychnine poisoning or natural causes was immediately DISPUTED: the Honolulu coroner's jury and chemist R. A. Duncan found that she had been murdered with strychnine placed in her bicarbonate of soda, while Stanford president David Starr Jordan sailed to Hawaii, retained physician Dr. Ernest Waterhouse, and announced to the press that she had died a natural death.[686]

The Spire and the Plank (1906–1913)

William James came to Stanford anyway, arriving in January 1906, nearly sixty-four and tired of lecturing.[687] On January 17, 1906, he told his wife Alice why: "If I can only get through these next 4 months and pocket the $5,000 I shall be the happiest man alive."[688] Five thousand dollars for one term was roughly $186,000 in modern money—more than sixteen years of the $25-a-month wage Hoover had earned typing for Branner. James was being paid a young engineer's youth for four months of talk.

His only class was a history of philosophy. Three hundred students enrolled, and another hundred and fifty visitors sat in. He wrote that they did not know what hypothesis and concept meant, and that he found it hard to aim too low. One of his few mentions of the woman who had wanted him there was a thank-you: the memorial holiday for Jane Stanford's death let him cancel a class.[689]

On April 18, 1906, at 5:12 in the morning, the United States Geological Survey marked a foreshock. The main break came twenty to twenty-five seconds later and tore nearly three hundred miles of the San Andreas Fault. William James was in a campus bedroom. He timed his room at 5:13.[690]

"My room was shaken like a rat by a terrier," James wrote home that week.[691] The wooden houses held. The chimneys came down. The higher university buildings, he wrote, were "largely piles of ruins."[692]

At the center of the Quadrangle, the crossing of Memorial Church moved by itself. The eighty-foot spire fell through the chancel roof. The mosaic on the north face blew out into the Quad. The apostles commissioned for the altar were smashed. Hundreds of feet of arcade lay flat. The spire was not put back.[693]

The same morning, in San Francisco, a different kind of monument burned. The golden spike from Promontory sat in the museum at Stanford, but the polished laurel tie it had been driven into burned in the San Francisco fire that followed the quake.[694] The object that marked the ceremony survived. The object the workers built toward did not.

On Nob Hill, the Stanford mansion went the next day, April 19, as the Grace Cathedral nearby also burned.[695] Ah Wing, the family's old cook and the caretaker on the hill, locked the door, walked to Jones Street, looked back, and saw the house in flame.[696] The fire finished what the fault had started.

The other man moving through the city that morning was thirty-six-year-old A. P. Giannini. What he hauled out of North Beach—and how much—is DISPUTED across three accounts. The popular legend, told by *Time* magazine and UC Berkeley's history of Giannini Hall, holds that Giannini was thrown from his bed in San Mateo, borrowed a produce wagon, drove into the city, pulled about $2 million in gold, coins, and securities out of the Bank of Italy—roughly $74 million today—hid the money under vegetables, and rolled past the looters. The Office of the Comptroller of the Currency, the federal office that charters national banks, records the same morning with a much smaller figure: $80,000 in coins—roughly $3 million today—hidden under crates of oranges, followed by loans on a handshake from a wharf. The scholarly biography by Marquis James and Bessie Rowland James, *Biography of a Bank*, says Giannini used two of his stepfather's produce wagons, loaded with all the cash, gold, and records they would hold, and set up a wooden plank across two barrels on the Washington Street wharf within two days.[697] The plank, the wagon, and the loans to people with nothing left to pledge are in every serious account. The $2 million is the legend. The $80,000 is the comptroller.

A bank is a gate if it only lends to people who already have collateral. Giannini did the opposite, and he gave it a shape. Character lending is a loan on the person, not the pledge—a face, a signature, a handshake. Branch

banking is one bank with many offices, instead of a separate little bank in every town.

Back at Stanford, the earthquake let William James go. He walked into San Francisco with a colleague looking for the colleague's sister. A whole population stood in the streets with whatever they had pulled from houses about to burn. James called the fires "most beautiful in the effulgent sunshine."[698] By early May he was back in Cambridge. Before he left, on April 19, 1906, he wrote trustee Horace Davis that the faculty did not trust the people governing the university, and that Mrs. Stanford's death had been a chance for a new start which nobody had taken. On May 20, President David Starr Jordan answered Davis that James was not speaking for the responsible members of the faculty.[699]

Jordan was a great ichthyologist, the first president of the university, a pacifist at real cost, and a eugenicist. In 1907 he published *The Human Harvest: A Study of the Decay of Races Through the Survival of the Unfit*, and in 1912 he followed it with essays in *World's Work*—"Foreclosing the Mortgage on War" and "The Perennial Bogey of War"—arguing that modern warfare bred down the stock by killing off the strongest young men.[700] He loved the word *bionomics*—the claim that one science could cover heredity, history, economics, and ethics, and then tell a society what to do.[701] Eugenics was the idea that a nation should breed human beings the way a farmer breeds livestock, encouraging the births it calls fit and discouraging or preventing the births it calls unfit. In those decades this was mainstream progressive science. University presidents held it. So did public health officers. If poverty and what they called feeblemindedness were mostly inherited, then the hierarchy on the ground was nature.

California wrote it into law. In 1909 the state authorized the medical superintendents of state institutions to operate on inmates when, in the superintendent's judgment, it would improve the person's physical, mental, or moral condition [UNVERIFIED: exact statutory text and citation for the 1909 California asexualization act not in provided excerpts]. No consent from the patient was required, and no court order. Over the following decades California sterilized more people under that authority than any other state; later counts, analyzed in detail by historian Alexandra Minna Stern, put the state's share at roughly a third of the recorded sterilizations in the United States, though the counts do not agree with each other because the people it was done to were not carefully counted [UNVERIFIED: exact sterilization counts and methodological range from Stern not in provided excerpts].[702]

That same year, 1909, W. Cameron Forbes—of the Boston Forbes family that had fought western railroad men over rates in the 1870s—became the civilian governor-general of the Philippines, serving until 1913.[703] Boston, Manila, and the column in Union Square were one circuit. And in 1909, Herbert Hoover published *The Principles of Mining*, followed in 1912 by his and Lou Henry Hoover's landmark English translation of Georgius Agricola's sixteenth-century Latin mining treatise, *De Re Metallica*.[704]

In 1910, a psychologist named Lewis Madison Terman arrived at Stanford with his wife, Anna, and two small children, Fred and Helen. Terman's young adulthood had been marred by a long and debilitating course of tuberculosis, requiring the move to California's mild climate in the first place.[705] Fred, the son, came west because his father could not stay east and

breathe. At Stanford, where Thomas Welton Stanford's bequest was also funding a fellowship in psychical research that brought John Edgar Coover to run experiments on the mind in 1912, Lewis Terman set out to measure human intelligence by a single number.[706]

Down the coast, another group of outsiders was arriving for the sun and the payroll. On October 27, 1911, the Nestor Film Company opened the first permanent studio in Hollywood, at the northwest corner of Sunset Boulevard and Gower Street. The building had been the Blondeau Tavern before a local liquor law hurt the trade. David Horsley, who had been making pictures in Bayonne, New Jersey, leased it. Behind the tavern stood a wooden stage about twenty by forty feet, with muslin overhead to soften the sun. The actors brought their own clothes. The carpenters brought their own tools.[707]

Horsley and the other independents had left New Jersey in part because Thomas Edison's lawyers were winning. Edison and a handful of licensed companies held the patents on cameras and projectors through the Motion Picture Patents Company, the Trust. The independents went west to get far from the process servers. Nestor's first California picture was a one-reel western, about 950 feet long, called *The Law of the Range*, released on December 13, 1911. The advertisement called it a "rousing Western foto-drama… taken in the West." The trade-paper reviewer in *Moving Picture World* wrote: "The story is interesting, but the photographs are not very good."[708]

In the autumn of 1912, Herbert Hoover joined the board of trustees of Leland Stanford Junior University. A trustee legally holds an institution for somebody else—at Stanford, written down in 1885 as the children of California. The board hires the president, watches the endowment, and decides what the founder's forever clauses still mean. As biographer Kenneth Whyte wrote, "He joined the Stanford board of trustees in the autumn of 1912 in circumstances as cloudy as the awarding of his geology degree seventeen years earlier."[709]

The cloud was a residency rule. Trustees were supposed to live close enough to do the work. Hoover escaped it by claiming he had immediate plans to move to San Francisco, though he knew he was at least a year away from leaving London. He still called the university "the best place in the world."[710] He wanted his hand on it. Once he was on the board, he said the place was deficient in a very great proportion of the departments, and that too many of the professors were not good enough—and by January 2, 1913, he was drafting a formal memorandum on Stanford's academic and financial positions.[711] Unlike William James, who had come for $5,000 and told the truth on the way out, Hoover came onto the board while still living in London and stayed for the rest of his life.

In 1913, David Starr Jordan left the Stanford presidency, though the bionomics he had sheltered did not leave with him. That same year, despite the lobbying of Secretary of State William Jennings Bryan and the conflicted misgivings of California progressives like Chester Rowell, California passed the Alien Land Law, telling first-generation Japanese immigrants—*issei*, born in Japan, distinct from their American-born citizen children, the *nisei*—that they could not own farmland.[712]

In December 1913, Cecil B. DeMille got on a train heading west to shoot *The Squaw Man*. The plan was Flagstaff, Arizona. Why he did not stay in Flagstaff is DISPUTED. The story DeMille later told was that Flagstaff was colder than when he left New York, so everybody got back on the train and rode all the way to California—a version public television's *Lost LA* notes is widely taken to be apocryphal. A different memory, kept by the American Film Institute's historians, has DeMille saying he got to Flagstaff, saw high mountains, and wanted plains with mountains in the distance; the star, Dustin Farnum, said they ought to go where the other companies were. They reached Los Angeles on December 20. On Christmas Eve, DeMille signed a lease on a barn at Selma and Vine. Shooting started December 29. To save $3, DeMille played a faro dealer himself.[713]

Los Angeles was an open-shop town, where men worked without joining a union. When the studios needed carpenters and electricians to build a street that looked like Wyoming, the wage was the point. On *The Squaw Man*, DeMille noted that carpenters and other hands cost 25 to 50 percent less than on the East Coast.[714] A carpenter in Los Angeles in 1913 worked at three-quarters of a New York wage, or at half.

Bionomics and Bread Lines (1914–1928)

By the summer of 1914, Herbert Hoover was a partner on London Wall, two doors from the rooms where he was about to try to feed a country.[715] On June 28, 1914, a Serbian nationalist killed the Archduke Franz Ferdinand in Sarajevo. Hoover, in London, told himself it was just another Balkan lapse into barbarism and went back to work.[716] He was wrong, and he was in the right city to find out how wrong.

On the afternoon of August 4, 1914, a few hours after German troops crossed into neutral Belgium, Hoover walked a little more than a city block to the American consulate.[717] That walk was the hinge: a mining engineer with an office on London Wall stepping into the room where he would take on the feeding of a nation. Belgium had about 7.5 million people and grew only about 20 percent of the food it ate.[718] The German army took the harvest, and the British blockade, meant to starve Germany, also closed the occupied country. Between them, a nation that imported four pounds of every five it ate began to beg.[719]

Hoover's first errand was small. An American engineer named Millard Shaler had 1,500 tons of cereals stuck on a dock.[720] Hoover saw that 1,500 tons would not touch a country. When the crisis hit, the skill that mattered was the mining engineer's list: survey, account, price the risk, move food, talk to governments that hate each other. He had practiced it underground. He built the Commission for the Relief of Belgium, the CRB—a private, neutral machine in London, buying food around the world, landing it at Rotterdam, moving it by canal, and handing it to Belgians to distribute.[721] He wanted, in his own words, the strongest centralization and an effective monopoly: absolute command of his half, no knitting bee.[722]

That autumn, Hoover sent a clipped cable from London: "Over one million people on bread line in Belgium at present moment with supplies estimated to last from one to three weeks. We shall require upwards of twenty thousand tons foodstuffs monthly."[723]

He also ran a campaign he named himself: “enormous propaganda.”[724] He browbeat American newspapers and governors, hinting to one state that another state had already given more. He told reporters to go see the famine and to play up what he called the “detailed personal horror stuff.”[725] He sent a motion-picture crew to film the food lines in Brussels. He hired famous writers. George Bernard Shaw sent a sentence: “one Belgian goes without a full meal whilst thousands of lapdogs are living in luxury from Canterbury to San Francisco.”[726]

The first gift ship, carrying 3,500 tons of beans, bacon, flour, and rice—a sixth of a month’s requirement—reached Holland on November 21, 1914.[727] Hoover got two great British shipping lines to carry the food free, along with the piers and the telegraph.[728] He treated charity as a logistics firm. You cannot move 20,000 tons a month on pity alone. The ships sail on a schedule, so the story has to sail on a schedule.

While Hoover was moving grain into Rotterdam, Lewis Madison Terman was finishing his book at Stanford. In 1916 he published *The Measurement of Intelligence*.[729] Inside it was the test Alfred Binet had built in France in 1905 to find children who were falling behind in school so they could be helped.[730] Binet had warned against using it as a measure of inborn worth.[731] Terman and his Stanford team did what Binet had warned against: they translated the items, reset the age norms for American schools, and forced the scores into a curve whose average would always be 100.[732] Mental age divided by actual age, times 100: a single number for a whole child, which Terman called the Stanford-Binet intelligence quotient, or IQ.[733] A child could finish in about fifty minutes. Terman believed general intelligence was an “original endowment” dictated by genetics alone, unchanged by school or hard work.[734] Once a system has a single number, it can sort a school, an army, an immigration desk, or a sterilization ward and say it is only following the measurement.

In 1917, when the United States entered the war, Terman helped design the Army’s intelligence tests.[735] More than 1.7 million draftees took them.[736] A campus instrument became a government instrument in a year.

That same year, President Woodrow Wilson made Hoover the United States Food Administrator.[737] His name became a verb: to *Hooverize* meant to clean your plate, to skip the meat, to waste nothing.[738] Reporters expected a giant and found an unobtrusive man at a desk. In California, by 1918, A. P. Giannini’s Bank of Italy had become the first bank with branches across a whole state.[739]

In 1919, with the war over, Hoover was in Paris advising Wilson at the peace conference.[740] While feeding Belgium, he had conceived of gathering the paper of the war. In June 1919 he sent a telegram to Stanford’s president, Ray Lyman Wilbur—a cable the Hoover Institution still calls its founding document.[741] Hoover offered $50,000, roughly $964,000 today, and told Wilbur to send Professor Ephraim D. Adams to Paris to start collecting.[742]

Adams went, joined by Ralph Lutz and Frank Golder.[743] They gathered pamphlets, newspapers, posters, and government documents across Europe, adding them to the files of the Commission for the Relief of Belgium, the

United States Food Administration, and the American Relief Administration.[744] That same year, 1919, a student named John Steinbeck enrolled at Stanford.[745]

In 1921 [UNVERIFIED: exact start year 1921 not in the Episode 5 review or excerpts], Terman began his long study of gifted children, *Genetic Studies of Genius*, starting with a core of 643 California children who scored at or above 135 on his test.[746] He pledged not to publish their names. He enrolled his own son and daughter, Fred and Helen.[747]

He missed one. The study passed over the only child of a Stanford engineering professor: William Shockley.[748] Shockley scored 129, six points under the line Terman was using for genius—a score that ranked him, in Malcolm Harris's phrase, "slightly subgenius."[749]

In 1922, Hoover's collection was renamed the Hoover War Library.[750] That year it took in 40,000 documents, housed inside the Stanford library.[751] A library is a ledger: whoever keeps the record owns the ground, and Hoover had put the record of a war on his own campus under his own name. In 1925, John Steinbeck left that campus without a degree, having accumulated ninety-three units across six years and used the place as a library before going to ranch work.[752] The pipeline produces presidents; by accident, it also produces the witnesses who write them down, though only one of those outputs gets a tower. And in 1928, A. P. Giannini gave $1.5 million—roughly $29 million today—to found the Giannini Foundation of Agricultural Economics at UC Berkeley and build Giannini Hall.[753] The man who had left school at thirteen and lent on a face put his name on the public university, not on Stanford.[754]

Hooverville and the Garage (1928–1939)

In November 1928, Herbert Hoover was elected president of the United States.[755] He had been secretary of commerce. He had been the man who Hooverized. He believed, with a mining engineer's faith, that a problem of production and distribution could be administered.

The stock market broke in the autumn of 1929.[756] What followed was a collapse of work. The Bureau of Labor Statistics did not begin its modern monthly unemployment survey until 1940, but its later retrospective estimate for 1933 put unemployment at 24.9 percent: 12,830,000 people, about one worker in four, with March 1933 as the worst month and farmers losing their land often not even in the count.[757] Irving Bernstein, writing for the Department of Labor, noted that United States Steel had about 225,000 full-time employees in 1929; by April 1, 1933, the full-time number was zero.[758] Whoever was left was part-time.

In November 1930, Giannini renamed the Bank of Italy the Bank of America.[759] That same year, the word *Hooverville* first showed up in print, credited to Democratic National Committee publicity director Charles Michelson.[760] People who had lost their rooms built camps of cardboard, tar paper, tin, and packing crates, and named them for the president.[761]

In October 1931, an unemployed logger named Jesse Jackson and about twenty men put up shacks on the tidal flats south of Pioneer Square in Seattle, where a shipyard had been.[762] The camp grew to as many as 1,200 people on about nine acres—a few city blocks, named for a man who had moved 20,000 tons of food a month.[763] The city tried to burn it. The men rebuilt it, and the camp lasted until 1941, when the war wanted the land back for ships.[764] Belgium could be treated as logistics, because the food existed somewhere else. A depression in the country that grew the food was a question of who owned the work. You cannot employ a quarter of a nation by being competent.

On July 28, 1932, the limit had a photograph. World War veterans—the Bonus Army—had camped in Washington asking for early payment of a bonus the law had promised them later.[765] Troops under General Douglas MacArthur, the son of the general who had toasted the Army in San Francisco in 1903, drove them out with cavalry and tanks, burning the camps.[766] DISPUTED: How much of the order to cross the Anacostia River Hoover had given, and how much MacArthur decided once he was over the bridge, remains argued by historians.[767] The record shows the eviction, the general, and the president who took the blow.

In November 1932, Hoover lost the presidential election to Franklin Delano Roosevelt, the grandson of Warren Delano Jr., a partner in Russell and Company, the American firm that had made a great part of its China fortune in opium.[768] After the defeat, Hoover went back to Palo Alto. He sat in his study handling as many as 1,500 letters a day, walking over to the Hoover War Library to watch the anti-Hoover literature accumulate on shelves he had paid for.[769] He spoke of the relief of no longer making ten vital decisions a day.[770] The archive held the verb and the insult in one building. He had built a machine for keeping the record, and the record kept him.

While Hoover watched his library fill, Lewis Terman's son Fred was teaching electrical engineering across the campus. In 1934, two of Fred Terman's students, William Hewlett and David Packard, finished their degrees.[771] Packard—six foot five, a high school class president, an athlete, the kind of man David Starr Jordan had liked to imagine as the Stanford type—went to General Electric in Schenectady, New York.[772] Hewlett, who had impressed the admissions office less, went on to graduate work at Stanford and at MIT, and then came back.[773] Terman took both. The eugenic dream of the right stock and the engineering dream of the right circuit were not the same dream, but they passed through the same family: Lewis measured the boy; Fred deployed him. Terman told Hewlett and Packard to make a run for it in California instead of becoming permanent employees of a firm back east.[774]

In 1937, Giannini's Bank of America lent Walt Disney the money to finish *Snow White and the Seven Dwarfs*, and the bank stayed with the studio through *Fantasia*.[775] In the late 1930s, at Stanford, brothers Russell and Sigurd Varian worked with physicist William Hansen to build the klystron, a vacuum tube that generated high-power microwaves.[776] In 1938, Fred Terman arranged a Stanford fellowship so David Packard could return from Schenectady to work on vacuum tubes with Russell Varian.[777]

Packard took a leave from General Electric. He and his wife, Lucile, drove west with a used Sears drill press in the car.[778] In September 1938, the Packards moved into the downstairs flat at 367 Addison Avenue in Palo Alto—a house built about 1905 for Dr. John Spencer, a physician who became mayor—paying $45 a month in rent, roughly $1,100 a month today.[779] Hewlett moved into a shed behind the house, while the landlady, Ione Spencer, lived upstairs in 369.[780] They chose the house because it had a twelve-by-eighteen-foot garage in the back.[781] Lucile Packard went back to work in the Stanford registrar's office, and her paycheck was the steady money; in the evenings, she kept the books.[782]

Hewlett and Packard began part-time work in the garage with $538 in working capital—roughly $12,800 today—made up of cash plus the value of the drill press.[783] Their first real product was an audio oscillator, the Model 200A, a box that made a clean tone.[784] In 1938, a sound engineer from Walt Disney's studio heard it and asked for changes.[785] The modifications produced the Model 200B, and Disney ordered eight of them, at $71.50 each, to test the sound system for *Fantasia*.[786] On January 1, 1939, the two men formalized their partnership in the garage.[787] A coin toss decided the order of the names: Hewlett before Packard.[788] Terman had not written them a check for a million dollars—venture capital, as a profession, did not exist yet. What he wrote was permission, a fellowship, a problem to work on, and a sentence: stay here.

Total War and the Lease (1941–1945)

On December 7, 1941, Japanese planes attacked Pearl Harbor. About three weeks later, Frederick Terman was called east to Cambridge, Massachusetts, to run the Radio Research Laboratory at Harvard.[789]

The Harvard laboratory was the other half of radar: not how to see the enemy, but how to keep the enemy from seeing you.[790] While the Varians' klystron tube gave Allied radar the microwaves to see in the dark, Terman's teams worked with British scientists on "window," or "chaff"—strips of metal foil dropped from airplanes so a radar screen filled with ghosts.[791] Terman traveled to England to coordinate the teams.[792] At Normandy, Allied planes dropped enough chaff and jammed enough signals to paint ghost fleets on German screens.[793] Terman did not storm a beach; he helped decide what the other side's instruments would believe. A laboratory wearing a university's name was taking federal money to do work the public could not read—the contractor state in embryo.

In 1942, Philip Morse called the other Stanford son—William Shockley, who had grown up in Palo Alto, gone through Caltech and MIT, and joined Bell Laboratories—into anti-submarine operations research as research director.[794] Shockley solved his first problem in days. Ship-dropped depth charges were set to explode seventy-five feet down, which worked if a submarine had time to dive. The same charges, dropped from airplanes onto boats still near the surface, exploded too deep to hurt them. Shockley had the airplane charges reset to thirty-five feet. Within two months, successful

anti-submarine attacks rose by a factor of five.[795] A six-point miss on a childhood IQ test became a fivefold increase in submarines hit, because a man moved a fuse forty feet.

Seventy-four days after Pearl Harbor, on February 19, 1942, President Franklin Roosevelt signed Executive Order 9066.[796] The order let the secretary of war mark military areas and exclude any or all persons from them. It did not say the word *Japanese*. It did not have to. Lieutenant General John DeWitt drew the zones and ordered the removal, and Congress passed Public Law 503, making disobedience a crime punishable by up to a year in prison and a $5,000 fine—roughly $100,000 today.[797]

About 122,000 people of Japanese ancestry on the West Coast, most of them American citizens, were forced into guarded camps the government called relocation centers.[798] Neither the buses nor the camps distinguished carefully between *Issei*, the first generation born in Japan whom the California Alien Land Law of 1913 had already barred from owning farmland, and *Nisei*, the second generation born in the United States as citizens.[799] Not one person of Japanese ancestry in the United States was convicted of a serious act of espionage or sabotage during the war.[800] One widely repeated estimate puts the property loss—orchards, nurseries, shops sold in days—at about $400 million then, roughly $8.2 billion today [UNVERIFIED: widely repeated figure not pinned to a primary Commission on Wartime Relocation and Internment of Civilians page in the review].[801] First the law forbade ownership; then the army forced a sale; then the state called it military necessity.

In Palo Alto, local historians counted about 144 Japanese Americans.[802] In March 1942, they registered their property at the police station on Bryant Street. In April, an 8:00 p.m. curfew was imposed on them alone. Before the buses came, Arthur Okado, president of the Palo Alto Japanese American Association, published a statement in the *Palo Alto Times*: “Without reservation, we are loyal to this, our country, the United States of America” [UNVERIFIED: quoted via secondary local-history pages without re-opening the original *Palo Alto Times* issue].[803] The association had bought about $2,700 in defense bonds—roughly $55,000 today—and seven of their men were already in the Army.[804]

On May 19, DeWitt ordered Palo Alto’s Japanese Americans out.[805] On May 26, 1942, 144 people boarded buses for the Santa Anita racetrack, renamed a reception center, taking only what they could carry and no pets, while friends passed candy and oranges.[806] By the middle of June, not one Japanese American remained in Palo Alto.[807] Ten weeks after Okado’s statement, the FBI arrested him; he was later cleared and came home in February 1945.[808] Most of the Palo Altans were sent on from Santa Anita to the camp at Heart Mountain, Wyoming, including fifty-nine students from the Palo Alto schools.[809] Cherry Ishimatsu, of Palo Alto High School, later remembered the scene inside Heart Mountain: “Here we are, all prisoners behind barbed wire fences and a military guard surrounding us, and here we have young people doing a parade with their American flags flying” [UNVERIFIED: quoted via secondary local-history pages without re-opening a primary Densho transcript].[810] In Cambridge that same season, a Stanford professor helped drop metal foil so a radar set would lie, and the lie

saved lives on a beach in France; in Palo Alto, a city registered its neighbors at a police desk and put them on a bus to a racetrack. Both were the contractor state—one contract with a laboratory, the other with a general.

In 1943, Lewis Terman retired from teaching at Stanford.[811] In July 1945, as the war neared its end, Vannevar Bush delivered his report *Science, the Endless Frontier* to the president, arguing that because basic research is where practical power comes from—and because private firms will underfund it when they cannot capture all the benefit—the federal government should fund it permanently through universities, with scientists deciding where the money goes.[812]

The war in Europe had ended in May 1945; the war in the Pacific ended in August, with the surrender signed in September. As the war ended, 1945 into 1946, Frederick Terman came home to Palo Alto.[813]

He returned to a small department and a university that had spent the war lending its best people to other men's buildings.[814] MIT had the radar lab; Harvard had Terman; Berkeley had a hand in the atomic bomb. Stanford had individuals. Terman had learned in Cambridge how federal money moved, what a proposal had to look like, and how to run classified rooms where most of the staff could not discuss most of the work.[815] He had also learned that a government that has discovered science as a strategic asset does not give it back. Bush wrote the theory in July; Terman brought the practice west in his head. His diagnosis, repeated for twenty years, was that Stanford had been the wrong shape: federal money goes to places already visibly excellent in the fields the government wants, and to men the decision-makers already know.[816] Adequate at many things meant invisible when the contracts were written.

Four years after Terman returned, in 1949, A. P. Giannini died; by then the Comptroller of the Currency could call Bank of America the largest bank in the world, with more than five hundred branches and about $6 billion in assets, roughly $84 billion today.[817] In 1956, Lewis Terman died—the same year the boy his study had ranked slightly subgenius, William Shockley, shared the Nobel Prize in Physics for the transistor.[818]

When Fred Terman stepped off the train in Palo Alto at the end of the war, the land was waiting, and the land had a lock on it. On November 11, 1885, Leland Stanford's Founding Grant had told the trustees they had no power to sell or convey the farm's acres.[819] For sixty years that clause had frozen a fortune in dirt—horses, then orchards, then a war that barely touched the ground.

Terman looked at the Founding Grant's sentence and saw a door.[820] The grant said the land could not be sold. It did not say the land could not be leased. Under a lease, the university keeps the title, the tenant pays rent and builds the laboratory and hires the graduates, the appreciation stays with the landlord, and the landlord chooses the neighbors.[821] Do not sell the farm. Rent it to the future.

Part 6: The Park

01. The Yards and the Stalls (1937–1946)

July 7, 1937. The Marco Polo Bridge outside Beijing. The war that the postwar peninsula inherited did not begin at Pearl Harbor. It began in Asia, with the Second Sino-Japanese War—not the first Sino-Japanese war of 1894–1895—and the fighting had already run four years before the United States entered it.[822] Rana Mitter, a historian of China's war, puts the Chinese dead in a range of fourteen to twenty million.[823]

When the United States entered the war, the San Francisco Bay built the ships. Henry Kaiser's four yards at Richmond turned out 747 vessels, 519 of them Liberty ships, according to the National Park Service.[824] Richmond grew from a town of about twenty thousand people to more than one hundred thousand.[825] One hull, the SS *Robert E. Peary*, went from keel to launch in four days, fifteen hours, and twenty-nine minutes, from November 8 to November 12, 1942.[826] Four days for one hull. Across the water, Bechtel's partner yards and Bechtel's Pacific bases, stretching from Alameda out across the islands, were the work of a different company on the same bay.[827]

On January 3, 1942, on the island fortress of Corregidor, President Manuel Quezon of the Philippines signed Executive Order No. 1, awarding General Douglas MacArthur five hundred thousand dollars—about ten million dollars today—along with smaller sums for his aides, an order the historian Carol Petillo uncovered in 1979.[828] That spring, in Oakland, the city council stitched old streets into a single thoroughfare by council actions on March 19 and April 8, 1942, and put up the signs for MacArthur Boulevard on April 11 while the general was still in the Pacific (the Interstate 580 freeway would not receive the name until a 1968 resolution).[829]

Seventeen days after the Oakland street signs went up, on El Camino Real in San Bruno, a horse track became a prison.

The Tanforan Assembly Center operated from April 28 to October 13, 1942.[830] The track had been named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, the grantee of Rancho Buri Buri; the stalls stood on land from that Mexican grant, beside the old royal road.[831] While the buses from Palo Alto had gone south to a different racetrack at Santa Anita, Tanforan was the northern assembly center.[832] Densho, the Japanese American history project, counts 8,033 people held behind its fences.[833] Sixty-four percent of them were citizens of the United States.[834] About 3,700 people lived in horse stalls, three to six people in a stall built for one horse.[835]

The artist Miné Okubo was held there. "I was given a number and placed in a horse stall," she wrote.[836]

Fred Korematsu was held there too.[837] Down the peninsula, Stanford sent books to the camp library.[838] Converting the track had cost $1,188,000, about twenty-four million dollars today.[839] Feeding the prisoners, Densho records, cost thirty-seven cents a day for each person—about seven dollars

and sixty cents today—while the Army spent fifty cents a day to feed a soldier.[840] In the same year an American general was paid a fortune on a fortress, Bay Area families were numbered into stalls at thirty-seven cents a day. Both were the war. They were not the same experience of it.

At Stanford, the war was recorded from a shaft of stone that had just gone up. Arthur Brown Jr. had designed the Hoover Tower at 285 feet, and in 1938 Herbert Hoover had settled on six hundred thousand dollars to build it, about fourteen million dollars today.[841] Its bells came from the Belgian pavilion at the 1939 World's Fair, the great bell cast with the line: "For Peace Alone Do I Ring."[842] The university dedicated the tower on June 20, 1941, with Hoover in the room, less than six months before Pearl Harbor.[843] During the war the tower served as a listening post, recording broadcasts from the Far East.[844]

In January 1945, while the war in Europe was still on and Frederick Terman was still the man Harvard had borrowed to run its Radio Research Laboratory, Terman talked Stanford's president, Donald Tresidder, out of twenty-five thousand dollars—about $465,000 today.[845] Malcolm Harris is particular about the name Terman chose. Not a microwave laboratory. Not the Stanford microwave laboratory. The Microwave Laboratory.[846] Twenty-five thousand dollars was a drop, Harris writes, and Terman knew where the spigot was: the Office of Naval Research.[847] A university that had spent the war lending its people to other people's buildings was about to have a building of its own, aimed at the microwave tubes—the klystrons—that Russell and Sigurd Varian had already taught the military to want.[848]

That August, on V-J Day, a young Navy radio technician named Douglas Engelbart shipped out for the Philippines. On the way, he read Vannevar Bush's *Atlantic* essay, reprinted in *Life*, about a machine that could store and link human records.[849]

Across the Pacific, Douglas MacArthur began the occupation of Japan. His father, Arthur MacArthur Jr., had been military governor of the Philippines from May 5, 1900, to July 4, 1901, had clashed with William Howard Taft, and had left the landlords where they were; the son had served there as a military adviser and field marshal in the 1930s, not a civil reformer.[850] In Japan, as supreme commander, the younger MacArthur broke up the tenant farms.[851] Between November 1945 and August 1950, tenant-farmed land in Japan fell from 45.9 percent to 9.9 percent.[852] By July 1950, about 4.76 million acres had changed hands—an area on the order of six hundred Stanford farms.[853]

On March 17, 1946, at LaGuardia Field in New York, Herbert Hoover boarded a military C-54.[854] He was seventy-one years old. President Harry Truman had asked him to survey the postwar famine.[855] Hoover told reporters he had been called back like an "old family doctor."[856] Over five weeks he flew about thirty-five thousand miles across twenty-five countries.[857] In Germany he walked the ruins of the chancellery offices and the bunker; in Vienna he rode in Hitler's Mercedes.[858] That same year, the Hoover Institution was offered what it describes as "the only original portions" of the Joseph Goebbels diaries: handwritten pages from 1925 and 1926, and typed pages from 1942 and 1943 (the larger Nazi Party archive at

the Hoover Library is a microfilm copy; the paper originals went to the German Bundesarchiv).[859] A bell for peace, and the diary of the man who ran the opposite, sat in the same shaft.

In 1946, back on the peninsula, Terman and the trustees founded the Stanford Research Institute (the exact November 6, 1946 filing date of the trustees' articles of incorporation remains [UNVERIFIED]).[860] Terman wanted the institute to take the big military contracts.[861] Sally Denton's history of the Bechtel Corporation puts Steve Bechtel Sr., the builder, among SRI's founding directors.[862] Tresidder, a Hoover man, feared a university that lived on government money.[863] Harris records that Tresidder fired the institute's director rather than let one Navy contract turn SRI into a public charge, and then died before he could name a replacement.[864]

The man who had feared the government's check was gone.

02. Across the Creek (1948–1951)

San Francisquito Creek is a ditch with a county line in it. Palo Alto, in Santa Clara County, sits on the Stanford side. East Palo Alto, in San Mateo County, sits on the other, across the water and across Highway 101. The name says east; the map says north.[865]

The name itself had been a compromise. East Palo Alto's general plan records an informal election on December 12, 1925, in an area that had been called Ravenswood and Runnymede.[866] "East Palo Alto" won the vote. There was talk, even then, of annexing to Palo Alto, but the enabling law never arrived.[867] The community kept a name that sounded like a neighborhood of the rich city, and a government that was not.

After the war, the federal government, through the Federal Housing Administration, became a superbank for private mortgages.[868] The loans were designed to cost the Treasury nothing, which meant the government only backed borrowers and neighborhoods it called solid.[869] The old Home Owners' Loan Corporation maps had colored districts by investability, using red for Black neighborhoods—the practice that took the name redlining.[870] A red line on a map was a refusal of a loan, drawn in advance of the applicant.

Restrictive covenants did the same work in the deed, binding a buyer to sell later only to white buyers. In 1948, in *Shelley v. Kraemer*, the Supreme Court ruled that courts could not enforce racial covenants.[871] The FHA did not therefore start insuring Black buyers. As Harris writes, following Richard Rothstein's *The Color of Law*, the agency did not insure Black buyers under any circumstances; it brought the red line along.[872]

In the late 1940s, the Peninsula Housing Association of Palo Alto, a group of middle-class and working-class residents, bought an undeveloped 260-acre ranch near Stanford and planned a cooperative subdivision: hundreds of houses and common spaces.[873] Of the first 150 families, Harris writes, 2 percent were Black.[874] Two percent of 150 is three families.

Three families were enough. The federal insurers would not back the construction loans. Banks, including Bank of America, would not make the loans the FHA refused.[875] The association gave up and sold the land to a developer who accepted the condition: whites only.[876] The subdivision is called Ladera. It is still there. The cooperative that tried to include Black families is not.

Even the new electronics payroll could not cross the line. Art Fong, a Chinese American engineer recruited to Hewlett-Packard, came home from the war and could not find a rental in Palo Alto.[877]

In 1950, a few miles down the bay at Moffett Field, the former Navy radio technician Douglas Engelbart was working as an engineer at the Ames Aeronautical Laboratory.[878] Thinking back to the Vannevar Bush essay he had read on the troopship, he decided the rest of his life would go toward building a screen that made hard problems solvable. Pretty much everyone, Harris notes, thought the idea was nuts.[879]

On April 17, 1951, at the San Francisco airport, Douglas MacArthur came down the stairs of his plane into the floodlights.[880] Six days earlier, on April 11, President Truman had relieved him of command in Korea.[881] MacArthur had not stood in the continental United States in fourteen years.[882] Governor Earl Warren met the plane and took his arm; signs in the crowd read “MacArthur for President.”[883] The next morning, on the steps of San Francisco City Hall, MacArthur renounced any run for political office (though on July 7, 1952, he would deliver the keynote address at the Republican National Convention and allow his name to be placed in nomination before Dwight Eisenhower won on the first ballot).[884]

Five months later, on September 8, 1951, the peace of the Pacific was signed a short walk from City Hall without him.[885] At the War Memorial Opera House, Japan and forty-eight other nations signed the Treaty of San Francisco, while the Soviet Union, Poland, and Czechoslovakia refused.[886] That same day, at the Presidio, the United States and Japan signed the security treaty that kept American troops in Japan.[887] MacArthur’s name was on neither document.[888]

The Korean War was already remaking the federal budget and the Stanford campus. In 1951, the Defense Department’s budget rose from thirteen billion dollars to fifty-eight billion—in today’s money, from about $167 billion to about $747 billion.[889] While American bombers leveled cities in North Korea, Harris writes, military support for electronics research at Stanford tripled.[890] In the early 1950s, with money from the Office of Naval Research, Stanford put up the Applied Electronics Laboratory to house the postwar offspring of the wartime Radio Research Laboratory that Terman had run at Harvard.[891] Its director was Bill Rambo, one of Terman’s men from Cambridge.[892]

Meanwhile, across Highway 101 from Palo Alto, speculators found a second tool to remake the map: blockbusting. Into the 1950s, after a white resident on the eastern side of the highway sold a house to a Black family, real estate speculators went door to door warning white neighbors that the block was changing and prices would fall, urging them to sell immediately.[893] The speculators bought at fire-sale prices and resold at a premium to Black

workers starved for houses they were allowed to buy. Within six years, Rothstein and Harris record, East Palo Alto went from exclusively white to 82 percent Black.[894] Palo Alto zoned less than 10 percent of its housing land for multifamily buildings and set a minimum size for a single-family lot.[895] Density, which is how working people share a mortgage, was planned out of the city next to the jobs.

03. The Lease (1953–1955)

In 1953, along an old lumber road that William Page, a New York timber man, had cut in the early 1860s out to a mill near today's California Avenue in Mayfield, a company moved onto Stanford land under a piece of paper signed with the university (sources differ on whether the Stanford Industrial Park was authorized in 1951 before Varian Associates moved into its building in 1953; the exact 1951 board authorization date remains [UNVERIFIED]).[896] The paper was not a deed.

The Founding Grant of November 11, 1885, had forbidden the trustees to sell or convey the Palo Alto farm.[897] It did not forbid a ground lease. Under a ground lease, the university kept the title; the tenant paid rent, built a laboratory, and hired the graduates, and when the term ended the buildings and the ground remained the landlord's to lease again.[898] A sale would have been a goodbye. A lease brought the tenant back, and the deed never changes hands.

The first lease at the Stanford Industrial Park went to the research department of Varian Associates, the firm founded by Russell and Sigurd Varian, in 1953.[899] It was no surprise, Harris writes, because Frederick Terman was both an investor in Varian and a member of its board.[900] The man telling the university what its land was for also owned a piece of the tenant.

In 1954, Stanford's president, J. E. Wallace Sterling, promoted Terman to provost.[901] As the historian Rebecca Lowen noted, Sterling was sufficiently anticommunist and sufficiently anti-Roosevelt to please Herbert Hoover.[902] The man who knew how to write the defense proposal was now in charge of the campus.

After Varian, Harris writes, the park along Page Mill Road filled with companies Terman already knew.[903] Hewlett-Packard—whose founders had started in an Addison Avenue garage with $538 and a drill press—put its headquarters on the leased land.[904] General Electric came. Eastman Kodak came.[905] The park could recruit a professor the way it recruited a tenant: Carl Djerassi, a co-inventor of the birth control pill, came to teach chemistry at Stanford and brought his company, Syntex, along behind him.[906] Even the federal government took a piece of the farm's edge when Stanford ceded 87.5 acres on Miranda Avenue in 1955 for the Palo Alto Veterans Administration hospital.[907]

In 1954, Terman added a door for the people inside the park's buildings: the Honors Cooperative Program.[908] A company in the park could send its employees to Stanford for advanced classes, ahead of the ordinary admissions line, if it paid double tuition. Harris records that the companies

paid it.[909] A student finished a class in Terman's engineering school, crossed Page Mill Road, and badged into a firm that had paid double her tuition, where her professor might sit on the board or consult on weekdays. As Harris summarizes the loop: the government sponsored microwave research at Stanford, Stanford professors consulted for Varian, Varian hired Stanford students trained to make microwave tubes, and Varian sold the tubes to the government.[910]

Arthur Rock, an investment banker who arrived from New York a few years later, put the arrangement in two sentences during a 2007 interview at the Computer History Museum.

"They allowed their professors to work part-time, and this was unknown in the East," Rock said. "And also, they had all this land, and they allowed these companies to rent or lease space from them."[911]

At MIT and Princeton, in Rock's account, professors were not allowed to start companies on the side.[912] At Stanford, talent was allowed to leave the classroom on Tuesdays, and the land was allowed to leave the orchard without leaving the deed.

04. The Eight (1955–1965)

In 1955, William Shockley left Bell Laboratories to start a company.[913]

Three decades earlier, as a boy in Palo Alto, Shockley had been tested by Frederick Terman's father, the psychologist Lewis Terman, whose genius study had passed the boy over with an IQ score of 129—slightly subgenius.[914] During the war, Shockley had helped move the Navy's depth-charge fuse setting from seventy-five feet to thirty-five; after it, his treatise on electrons and holes became the semiconductor field's textbook.[915] His widowed mother still lived in Palo Alto. When Beckman Instruments of Los Angeles agreed to finance his new venture, Shockley put the Shockley Semiconductor Laboratory near her, opening in February 1956 in Mountain View, the next town down the peninsula.[916]

In 1956, Mountain View was orchards, wartime housing, and the edge of Moffett Field. There was no chip industry to join and no supply chain. The supply chain was the young physicists and engineers Shockley recruited one by one from Eastern laboratories, who boarded trains because the man who wrote the textbook had asked them.[917]

In November 1956, the Nobel Prize in Physics was awarded to Shockley, John Bardeen, and Walter Brattain for the transistor.[918] Where a vacuum tube switched current inside hot glass with a filament that burned out, a transistor switched current inside a solid crystal, and while Bardeen and Brattain remained identified with the first point-contact device, Shockley's name sat on the junction transistor that could actually be manufactured.[919]

He had the standing to recruit the best young people in the trade. He did not have the temperament to keep them. Jay Last, one of the recruits, recalled later that Shockley had no business sense at all.[920] With the Nobel Prize secured, Harris writes, Shockley's ego was at its height; he compared the young scientists to one another and to himself, and treated the staff as

lackeys.[921] Then he changed the work. Obsessed with a four-layer diode of his own design, he forbade the staff from growing the silicon transistors they had been hired to make.[922]

By the summer of 1957, seven of the young men—Julius Blank, Victor Grinich, Jean Hoerni, Eugene Kleiner, Jay Last, Gordon Moore, and Sheldon Roberts—had decided to leave together.[923] Robert Noyce, a preacher's son, joined them at the last moment to make eight; Last was twenty-seven, Kleiner a Jewish refugee of about thirty-two, and Moore the only one from Northern California.[924]

At first, Last recalled, they simply hoped an existing company would hire them as a group so they could keep making silicon transistors.[925] Eugene Kleiner wrote to his father's contacts at the Wall Street firm Hayden, Stone & Co. The letter reached a junior banker named Arthur Rock, who flew out to California with his boss, Bud Coyle.[926] Rock's advice was the one that felt foreign: do not join a company; start one.[927]

"There wasn't any venture capital really at that time," Last remembered.[928]

Rock and the group went down a list of established corporations, looking for a backer. The exact count is disputed between the participants' memories: in Rock's recollection, they approached thirty-five companies and all thirty-five turned them down, unwilling to set up a subsidiary that might upset their own employees; in Last's recollection, the list was about twenty or thirty companies, all saying no, with corporate lawyers stopping the ones that came close.[929] They were about to give up when someone suggested Sherman Fairchild, an heir, inventor, pilot, and the largest individual shareholder in IBM.[930]

Fairchild Camera and Instrument agreed to put in $1.5 million—about $17.9 million today.[931] In Rock's recollection, the eight founders took about 80 percent of the equity and Hayden, Stone took 20 percent, a split Rock later told a room of venture capitalists they could thank him for inventing.[932]

In September 1957, the eight men left Shockley.[933] A few weeks later, in October 1957, the Soviet Union launched Sputnik, and the United States suddenly wanted tough, reliable silicon transistors for a space race.[934] Shockley's business plan had just been vindicated, and his building was empty of the men who could execute it. Fairchild Semiconductor set up shop in Mountain View, in the same town as the laboratory they had left, and within a year was shipping circuits to IBM.[935]

What Shockley called them is also disputed. Harris writes that Shockley tarred the group as "the traitorous eight."[936] Shockley's wife, Emmy, later insisted her husband never used the phrase, and Jay Last agreed: "This was some newspaper guy way after the fact," Last said, noting he had not heard the term for at least ten or fifteen years after the walkout, while Rock years later slipped and said "treacherous."[937]

Whatever they were called, their walkout was legal because of a California statute written eighty-five years earlier. California Business and Professions Code Section 16600, descended from Civil Code Section 1673 of 1872 and recodified in 1941, voids any contract that restrains a person from engaging

in a lawful profession.[938] On the East Coast, along Boston's Route 128, a noncompete clause could tie a departing engineer up in court.[939] In Mountain View, eight men could walk out on a Friday, keep their houses and their children's schools, and open a competing shop down the street on Monday.

While the chipmakers settled into Mountain View, the missile builders arrived next door. In 1955, Lockheed Aircraft had laid plans to move its expanding missiles division from Los Angeles to Sunnyvale, next to the Ames wind tunnels at Moffett Field, which had been built with one of the region's rare large prewar defense contracts in 1931 under President Hoover.[940] In 1956, with prodding from Terman, Lockheed opened a research complex inside the Stanford Industrial Park and a manufacturing complex in Sunnyvale (the exact month in 1956 when Lockheed opened the facilities remains [UNVERIFIED]).[941] According to Steve Blank's history of the division, Lockheed's local missiles payroll grew from zero in 1956 to more than 28,000 employees by 1965.[942] It was a contract economy that could fill a county and empty a profession in the same decade: when a defense cycle turned in 1963 and 1964, Harris records, some 30,000 California engineers were laid off statewide.[943]

Inside the university, in 1959, the trustees set the Hoover Institution apart as an independent center within Stanford, under Herbert Hoover's charge that its purpose was to demonstrate that private enterprise could best serve the people.[944] On January 17, 1961, leaving the presidency, Dwight Eisenhower went before the cameras to name the machinery that now paid the bills on both sides of El Camino Real.[945]

"In the councils of government, we must guard against the acquisition of unwarranted influence, whether sought or unsought, by the military-industrial complex," Eisenhower said.[946]

By 1967, as Rebecca Lowen and Malcolm Harris record, Stanford trailed only MIT and Caltech among American universities in total defense contracts.[947]

05. The Demo and the Contract (1959–1968)

Around the turn of the 1960s—published accounts split the date between 1959 and 1961 [DISPUTED]—a Stanford creative writing fellow named Ken Kesey took a night job and volunteered for a government drug study at the Menlo Park Veterans Administration hospital on Willow Road.[948] He was paid about seventy-five dollars a session, on the order of eight hundred dollars today.[949] Historians tie the Willow Road trials to Central Intelligence Agency funding, though careful accounts do not file those trials under the name MKUltra, and the Willow Road facility was distinct from the Palo Alto Veterans Administration hospital on Miranda Avenue, for which Stanford had ceded 87.5 acres of university land in 1955.[950]

A few miles away, at the Stanford Research Institute in Menlo Park, Douglas Engelbart was building his screen. A former Navy radio technician who had shipped out for the Philippines on V-J Day—reading Vannevar Bush's *Atlantic* essay, reprinted in *Life*, on the voyage—Engelbart had worked as an engineer at the Ames Aeronautical Laboratory at Moffett Field

in 1950 before deciding to spend his career on a display that could help human beings solve complex problems.[951] At the time, almost everyone thought the idea was “nuts.”[952] SRI hired him, and a small Air Force grant allowed him to write down his framework for augmenting the human intellect, joined by roughly $80,000 from NASA for his NLS (oNLine System) work and subsequent backing from J. C. R. Licklider and Bob Taylor at the Pentagon’s Advanced Research Projects Agency (ARPA).[953] In 1964, at SRI, Engelbart and Bill English built the prototype of a wooden handheld pointing device they called a mouse.[954]

Out in the surrounding towns, the two worlds kept crossing the same streets. On May 5, 1965, a band called the Warlocks played at Magoo’s Pizza in Menlo Park; on December 4, 1965, at an Acid Test in San Jose, the same musicians played for the first time under the name the Grateful Dead, and the following month, in January 1966, a Stanford-educated biologist and photographer named Stewart Brand co-produced the Trips Festival in San Francisco.[955] At SRI, meanwhile, a veteran of the wartime Office of Strategic Services named Al Hubbard turned up at an institute project called Alternative Futures, wearing khakis, a badge, and a pistol, guiding LSD trips and gathering opposition data on the New Left—though as Malcolm Harris notes, drawing on historians Martin A. Lee and Bruce Shlain, Hubbard was not a CIA operative as such.[956] Despite later theories, mostly written about Laurel Canyon in Los Angeles, that intelligence agencies manufactured the counterculture, nobody has shown that the Grateful Dead were a government project.[957] On the peninsula, the free-love surface and the Pentagon payroll simply shared a zip code.

Dual-use meant one letterhead and two products. In 1965, according to SRI’s own institutional history, the Stanford campus first learned that the institute held two chemical-warfare contracts; in 1967, students learned of SRI contracts tied to counterinsurgency in Southeast Asia and began picketing.[958] SRI maintained that the chemical work was almost exclusively defensive and that many of the protesters’ demands were unfounded; the students answered that “defensive” did not excuse the work, and that the university’s trustees owned the institute.[959] In 1967, too, the Chicago economist Milton Friedman, who would later make the Hoover Institution his home, first met California’s new governor, Ronald Reagan.[960] By 1968, Harris writes, SRI had helped develop the CIA’s strategic-hamlet plan in Vietnam, and its counterinsurgency work ran into the millions of dollars.[961] (Inside the Pentagon’s wider Vietnam apparatus, as historian Sharon Weinberger later documented, ARPA’s William Godel and his Project Agile team had also laid early groundwork in the summer of 1961 for the strategic-hamlet program and funded counterinsurgency field studies across Vietnam and Thailand.[962])

During the academic year of 1968, inside the university’s finance office, another mechanism was quietly taking shape. Since January 1, 1956, Stanford had handled faculty inventions through an outside agreement with the nonprofit Research Corporation, under which an inventor kept $100 plus 16.66 percent of gross receipts and Stanford split the net fifty-fifty.[963] In June and October 1968, a contract administrator named Niels Reimers and his colleague Frank Newman began drafting notes for an in-house patent licensing program.[964] On July 19, 1968, Newman wrote to Kenneth M. Cuthbertson, Stanford’s vice president for finance, proposing that $15,000

be set aside for outside attorneys' fees to launch a pilot office that would split net royalties one-third to the inventor, one-third to the academic department, and one-third to the university: "We can operate on a low-risk yet effective basis. . . . The upside chance for gain may be considerable."[965]

On December 9, 1968, in Brooks Hall in San Francisco, Engelbart walked onto the stage of the Fall Joint Computer Conference.[966]

The name "Mother of All Demos" came later.[967] On the day, according to the Smithsonian's account, he had asked for the slot in March and been given ninety minutes in a hall that held two thousand people.[968] He said later he was "nervous as hell."[969] A truck had hauled his gear over Skyline Boulevard from Menlo Park, and a projector borrowed from NASA threw his screen onto the wall behind him.[970] The stage lights were so bright he could not see the audience.[971] His console was linked over telephone lines to a computer thirty miles south at SRI, and to colleagues in Menlo Park whose faces joined the text on the screen.[972] With the mouse under his hand, he typed a word, typed it again, and showed a room of engineers accustomed to paper punch cards that text could be pointed at, edited, copied, and rearranged in real time.[973]

"If in your office, you, as an intellectual worker, were supplied with a computer display backed up by a computer that was alive for you all day, and was instantly responsive to every action you have, how much value could you derive from that?" Engelbart asked the hall.[974]

Behind one of the cameras stood Stewart Brand, whom Engelbart had brought onto the stage crew; Bill Paxton, one of Engelbart's engineers, remembered the video feedback loop on the big screen as psychedelic.[975]

"I've never seen anything that was so dangerous, and undertaken with such bravado," Brand said later.[976]

Harris puts the cost of staging that single day at $175,000, about $1.7 million today.[977] The Defense Department's Advanced Research Projects Agency and the Air Force paid the bill; an ARPA contract cover from the following spring lists an order of about $1.5 million—roughly $14.5 million today—for the study of computer-augmented management.[978] In Menlo Park in 1968, the landlord of the mouse and the landlord of the counterinsurgency study shared a letterhead.

Four months later, the students stopped asking from the sidewalk.

06. April Third (1969–1973)

On April 3, 1969, Stanford students met on campus and took the calendar date for their name.[979] According to the university news service's report the next day, they voted to demand an end to all classified research, all research in chemical and biological warfare and counterinsurgency, and all studies used against the people of Vietnam, Laos, and Thailand, calling on the Board of Trustees to halt the work at the Stanford Research Institute when the board met on April 8.[980]

By late 1968, SRI's history records, many campus critics had wanted the university to cut SRI loose, only to realize that an independent institute would simply keep the military contracts with nobody left on campus to answer for them.[981] The students now demanded closer university control to abolish the war work, while SRI's president, Charles Anderson, and much of his staff wanted independence so no campus meeting could tell them which contracts were moral.[982]

On the evening of April 8, 1969, students walked into the Applied Electronics Laboratory and occupied it.[983] On April 10, students packing Dinkelspiel Auditorium voted the trustees' response unacceptable and endorsed the sit-in.[984]

The Applied Electronics Laboratory was Frederick Terman's building—a postwar campus version of the Radio Research Laboratory he had run at Harvard, built in the early 1950s with the Office of Naval Research to house the offspring of wartime electronics.[985] Bill Rambo, the laboratory's director and one of Terman's veterans from Cambridge, protested that the building had a sinister reputation that was not fully deserved, and that most of its rooms were freely accessible.[986] The students were not interested in the doors; they were interested in the contracts. They held the building until April 18—nine days in historian Leslie Berlin's count, "more than a week" in Malcolm Harris's phrase [DISPUTED].[987] Using the laboratory's own presses and hundreds of thousands of its blank sheets, they printed a daily newsletter: the first issue, on April 10, was titled *Unclassified*, and the issues that followed were titled *Declassified*.[988] (One early flyer misnamed the building the "Atomic Energy Lab," an accidental joke on a campus where atomic energy was one of the few military sciences Terman had not led.[989])

On April 14, when faculty and staff walked through the corridors to identify students for disciplinary hearings, every demonstrator gave the same name: "The April Third Movement."[990] Stanford's new president, Kenneth Pitzer, in his first year in office, sent written notice that the occupiers were violating university policy by stopping the ordinary work of the lab, though unlike a subsequent sit-in at Encina Hall at the start of May, he did not call the sheriff to clear the laboratory.[991] On April 15, after an internal debate, the occupiers reaffirmed that they would not tamper with classified files; they had come to stop the work, not to steal it.[992]

Inside the campus argument over whether the electronics work was merely "defensive," the Stanford English professor H. Bruce Franklin answered a provost in words Harris records:

"Those are electronic countermeasures, used by our bombers to jam enemy radar. If you think that's defensive, you're out of your mind."[993]

In the occupation newsletter, a song written in the voice of an institute researcher put the distance between the bench and the war into four lines:

"I have never killed a man. I have never seen him die. These formulas: they could be for jelly. I don't think I know."[994]

On April 17, the day before the students walked out of the laboratory, one occupier told the university news service where the circuit led next:

“If we are successful here the next step will be to move to the Stanford Industrial Park.”[995]

A poster from those weeks called for a “Strike Against the Stanford Empire.”[996] On Friday, May 16, 1969—the morning after police shot demonstrators at People’s Park in Berkeley, where 482 marchers were arrested in a single day and nearly a thousand between May 15 and May 24—a flyer went out at Stanford under the banner “SMASH WAR RESEARCH.”[997] It printed a map linking the Stanford Research Institute, the Hoover Institution, the Graduate School of Business, Varian, Hewlett-Packard, Lockheed, and IBM, captioned: “Since World War II—How to Run an Empire from Palo Alto.”[998] That morning, about three hundred protesters blocked the entrance to the Stanford Industrial Park at Oregon Expressway and Hanover Street, parking a school bus across the road and letting the air out of its tires.[999] About one hundred fifty police officers met them.[1000] Around eleven o’clock in the morning, the police fired tear gas and made fifteen arrests.[1001]

That same day, May 16, 1969, Provost Richard W. Lyman sent a memorandum to department chairmen and principal investigators regarding the university’s new patent licensing program; five days later, on May 21, Niels Reimers sent chemistry professor William S. Johnson a royalty distribution notice on a patent application for the “Synthesis of Juvenile Hormones.”[1002] By the start of 1970, Reimers had opened Stanford’s Office of Technology Licensing, charging a flat 15 percent administrative fee against gross royalty income to turn campus laboratories into a self-funding patent engine.[1003]

Meanwhile, in the autumn of 1969, the Pentagon’s network went live on the peninsula: the first segment of the ARPANET connected Leonard Kleinrock’s laboratory at UCLA to Douglas Engelbart’s group at the Stanford Research Institute, uttering its first two-letter transmission—“Lo”—before the system crashed.[1004] And before the year was out, on December 12, 1969, two young filmmakers named Francis Ford Coppola and George Lucas opened a company called American Zoetrope in a warehouse on Folsom Street in San Francisco, a northern outpost of the new screen culture that would soon meet the peninsula’s engineers.[1005]

At Stanford, the trustees answered the occupiers not by stopping the contracts, but by severing the link. On March 31, 1970, according to SRI’s institutional history and historian Stuart W. Leslie’s account, the Stanford Research Institute was formally separated from the university—sold, in effect, to itself.[1006] (A campus study committee the previous spring had debated valuations of the institute ranging from a book value of around $25 million to estimates of $45 million to $58 million [UNVERIFIED].) Under the divestiture agreement, SRI agreed to pay Stanford about 1 percent of its gross receipts until $25 million had been paid, and half a percent thereafter, while retaining the right to use the Stanford name for a term of years.[1007] By 1989, more than $25 million had gone back to the university; SRI staff took to calling the annual payment “alimony,” and in some years it exceeded the institute’s net profit.[1008] Campus ROTC programs followed the institute out the door: Air Force ROTC at Stanford was terminated in June 1971, and the Army and Navy ROTC programs concluded in June 1973.[1009]

The morning after March 31, 1970, the same scientists sat at the same desks in Menlo Park, working on the same contracts. A student could no longer point at SRI and tell the trustees, *You own this.* The trustees had traded governance for a royalty. Down on Page Mill Road, Varian, Hewlett-Packard, and Lockheed remained tenants on the leased orchard, and the Honors Cooperative Program still collected double tuition.[1010]

The rebellion did not stop the machine. It made the machine look less like a weapons lab.

Three years later, in 1973, the Hoover economist Milton Friedman spent a day in a small private plane hopping across California with Governor Ronald Reagan, barnstorming for Proposition 1, a ballot measure to cap state spending in the California Constitution.[1011] The proposition narrowly lost, but by the last stop of the flight Friedman had made up his mind about the man beside him.[1012]

"I was able to give an enthusiastic yes to a reporter's question as to whether I would support Reagan for president," Friedman recalled [UNVERIFIED].[1013]

Part 7: The Road

1969–2002

The Ticket Booth (1969–1974)

December 12, 1969. San Francisco. In a warehouse at 827 Folsom Street, south of Market, Francis Ford Coppola and George Lucas open a film company and call it American Zoetrope.[1014]

The word is Greek before it is English. *Zoe*, life. *Tropos*, a turn. A zoetrope is a nineteenth-century toy, a drum with slits, spun so that a strip of still drawings looks alive. They were naming a machine that turns still pictures into a world.[1015] They had driven up from Los Angeles with a crew of film-school friends, among them the editor and sound designer Walter Murch. None of them, Murch said, had family in the industry [UNVERIFIED].[1016] They wanted American movies made under the influence of the films they loved from everywhere else.

Down the Peninsula, in the hills above the Stanford Quad, an economist named George Shultz had just finished a fellowship year. He had taught at M.I.T. and then at the University of Chicago, where he became dean of the business school. In 1968 and 1969 he was a fellow at the Center for Advanced Study in the Behavioral Sciences on the Stanford campus. Then Richard Nixon put him in the cabinet three times: secretary of labor, the first director of the Office of Management and Budget, and secretary of the Treasury.[1017]

Down on the flat of the campus, the university was building a different kind of office. Since January 1, 1956, Stanford had relied on an outside agreement with the Research Corporation to handle campus patents, under

which an inventor received $100 and 16.66 percent of gross receipts while Stanford took half the net.[1018] The story of the university's own operation begins in the summer of 1968 with a man named Niels Reimers and an administrator named Frank Newman. On July 19, 1968, Newman wrote to Stanford's vice president for finance, Kenneth M. Cuthbertson, proposing that $15,000 be set aside for outside attorneys' fees to run a low-risk pilot program: take what a professor invents, file a patent, and license the patent to a company in exchange for a royalty, splitting the net proceeds one-third to the inventor, one-third to the department, and one-third to the university ("the upside chance for gain may be considerable," Newman wrote).[1019] During the pilot period, Reimers and Newman charged a flat 15 percent administrative fee, plus out-of-pocket expenses, against gross royalty income to cover costs, before Newman bowed out at the end of 1969 to work on federal education policy.[1020] In the program's first eighteen months, when only five inventions had been licensed, Reimers reported that three petitions to federal sponsors—two to the Department of Health, Education, and Welfare and one to the National Science Foundation—for title to campus inventions were initially denied before being granted after meetings with agency officials in Washington.[1021] By the summer of 1970, Stanford made the Office of Technology Licensing permanent.[1022] Reimers and an assistant, Sally Hines (joined in 1971 by an associate, John Poitras), worked out of a trailer behind Encina Hall decorated with ferns and memorabilia.[1023] On the wall, in capital letters, a sign said MAKE IT HAPPEN. On Friday afternoons they filed the week's paper over popcorn and a beer called Henry Weinhard's. They called the ritual Management by Beer.[1024]

A patent is a temporary monopoly. The government gives the holder the exclusive right to an invention for a term of years, and in exchange the inventor publishes how it works, so that when the term ends anyone can use it. A license is the permission slip. The university keeps the patent and rents the right to use it. A royalty is the rent, usually a percentage of sales. Equity is different. Equity is a slice of the company itself.

On January 11, 1971, six months after Reimers's office became permanent, a weekly trade paper called *Electronic News* landed on desks from Boston to Sunnyvale. The headline, over a series by a columnist named Don C. Hoefler, read "Silicon Valley U.S.A."[1025] It was the first time those two words appeared in print as the name of a place. Hoefler was writing about semiconductors, the chips cut from silicon, the second most common element in the earth's crust. He was not writing about social networks or search engines. Those did not exist. He was naming a factory district. The name outlived the factories, though it caught on slowly: as late as December 1979, local papers on the Peninsula still called it "the Santa Clara Valley electronics industry," and the phrase "Silicon Valley" did not appear in the *New York Times* until November 11, 1975.[1026]

In 1972, north in the city, Coppola and Lucas's American Zoetrope moved into the Sentinel Building at 916 Kearny Street in North Beach, the old Italian neighborhood where A. P. Giannini had opened a bank in a remodeled saloon in 1904 [UNVERIFIED].[1027] By November 15, 1972, Stanford's *Campus Report* noted that Reimers's technology licensing program had earned over $200,000, most of it from upfront license fees rather than royalties on marketed goods.[1028]

That same year, in Menlo Park, a partnership opened its doors on a suburban road running west from the freeway, past the edge of the university, and up toward the coastal hills. The buildings on Sand Hill Road were low. They sat back among oak trees, with more parking than a visitor could explain. There was nothing on the road that looked like power. That is the point of a gate that wants to be mistaken for geography.

At 3000 Sand Hill Road, the partners were Eugene Kleiner and Tom Perkins. Kleiner had a private office in the building before the firm even had a name. The partnership began in 1972 with an $8 million fund—about $64 million in today's money—one of the largest in a business that was still small enough to fit in a handful of rooms.[1029] Half of it came from one man, Henry Hillman, a Pittsburgh industrialist. The other half came from individuals, trust funds, and institutions, among them an insurance company and Rockefeller University.[1030]

A gatekeeper is not the person who does the work. A gatekeeper is the person who decides who is allowed to pass. A century earlier, the railroad through this same valley decided which towns could ship their wheat. The charge was called a freight rate, and the men who set it said it was whatever the traffic would bear. Sand Hill Road was about to do a quieter version of the same job. The thing being moved was no longer wheat. It was a company that did not exist yet.

Kleiner was not a banker. He was one of the eight young researchers who had walked out of William Shockley's laboratory in 1957 and started Fairchild Semiconductor. As a boy in Vienna he had watched Nazis force his father to open the family safe. A police officer who knew the family pulled the father out of a line bound for the camps. Kleiner never finished high school. He volunteered for the American army, he said, because he had a personal thing to settle, and the G.I. Bill put him through engineering school.[1031] Perkins was different. He had started a laser firm acquired by Spectra-Physics and made Hewlett-Packard's computer division the biggest in that company, and he had the polish of a man who had studied how a businessman was supposed to look. At Harvard Business School he had sat in the class of Georges Doriot, the French professor who taught a generation of Americans that venture investing was a craft.[1032]

The same year, 1972, Don Valentine founded a firm called Sequoia Capital. Valentine had worked at Fairchild for eight years before leaving in 1967 with Charlie Sporck and Floyd Kvamme to revive National Semiconductor.[1033] Sequoia moved onto Sand Hill Road while other people's money was still learning the address.[1034] The operators had become the financiers. They already knew the people worth funding, because they had been those people's bosses.

Also in 1972, Nolan Bushnell, Ted Dabney, and an engineer named Al Alcorn—who had worked at Ampex in Redwood City andNutting Associates at 500 Logue Avenue—launched a video-game company called Atari.[1035] Using a circuit of fewer than twenty fifty-cent chips to move a spot of light across a television screen, Alcorn built a table-tennis game called *Pong* that taught a generation to lose quarters in a bar.[1036] By 1973 Atari had sold nearly 7,000 *Pong* cabinets (while total volume including copies hit 45,000 units), and Valentine took 12,500 shares of Atari stock in exchange for advising the company on organization and marketing and helping it raise $2 million.[1037]

Then Perkins did something that complicates the postcard. Shortly before a young associate named Robert Swanson joined the firm, Kleiner and Perkins left 3000 Sand Hill Road. They moved to the twenty-ninth floor of Two Embarcadero Center in San Francisco, looking out at the bay. Perkins did not want the neighbors to see whom he took to lunch.

"We didn't want to be considered part of the flock," Perkins said. "Eagles don't flock, was our joke."[1038]

The joke lost. The other firms stayed on the road. Within a decade the low buildings among the oaks were the largest concentration of venture capitalists in the valley, and, as Leslie Berlin wrote, perhaps in the world.[1039] You can move your own office to a skyscraper. You cannot move the gate, once everyone else has agreed that the gate is a road.

In 1974 Congress passed a pension law with a stern name, the Employee Retirement Income Security Act, and a sterner idea inside it: a prudent person does not gamble with workers' retirement. Fund managers read that as a ban on anything speculative. Venture capital is speculative by design. Most of the companies fail.[1040]

That same year, George Shultz left the Treasury Department and went to Bechtel. Bechtel was a San Francisco engineering and construction company, private, vast, and in the habit of building the things governments decided to want: pipelines, power plants, cities in the desert. Shultz became an executive, and then president. While he was there, he joined the Stanford faculty part time and taught management and public policy. Shultz and his wife lived in a modest house on the campus, and he drove an Oldsmobile when his rank entitled him to a Cadillac.[1041] The professorial manner was not a costume he put on later. His colleagues told the historian Laton McCartney that in a company where engineers shouted, George spoke beautifully—logical, calm, the collected professor.[1042] It worked on them.

Back in the trailer behind Encina Hall, the pilot program was now a four-year-old official operation called the Office of Technology Licensing.[1043] On May 20, 1974, the *New York Times* ran a story by Victor K. McElheny under the headline "Animal Gene Shifted to Bacteria; Aid Seen to Medicine and Farm."[1044] That summer, Niels Reimers sat at his desk going through the mail.

Patenting Life (1974–1980)

Two scientists had done something that would split the century. In 1973 Stanley Cohen at Stanford and Herbert Boyer at the University of California, San Francisco—together with Annie C. Y. Chang and Robert B. Helling—had published a paper in the *Proceedings of the National Academy of Sciences* showing how to cut a gene out of one organism and stitch it into a bacterial plasmid in the DNA of another.[1045] DNA is the molecule that carries heredity, the long instruction sheet inside a cell. Their method let a person rewrite a line of that sheet. In the summer of 1974, Reimers decided the method should be patented, and he began the fight to file.[1046] On July 29, 1974, after speaking with Cohen, Reimers noted for his file: "Herb Boyer is willing to cooperate, also does not want personal gain."[1047]

The fight started immediately. The plan was criticized for its potential to undermine the university's basic mission, which was to increase the world's stock of knowledge, not to fence it.[1048] Faculty said the claim reached too far. Even Stanford biochemist Paul Berg, whose own work with Janet Mertz had helped lay the foundation for gene splicing, questioned the move at a campus meeting with Reimers, Cohen, and Joshua Lederberg.[1049] Reimers hired patent attorney Bertram Rowland of Flehr, Hohbach, Test, Albritton & Herbert to draft the parent application—filed on November 4, 1974—and negotiated an inter-institutional agreement with the University of California, signed August 29, 1975, to split the royalties.[1050] Meanwhile, the press ran nightmares. In February 1975, 150 scientists—organized in part by molecular biologist Maxine Singer—met at Asilomar, on the Monterey coast, and agreed to regulate themselves.[1051] The gesture advertised the fear instead of calming it. In June 1976, the mayor of Cambridge, Massachusetts, Alfred E. Vellucci, declared "Test Tube Laboratory Fine, but Build It Somewhere Else," before asking the National Academy of Sciences in May 1977 whether a strange orange-eyed creature supposedly sighted nearby could be connected to recombinant DNA experiments—and vowing he would be damned if his city was going to be diseased over academic freedom.[1052]

In 1975, at Kleiner Perkins, Robert Swanson lost his job. Swanson was twenty-eight. He had talked his way through an M.I.T. chemistry degree and a management degree in five years, opened a Citicorp venture office in San Francisco, and been hired by Eugene Kleiner, who saw a version of his own nerve.[1053] Swanson lasted less than two years. The break came over a company called Cetus, an early firm where Swanson served on the board. At a long lunch, a Nobel laureate mentioned recombinant DNA as something Cetus might explore. Nobody else at the table wanted it, including Tom Perkins, who had come to the lunch to show how worried the firm was. Swanson still thought he had a job. Perkins told him the partners would like to have just the two of them working there. Swanson could stay on the payroll until the end of 1975.[1054]

"Maybe I'm not very good," Swanson said quietly to himself.[1055]

He had time, a Ping-Pong table that doubled as a desk, and a stack of cheap meals. He cold-called the scientists who had been at Asilomar. He had never heard of Cohen, Boyer, or Reimers. Boyer, unusually for a biologist in 1976, was willing to talk about a company. His cats were named Watson and Crick.[1056] Most of his colleagues would not have taken the call on a university phone. Arthur Levinson, who would later run Genentech and sit on the boards of Apple and Google, remembered that as a young biochemist he walked down the street to a pay phone if he needed to speak to a company.[1057] A venture capitalist named Brook Byers—who joined Kleiner Perkins after Swanson left—compared a scientist thinking about business to Bob Dylan going electric. People wondered what was up, and whether bad would come of it.[1058]

Swanson asked Kleiner and Perkins to pay him a salary while he incubated the company, the way they had done for the founders of Tandem Computers. They said there was no room. The fund was barely making money. So in 1976 Swanson and Boyer started it themselves, drawing up a six-page "Genentech Outline for Discussion" for Kleiner & Perkins on April 1, 1976, and writing to Stanford and the University of California on April 19 to

inquire about a license.[1059] Swanson's first proposed name was Herbob, a mash of Herbert and Bob. Boyer countered with a word built from genetic engineering technology: Genentech.[1060]

Across the valley, another company was forming. Apple had been started in 1976 by Steve Jobs—who had worked at Atari from September 1974 through December 1975—and Steve Wozniak, who had demonstrated his circuits at meetings of the Homebrew Computer Club in Menlo Park.[1061] It was saved as a business later that year by Mike Markkula, a former Fairchild and Intel marketing manager who had retired as a multimillionaire at thirty-two after his Intel stock options split and rose from a $6.22 strike price to about $45 a share.[1062] Markkula helped write Apple's business plan in November 1976, wrote the checks, brought in former National Semiconductor executive Mike Scott as president, hired Regis McKenna's agency to design the rainbow-striped bitten-apple logo, and cared about how the thing looked.[1063]

The money that made Markkula's instructions matter came, in part, from the East. In the fall of 1977 Markkula and Jobs flew overnight to New York. They changed into new blue suits in a men's room at 30 Rockefeller Plaza and rode the elevator to the fifty-sixth floor, to the offices of Venrock, the venture firm of the Rockefeller family.[1064] The partner they had to convince was Peter Crisp. Jobs talked about a computer in every living room. The partners, by their own later account, barely heard the specifics. The whole territory was too speculative to take literally. They stepped into the hall.

"We went out in the hall, four or five of us," Crisp recalled, "we looked at each other, and we shrugged our shoulders and said, What the hell?"[1065]

They put in $300,000—about $1.7 million in today's money—for 10 percent of Apple. The deal valued the company at $3 million, or about $16 million today.[1066] Crisp later said people gave Venrock too much credit for being smart. Arthur Rock, the financier who had backed Intel, took a $50,000 slice of Venrock's own allocation, because Crisp owed him a favor from 1968.[1067]

By the end of 1977, back at Stanford, Niels Reimers's clipping file on the recombinant DNA fight was three inches thick. In March 1977 protesters led by Jeremy Rifkin's People's Business Commission had disrupted a National Academy of Sciences forum in Washington, chanting against patents on life.[1068] Reimers kept going. He kept thinking, he said later, about the potential for good: drugs, hormones, vaccines that did not depend on a weakened virus.[1069]

In 1978 Congress cut the tax on capital gains and eased the pension rules. On January 4, 1979, after years of argument and three earlier filings, Bertram Rowland filed Continuation-in-Part Application Serial No. 1,000,021—the patent application that would finally issue for Cohen and Boyer's method—with the United States Patent and Trademark Office.[1070] That same year, George Lucas's company, Lucasfilm—which he had built after leaving the Folsom Street experiment—started a computer division to drag high technology into movies.[1071]

And in 1979 the Labor Department clarified the pension point that mattered. Under ERISA, prudence would be judged on the whole portfolio, not on each wild bet inside it. A pension fund could put a small slice into companies that might go to zero. Within a year, venture capitalists had more than half a billion new dollars to invest—about $2.3 billion in today's money.[1072] The retirement savings of teachers, truck drivers, and clerks had become the raw material of a road in Menlo Park. The beneficiaries were not asked.

In December 1979, at Apple, Mike Markkula wrote a one-page creed called "The Apple Marketing Philosophy." It had three words at the top: Empathy, Focus, Impute. Under the third, he wrote: "Our image is the combined result of everything the customer sees, hears, or feels from Apple."[1073]

It is also a description of a university. The sandstone, the red tile, the church at the center of the Quad, the tour guide walking backward: those are an image, and the image is doing work. Markkula called the work "impute." If a company presents a product in a slipshod way, he wrote, people will perceive it as slipshod. If it presents it in a creative, professional way, it imputes the qualities it wants them to feel.[1074] You do not argue them into it. You surround them.

Catholic California had put saints on the land: San Francisco, Santa Clara, the missions, the royal road. The fortunes that arrived after the gold rush were largely Protestant, and then, in the late twentieth century, frankly secular. They did not build missions. They built companies that asked for the feelings a mission used to ask for: belief, pilgrimage, a story about the future in which the customer is already late.

By 1980, at Atari, the memo had won in a different way. In 1976 Bushnell and Valentine had sold Atari to Steve Ross's Warner Communications for $28 million, and by the end of 1978 Warner had installed a former Burlington Industries textile executive named Ray Kassar as chief executive and forced Bushnell out.[1075] In 1980 Al Alcorn wanted to build a new kind of game system, one that would throw a hologram behind the game. Kassar told him to talk to marketing. The marketing manager told him to write a business plan.

"We didn't need a business plan for Pong or the VCS," Alcorn said.[1076]

He needed one now. The Video Computer System—engineered around a custom chip code-named Stella—was the box that had brought Atari's games into the living room, capturing over 80 percent of the home video-game market.[1077] Both *Pong* and the VCS had been built by people who shipped first and wrote the memo second. Kassar nodded at prototypes and released almost none of them. After four key programmers left to start Activision because Kassar refused to give them royalties or screen credit, Atari's cartridge releases for the 2600 dropped from thirteen in 1978 and twelve in 1979 to six in 1980.[1078] Alcorn's hologram machine, called Cosmos, drew thousands of orders at the January 1981 Consumer Electronics Show and was never sold. He had four words for what had happened to the company: all creativity had ceased.[1079]

On October 14, 1980—four months after the Supreme Court ruled in *Diamond v. Chakrabarty* that a genetically modified living organism could be patented—Genentech, underwritten by Hambrecht & Quist, began

trading on the NASDAQ exchange under the symbol GENE.[1080] The opening price was $35 a share. In the first twenty minutes it hit $89, the fastest first-day gain Wall Street had seen. At the close the company was valued at $532 million, about $2.2 billion in today's money.[1081] Swanson and Boyer had each put in $500 at the start. Each of those stakes was worth about $65 million by the closing bell. The slice Kleiner and Perkins had eventually bought, for $100,000 and then another $100,000, was worth the same.[1082] The firm that had fired Swanson was now rich because of him. Two months later, on December 12, 1980, Apple Computer went public as well.[1083]

Seven weeks after Genentech's offering, on December 2, 1980, the United States Patent and Trademark Office issued Patent 4,237,224, "Process for producing biologically functional molecular chimeras."[1084] A chimera, in the old stories, is an animal sewn together from different animals. In the patent, it was Cohen and Boyer's method. Stanford and the University of California owned it.

Ten days after the patent issued, on December 12, 1980, President Jimmy Carter signed the Patent and Trademark Law Amendments Act (Public Law 96-517).[1085] Everyone called it Bayh-Dole, after the two senators, Birch Bayh of Indiana and Bob Dole of Kansas. If a university, a small business, or a nonprofit made an invention using federal research money, the institution could elect to own the patent. Before this, the federal agency that had paid for the work generally kept the rights, and a great many inventions sat on a shelf because a company would not spend a decade turning an idea into a product it was not allowed to own. Bayh-Dole flipped the default. The university took title. The university had to file for the patent, had to try to license it, had to share the royalties with the human inventor, and had to give the government a license to use the invention for government purposes. The taxpayer funded the science. The university owned the paper. A company rented the paper and built the product.

Stanford had not waited for the law. Reimers had been running the office for a decade, and working alongside Norman J. Latker, the patent branch chief at the Department of Health, Education, and Welfare, he had already fought for Institutional Patent Agreements and the right to hold title on particular inventions.[1086] As the historian Doogab Yi later summarized the argument Reimers and his allies made: "They claimed that the 'public' ownership of inventions was fundamentally at odds with the promotion of the public interest through development and licensing."[1087] Bayh-Dole made the Stanford arrangement the national rule for every campus that took federal money. A trailer behind Encina Hall had become a clause in the United States Code.

The Bechtel Cabinet and the Myth Machines (1981–1995)

In 1981 Caspar Weinberger, who had served in Nixon's cabinet and then as a Bechtel executive, became Ronald Reagan's secretary of defense. In June 1982 Reagan asked George Shultz to be secretary of state. He was sworn in on July 16.[1088]

Senators asked, at Shultz's confirmation, whether a man who had just run Bechtel could conduct the foreign policy of a country in which Bechtel had business. He was confirmed anyway. Sally Denton, writing later about the company's nuclear work, recorded a line from those hearings:

"Private interests have no business with nuclear weapons," Shultz said.[1089]

Denton set the sentence down in order to watch him walk away from it years afterward, when the laboratories that design those weapons were being handed to private managers and Bechtel was in line.[1090] A revolving door is simpler than the scandal sheets make it. A revolving door is a career that passes through a public office and a private firm that does business with that office, in either order, more than once. The person does not have to fix a contract on the way through. The knowledge, the phone numbers, and the benefit of the doubt travel with the body. Two of the offices that buy the largest things a government buys, diplomacy and weapons, were now held by men who had just been signing Bechtel's paper. At the Bohemian Grove, a private encampment in the redwoods north of the city, both the Bechtels and Shultz were regulars.[1091] The network fit in a camp, a campus, and a company. It did not need a conspiracy. It needed a small enough number of dining rooms.

Back on the campus, in August 1981, Reimers placed advertisements in *Science*, *Nature*, and the *Wall Street Journal* offering nonexclusive licenses to the Cohen-Boyer patent for a $10,000 fee and royalties as low as 0.5 to 1 percent on end products, provided companies signed by December 1, 1981—retaining the Washington patent firm Finnegan, Henderson, Farabow & Garrett (which he privately nicknamed "Bruno, Boris, and Brunhilde") and sweating until the eleventh hour before Genentech signed.[1092] In 1983, Stanford earned more from technology licensing than any other university in the country: $2.5 million, about $8 million in today's money. Almost 60 percent of that year's income came from the recombinant DNA patent.[1093] When Reimers had started, there were a handful of such offices in the country. After the chimera patent and Bayh-Dole, they were simply how a university did business.

Yet the office was still missing the biggest upside when a campus invention turned into a company. In March 1980, a Stanford graduate student named Andy Bechtolsheim—who had worked as a no-fee consultant at Xerox's Palo Alto Research Center, a mile from campus—filed an invention disclosure with the Office of Technology Licensing for a workstation built from off-the-shelf parts for the Stanford University Network, or SUN.[1094] After a campus review deemed the design only "marginally patentable," Stanford released the rights to Bechtolsheim, who co-founded Sun Microsystems in 1982.[1095] Reimers spent the 1980s arguing that Stanford should take stock in startups, not only cash royalties, because a fast company might design the licensed invention out of its product before a royalty check ever arrived. He estimated that by refusing equity the office had left more than $50 million on the table.[1096]

By 1984, the fourteen investments in Kleiner Perkins's first fund showed a combined profit of $208 million, about $670 million today. Ninety-five percent of the profit came from two companies, Tandem and Genentech.[1097] Without those two, the fund would still have beaten the stock market. With them, it became the story the industry tells itself. A power law is a distribution in which the winners do not win a little more than the losers;

they win so much, and so increasingly, that the middle barely matters. Most venture bets go to zero. A handful return the fund. On Sand Hill Road, a company that might decently double was a waste of a seat.

That same year, 1984, Francis Ford Coppola's brother August Coppola—Nicolas Cage's father—became dean of the College of Creative Arts at San Francisco State.[1098]

Across the bay, George Lucas needed cash. He put the computer division of Lucasfilm up for sale. Steve Jobs, pushed out of his operational role at Apple in the spring of 1985 after a boardroom clash with chief executive John Sculley and looking for a next act, bought it.[1099] The popular sentence is "Jobs bought Pixar." The record is more precise, and Alvy Ray Smith, who was there, kept the paperwork. On January 30, 1986, Jobs wrote a certified check for $5 million. Smith and Ed Catmull signed it over to Lucasfilm. Jobs then put another $5 million into the new company to capitalize it: $10 million in all, or about $30 million in today's money.[1100]

The first press release, dated February 10, 1986, was coy. It said Jobs had paid an "undisclosed sum in the millions."[1101] It also said the new firm would sell a graphics computer for about $125,000, a machine faster at pictures than a $6 million supercomputer. About forty employees came along, John Lasseter among them. They owned 30 percent. Jobs owned 70 percent, and he was chairman, not the founder in the way the brochures later implied. Catmull was president. Smith was vice president. Jobs was the investor.[1102]

That autumn, on November 5, 1986, the state of California renamed the tunnel where the road north out of the city ducks under a hill in the Presidio toward the Golden Gate. For years the official name was the Presidio Tunnel; before that, locals called it the Funston Avenue tunnel, after the approach. Senate Concurrent Resolution 86, introduced by San Francisco's state senator Milton Marks, added the words General Douglas MacArthur Tunnel, and Caltrans still carries both names on its list of structures.[1103]

The first name was Douglas, not Arthur. Arthur MacArthur was the father, a general who commanded the Department of the Pacific, and who, on the night of May 14, 1903, when Theodore Roosevelt dedicated the column in Union Square, stood up and toasted the Army. That column is not his; it belongs to Admiral George Dewey and the battle of Manila Bay. The son is the one on the tunnel: Douglas MacArthur, who commanded in the Philippines, who left Corregidor in 1942 and said he would return, who ran the occupation of Japan, and whom President Truman fired in 1951.[1104] The Peninsula and the city have a habit of welding a famous last name to the wrong man.

Seven months later, on June 9, 1987, a monthly tabloid appeared on the Stanford campus, aimed at conservative readers and edited by a philosophy undergraduate named Peter Thiel.[1105] He had founded *The Stanford Review* that spring with Norman Book, a friend from high school who was about to enroll. The masthead of the first issue listed twelve names. Thiel was editor in chief. All twelve were men—a pattern that held for the rest of Thiel's career.[1106]

The Stanford Review was a student paper, a place where young people tried on sentences they might later be paid to mean. Nobody was being bribed to print the *Review*. What a paper can be, even a small one, is a seedbed. It sorts people. It teaches them who the enemy is supposed to be, and it gives them the experience of publishing the sort of sentence that makes a room go quiet. Thiel was a student with a grievance against the university's culture, against what he and his writers treated as a closed, moralizing elite.[1107] The method of answering it was to build a counter-elite in the same zip code, using the same credential. The names on those early mastheads—Keith Rabois, David Sacks, Norman Book, and a half dozen younger staffers—would turn up again.[1108]

On January 20, 1989, George Shultz finished his term as secretary of state—six and a half years, the longest stretch in the job since Dean Rusk—and returned to Stanford as a professor of international economics at the Graduate School of Business and a distinguished fellow at the Hoover Institution.[1109] That same year, 1989, Stanford finally let Niels Reimers's licensing office start taking stock in companies.[1110] By the time Reimers stepped down as director in March 1991, the Office of Technology Licensing was grossing $24 million a year.[1111]

In 1990 Robert Swanson stepped down as chief executive of Genentech after the Swiss drug company Roche bought a majority stake.[1112]

In 1991, at San Francisco State, Dean August Coppola held what he called a skybreaking instead of a groundbreaking for the new Fine Arts Building. He had persuaded the state legislature to put up $15 million for the building in the middle of a budget crunch—six times what Stanford had earned from all its licenses in 1983.[1113] Alumni tossed beans into a hole. Coppola set a glass model of the building in the hole, poured wine over it, and said they would see how the dreams grew.

"The idea is to look up, rather than down," Coppola said. "Look up to the sky, the clouds, teach young people to dream."[1114]

Coppola stepped down as dean in 1992, and the building opened in 1993 with labs, a sound stage, and a 150-seat theater bearing his name.[1115]

Meanwhile, the federal government was handing over the physical pipes of the internet to private carriers. In 1986 the National Science Foundation had launched NSFNET to link university supercomputing centers, upgrading the backbone to T-1 lines in 1987 and T-3 lines in 1989 under the direction of Stephen Wolff.[1116] In 1990 an IBM and MCI consortium formed Advanced Network and Services (ANS), collecting $9.3 million a year in federal funds while opening the backbone to commercial traffic—a quiet privatization exposed at congressional hearings in March 1992.[1117] Following Senator Al Gore's High Performance Computing Act of 1991 (which provided $495 million for high-speed networking research and yielded the Mosaic web browser at the University of Illinois, quickly spun off as Netscape in Mountain View), the National Science Foundation retired the public NSFNET backbone in 1993–1995 in favor of private commercial providers.[1118] In San Francisco, Louis Rossetto Jr. and Kevin Kelly launched *Wired* magazine in January 1993—backed by Nicholas

Negroponte of the MIT Media Lab and inspired by Menlo Park's Stewart Brand—turning the deregulation of telecommunications and the rise of the internet into a utopian lifestyle creed.[1119]

In November 1995 Pixar released *Toy Story*. A week later the company went public, in what *Variety* called the largest initial public offering of the year.[1120] The graphics computer had not been the fortune. The fortune was the nineteenth-century toy, scaled up. Hollywood had not moved to the Bay Area. The Bay Area had learned to manufacture myth in software, and was about to sell the factory back to Hollywood.

BackRub (1995–1998)

That same year, 1995, a young entrepreneur named Elon Musk was in California. Whether Musk ever enrolled at Stanford is DISPUTED. As the popular story goes, Musk enrolled in a Stanford doctoral program in physics and materials science and dropped out after two days because the internet was more interesting.[1121] When a lawyer later asked the university for the file, however, the graduate admissions office wrote back that, based on the information given, it was unable to locate a record for Elon Musk. Physics professors who were teaching at Stanford in 1995 did not remember him, and some of his early detractors argued that the admission itself was a story told to decorate a young entrepreneur.[1122] The documentary record on the enrollment is thinner than the legend. What the record does show is that he built a first company, Zip2.[1123]

In the middle of the 1990s, a dormitory room at Stanford was an operations center. Larry Page, a graduate student whose adviser was Terry Winograd inside the National Science Foundation–backed Stanford Digital Libraries Project, had stuffed a mismatched collection of computers into a cabinet he built out of Lego bricks.[1124] The project running on them was called BackRub. It was a search engine. The name was a small joke about the way the system crawled the web by following links backward, looking at who pointed at whom. Page also named the ranking math after himself: PageRank. People assumed it meant a web page. It was a sly vanity.[1125]

The problem was mechanical before it was philosophical. The campus network into the Gates building could carry about 10 megabits a second. The university's main line, a T3, could carry about 45.[1126] Scott Hassan, an early collaborator, remembered what happened when the team found a switch in the basement that had been set wrong.

"We were using all the bandwidth of the network," Hassan said. "And this was from a single machine."[1127]

At times the crawler consumed half of Stanford's entire internet capacity.[1128] The university let the project run. Page's professors were using it to answer questions. So were their friends. A piece of student work was eating the institution's pipe, and the institution's response was to become a customer.[1129]

In January 1997 Page wrote a plan down in emails to a venture capitalist named Vinod Khosla, who had money in a search company called Excite. Excite would buy BackRub. Page, and only Page, would go work there for

seven months, long enough to install it, and then he would leave in time for the fall term and his doctorate. He claimed the technology would lift Excite's traffic by 10 percent, which he translated into $130,000 a day in extra advertising, or $47 million a year. The price for all of this was $1.6 million—about $3.3 million in today's money. Of that, $300,000 (about $625,000 today) was for Stanford, because Page and Sergey Brin had done the work as graduate students and the university owned the patent. Another $200,000 was salary, $400,000 was a bonus if he installed it in three months, and $700,000 was Excite stock.[1130]

Khosla came back at about $750,000 total.[1131] The deal died in a conference room. Excite's chief executive, George Bell, a magazine executive the investors had installed as adult supervision, watched a bake-off. The query was the single word "internet." Excite's first results, Hassan remembered, were Chinese pages where the English word stuck out of a block of characters. BackRub's first results told the user how to use a browser. Bell was upset. If people found the answer immediately, they would leave the site, and his advertising depended on them staying. Stickiness, in the dialect of that year, was the virtue.[1132]

"He told us he wanted Excite's search engine to be eighty percent as good as the other search engines," Hassan recalled.[1133]

Hassan and the others looked at each other. The buyers did not want a better product. They wanted a product worse than the one they were being offered, because a worse product would keep the visitor confused, and a confused visitor was a longer visit. Page and Brin left with the patent still in the university's name.[1134] (On January 29, 1998, Brin and Page published their technical report "The PageRank Citation Ranking: Bringing Order to the Web" through the Stanford InfoLab, and that same year presented "The Anatomy of a Large-Scale Hypertextual Web Search Engine," whose Appendix A argued that advertising-funded search engines would be inherently biased against consumers.[1135])

In December 1997, Stanford and the University of California's recombinant DNA patents expired. Over the life of those patents, 468 companies had taken a license.[1136] Licensees had paid $10,000 a year until they had products on the market, and then royalties starting at 1 percent (dropping to as low as 0.5 percent on end-product sales over $10 million, with higher rates of 1 to 10 percent on bulk and basic genetic products).[1137] Because Stanley Cohen had been consulting for a company that would compete with Genentech, an exclusive deal had been impossible, so the universities had licensed the method widely on purpose, after Cohen himself urged Niels Reimers to license nonexclusively.[1138] A license, for a young biotech firm, had also been a permission slip saying a famous university had looked at its science and not thrown it out.[1139]

In 1998 NationsBank, of Charlotte, North Carolina, bought BankAmerica, the San Francisco company that still carried A. P. Giannini's name. The merged bank kept the name Bank of America and put the headquarters in Charlotte.[1140] The bank that had lent on a face and a signature from a North Beach wharf after the 1906 fire was now run from North Carolina. Giannini Hall was still at Berkeley; his bank was not in his city.[1141]

The jobs on the Peninsula stayed, while the bedrooms did not keep up. Henry George, watching the railroad do this to wheat towns a century earlier, had a name for who captured the gain: the person who held the land [UNVERIFIED].[1142] Stanford, by its November 11, 1885 Founding Grant, could not sell the farm. It could lease it.[1143] And on Sand Hill Road, four minutes from the campus that supplied the people being judged, a company that could not get a meeting in one of the low buildings did not merely lack money. It lacked the introductions to customers, the executives the partners knew by their first names, and the next round of capital.[1144]

On September 4, 1998, after receiving an early investment from Sun Microsystems cofounder Andy Bechtolsheim—who had designed the original Stanford University Network workstation as a Stanford graduate student in 1980—Page and Brin filed the incorporation papers for Google and moved off campus.[1145] Sergey Brin's girlfriend knew a manager at Intel, Susan Wojcicki, who had just bought a house on Santa Margarita Street in Menlo Park with her husband for $615,000, about $1.3 million in today's money. To help pay the mortgage, the couple rented Google the garage and several rooms for $1,700 a month, or about $3,500 today.[1146] Page and Brin hired a fellow student, Craig Silverstein, who had offered to show them how to compress the links so the system would fit in memory—an offer he later said was a way to get his foot in the door—and they kept a hand on campus that fall, coteaching a class.[1147]

Stanford licensed PageRank to Google for stock: 1.8 million shares.[1148] Not a royalty on each search, but a piece of the firm. Many search companies already existed, and most of the companies the Office of Technology Licensing called had been unimpressed. The licensing associate who sat with Page and Brin thought a startup was the right licensee.[1149]

At a party, Kevin Kelly, the writer and editor who had helped start *Wired*, asked Page why the world needed another search engine.

"We are not really interested in search," Page said. "We are making an AI."[1150]

The Network (1999–2002)

In November 1999 Genentech agreed to pay $200 million to the University of California to settle a patent dispute, with both sides stating that the settlement was not an admission that Genentech had infringed UC's patent.[1151] The following month, in December 1999, Genentech cofounder Robert Swanson died of brain cancer at fifty-two. As Arthur Levinson later observed, the man who spent his life turning recombinant DNA technology into drugs that saved other people did not get a drug that saved him.[1152]

At Thanksgiving in 1999, a group of young men left an office in Palo Alto and walked to an automated teller machine. One of them put in a debit card from a company he had named X.com. He typed a number. The machine thought, and then it produced bills.

"Elon was very, very happy with that," a colleague remembered.[1153]

Elon Musk was not yet thirty. He had sold Zip2 and put the money into a financial website that wanted to be a bank without the marble. The card working at a machine near the office was the product, made physical: money had come out of a wall because a database said it could. (A few years later, in a 2003 talk at Stanford, Musk compressed the idea: “Money is low bandwidth. You don’t need some sort of big infrastructure improvement to do things with it. It’s really just an entry in a database.”[1154])

X.com collided with another Palo Alto company, Confinity, where Peter Thiel and Max Levchin were trying to beam money between handheld computers. Levchin had come from Kiev; as a boy he wrote programs longhand, on paper, because he did not yet have a machine that could run them.[1155] The merged firm took the name customers actually used: PayPal.

The internal war was bitter. In 2000 Musk was pushed out of the chief executive’s chair [UNVERIFIED: exact year 2000;podcast script says “two years earlier” than October 2002].[1156] Thiel’s circle, several of them alumni of *The Stanford Review*—including Keith Rabois, Norman Book, and David Sacks, who became chief operating officer—resented the X.com rebranding and the software rewrites.[1157] Fraud leaked through the system faster than they could patch it.[1158]

What saved the company was a network effect—when a service becomes more useful to each person because other people are already using it. PayPal found its road on eBay, the auction site, where strangers were already trading with strangers and needed a way to pay a person they would never meet. Even though eBay was pushing a payment service of its own, called Billpoint, users chose PayPal’s ledger.[1159]

By the end of 2001, the expired Cohen-Boyer recombinant DNA patents had brought Stanford and the University of California about $255 million in licensing revenue—$254,763,248 in total royalties through fiscal year 1998–1999, or about $480 million in today’s money—90 percent of it from royalties on actual products, and the other 10 percent from the fees companies paid just to be allowed in the room.[1160] After a $38,214,630 administrative fee and $1,801,761 in expenses, Stanford and the University of California each received $107,373,658, with Stanford splitting its share evenly in thirds among the inventor, the department, and the university.[1161]

By the summer of 2002, about 60 percent of PayPal’s business was happening on eBay, which had 46 million users. On July 8, 2002, under a merger agreement dated the day before, eBay announced it would buy PayPal. Each share of PayPal would become 0.39 shares of eBay. On eBay’s stock price of July 5, the deal was worth $1.5 billion, or about $2.8 billion in today’s money.[1162]

The sale closed on October 3, 2002. PayPal’s stockholders approved it. An eBay executive took the payments job, and Billpoint would be phased out while PayPal kept its name. The same day, Peter Thiel resigned as chief executive and wrote the company a note.[1163] Effective at the close of the markets, he wrote, this was his last day. In founding PayPal, he and Max Levchin had started by hiring their friends, and then their friends’ friends, moving out in concentric circles. He called those friendships the lasting testimony.[1164]

“I have always known that people are the most valuable part of any business,” Thiel wrote, “and today I’m more certain than ever that this is the case.”[1165]

Levchin was still the engineer who had built the system that fought the fraud. Reid Hoffman, whom Thiel had made chief operating officer against the advice of his own board, had already spent a weekend at his grandparents’ house in Gualala trying to imagine what came after a collapse; on the third day he had floated a network of professional names that the world would later call LinkedIn.[1166] David Sacks had been left in charge while Thiel traveled, and then learned he would not run the company he had been running. Musk, pushed out of that chair two years earlier, was already putting what he had into a car, and into rockets. Jack Selby, who went with Thiel, said they were trading again before the month was out. Ken Howery went too.[1167] By the time Thiel had an office of his own, it was on Sand Hill Road.[1168]

In August 2004, Google would go public, and Stanford would sell 184,207 of its 1.8 million shares in the offering for $15.7 million, disposing of the rest in early 2005 under a university policy that the Stanford Management Company sell public equities from license agreements as soon as allowed, for institutional conflict-of-interest and insider-trading reasons; as licensing director Katherine Ku confirmed to Lisa M. Krieger of the *San Jose Mercury News* on December 1, 2005, when the stock closed at $404.91 and the held stake would have topped $700 million, Stanford’s total was $336 million—on the order of three-quarters of Leland and Jane Stanford’s 1885 founding gift of $20 million (about $744 million today).[1169] In January 2006 the Walt Disney Company would announce the purchase of Pixar for about $7.4 billion, or $12 billion in today’s money;[1170] in 2007 *Fortune* would photograph Thiel’s circle dressed as a crime family and name them the PayPal Mafia, leaving out the engineers who had stayed and argued about passwords and hated the khaki;[1171] in March 2009 Roche would agree to buy the rest of Genentech for $46.8 billion ($95 per share after six stock splits between 1983 and 2004), making every $35 share from October 1980 worth $4,560;[1172] and by 2016 and fiscal year 2017 Stanford’s Office of Technology Licensing—which had taken equity in 121 companies by 2015—would count nearly $2 billion (about $1.91 billion through fiscal year 2017) in cumulative licensing income, almost all of it from three inventions out of roughly 12,000 evaluated: recombinant DNA, a set of antibody patents, and Google.[1173]

On October 3, 2002, the young men who had been trying to throw one another out of the PayPal building walked out of it in the same week, holding stock in the buyer. Sand Hill had been the gate for the companies. This was the gate for the people. The product was sold. The address book was not.

The machine does not stop when the company is bought. It changes desks.

Part 8: The Open Era

2002–2026

1. The Seeing Stone (2000–2008)

Across the Pacific, as the century turned, a new search market was taking shape. It was not a Stanford export, and the record has to be exact. On May 17, 1994, a link between a Chinese physics institute—the Institute of High Energy Physics—and Stanford's SLAC laboratory had given that institute a full internet connection; China dates its own full-function connection to April 20 of that year, through a different line.[1174] Stanford was in the first week of the Chinese internet. It was not the whole invention of it.

Robin Li did not go to Stanford. He did not go to Berkeley. He did not work at Google. He studied information management at Peking University, and computer science at the State University of New York at Buffalo, finishing in 1994.[1175] At a Dow Jones company in New Jersey, IDD Information Services, he built a link-ranking system called RankDex, and he filed a patent—U.S. Patent 5,920,859—on February 5, 1997. It was granted in 1999 and assigned to his employer.[1176] Stanford's patent on PageRank—Larry Page's U.S. Patent 6,285,999, assigned to the Leland Stanford Junior University—discusses that system as prior art.[1177] The word is *cited*. It is not copied.

The person who carried Li into the Valley's mythology was his co-founder, Eric Xu, who held a biology doctorate from Texas A&M and had been a postdoc at Berkeley.[1178] Xu made a documentary for Chinese state television, a journey into Silicon Valley, and he took Li to meet Jerry Yang, the founder of Yahoo. Xu's line, later, was that he was inspired by seeing an ethnic Chinese person build a company that powerful, and that he was sure Li was inspired too.[1179] At a screening of the film at Stanford, Li's wife told Xu she wanted her husband to be an internet entrepreneur.[1180] The seed money, $1.2 million, came half from a firm called Peninsula Capital and half from Integrity Partners—$600,000 each.[1181] Bloomberg lists Greg Penner, a Stanford Graduate School of Business alumnus, among the seed investors.[1182] Li had trained in Buffalo and then in New Jersey. He had not trained on this bay. Xu had. They founded the company in January 2000 and built it in Beijing. The name is Baidu.[1183]

Google arrived in the Chinese language in 2000, was blocked for two weeks in September 2002, and by the end of that year had about a quarter of the market.[1184]

On the other coast of the United States, in February 2004, Mark Zuckerberg and his roommates in a dormitory at Harvard put a site online and called it Thefacebook.[1185] It is the one eastern beat of the origin, not of the bloodline. By summer the company was in Palo Alto, where it would later leave its University Avenue headquarters behind and move to the former Sun Microsystems campus in Menlo Park.[1186] It did not stay a college directory. The product became the habit, and the habit became the crop.

The attention economy is a market in which the thing being sold is not the song, or the post, or the game. It is the time you spend, and the prediction of what you will do next. Shoshana Zuboff gave the harsher name, surveillance capitalism.[1187] In plain words: the service looks free because you are not

the customer. The customer is whoever pays to know what you are likely to do, and then to nudge you toward doing it. A recommendation is not a suggestion in the old sense. It is a bid.

Why a platform can run that market without owning what its users post is a statute from 1996. In 1995, in *Stratton Oakmont, Inc. v. Prodigy Services Co.*, a New York court held the online service Prodigy liable as a publisher, because Prodigy moderated its boards.[1188] Moderate, and you own the words. Leave them alone, and you might not. Representatives Chris Cox, a California Republican, and Ron Wyden, an Oregon Democrat, wrote the answer into the Telecommunications Act of 1996.[1189] Twenty-six words: "No provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information content provider."[1190] That is Section 230. A railroad act, a century earlier, gave a company the right of way. This sentence gave a platform the right not to be treated as the speaker of what its users post.

In 2005 Congress built a different shield for a different industry. It passed the Protection of Lawful Commerce in Arms Act.[1191] The law makes gunmakers nearly impossible to sue for what a criminal does with a gun. It has an exception: if the maker knowingly violates a statute applicable to the sale or marketing of the product—a predicate exception—the shield drops.[1192]

June 12, 2005. Stanford's commencement.[1193] Steve Jobs did not go to the university. He stood up anyway, and he told the students about the day he was fired from the company he had founded. "The focus of my entire adult life was gone, and it was devastating," Jobs said.[1194] The sentence was about Apple in 1985. The use of it, on this lawn, was the cult energy named in an earlier Apple marketing memo. Mike Markkula had called the process by which a customer develops a sense of a company's character "imputing"—writing that a company's image is everything the customer sees, hears, or feels [UNVERIFIED: exact wording of memo quote].[1195] A founder at a university he did not attend, telling a story of exile and return, was that philosophy performed as a sermon. He had the honesty, at least, to describe the firing as a devastation, not as a brand.

A different kind of company was taking shape in Palo Alto. Palantir. In Elvish, *palantír*: a seeing stone. In Tolkien's story the stones are powerful, and they are not unambiguously good. Sauron uses them to spy, to talk to conspirators, and to bend people who do not know the stones are dangerous to hold. Peter Thiel named the company after that object.[1196]

The founding myth is cleaner than the recruiting. Thiel drove to the University Club in Palo Alto one morning to see Alex Karp, a man he knew from Stanford Law School.[1197] Karp thought of himself as a liberal in a circle of conservatives. The early team was the student paper come due. Stephen Cohen had edited the *Stanford Review*. Joe Lonsdale had edited it before him, and had interned at PayPal. Nathan Gettings had worked on fraud at PayPal.[1198] Sequoia's Michael Moritz passed. Kleiner Perkins passed.[1199] The check that mattered, in 2005, was $2 million from In-Q-Tel, the Central Intelligence Agency's own venture fund, against a rough prototype—about $3.4 million in 2026 money.[1200] Two years earlier, In-Q-Tel had backed the geospatial-imaging startup Keyhole, whose EarthViewer software supported U.S. troops in Iraq in 2003 before Google acquired the

company to build Google Earth; in 2005, a year after Google went public, In-Q-Tel sold 5,636 shares of Google stock for $2.18 million.[1201] On Sand Hill Road, the venture firms declined Palantir's product. The state's venture fund bought a way of seeing.

Call it software as state capacity. Not a social network. A tool a government uses when it wants to know who connects to whom, and it would rather rent the tool than build it. The company later moved into Facebook's old headquarters on University Avenue in Palo Alto, and it added a log, a system it called Gotham, that recorded each search, so an analyst could be audited.[1202] Karp's pitch was that the privacy trail would make the mining acceptable. Max Chafkin records a former engineer who sat in meetings where, right after the privacy talk, a government client would suggest looking up an ex-girlfriend.[1203] The employees would remind the client that the search was logged, and then let the search happen. A log is not a refusal. It is a receipt.

In 2006, Google opened Google.cn in Beijing under Kai-Fu Lee, who had gone to Columbia and Carnegie Mellon, not to Stanford, and who had run research for Microsoft in Beijing.[1204] The company censored results in-house, and it put up a disclaimer. Eric Schmidt, the chief executive, described the decision as an "evil scale."[1205] They were not wild about the restrictions, he said, and they decided that not serving the users at all was the worse evil. Baidu's answer was an advertisement in which a Chinese poet defeats an American in a top hat, and the crowd shouts the last line: "Baidu understands Chinese better!"[1206]

In December 2007, back in California, a blog called Valleywag, part of Gawker Media, published a post by Owen Thomas. The headline was a taunt: "Peter Thiel is totally gay, people."[1207] Nick Denton, Gawker's founder, had suggested the story.[1208] Thiel is a Stanford graduate, twice—bachelor's and law—and the co-founder of the *Stanford Review*, the conservative student paper where he and David Sacks had honed the arguments they published in their 1995 book *The Diversity Myth: Multiculturalism and the Politics of Intolerance at Stanford*.[1209] He did not sue over the Valleywag headline. He began looking for someone else's suit to fund, in secret.[1210]

In 2008, an international businessman sat with China's deputy minister of culture.[1211] The official opened with a question: "Who is the CEO of Hollywood?"[1212]

There wasn't one.

That same year, on Sand Hill Road, Sequoia Capital—the firm Don Valentine had started—hired two investors away from the Stanford endowment.[1213] The job was a new business called Heritage, first to manage the partners' own wealth, then the wealth of the founders Sequoia had backed. Sebastian Mallaby does not give their names.[1214] The fact is the direction of the hire. The university's money managers were worth recruiting onto the road.

2. The Bus and the Loop (2009–2016)

In 2009, Remington, through its parent, Freedom Group, made a confidential product-placement deal with Activision.[1215] The gun was Remington's Adaptive Combat Rifle. The game was *Call of Duty: Modern Warfare 2*. No money changed hands. Both sides agreed to keep it secret. The documents came out years later, in the Sandy Hook litigation, and the *Wall Street Journal* reported them.[1216]

One memo read: "A primary means for young potential shooters to come into contact with firearms and ammunition is through virtual gaming scenarios."[1217] Another said that people seek out the brands of the guns they have seen, and that a lack of direct branding helps shield the company from looking like an endorsement. Executives wrote about winning their "fair share of these young consumers."[1218] A former vice president, John Trull, told the *Journal* that the executives did not understand multiplayer team deathmatches when they made the deal.[1219] The memo is real. The understanding may have been thin. Both of those sentences can be true.

By late 2009, in China, Baidu had about 58 percent of the search market and Google about 36 percent.[1220] What pushed Google out was not the ad. It was a hack, Operation Aurora, that reached into the Gmail accounts of human-rights activists, including a twenty-year-old Tibetan activist at Stanford, during Matt Sheehan's last year on the campus.[1221] On January 12, 2010, Google posted "A New Approach to China," and said it would stop censoring.[1222] In March 2010 it redirected the Chinese service to Hong Kong. By 2014 the services were blocked.[1223] The 2010 exit "cleared the way for Baidu" to own Chinese search.[1224]

In 2012, YouTube began optimizing its recommendation system for watch time.[1225] Think of a television network that is paid only when you do not change the channel. The rational program is the one you cannot leave. On December 14, 2012, at Sandy Hook Elementary School in Newtown, Connecticut, twenty first graders and six educators were killed.[1226] On the radio, Alex Jones said the Sandy Hook murders were a hoax, staged with crisis actors, to tighten the gun laws.[1227]

October 2013. A parking lot on El Camino Real, in Palo Alto.[1228] El Camino Real, the royal road. A fourteen-seat Mercedes bus idled at the curb, leather seats inside, purple lights in the ceiling, the name of a brokerage, DeLeon Realty, on the side. Matt Sheehan was on it, a Palo Alto native home from China with a broken ankle.[1229] Between 2011 and 2013, China had poured more cement than the United States used in the entire twentieth century.[1230] Now the money was here. The men in the lot smoked and complained, in Mandarin, about business at home. At ten they stubbed the cigarettes and climbed on. Kim Heng, who ran the China work for the firm, stood in the aisle and spoke mostly Mandarin, with English dropped in for the street names.[1231]

The brochure was the document. A photograph of the Great Wall came down the page and became the Stanford campus, and Hoover Tower, the stack at the middle of the farm, became one more guard post on the wall. The wall

continued, on the paper, into the Golden Gate Bridge.[1232] A university tower, drawn as a Chinese fortification, sold to families who had come to buy a house near the schools.

The second stop was a mansion in Menlo Park. Marble stair, limestone patio, a pool. One of the men joked. First the Mandarin, which Sheehan renders as an obscenity—*wo cao*—and then the English: “It’s such a beautiful place. Why didn’t Bo Xilai come here to buy his villa?”[1233]

Bo Xilai was a senior Chinese official who had fallen, in 2012, in a corruption scandal.[1234] The joke landed because the bus knew what the house was for. Sheehan’s phrase for the use is “the new Swiss bank account.”[1235] A house in Palo Alto held value outside Chinese politics, outside a currency, and, for some buyers, outside the question of how the money was made. Fathers, he writes, often skipped the green card so they would not owe American tax on worldwide income. Mothers and children stayed. Or the house sat empty except for the vacations.[1236]

Between 2011 and 2015, according to the National Association of Realtors figures Sheehan cites, Chinese residential purchases in the United States more than quadrupled, to $28.6 billion—about eighty-five times what Stanford got for its $336 million sale of Google shares.[1237] In 2013, 53 percent of the reported Chinese purchases were in California. In Palo Alto, during 2012 and 2013, the median house price rose 50 percent, to nearly $2 million, and it had done that rise in two years; by 2015, the median home price in Palo Alto reached $2.5 million.[1238] Some local brokers estimated that 15 to 25 percent of the town’s sales were going to Chinese buyers. One broker told Sheehan he had just sold a house to a contest of four people from China, and none of them had seen it.[1239]

The legal door for a number of the families was a visa called EB-5. At the time, $500,000 into a project in a high-unemployment area—about $700,000 in 2026 money—if the project created ten jobs, put a person in line for a green card. In 2014, 9,128 of those visas went to Chinese applicants, more than 85 percent of the program.[1240] The university’s town had become a store of value, and the federal immigration code had become part of the financing.

Sheehan’s own family is the turn. His great-grandfather, Tommy Sheehan, a San Francisco longshoreman, used to drive a priest named Peter C. Yorke around the city. In 1901 Yorke had keynoted the California Chinese Exclusion Convention: “Their thoughts are not our thoughts; their blood is not our blood; their outlook is not our outlook.”[1241] One century later the great-grandson sat on a bus that sold Palo Alto to Chinese families, and the brochure put the university’s tower on the Great Wall. The priest’s sentence and the brochure are the same road, read in opposite directions.

Hold the older photograph against a newer one. In 1869, at the golden spike, the picture of the finished transcontinental railroad has engineers and white workers. Not a single Chinese face. The men who had done the most dangerous work on the mountain are not in the frame. In 2013, when a new span of the Bay Bridge opened between San Francisco and Oakland, the mayors were Ed Lee and Jean Quan, both Chinese American, both the children of working people. Quan later said the posing was a joke between them: “This time the Chinese were in the middle.”[1242] Neither photograph is the bus. The bus is the money, which does not pose. It buys.

The loop closed in both directions. In May 2014 Baidu hired Andrew Ng, a Berkeley doctorate who had helped build deep learning at Stanford and had led Google Brain, the project that trained a network to recognize cats in YouTube videos.[1243] Baidu put his lab less than a mile from Google's own campus in Sunnyvale. The company, Sheehan notes, was worth about $60 billion—about $85 billion in 2026 money.[1244] The firm that had won the home market came back and hired the person who had taught the machines to see.

By August 2014, Christopher Wylie writes, two months after they launched a harvesting app built by a Cambridge researcher named Aleksandr Kogan, Cambridge Analytica had collected Facebook data on more than 87 million people.[1245] Facebook later confirmed that data on up to 87 million users had been improperly shared. Eighty-seven million is more than twice the population of California. The word for what the firm said it was selling is psychographics: not your age and your zip code, but your personality, inferred from what you liked, and then a message aimed at that inference. Wylie, who helped build it and then described it in his book *Mindf*ck*, wrote the sentence that belongs to this university: "And if their colleagues at universities like Harvard or Stanford were also getting interested in our work, I thought that surely we must be onto something."[1246]

That is legitimation. A famous campus does not have to audit the work. It only has to be nearby, and the nearby is read as a permission slip. Wylie also writes that staff from Palantir, Peter Thiel's data company, asked about access to the data set and pointed at a possible legal loophole, at a meeting in Palantir's Soho Square office in London in the summer of 2014 [DISPUTED: attributed to Wylie's memoir; not a court finding].[1247] That is his account. It is not a court's.

That same year, 2014, in West Virginia, a young man named Caleb Cain, who had dropped out of college and thought of himself as liberal, went looking on YouTube for self-help videos.[1248] He found a host named Stefan Molyneux. "He was willing to address young men's issues directly, in a way I'd never heard before," Cain later told *New York Times* reporter Kevin Roose.[1249] Recommendations pulled him toward the far right; he later said the videos made him feel as if they were giving him power and respect and authority.[1250]

In Beijing, meanwhile, an ex-Googler named Li Zhifei, a Johns Hopkins doctorate, had gone home and founded an artificial intelligence company called Mobvoi.[1251] Sheehan noticed two glass arrows set into the office floor. One read *California* and pointed at the desks. The other read *Silicon Valley* and pointed at a pair of chairs. Li told him the construction workers could not read the English, and had installed the arrows backward. Google, blocked in the country, invested about $40 million to $45 million in him in 2015—what Sheehan calls "the first direct investment by the parent company into a Chinese start-up since it left."[1252] The arrows pointed the wrong way. The money did not.

At the same time, the Stanford aura was failing in public at a blood-testing company. Elizabeth Holmes had started as a Stanford undergraduate. Sebastian Mallaby writes that she persuaded the "dean of the engineering school" to vouch for her, and that she recruited "elder statesmen from Stanford's Hoover Institution" onto the board.[1253] Former secretary of state

George Shultz sat among them. His presence was a license. By the middle of the decade the company, Theranos, was valued at $9 billion—about $12.7 billion in 2026 money.[1254] It went to zero.

Mallaby's line on the money complicates the postcard: "almost none of the money that Holmes raised came from Sand Hill Road practitioners."[1255] The Walton family put in $150 million. Rupert Murdoch put in $121 million. The DeVos family and the Cox family put in $100 million each. A venture investor named Tim Draper made a modest angel bet because of a family connection.[1256] His father, William Draper III, had been at Yale with George H. W. Bush, and in Skull and Bones.[1257] The Ivy roots of the Valley are another story. What the outsiders bought was the credential. Holmes held shares with 100 votes each, the founder-friendliness pushed past Google and Facebook. They "faked it until they could make it."[1258] She faked a machine that did not work. The board of elders made the fake look like a Stanford result. A Hoover letterhead is cheaper to read than a lab notebook. The letterhead was the product.

In March 2016, Peter Thiel's secret project reached a courtroom in Pinellas County, Florida.[1259] Terry Bollea, known as Hulk Hogan, had a sex tape, and Gawker had published it. A Florida jury awarded $140 million: $115 million in compensatory damages, then $25 million in punitive damages.[1260] Thiel later told the *New York Times* he had spent about $10 million funding the case. "It's less about revenge and more about specific deterrence," he said.[1261]

Specific deterrence is a lawyer's phrase. You punish one person so that person, and the people watching, do not do the thing again. To his friends, Chafkin records, Thiel called it "the most philanthropic thing I've ever done."[1262] Gawker went bankrupt. Univision bought its assets for $135 million, and the flagship site shut down.[1263] The company later settled the Bollea case for $31 million.[1264] The jury's number and the check that cleared are different numbers, and both belong in the record. A man who had learned to publish in a dormitory newsroom had funded a wrestler's lawsuit until a newsroom died. The railroad, a century earlier, put reporters on a payroll. This was the later version. You do not buy the copy. You buy the plaintiff.

3. Four Companies (2017–2021)

In Hollywood, Wang Jianlin of Dalian Wanda had bid $4.9 billion for 49 percent of Paramount, and had been turned down.[1265] In 2017 Beijing's capital controls, followed by a ban on what it called "irrational" overseas investment in entertainment, stopped the rest of that shopping.[1266] Yet in January 2018 the Chinese conglomerate Tencent bought just under 10 percent of Skydance—the production company run by Larry Ellison's son, David Ellison—at about a $1.5 billion valuation, with co-financing and the right to distribute in China.[1267]

That same year, 2018, *The Intercept* reported a secret Google project called Dragonfly: a censored Chinese search app that would require a sign-in and would share data with a Chinese partner.[1268] More than 740 Google employees signed a letter of protest. The letter said the objection was not

about China; it was about tools that help the powerful oppress the vulnerable, wherever they are. Within about a month, Sheehan writes, the company "effectively ended" the work.[1269] Also in 2018, President Donald Trump said in private that "almost every student that comes over to this country is a spy."[1270]

August 6, 2018. In about a day, Apple removed most of the InfoWars podcasts, YouTube terminated Alex Jones's channel, and Facebook removed four of his pages.[1271] Spotify had already pulled the show. Twitter gave him a seven-day suspension, then banned him for good on September 6, 2018.[1272] Four companies, most of them on the West Coast, switched one man off.

On January 25, 2019, YouTube began reducing recommendations of what it called "borderline" content, at first in the United States, saying the change touched under 1 percent of videos.[1273] That December it claimed a 70 percent average drop, in the United States, in watch time of such content coming from recommendations to people who had not subscribed. YouTube denied that its system radicalizes people.[1274]

On March 19, 2019, the Connecticut Supreme Court held that the state's unfair-trade law—the Connecticut Unfair Trade Practices Act—could serve as a predicate exception to the 2005 federal shield protecting gunmakers.[1275] Remington had marketed its rifle in *Maxim* with the line: "Consider Your Man Card Reissued."[1276] Remington argued that its ads did not target at-risk youth and that nothing linked the ads to the Sandy Hook shooting. On November 12, 2019, the United States Supreme Court declined to review the Connecticut ruling.[1277] The door in the shield was open.

In July 2019, the Federal Trade Commission fined Facebook $5 billion, a record penalty, and imposed a sweeping new privacy order.[1278] Five billion dollars is about fifteen times what Stanford received when it sold its Google shares for $336 million.[1279] The fine did not move the ledger. It priced a scandal. The company kept the users, the prediction, and the headquarters in Menlo Park. That same month, in the July 2019 trailer for Skydance and Paramount's *Top Gun: Maverick*, the Taiwan and Japan flag patches on the flight jacket were swapped out while Tencent was a backer.[1280] Tencent later withdrew; when the film opened in 2022, it restored the flags and did not play in mainland China.[1281]

Also in 2019, at Stanford, Alex Stamos started the Stanford Internet Observatory, with Renée DiResta as research director, to study online networks and election rumors.[1282] On October 7, 2019, the university's rename of Serra Mall took effect, and on November 14 it was dedicated as Jane Stanford Way, making the university's official front-door address 450 Jane Stanford Way while leaving Junípero Serra's name on Serra Street running out to El Camino Real.[1283] In October 2020, David Starr Jordan's name came off Jordan Hall, which became Building 420.[1284]

On April 16, 2020, the *New York Times* released the first episode of Kevin Roose's podcast *Rabbit Hole*.[1285] Caleb Cain had handed Roose his whole YouTube history: more than 12,000 videos and 2,500 searches. In Cain's years on the platform, from 2014 to 2018, recommendations had driven more than 70 percent of watch time on YouTube.[1286] The same machine had

eventually carried Cain out: a change in the recommendations brought him creators on the left who used the same style, Natalie Wynn among them, and Cain became a creator himself.[1287] The chief executive Roose interviewed in the fourth episode of the series was Susan Wojcicki—the Intel manager who, in 1998, had rented her Menlo Park garage to the founders of Google for $1,700 a month.[1288] By 2020 she was running the largest video service on earth. The garage and the recommendation were the same peninsula, twenty-two years apart.

The academic research that followed complicated the story [DISPUTED: extent of algorithmic radicalization on YouTube]. A 2021 paper in the *Proceedings of the National Academy of Sciences* (Hosseinmardi et al.) found far-right consumption on YouTube small, stable, and matched to what people already did elsewhere, with no clear recommendation-driven rabbit hole.[1289] A 2023 paper in *Science Advances* (Chen, Nyhan et al.) found extremist channels concentrated among a small group of already resentful users, who arrived by subscriptions and outside links; people who had not subscribed rarely got those recommendations.[1290] A 2024 study in *PNAS* (Hosseinmardi et al.), using bots, found that recommendations had a mild moderating effect compared with what users chose for themselves.[1291] Most of that data came from after the January 2019 changes, whereas Cain's descent was from 2014 to 2018.[1292] The rabbit hole can be real for a few and a myth for most.

The microphone that the four companies had switched off in 2018, meanwhile, found another plug. On October 27, 2020, months into a Spotify licensing deal reported at about $100 million, Alex Jones sat as a guest on Joe Rogan's podcast, alongside comedian Tim Dillon.[1293] Spotify employees protested on the company's internal chat. The general counsel's guidance, as reported, was that Spotify was not going to ban specific individuals from being guests on other people's shows. YouTube's position was that it bans channels, not guests.[1294] Rogan said he fact-checked every crazy thing Jones said. He pushed back on some claims, and Spotify said it did not edit him.[1295] The economic fact was plainer than the guilt: the companies that had switched Jones off were not the only companies that could carry a voice, and the one that paid the most for a single voice decided a guest was not a channel.

In 2021, a Connecticut judge found Jones liable by default in the Sandy Hook defamation suits for discovery violations.[1296] And on March 12, 2021, Peter Thiel made his first contribution—$10 million—to a political committee called Protect Ohio Values, formed to back his former employee J. D. Vance for the United States Senate.[1297]

4. The Freshman and the Sink (2022–2024)

On February 15, 2022, nine Sandy Hook families settled with Remington for $73 million, paid by Remington's insurers—the maximum the policies covered—after the company's second bankruptcy.[1298] In April 2022, Peter Thiel gave two more checks to Protect Ohio Values: $3.5 million on April 18, and $1.5 million on April 25, bringing his total to $15 million (following his $10 million contribution on March 12, 2021) for J. D. Vance's Senate race.[1299]

May 16, 2022. Just after midnight. A young man in Uvalde, Texas, named Salvador Ramos turned eighteen. The families who would later sue alleged that twenty-three minutes later he bought a rifle, a Daniel Defense DDM4V7. Eight days after that, on May 24, 2022, at Robb Elementary School, nineteen children and two teachers were killed.[1300]

That summer, *Top Gun: Maverick* opened in theaters with the Taiwan and Japan flags restored to the flight jacket; it did not open in mainland China.[1301] In August 2022, a Texas jury awarded the parents of six-year-old Jesse Lewis about $49 million to $50 million against Alex Jones, after Jones conceded at trial that the Sandy Hook shooting was “100 percent real” and parents testified that his followers had sent death and rape threats and harassed them in person. In Connecticut, where a judge had already found Jones liable by default in 2021 for discovery violations, a jury subsequently awarded the families $964 million against Jones, with the judge adding $473 million more. Jones filed for personal bankruptcy in late 2022.[1302]

On October 7, 2022, the United States government blocked the sale to China of the most powerful class of artificial intelligence chips—Nvidia’s A100 and H100—and of the tools that make them. An export control is a freight rate for a machine: whoever writes it decides who gets to multiply. Nvidia’s answer was a new pair of chips, the A800 and the H800, drawn to sit just outside the line.[1303]

October 27, 2022. Elon Musk completed the purchase of Twitter for $44 billion. When Musk was assembling the financing, Faiz Siddiqui records, Larry Ellison had been asked by text: “Any interest in participating in the Twitter deal?” Ellison replied: “Yes, of course,” and put in about $1 billion.[1304] On the day the deal closed, Musk walked into the headquarters carrying a porcelain sink, so that he could say, “let that sink in.”[1305] He fired top executives the same day. A post went out: “The bird is freed.”[1306] Forty-four billion dollars was a little more than Stanford’s entire endowment three years later ($40.8 billion).[1307] He had bought a microphone for more than the university whose file on him, when a lawyer later asked, could not be found [UNVERIFIED]. The price is the part that is settled. Days later, in November 2022, J. D. Vance won his Senate seat in Ohio.[1308]

November 29, 2022. A Stanford freshman named Theo Baker published, in *The Stanford Daily*, the first article on questions about Stanford president Marc Tessier-Lavigne’s scientific papers.[1309] It was not Baker’s first article; a piece on campus social life had run on October 24. This one was the one that moved the board. He had been tipped, on November 1, toward PubPeer, a website where scientists comment on one another’s papers. He took the images to forensic readers, among them Elisabeth Bik. Then he called his editor, Sam Catania.[1310]

“I called Sam, the Daily editor,” Baker wrote. “‘Holy shit,’ he said. ‘We need to get going. How quickly can you write this?’”[1311]

The Board of Trustees announced a review the same day.[1312] The report arrived on July 17, 2023, from an outside scientific panel, transmitted by a lawyer, Mark Filip, in a letter addressed to the special committee at 365 Lasuen Street.[1313] The findings have to be stated the way they were written, because the slogan is cruder than the page. After reviewing more than 50,000 documents across twelve papers, the panel concluded that Tessier-

Lavigne did not personally engage in research misconduct—the federal term for fabrication, falsification, or plagiarism. It did not find that he directed the manipulation, or that he knew of it when it happened. It also found that several papers exhibited manipulation of research data, and that in at least four of the five papers on which he was a principal author there was apparent manipulation by others in his lab. Each of those five, the panel wrote, had "serious flaws in the presentation of research data."[1314] The board, two days later, added its own sentence: when the problems surfaced, he took "insufficient steps to correct the scientific record."[1315]

On July 19, 2023, Tessier-Lavigne resigned the presidency, effective August 31. Richard Saller became interim president on September 1. Tessier-Lavigne remained a tenured professor of biology. His letter to the campus said the report refuted the allegations of fraud and misconduct made against him, and that he was stepping down for the good of the university. Both the letter and the panel report are public.[1316] A freshman and a student paper had started a process that ended a presidency. The president left the office. He kept the chair. That was what governance looked like when it worked, and the limit of what it worked to do: it did not find a forger in the president's own hand; it found a lab, a record left uncorrected, and an office that could not survive the finding.

Down the hill from the campus, on Sand Hill Road, the Rosewood Hotel remained what Baker called in his book "the modern clubhouse for Silicon Valley."[1317] It sits on land Stanford owns. A night starts in four figures. Baker's verdict was that the epicenter is a row of dull buildings whose advantage is being close to the campus down the hill. Under the hotel, the tenant may own the building. The university owns the dirt.[1318] The bank A. P. Giannini founded had left for Charlotte in 1998.[1319] The clubhouse stayed on the farm.

In October 2023, the United States closed the export-control loopholes on Nvidia's chips; Nvidia answered with the H20.[1320] In February 2024, Joe Rogan renewed his deal with Spotify for about $250 million, according to *The Wall Street Journal*, and the new agreement was no longer exclusive.[1321]

May 24, 2024. On the two-year anniversary of the Robb Elementary shooting, Connecticut lawyer Josh Koskoff—the same lawyer who had sued Remington for the Sandy Hook families—filed two lawsuits. One, *Mata v. Meta*, was filed in Los Angeles Superior Court for about 45 family members against Meta, parent of Instagram and Facebook, and Activision Blizzard, maker of *Call of Duty*. The other, *Rubio v. Daniel Defense*, was filed in Uvalde County for 31 family members against the rifle manufacturer, and was later removed to federal court. Both complaints were filings, not findings [DISPUTED]. The Los Angeles complaint alleged that the companies raked in billions from the comfort of "Menlo Park and Santa Monica."[1322]

"Just 23 minutes after midnight on his 18th birthday, the Uvalde shooter bought an AR-15 made by a company with a market share of less than 1 percent," Koskoff said. "This three-headed monster knowingly exposed him to the weapon, conditioned him to see it as a tool to solve his problems, and trained him to use it."[1323]

That was the families' theory. The complaint alleged that in November 2021 Ramos had downloaded *Call of Duty: Modern Warfare*, and that the rifle appeared on the game's opening title page. It alleged that Daniel Defense's Instagram called a rifle with a holographic sight "totally murdered out," and captioned a photo of soldiers "Hunters Hunt." It alleged that within weeks he was back on the company's website, and that Instagram's own rules prohibited firearm advertising, and that Meta enforced other rules and not this one. Activision's lawyers answered that a war game is speech, and that the company owed no duty to stop a third person's violence. None of these claims had been found by a jury.[1324]

In June 2024, the Stanford Internet Observatory—started in 2019 by Alex Stamos, with Renée DiResta as research director—became the fight on campus [DISPUTED]. Having published on election rumors, it had drawn lawsuits and subpoenas from the House Judiciary Committee. *Platformer* reported that the observatory was being dismantled: Alex Stamos had left, Renée DiResta's contract was not renewed, and staff were told to look for jobs. Stanford disputed the verb, stating that the work would continue under new leadership. Both statements are on the record. The office had become politically expensive, and the expense was not in dispute.[1325]

On July 7, 2024, David Ellison's Skydance signed an agreement to acquire Paramount.[1326] In October 2024—on October 25—Joe Rogan recorded a three-hour interview with Donald Trump in Austin. On November 4, 2024, after posting an interview with Elon Musk, Rogan endorsed Trump, saying Musk had made "the most compelling case."[1327] Across the 2024 election cycle, as Faiz Siddiqui tallied it in December of that year, Musk spent more than $275 million through a super PAC called America PAC toward Trump and other Republicans—about eighteen times Thiel's $15 million for Vance.[1328] Trump and Vance won.

Later in November 2024, the satirical newspaper *The Onion*, backed by Sandy Hook families, won a bankruptcy auction for the company that owned Alex Jones's InfoWars. A federal bankruptcy judge threw the result out over process and sent the families to state court.[1329] And in December 2024, federal export controls reached high-bandwidth memory, packaging, and chipmaking equipment.[1330]

5. The Arithmetic and the Studio (2025–2026)

January 20, 2025. Washington. The presidential inauguration was supposed to be outdoors, but the cold moved it inside, into the Capitol Rotunda, and the move scrambled the seating. What the photographs show, and what the Associated Press and *The Guardian* reported, was a row of technology executives seated with the president's family, ahead of several cabinet nominees: Elon Musk, Mark Zuckerberg, Jeff Bezos, and Sundar Pichai of Google. Tim Cook was in that company of guests. The head of TikTok was in the room, the Associated Press noted, beside the nominee to lead the intelligence community. The weather is why the chart changed. The pictures are what remain. Proximity is the evidence. It does not require a theory of anyone's heart.[1331]

That same day, an executive order renamed the United States Digital Service as the United States DOGE Service—the Department of Government Efficiency. It was not a cabinet department. It was a renamed office in the

Executive Office of the President, plus a temporary organization scheduled to end on July 4, 2026. Agency heads were told to give it access to unclassified records and software. Musk's role was a special government employee, a category the law limits to about 130 days in a year.[1332]

The chokepoint now was the arithmetic. You could have the data, the graduate students, and the model on a whiteboard, and still be stopped by whoever owned the machines that did the multiplying. For its fiscal year ended January 26, 2025, Nvidia reported audited revenue of $130.5 billion, up 114 percent from $60.9 billion the year before. The data-center line—the machines sold for artificial intelligence work—was $115.2 billion. One hundred thirty point five billion dollars was a little more than three times Stanford's endowment.[1333] Larry Page, at a party in the late 1990s, had said he was not really interested in search; he was making an AI. The training ground he described now had a toll booth, and the booth was a chip. A headline number in the hundreds of billions announced for new data centers was a plan, not money spent. The gate had moved from the road to the chip.[1334]

Also in January 2025, OpenAI told the *Financial Times* it had seen evidence that a Chinese lab, DeepSeek, had used distillation on OpenAI's models, against OpenAI's terms—training a new model on the answers of an older one, so the student copied the teacher's results without the teacher's data [DISPUTED]. OpenAI did not publish the evidence. A House committee later called unauthorized distillation "highly likely," relying on what OpenAI had provided.[1335]

In April 2025, selling Nvidia's H20 chip into China required a federal license, which Nvidia's own Form 8-K filing from April 9, 2025, treated as a ban in practice.[1336] On May 28, 2025, the White House confirmed to Reuters that Elon Musk's off-boarding from the government would begin that night. He had led an advisory wrecking crew, and the 130-day clock ran out. The companies remained his. The job did not. Untouchability is the distance between the person and the instrument: when the temporary chair ends, the corporations are where he left them. The statute makes the advisor temporary. The cap table does not.[1337]

In July 2025, the Skydance–Paramount merger cleared its path. On July 2, 2025, Paramount paid $16 million to settle Trump's lawsuit over a *60 Minutes* broadcast. On July 17, days after CBS host Stephen Colbert called the settlement, in his words, "a big fat bribe," CBS canceled *The Late Show*, announcing that its last season would end in May 2026. On July 24, 2025, the Federal Communications Commission approved the Paramount acquisition, valued at about $8 billion. It closed on August 7, 2025, as Paramount Skydance, at a figure Reuters put at $8.4 billion.[1338] That same month, in August 2025, federal licenses for Nvidia's H20 chip resumed.[1339]

On August 31, 2025, Stanford closed its fiscal year with an endowment of $40.8 billion—about fifty-five times the founding gift, counted in today's money. In that fiscal year it paid out $1.9 billion. Undergraduate tuition in the same report was $65,127, with nearly half of undergraduates receiving need-based aid.[1340]

On September 5, 2025, Judge William F. Highberger of Los Angeles Superior Court dismissed Meta from the Uvalde families' lawsuit [DISPUTED]. He wrote that holding the company liable would impose

“limitless liability” on a creator or a disseminator for the violent acts of others anywhere in the world. Meta’s defense firm, Gibson Dunn, later summarized the ruling this way: the shooter was the proximate cause, and Meta’s conduct was “so far attenuated” from the harm that it was only “negligible or theoretical.” The same day, Highberger denied Activision’s anti-SLAPP motion to throw the case out as an attack on free speech; Activision then argued it owed no duty, on the same ground as Meta. Highberger reaffirmed Meta’s dismissal in December 2025, and the families appealed. The families had argued that Menlo Park built the hallway that led a teenager to a rifle. The court, so far, said a hallway is not a cause.[1341]

On October 6, 2025, Paramount Skydance bought *The Free Press*, reported at about $150 million, and named its founder, Bari Weiss, editor in chief of CBS News, reporting directly to David Ellison.[1342] On October 14, 2025, the United States Supreme Court refused to hear Alex Jones’s appeal of the Connecticut Sandy Hook judgment, which appeals had brought to about $1.25 billion.[1343] On November 17, 2025, in federal court in the Central District of California, a San Jose resident named Chenguang Gong was sentenced to 46 months in prison for taking missile-detection infrared sensor technology, after a run of applications to Chinese talent programs between 2014 and 2022.[1344]

On December 5, 2025, Netflix—the company Reed Hastings, Stanford master’s in computer science (in artificial intelligence), class of 1988, had founded with Marc Randolph on August 29, 1997, in Scotts Valley before moving its headquarters to Los Gatos—agreed to buy Warner Bros., HBO, and HBO Max at an enterprise value of $82.7 billion.[1345] Days later, on December 8, 2025, Reuters reported that the president allowed Nvidia to sell a newer chip, the H200, into China if the United States government took a 25 percent cut of the sale.[1346]

January 29, 2026. A federal courtroom in San Francisco. Linwei Ding, a former Google engineer who had come to the United States in 2010 for a master’s degree, was convicted on seven counts of economic espionage and seven counts of stealing trade secrets. The government said he had uploaded more than 500 files—more than 2,000 pages—of Google’s AI infrastructure. While Ding was in China, an intern back in Mountain View had swiped Ding’s badge so the building would record him as at his desk. The exhibit built for the jury was Ding’s own pitch for a company he called Zhisuan: “We have experience with Google’s ten-thousand-card computational power platform; we just need to replicate and upgrade it.” The Department of Justice called the verdict the first AI-related economic-espionage conviction.[1347]

Two weeks later, on February 12, 2026, OpenAI sent a memo to the House committee alleging that DeepSeek had continued its distillation practices through obfuscated routers and resellers [DISPUTED]. No court had found this; it remained an allegation.[1348] And on February 26, 2026, the Warner Bros. Discovery board declared Paramount Skydance’s takeover bid superior to Netflix’s. Netflix did not raise.[1349]

In April 2026, *The Onion* announced a new deal to license and then buy the InfoWars brand under a Texas state court receiver order; its chief executive said the Sandy Hook families had not received a penny, and a Texas appeals

court stayed the transfer while Jones appealed.[1350] By May 2026, about ten Chinese firms were cleared to buy Nvidia's H200 chips.[1351]

Then, on August 20, 2026, U.S. District Judge Vince Chhabria vacated all seven espionage counts against Linwei Ding, ruling there was insufficient evidence that Ding knew or intended that what he did would benefit the Chinese government. The seven trade-secret counts stood. On September 1, 2026, Ding was sentenced to twelve months minus a day in custody, a $25,000 fine, and restitution of more than $189,000.[1352] Matt Sheehan had noted the caution that belongs next to such cases: a string of Chinese American researchers had been charged with espionage only to have the cases prove unfounded, and in 2018 the president had remarked in private that "almost every student that comes over to this country is a spy."[1353] The Ding case was what those anxieties collapsed into when a court did the sorting. The theft was proved. The spy charge was not. A badge swipe is not a nationality.

Which left the argument people repeated across the Valley: that China can manufacture, but cannot make California [DISPUTED]. The sources line up on both sides. The case for a gap comes from the scholar Jeffrey Ding, no relation to the engineer, who argues that China's weakness is diffusion—the spread of a technology through the whole economy—more than the moment of invention, while a RAND study finds the links from Chinese research to Chinese industry weaker than in the United States. The case against the slogan is a measurement. In September 2024 the Information Technology and Innovation Foundation put China ahead or level in nuclear power, electric vehicles, and batteries; near the frontier in robotics, artificial intelligence, and quantum; and behind in semiconductors and biopharma. Its 2026 Hamilton Index had China leading seven of ten advanced industries. Dan Wang, in his book *Breakneck*, as rendered in a *Foreign Affairs* review, reversed the premise: China is an engineering state building at breakneck speed, set against the United States' lawyerly society blocking everything it can, good and bad. And Sheehan noted the rhyme with the 1980s, when Americans said Japan could copy and could not invent, while Japanese investment kept "generating jobs, income, and even innovation."[1354] China builds at a scale California no longer can. Whether it can grow the loop on this peninsula—risk capital, an open university, and immigrant talent on the same road—is the question that remains open.

In Texas, on August 21, 2026, the state's Third Court of Appeals cut the Sandy Hook parents' $49 million jury award against Alex Jones to about $6 million—$4.1 million in compensatory damages plus about $1.5 million in punitive damages. Jones said he would appeal to the Texas Supreme Court; the families' lawyer, Mark Bankston, called the cut "irrelevant." As of August 2026, Jones had paid nothing on either state's judgment,[1355] and on September 9, 2026, the Texas Supreme Court refused to lift the stay on the sale of InfoWars to *The Onion*. Jones had already left the InfoWars name behind.[1356]

October 6, 2026. Justice Elena Kagan denied a last-minute petition, and Paramount Skydance's purchase of Warner Bros. Discovery closed. The combined company was renamed Skydance, with David Ellison and Ynon Kreiz as co-chief executives. The Ellisons and RedBird held all the voting shares. Forty-seven billion dollars of the new equity came from the Ellisons, RedBird, and the sovereign wealth funds of Saudi Arabia, Abu Dhabi, and Qatar.[1357]

Tie it to the start. At Ampex, Larry Ellison had worked on a failed CIA contract called Project Oracle. Three employees left and started a company in 1977, Larry Ellison with Bob Miner and Ed Oates; on the third name, they called it Oracle, and its first customer was the CIA. In the Reagan years, Malcolm Harris writes, the agency mailed checks in plain white envelopes and did not mind that the first versions did not work; an engineer recalled, “We bought this thing knowing damn well it wasn’t going to work. We’re buying an idea.”[1358] A century before that, in January 1880, Eadweard Muybridge had shown Leland Stanford’s horses moving on a zoopraxiscope at the Stanford mansion in San Francisco—“The age of movies began in California, not in Hollywood but in the Bay,” Harris writes[1359]—and in 1914 Cecil B. DeMille had shot a western, *The Squaw Man*, in a barn at Selma and Vine for the Jesse L. Lasky Feature Play Company, which became Paramount.[1360] Now it sat in one portfolio: one family, a database bought as an idea, and then CBS, CNN, HBO, Paramount, and Warner Bros.

Rail controlled movement. Stanford controls succession.

It is. And it connects to me. I did not inherit a railroad, or a fund, or a studio. I inherited the habit of treating the names as scenery. Serra. Jordan. Terman. Page Mill. Sand Hill. Once you hear them as a succession, you cannot go back to hearing them as scenery. The walk after this is only the sidewalk, in the order the names were laid down.

6. Coda: The Street Names

Dusk on the farm. The oaks go black first. The signs are already lit, which is the fact this walk is heading toward, and it will not be hurried. Read them in the order the layers were laid down. You can do it on foot in an afternoon. Four centuries of proprietors, one sidewalk.

Gálvez. José de Gálvez, the inspector general who sent the expedition north. The administrator, not the priest who walked. The people already living on this ground do not have a sign at this corner.[1361]

Serra. Junípero Serra’s name came off the mall across the front of the campus. Serra Mall became Jane Stanford Way. The street that kept his name is the one that runs out.[1362]

Lasuen. Fermín de Lasuén, Serra’s successor, who carried the missions farther up the coast, including La Purísima. The scientific panel’s letter, in 2023, was addressed to 365 Lasuen Street.[13631364]

El Camino Real. The royal road, still the main street of the Peninsula. In October 2013 the buyers’ bus left from a lot on this road, purple lights in the ceiling, a brochure in the aisle.[1365]

Alvarado. Juan Bautista Alvarado, the governor who in 1836 led a revolt against Mexico City’s grip on California. David Starr Jordan, the first president, put the name on the row.[1366]

Stanford Avenue. The farm, the railroad fortune, the grant that could not be sold. The family could lease it. They could not cash it out.[1367]

Jane Stanford Way. The rename of Serra Mall took effect on October 7, 2019, and the university dedicated it on November 14. The official address became 450 Jane Stanford Way. Serra Street, from Campus Drive East out to El Camino, kept the old name. The co-founder is on the front door. The priest is on the exit.[1368]

Jordan Hall. The first president's name came off the building in October 2020. It is Building 420 now.[1369]

Hoover Tower. The archive, and the institution where a secretary of state came back to sit, and then sat on a blood-testing board. On the 2013 brochure, this tower is a guard post on the Great Wall.[1370]

Terman. Frederick Terman's engineering buildings, at the edge of the farm he helped lease to industry. The research park is the lease, made architectural.[1371]

Page Mill. William Page's lumber road, before it was the spine of the research park. It still runs from the timber to the chip.[1372]

Sand Hill. Four minutes from the Quad. The gate. The Rosewood, on the university's land, calling itself the modern clubhouse. A company that cannot get a meeting in one of the low buildings does not merely lack money. It lacks the next introduction.[13731374]

Stand where the royal road meets the farm and the order is the order of the history. A mission administrator. A priest. A governor. A railroad family. A co-founder, put on the front door in 2019, while the priest was left on the road out. A tower that a brochure redrew as a wall. A lumber road that became a chip road. A venture road that became a clubhouse on leased ground. The signs do not argue. They accumulate.

The golden-spike photograph is the same kind of document, made a century and a half before two mayors stood in the middle of a new bridge and said, "This time the Chinese were in the middle."[1375] Who is in the frame. Who is not. A street sign is that choice, cast in metal and lifted out of reach.

And the names that are not here. The Muwekma Ohlone, whose land the campus occupies—little on the map, and what recognition exists is recent and small. The Chinese workers of the Central Pacific, absent from the golden-spike photograph, and absent from any building at the scale of the work they did. The people sterilized under the 1909 law: nothing. The farm workers: Cesar Chavez has streets in California, real ones, and they are mostly in the Central Valley and in city neighborhoods, not on this walk. The list is the record. What a place commemorates is what it decided to remember, and this peninsula decided to remember the people who held the title.[1376]

Read a sign as a document and it answers four questions: Who. When. For what. And who decided. British streets carry empire. Parisian streets carry writers. Most American streets carry presidents and trees. This peninsula carries the people who held the title, in the order they held it.

Subliminal does not mean hidden. It means beneath the threshold at which something is consciously registered. The street signs are twelve feet up and lit at night. It was never a secret. It was only subliminal.

Endnotes

Part 1

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1. Santa Clara University Ohlone heritage report and Stanford University Jasper Ridge chronology, as cited in Episode 1 Script (“02. The Broken Link”) and Episode 1 Review Log (“Puichon; Jasper Ridge 5,000+ years; Muwekma through baptismal books”).↩

2. Santa Clara University Ohlone heritage report and Stanford University Jasper Ridge chronology, as cited in Episode 1 Script (“02. The Broken Link”) and Episode 1 Review Log (“Puichon; Jasper Ridge 5,000+ years; Muwekma through baptismal books”).↩

3. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023), ch. 1.1 (`harris-palo-alto`), as quoted in Episode 1 Script (“02. The Broken Link”) and Episode 1 Review Log: “in response to abundance rather than scarcity” and “social luxury.”↩

4. Santa Clara University Ohlone heritage report, as cited in Episode 1 Script (“02. The Broken Link”) and Episode 1 Review Log (“~500 groups, ~100 languages, ~40,000 in the Bay Area… spoken as historians’ estimate”).↩

5. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023), introduction and ch. 1.1 (`harris-palo-alto`), as quoted in Episode 1 Script (“02. The Broken Link”) and Episode 1 Review Log: “To speak of the Ohlone is to speak of the broken link between the Bay Area’s indigenous people and the Bay Area itself.”↩

6. Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Jesuit expulsion 1767; Loreto Feb. 1768; Franciscans March 1768; Portolá and Serra sent north”).↩

7. United States Conference of Catholic Bishops (USCCB) and Library of Congress authority record, as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Serra dates, chair, death, burial, nine missions”).↩

8. Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Jesuit expulsion 1767; Loreto Feb. 1768; Franciscans March 1768; Portolá and Serra sent north”).↩

9. Juan Crespí diary, October 1769, and James Clifford Shirley, *The Redwoods of Coast and Sierra*, as cited in Episode 1 Script (“01. The Tall Stick”) and Episode 1 Review Log (“Crespí redwood description near Watsonville, Oct. 1769; this tree not named”).↩

10. Monterey County Historical Society and National Park Service (NPS) Sweeney Ridge page, as cited in Episode 1 Script (“01. The Tall Stick”) and Episode 1 Review Log (“64 men left San Diego July 14, 1769”).↩

11. Zoeth Skinner Eldredge, *The March of Portolá and the Discovery of the Bay of San Francisco*, quoting the expedition diary, and National Park Service (Golden Gate National Recreation Area), as cited in Episode 1 Script (“01. The Tall Stick”) and Episode 1 Review Log (“Nov. 4, 1769, feast of San Carlos; Mass; view from the summit”; “Sweeney Ridge is that viewpoint”); see also Marianne Babal, “The Top of the Peninsula: A History of Sweeney Ridge and the San Francisco Watershed Lands, San Mateo County” (San Francisco: Golden Gate National Recreation Area, National Park Service, 1990), in Gray Brechin, *Imperial San Francisco* (`brechin-imperial#69`).↩

12. Gaspar de Portolá, diary entry for November 5, 1769, and November 6–11, 1769 camp records, in Donald Eugene Smith and Frederick J. Teggart, eds., *Diary of Gaspar de Portolá During the California Expedition of 1769–1770* (1909), as cited in Episode 1 Script (“01. The Tall Stick”: “extended a great arm of the sea, sixteen to twenty leagues”), Episode 1 Review Log, and Open Issues.↩

13. Gaspar de Portolá, diary entry for November 5, 1769, and November 6–11, 1769 camp records, in Donald Eugene Smith and Frederick J. Teggart, eds., *Diary of Gaspar de Portolá During the California Expedition of 1769–1770* (1909), as cited in Episode 1 Script (“01. The Tall Stick”: “extended a great arm of the sea, sixteen to twenty leagues”), Episode 1 Review Log, and Open Issues.↩

14. Palo Alto city arborist report and Coombe land map for Leland Stanford, as cited in Episode 1 Script (“01. The Tall Stick” and “08. The Rancho”) and Episode 1 Review Log (“Tree ~110 ft, sprouted ~940, seal, *semper virens*, not on Stanford’s land”).↩

15. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023), introduction (`harris-palo-alto`), as cited in Episode 1 Script (“01. The Tall Stick”) and Episode 1 Review Log (“Harris repeats the five-days-under-the-tree version”).↩

16. Town histories and Monterey County Historical Society, as cited in Episode 1 Script (“04. The Royal Road” and “05. Saints on the Map”) and Episode 1 Review Log (“Portola Valley named for Portolá; founded Monterey June 3, 1770; left July 9”).↩

17. Francisco Palóu (1774) and Pedro Font, diary (March 1776), in Frederick J. Teggart, ed., and Palo Alto tree literature, as cited in Episode 1 Script (“01. The Tall Stick”), Episode 1 Review Log (“Palóu cross 1774; Font, 50 varas, March 1776”), and Open Issues.↩

18. Juan Manuel de Ayala, report in Zoeth Skinner Eldredge, *The March of Portolá and the Log of the San Carlos*, Angel Island Conservancy, and California Office of Historic Preservation (California Historical Landmark 236), as cited in Episode 1 Script (“04. The Royal Road”) and Episode 1 Review Log (“San Carlos entered the night of Aug. 5, 1775; 44 days; foot wound Apr. 3; Cañizares mapped the bay; left Sept. 18, Monterey Sept. 19”; “Ayala Cove; Isla de los Ángeles; Landmark 236, Aquatic Park, Beach and Larkin”).↩

19. Juan Manuel de Ayala, report in Zoeth Skinner Eldredge, *The March of Portolá and the Log of the San Carlos*, Angel Island Conservancy, and California Office of Historic Preservation (California Historical Landmark 236), as cited in Episode 1 Script (“04. The Royal Road”)

and Episode 1 Review Log (“San Carlos entered the night of Aug. 5, 1775; 44 days; foot wound Apr. 3; Cañizares mapped the bay; left Sept. 18, Monterey Sept. 19”; “Ayala Cove; Isla de los Ángeles; Landmark 236, Aquatic Park, Beach and Larkin”).↩

20. Francisco Palóu (1774) and Pedro Font, diary (March 1776), in Frederick J. Teggart, ed., and Palo Alto tree literature, as cited in Episode 1 Script (“01. The Tall Stick”), Episode 1 Review Log (“Palóu cross 1774; Font, 50 varas, March 1776”), and Open Issues.↩

21. Mission reference works, as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Santa Clara founded Jan. 12, 1777, at So-co-is-u-ka; eighth mission; first named for a woman; creek rejected for water”; “Six rebuilds”).↩

22. Episode 1 Script (“05. Saints on the Map”) and Episode 1 Review Log (“Mission Dolores from Nuestra Señora de los Dolores”).↩

23. Mission reference works, as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Santa Clara founded Jan. 12, 1777, at So-co-is-u-ka; eighth mission; first named for a woman; creek rejected for water”; “Six rebuilds”).↩

24. Mission reference works, as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Santa Clara founded Jan. 12, 1777, at So-co-is-u-ka; eighth mission; first named for a woman; creek rejected for water”; “Six rebuilds”).↩

25. Santa Clara University Ohlone heritage report, as cited in Episode 1 Script (“02. The Broken Link” and “03. What a Mission Was”) and Episode 1 Review Log (“Performed fake baptismal entry. No Early California Population Project page was pulled. The script describes the form and says it is the shape, not one person’s line”).↩

26. Santa Clara University Ohlone heritage project (“laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain’s colony in California”) and Richard Henry Dana Jr., *Two Years Before the Mast* (1840; “virtually slaves”), as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Four functions; neophyte; presidio; unpaid labor”).↩

27. Santa Clara University Ohlone heritage project (“laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain’s colony in California”) and Richard Henry Dana Jr., *Two Years Before the Mast* (1840; “virtually slaves”), as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Four functions; neophyte; presidio; unpaid labor”).↩

28. Santa Clara University Ohlone heritage project (“laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain’s colony in California”) and Richard Henry Dana Jr., *Two Years Before the Mast* (1840; “virtually slaves”), as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Four functions; neophyte; presidio; unpaid labor”).↩

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30. Santa Clara University Ohlone heritage project (“laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain’s colony in California”) and Richard Henry Dana Jr., *Two Years Before the Mast* (1840; “virtually slaves”), as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Four functions; neophyte; presidio; unpaid labor”).↩

31. Autry Museum of the American West, as cited in Episode 1 Script (“04. The Royal Road”) and Episode 1 Review Log (“No single Spanish El Camino Real; 20th-century construct”).↩

32. United States Conference of Catholic Bishops (USCCB) and Library of Congress authority record, as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Serra dates, chair, death, burial, nine missions”).↩

33. California State Parks, as cited in Episode 1 Script (“03. What a Mission Was” and “04. The Royal Road”) and Episode 1 Review Log (“Lasuén, nine missions, 1786–1798, including La Purísima 1787”; “21 missions, San Diego 1769 to Sonoma 1823, ~600 miles”).↩

34. Episode 1 Script (“09. Nine Days”) and Episode 1 Review Log (“Astor died March 29, 1848; estate ~$20 million; $400,000 library; Fort Astoria 1811”).↩

35. Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Independence 1821; Echeandía July 25, 1826; 1831 counts 18,683 vs 4,342; decree Aug. 17, 1833; Figueroa Aug. 9, 1834”).↩

36. Land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Peña grant 1841, 8,418 acres; sold to the Robles brothers 1847”).↩

37. Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Cinnabar shown to Robles, 1824; New Almaden by 1845”); see also Leonard William Ascher, “The Economic History of the New Almaden Quicksilver Mine, 1845–1863” (Ph.D. diss., University of California at Berkeley, 1934), in Gray Brechin, *Imperial San Francisco* (`brechin-imperial#69`).↩

38. Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Independence 1821; Echeandía July 25, 1826; 1831 counts 18,683 vs 4,342; decree Aug. 17, 1833; Figueroa Aug. 9, 1834”).↩

39. Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Independence 1821; Echeandía July 25, 1826; 1831 counts 18,683 vs 4,342; decree Aug. 17, 1833; Figueroa Aug. 9, 1834”).↩

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41. Dorothy Krell, Robert Jackson, and Stanford University facts page (`facts.stanford.edu`, Class of 2029), as cited in Episode 1 Script (“03. What a Mission Was”) and Episode 1 Review Log (“Baptisms 87,787; deaths 63,789 through 1832; Santa Clara 8,536 baptisms, 6,809 burials; peak 1,464 in 1827”; “San Gabriel, three of four children dead before age two; measles 1806 and 1828”; “Stanford first-years, fall 2025: 1,866”).↩

42. Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Independence 1821; Echeandía July 25, 1826; 1831 counts 18,683 vs 4,342; decree Aug. 17, 1833; Figueroa Aug. 9, 1834”).↩

43. Dorothy Krell, W. W. Robinson, and Robert Jackson, as cited in Episode 1 Script (“03. What a Mission Was” and “07. Secularization”) and Episode 1 Review Log (“Livestock Dec. 31, 1832: ~151,000 cattle, ~138,000 sheep; ~one-sixth of the land”; “Baptisms 87,787; deaths 63,789 through 1832; Santa Clara 8,536 baptisms, 6,809 burials”; “Santa Clara 1,125 (1832) to 291 (1839); Jackson”).↩

44. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023), introduction and ch. 1.1 (`harris-palo-alto`), as quoted in Episode 1 Script (“02. The Broken Link”) and Episode 1 Review Log: “To speak of the Ohlone is to speak of the broken link between the Bay Area’s indigenous people and the Bay Area itself.”↩

45. Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Independence 1821; Echeandía July 25, 1826; 1831 counts 18,683 vs 4,342; decree Aug. 17, 1833; Figueroa Aug. 9, 1834”).↩

46. Santa Clara University Ohlone heritage project (“were supposed to be held in trust by the Franciscans on behalf of Native people”), as cited in Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Trust language; four Native grants; Ulistac and Posolmi; small fraction”).↩

47. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

48. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

49. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1

Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

50. Josiah Royce, *California, from the Conquest in 1846 to the Second Vigilance Committee in San Francisco: A Study of American Character* (1886), and Federal Reserve Bank of Minneapolis CPI table, as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Hides at two dollars; ranchero passage; ‘mission Indians had in large part disappeared’”).↩

51. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

52. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

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54. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), as quoted in Episode 1 Script (“07. Secularization”: “virtually slaves” and “the harpies of the civil power”) and Episode 1 Review Log.↩

55. Land-case literature, as cited in Episode 1 Script (“08. The Rancho”), Episode 1 Review Log (“Soto grant is the Rinconada, ~2,229 acres, downtown Palo Alto, tree at the corner; campus is Buelna plus Peña”), and Open Issues.↩

56. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

57. Richard Henry Dana Jr., *Two Years Before the Mast* (1840), and Federal Reserve Bank of Minneapolis CPI table (1835 = 31; 2026 = 1004.8), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Dana quotes (bank notes, 75 cents, shoes twice around the Horn, 300 percent, Cordilleras, virtually slaves, harpies, hide-house, May 8 1836 cargo, bay as centre)”).↩

58. Robert Jackson and Mission Santa Clara *padrón* signed December 31, 1839, by Father Jesús María Vásquez del Mercado (Calisphere catalog), as cited in Episode 1 Script (“07. Secularization”) and Episode 1

Review Log (“Santa Clara 1,125 (1832) to 291 (1839); Jackson”; “Padrón signed Dec. 31, 1839, by Vásquez del Mercado”).↩

59. W. W. Robinson and land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Alvarado born Monterey 1809; 1836 revolt; Buelna grant 1839, 1,471 acres”; “Grant process: petition, diseño, título, juridical possession; up to 11 leagues”).↩

60. W. W. Robinson and land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Alvarado born Monterey 1809; 1836 revolt; Buelna grant 1839, 1,471 acres”; “Grant process: petition, diseño, título, juridical possession; up to 11 leagues”).↩

61. W. W. Robinson and land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Alvarado born Monterey 1809; 1836 revolt; Buelna grant 1839, 1,471 acres”; “Grant process: petition, diseño, título, juridical possession; up to 11 leagues”).↩

62. Land-case records, Randall Milliken ethnography, and San Mateo County parks history, as cited in Episode 1 Script (“05. Saints on the Map”) and Episode 1 Review Log (“Junipero Serra County Park on former Buri Buri land”; “Rancho Buri Buri 14,639 acres; name from Urebure”; “Buri Buri Park exists”).↩

63. Alexander Forbes, *California: A History of Upper and Lower California* (London, 1839), as cited in Episode 1 Script (“06. Silver and Hides”) and Episode 1 Review Log (“Forbes, *California*, London, 1839”).↩

64. W. W. Robinson and land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Alvarado born Monterey 1809; 1836 revolt; Buelna grant 1839, 1,471 acres”; “Grant process: petition, diseño, título, juridical possession; up to 11 leagues”).↩

65. Robert Jackson and Mission Santa Clara *padrón* signed December 31, 1839, by Father Jesús María Vásquez del Mercado (Calisphere catalog), as cited in Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Santa Clara 1,125 (1832) to 291 (1839); Jackson”; “Padrón signed Dec. 31, 1839, by Vásquez del Mercado”).↩

66. Santa Clara University Ohlone heritage project (“were supposed to be held in trust by the Franciscans on behalf of Native people”), as cited in Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Trust language; four Native grants; Ulistac and Posolmi; small fraction”).↩

67. Land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Peña grant 1841, 8,418 acres; sold to the Robles brothers 1847”).↩

68. Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Cinnabar shown to Robles, 1824; New Almaden by 1845”); see also Leonard William Ascher, “The Economic History of the New Almaden

Quicksilver Mine, 1845–1863” (Ph.D. diss., University of California at Berkeley, 1934), in Gray Brechin, *Imperial San Francisco* (`brechin-imperial#69`).↩

69. Federal Reserve Bank of Minneapolis CPI table (1845 = 28; 2026 = 1004.8), as cited in Episode 1 Script (“07. Secularization”) and Episode 1 Review Log (“Pico sold San Juan Capistrano to Forster for $710 in 1845”).↩

70. Land-case literature, as cited in Episode 1 Script (“08. The Rancho”), Episode 1 Review Log (“Soto grant is the Rinconada, ~2,229 acres, downtown Palo Alto, tree at the corner; campus is Buelna plus Peña”), and Open Issues.↩

71. California State Parks and Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023), ch. 1.1 (`harris-palo-alto`), as cited in Episode 1 Script (“09. Nine Days”) and Episode 1 Review Log (“Bear Flag June 14–July 9, 1846; Revere raised the U.S. flag”).↩

72. Gray Brechin, *Imperial San Francisco: Urban Power, Earthly Ruin* (University of California Press, 1999; `brechin-imperial`), John B. Frisbie, “John B. Frisbie Reminiscences” (Bancroft Library; `brechin-imperial#69`), and Ulysses S. Grant, *Personal Memoirs of U.S. Grant*, vol. 1 (New York: Charles L. Webster and Co., 1885), p. 53 (`brechin-imperial#34`, `brechin-imperial#69`), as cited in Episode 1 Script (“09. Nine Days”) and Episode 1 Review Log (“Bidwell ‘pretense’ / ‘Texas game’; Frisbie on the bay; Grant on the war”).↩

73. San Francisco Museum (`sfmuseum.org`) and California State Library Celebrate California note, as cited in Episode 1 Script (“05. Saints on the Map”) and Episode 1 Review Log (“Bartlett ordinance, Jan. 30, 1847; Benicia”).↩

74. Land-case summaries, as cited in Episode 1 Script (“08. The Rancho”) and Episode 1 Review Log (“Peña grant 1841, 8,418 acres; sold to the Robles brothers 1847”).↩

75. Nicholas Trist to Virginia Jefferson Randolph Trist, December 4, 1847, National Archives Milestone Documents, as quoted in Episode 1 Script (“09. Nine Days”) and Episode 1 Review Log (“Treaty Feb. 2, 1848; Trist letter Dec. 4, 1847”).↩

76. Podcast Ep. 1 script (“09. Nine Days”) and review log: Monday, January 24, 1848, at Coloma on the South Fork of the American River; James Marshall, a carpenter from New Jersey, building a sawmill for Swiss immigrant John Sutter on his eleven-square-league Mexican grant (nearly 50,000 acres, the legal maximum).↩

77. Josiah Royce (1886), cited in Podcast Ep. 1 script (“06. Silver and Hides” and “09. Nine Days”) and review log: Royce, born in a California mining town in 1855, warned in 1886 that the tale of Marshall’s discovery is simple in its main incident and tangled in its details, with the speeches later put in Marshall’s mouth disagreeing.↩

78. Treaty of Guadalupe Hidalgo (Feb. 2, 1848), National Archives milestone documents, cited in Podcast Ep. 1 script (“09. Nine Days”) and review log: signed Wednesday, February 2, 1848, at Guadalupe

Hidalgo by Nicholas Trist and Mexican commissioners Luis Gonzaga Cuevas, Bernardo Couto, and Miguel Atristain.↩

79. Nicholas Trist to his wife, December 4, 1847, National Archives milestone page, quoted in Podcast Ep. 1 script (“09. Nine Days”) and review log: “Knowing it to be the very last chance and impressed with the dreadful consequences to our country which cannot fail to attend the loss of that chance, I decided today at noon to attempt to make a treaty; the decision is altogether my own.”↩

80. Treaty of Guadalupe Hidalgo, Articles V and XII (Feb. 2, 1848), National Archives milestone page, quoted in Podcast Ep. 1 script (“09. Nine Days”) and review log: “In consideration of the extension acquired by the boundaries of the United States, as defined in the fifth article of the present treaty, the Government of the United States engages to pay to that of the Mexican Republic the sum of fifteen million of dollars.”↩

81. Podcast Ep. 1 script (“09. Nine Days”) and Gray Brechin, *Imperial San Francisco* (`brechin-imperial#34`, ch. 5, n. 85), noting the nine-day interval between January 24 and February 2, 1848.↩

82. National Archives milestone page on the Treaty of Guadalupe Hidalgo, cited in Podcast Ep. 1 script (“09. Nine Days”) and review log: transfer of more than 525,000 square miles (55 percent of Mexico) for $15 million plus up to $3.25 million in assumed debts, matching the $15 million nominal price of the 1803 Louisiana Purchase.↩

83. National Archives milestone page, cited in Podcast Ep. 1 script (“09. Nine Days”) and review log: Senate ratification on March 10, 1848, striking Article X; the review log notes that the exact Senate tally is disputed between the National Archives milestone page (34–14) and other official tellings (38–14).↩

84. Podcast Ep. 1 script (“09. Nine Days”) and review log on the deletion of Article X’s explicit guarantee for Mexican land grants.↩

85. *The Californian* (San Francisco), March 15, 1848, cited in Podcast Ep. 1 script (“09. Nine Days”) and review log.↩

86. Podcast Ep. 1 script (“09. Nine Days”) and review log: John Jacob Astor died March 29, 1848, in New York City, leaving an estate estimated at ~$20 million (roughly $770 million in 2026 consumer prices using the Minneapolis Fed CPI table, 1848 index = 26, 2026 index = 1004.8), including a $400,000 bequest for the Astor Library (later part of the New York Public Library), thirty-seven years after his Pacific Fur Company built Fort Astoria in 1811.↩

87. Gordon H. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`, 2019), citing Phillips, *Portsmouth Plaza*, p. 104, in Podcast Ep. 1 script (“09. Nine Days”) and review log.↩

88. Protocol of Querétaro (May 26, 1848), signed by Nathan Clifford, Ambrose Sevier, and Mexican foreign minister Luis de la Rosa (Papers of Abraham Lincoln / GAO excerpt), quoted in Podcast Ep. 1 script (“09. Nine Days”) and review log: the U.S. government “did not in any way intend to annul the grants of lands made by Mexico in the ceded territories,” preserving their legal value under Mexican law up to May 13, 1846, to be acknowledged before American tribunals.↩

89. President James K. Polk, Message to Congress, December 5, 1848, cited in Podcast Ep. 1 script (“09. Nine Days”) and review log.↩

90. Podcast Ep. 1 script (“08. The Rancho” and “10. Prove It”) and review log: Rafael Soto’s 1835 *Rinconada del Arroyo de San Francisquito* (~2,229 acres, downtown Palo Alto, with El Palo Alto at its northwest corner); Antonio Buelna’s 1839 *Rancho San Francisquito* (1,471 acres, held by his widow María Concepción Valencia in 1848); and José Peña’s 1841 *Rancho Rincón de San Francisquito* (8,418 acres, sold in 1847 to Secundino and Teodoro Robles).↩

91. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (`harris-palo-alto`, 2023), intro and ch. 1.1, quoted in Podcast Ep. 1 script (“02. The Broken Link” and “10. Prove It”) and review log: disease killed about half of Alta California’s ~300,000 Native people under Spain and Mexico, and “when the United States showed up, the rules changed.”↩

92. Podcast Ep. 1 script (“08. The Rancho” and “10. Prove It”) and review log, citing W. W. Robinson on Mexican land grants (*título*, *diseño*, juridical possession) versus American surveyed property lines and courts.↩

93. Santa Clara University Ohlone heritage report, cited in Podcast Ep. 1 script (“03. What a Mission Was”) and review log: 1767 expulsion of the Jesuits by King Carlos III of Spain and 1851 transfer of Mission Santa Clara from the Franciscans to the Jesuits to open Santa Clara College (now Santa Clara University).↩

94. California Land Act of 1851, Bancroft Library finding aid and David Hornbeck (1979), cited in Podcast Ep. 1 script (“10. Prove It”), review log, and Open Issues notes: both sources record 813 claims filed (with 582 patented per Bancroft), while the confirmation count is split between 514 and 604 depending on how appeals between the land commission and the federal courts are counted.↩

95. Podcast Ep. 1 script (“08. The Rancho” and “10. Prove It”) and review log: María Concepción Valencia filed her claim for Rancho San Francisquito in 1853 and received the final U.S. patent in 1868, after squatters settled and George Gordon bought out both the squatters and her family; the Robles brothers lost much of their rancho to lawyers’ fees and debts by the late 1850s.↩

96. Podcast Ep. 1 script (“10. Prove It”) and review log: Leland Stanford practiced law in Port Washington, Wisconsin, until an 1852 fire burned out his office, after which he followed his brothers to the California mining country and opened a store selling supplies to miners. ## Part 2↩

97. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Shilling, W N., 365-66”); Episode 2 Script & Review (`podcast/ep02-the-toll.md`): “To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer.”↩

98. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing CRS R47747 and California Secretary of State fact sheet); Source Inventory (`CARDINAL_3_87c0.txt` address chain: “Colton Hall, 570 Pacific

Street, Monterey"). The forty-eight delegates count is marked [UNVERIFIED] per Open Issues and Episode 2 Review ("Standard count of that convention; not re-opened in CRS this pass").↩

99. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Outline acquisition-gaps table: CA ~2 yr 7 mo; Oregon ~12 yr 8 mo; Nevada ~16 yr 9 mo; Utah ~48 yr; NM and AZ ~64 yr).↩

100. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing CRS R47747 and California Secretary of State fact sheet: Sept. 5, 1849 vote 28–8; signed Oct. 13; ratified Nov. 13, 1849, 12,061–811).↩

101. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Malcolm Harris, *Palo Alto* [harris-palo-alto], National Archives milestone for the Compromise of 1850, and CRS R47747 for Sept. 9, 1850 admission).↩

102. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: "Huntington, Collis… background of, 47-48"); Episode 2 Script & Review (`podcast/ep02-the-toll.md`).↩

103. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: "Lexington, 51-52; Sherman, William T…. on naval transport to California, 51-52, 53"); Episode 2 Review (citing Ambrose, pp. 51–52).↩

104. Richard Rayner, *The Associates: Four Capitalists Who Created California* (cited in winkler-corporations#54, ch. 4 n. 6, and Episode 2 Review, citing Rayner, ch. 2); Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: "Huntington, Solon, 48, 49").↩

105. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Ambrose, p. 49 for $3,000, walking at least twenty times, and "It was only twenty-four miles. I walked it," and Rayner, ch. 2 for the thirty-nine-mile figure).↩

106. Gordon H. Chang, *Ghosts of Gold Mountain* (chang-ghosts-gold-mountain, ch. 3); Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing 1894 Senate memorial addresses and research memo on the two Asa Whitneys: 1791–1874 car-wheel maker on the Mohawk and Hudson Railroad vs. 1797–1872 China merchant).↩

107. Asa Whitney, *A Project for a Railroad to the Pacific* (1849), p. 40, quoted in Gordon H. Chang, *Ghosts of Gold Mountain* (chang-ghosts-gold-mountain, ch. 3), via Episode 2 Script & Review: "Through us must be the route to Asia, and the change to our continent will be the last, the final change."↩

108. Gordon H. Chang, *Ghosts of Gold Mountain* (chang-ghosts-gold-mountain, ch. 3), and Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: "Benton, Thomas Hart, 49, 366"), via Episode 2 Script & Review.↩

109. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: "Crocker, Charles… westward overland journey led by, 44-47, 49, 53; Strobridge, James Harvey… discipline maintained by, 120, 164, 198"); Episode 2 Review (citing Ambrose, pp. 43–45, 163).↩

110. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3: “One of those present was Charles Crocker—now weighing 250 pounds—who was running a dry-goods store in town”; ambrose-nothing#0, Index: “Hopkins, Mark… background of, 53; in business, 53, 54, 62, 73, 103”); Episode 2 Review (citing Ambrose, pp. 49, 73 for 52 K Street).↩

111. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Outline dates and SF Museum pages: May 14, 1856 shooting of James King of William by James P. Casey at Montgomery and Washington; death May 20; hanging of Casey and Charles Cora on May 22 from Fort Gunnybags, 41 Sacramento Street).↩

112. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, contrasting the 1851 and 1856 San Francisco Committees of Vigilance).↩

113. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1: “In 1853, Congress had called for a survey of possible routes. Secretary of War Jefferson Davis, of Mississippi, sent out four teams of surveyors… Their work was published in eleven large volumes by the government… They did not explore the forty-second parallel”; ambrose-nothing#0, Index: “Gadsden Purchase, 59”); Episode 2 Review (citing Ambrose, pp. 30, 58–60); `references.bib` (`cprr-pacrr-survey-1853`, `senate-pacific-railway-act`).↩

114. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Judah, Theodore… Niagara Gorge Railroad, 55-56, 69; Practical Plan for Building the Pacific Railroad, A (Judah), 59-60; Sacramento Valley Railroad designed by, 56, 57-58”); Episode 2 Script & Review (citing Ambrose, pp. 58–60, 65–66).↩

115. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “August 13, 1859, was a hot day in Council Bluffs, Iowa… The Republican paper in town, the *Nonpareil*, gave Lincoln a warm welcome, saying that ‘the distinguished “Sucker” [Iowa slang for someone from Illinois] has yielded to the solicitations of our citizens and will speak on the political issues of the day at Concert Hall… Go and hear old Abe.’”↩

116. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “When he reached St. Joseph, Missouri, he could have taken the only line of railroad across the state to return to Illinois, but instead he had gone aboard a stern-wheel steamboat that toiled up the Missouri River for nearly two hundred miles to Council Bluffs… To his friend Pusey, Lincoln said, ‘Not one, but many railroads will center here—’”↩

117. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “In the audience was Grenville Mellen Dodge, a twenty-eight-year-old railroad engineer. The next day he joined a group of citizens who had gathered on the big porch of the Pacific House, a hotel, to hear Lincoln answer questions… W.H.M. Pusey, with whom the speaker was staying, recognized young Dodge. He pointed out Dodge to Lincoln and said that the young engineer knew more about railroads than any ‘two men in the country.’”↩

118. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “Lincoln sat down beside Dodge, crossed his long legs, swung his foot for a moment, put his big hand on Dodge’s forearm, and went

straight to the point: ‘Dodge, what’s the best route for a Pacific railroad to the West?’… Or, as Dodge later put it, ‘He shelled my woods completely and got all the information I’d collected.’”↩

119. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “Norman Judd, attorney for the M&M and a legal and political associate of Lincoln, borrowed the money from Lincoln to buy seventeen lots for $3,500, using the lots as collateral… The *Council Bluffs Bugle* was very suspicious. ‘It has been rumored that G. M. Dodge, in consequence of being so largely interested in the Riddle Tract, was bound to make his surveys in such manner as would insure his own investments.’”↩

120. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “Born on February 12, 1809… At age twenty-three, he had entered politics… Lincoln handed the IC a bill for $2,000. The railroad rejected it, claiming, ‘This is as much as Daniel Webster himself would have charged.’ Lincoln submitted a revised bill for $5,000. When the corporation refused to pay, he brought suit and won… ‘one of the foremost railroad lawyers in the West.’ In 1857, he was thus the natural choice to argue one of the most important cases about railroads. The Rock Island Bridge Company had built the first bridge across the Mississippi River…” Also Richard Rayner, *The Associates*, ch. 3, cited in Episode 2 Review for the “manifest destiny” argument in the *Effie Afton* bridge case.↩

121. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “On October 20, 1859, he and Anna had set out from San Francisco on the *Sonora*… One night, at dinner with Burch and Lane, Anna asked, ‘Would there be any advantage in establishing a Pacific Railway Exhibit on Capitol Hill?’… On December 6, 1859, he got an appointment with President James Buchanan… At Anna’s suggestion, he made the room into the Pacific Railroad Museum… He had it completed by January 14, 1860… Curtis proposed a generous government loan of $60 million… in the form of 5 percent thirty-year bonds.”↩

122. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1, and ambrose-nothing#15, ch. 3): “Together, they were able to get John Kasson to write the railroad plank for the Republican platform, calling for the government to support a transcontinental railroad: ‘A Railroad to the Pacific Ocean is imperatively demanded by the interests of the whole country; the Federal Government ought to render immediate and efficient aid in its construction.’ … [Judah] had to confess that ‘the debate on the slavery question, and other matters of little moment [sic], left us no time for the consideration of the Pacific Railroad Bill.’ Then he made up his expense account. He said the whole trip cost him $2,500, but he was charging the convention only $40, to cover his printing expenses.”↩

123. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “By this time, early October 1860… A letter arrived, from a Dutch Flat druggist, Daniel W. Strong, called ‘Doc’ by his customers. Doc had heard about Judah’s exploring and wanted to show him the old emigrant road from Donner Lake, which had been abandoned after the Donner tragedy in 1846-47… Everywhere else the Sierra crested twice, parallel ridges with a deep valley between them. Here there was only one crest…”↩

124. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Her husband ‘could not sleep or rest after they got into Dutch Flat and Strong’s store, till he had stretched his paper on the counter and made his figures thereon. Then, turning to Dr. Strong, he said for the first time, “Dr. I shall make my survey over this, the Donner Pass or the Dutch Flat route, above every other.”’ ‘The next morning,’ Strong wrote, Judah ‘said to me, “Give me some writing materials”—I produced some and he sat down and drew up what he called “articles of association” [for the Central Pacific Railroad] and he shoved them across the table to me and said, “sign for what you want”’”↩

125. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “At Cape Horn, for example, two miles above Illinoistown, the drops beneath the old unused wagon road Judah was following were as much as fifteen hundred feet.”↩

126. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Judah estimated the mileage from Sacramento to the Nevada state line to be 115 miles, meaning he and Strong needed subscriptions of $115,000… Strong worked on the mining communities of Dutch Flat, Illinoistown, Grass Valley, and Nevada City and brought in $46,500… He wrote a pamphlet, published November 1, 1860, entitled *Central Pacific Railroad Company of California*… The next evening [Nov. 15], he left Anna at the San Francisco Russ House… Judah went back to his wife and told her, ‘Pack your bag, for I am going up to Sacramento on the boat to-morrow…’”↩

127. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Starting the day after he arrived, Judah held several meetings at the St. Charles Hotel in Sacramento, on J Street… ‘We none of us knew anything about railroad building,’ Crocker later said, ‘but at the same time were enterprising men, and anxious to have a road built, and have it come to Sacramento.’ … According to Anna, what he told the assembled potential investors… was: ‘You are tradesmen here in Sacramento city…’”↩

128. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1): “Lincoln, on the train from Springfield as he headed east, had taken a turn at driving the locomotive. Dodge went to the inaugural and told his wife, ‘Old Abe delivered the greatest speech of the age. It is backbone all over.’ Then he got to the point: ‘It looks as though we can get all our measures through and then I’ll make tracks for home.’”↩

129. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing California Historical Landmark 623 for Union Square and Zoeth Skinner Eldredge, *The Beginnings of San Francisco*, Appendix B, for Sanchez Street and Judah Street).↩

130. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#4, ch. 1, and ambrose-nothing#15, ch. 3): “On June 28, 1861, with the Civil War already under way, the Central Pacific Railroad of California came into formal existence. It was incorporated with Stanford as president, Huntington vice-president, Hopkins treasurer, James Bailey secretary, Judah chief engineer, and with Stanford, Crocker, Bailey, Judah, Huntington, Hopkins, Strong, and Charles Marsh as directors.”↩

131. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Their combined wealth, according to Huntington, was $159,000. Neither Congress, the state of California, nor any syndicates of capitalists had put a single penny into the corporation. California’s first and greatest historian, Hubert Howe Bancroft, commented that for so ‘stupendous and hazardous an enterprise it appears an act of madness or of inspiration…’” Also Episode 2 Script & Review for the CPI and share-of-economy comparison.↩

132. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “On September 4, Stanford won election as governor.”↩

133. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Three weeks later, Judah placed before the partners his written report… maximum grades of but 105 feet to a mile… eighteen tunnels, mainly through the mountains towering above Donner Lake, would be driven with relative ease, even the longest, at 1,370 feet… Instead, on October 9, well pleased with Judah’s report, they adopted a resolution: ‘That Mr. T. D. Judah … proceed to Washington, on the steamer of the 11th Oct. inst., as the accredited agent of the Central Pacific Railroad Company of California…’ Back on the Pacific Coast, the principals of the Central Pacific—Stanford, Crocker, Huntington, and Hopkins—moved on their own, apparently without Judah’s knowledge, by drawing up on November 27, 1861, articles of association for the Dutch Flat and Donner Lake Wagon Road Company. Four hundred shares of stock at $1,000 each were issued, with Crocker as president and Hopkins as secretary and treasurer… Three toll gates were established—at Dutch Flat, Polley’s Station, and Donner Lake.”↩

134. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Ambrose, p. 73, USGS ARkStorm, CA fact cards, and NRHP Stanford House for the Jan. 10, 1862 inauguration by boat during the Great Flood, term to Dec. 10, 1863, and Central Pacific presidency 1861–1890).↩

135. Gordon H. Chang, *Ghosts of Gold Mountain* (chang-ghosts-gold-mountain, ch. 3), and Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Stanford, Leland… on Chinese immigrant workers, 150, 164, 243”), via Episode 2 Script & Review (citing Chang, ch. 3 for the Jan. 10, 1862 inaugural address and Ambrose, p. 149 for the campaign phrases).↩

136. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing series DOC / Gordon H. Chang on Moy Jin Kee, Dr. Yee Fung Cheung, and Jin Mun’s engraved gold ring).↩

137. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “A prominent historian of the railroad, Robert Russell of Western Michigan University, puts it as his opinion that because the Southerners were gone ‘Congress was enabled to enact Pacific railway legislation several years earlier than it otherwise could have done.’ … Grenville Dodge, then serving as a general in the Union Army, said that Lincoln told him and others that the road had to be built ‘not only as a military necessity, but as a means of holding the Pacific Coast to the Union.’” Also Gordon H. Chang, *Ghosts of Gold Mountain* (chang-ghosts-gold-mountain, ch. 3).↩

138. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “To add to Judah’s already overflowing pleasure and confidence, a fellow passenger was newly elected Representative Aaron A. Sargent of California… On arrival at New York, Judah worked up a report on the Sierra surveys of which he published a thousand copies… A few days before the session began, Senator James A. McDougall of California helped Judah draft a bill that followed, in general, the Curtis Bill… Huntington came to the capital right after Christmas to find Judah discouraged.”↩

139. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Despite these events, on January 21, 1862, Sargent —encouraged by Judah and Huntington—took a bold step when, in the midst of debate on another subject, he got the floor and spoke at length on Judah’s work… Judah complained that Sargent spoke before an empty hall, but the speech had its effect. Within a week, the House appointed a special subcommittee… He got an appointment as secretary of the Senate Pacific Railroad Committee (with Senator McDougall as chairman) and as clerk of the subcommittee on the Pacific railroad in the House, where Sargent was a member.” Also `references.bib` (`nps-pacific-railway`).↩

140. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “Stevens, who owned a foundry in Pennsylvania, insisted on a requirement that all rails and other ironwork be of American manufacture. Stevens declared he was for passing the law not because of the iron rails he could sell but because someday the Southerners would return to Congress ‘with the same arrogant, insolent dictation which we have cringed to for twenty years, forbidding the construction of any road that does not run along our southern border.’ … Representative Justin Morrill of Vermont… commented sourly that those who were putting up the capital were not interested in building a railroad west from the Missouri River, or in the Central Pacific, if it went farther east than the Nevada silver mines…”↩

141. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “On May 6, 1862… the House passed the Pacific Railroad Bill by a vote of 79 to 49… On May 23, 1862, Judah wrote to the editors of the *Sacramento Union*, ‘The Pacific Railroad is a fixed fact and you can govern yourselves accordingly.’ … the Senate passed the bill on June 20, by a vote of 35 to 5. The House concurred in the Senate amendments a few days later and sent the completed bill forward to the President. Lincoln signed it on July 1, even as Malvern Hill, the last battle of the Seven Days’ Battles in the Peninsula, was being fought, costing McClellan’s army a thousand dead and three thousand wounded and greatly depressing the nation.”↩

142. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “The Pacific Railroad Bill was complicated to an almost incomprehensible degree”; `references.bib` (`pacific-railroad-act-1862`, Act of July 1, 1862, 12 Stat. 489).↩

143. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “It called for the creation of a corporation, the Union Pacific (the name being a nice touch in 1862), that would build west from the Missouri River, while the Central Pacific would build east from Sacramento. Capital stock of the UP was to be a hundred thousand shares at $1,000 each, or $100 million. Both roads would have

a right of way of two hundred feet on both sides of the road over public lands and would be given five alternate sections (square miles) on each side per mile, or sixty-four hundred acres per mile.”↩

144. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “The railroad corporations would receive financial aid in the form of government bonds at $16,000 per mile for flat land, $32,000 for foothills, and $48,000 per mile for mountainous terrain after they had built forty miles approved by government commissioners… The corporations could get advance money in the form of 6 percent government bonds. This was a loan, not a gift; the bonds were to constitute a first mortgage on the railroads… The Central Pacific was required to complete fifty miles within two years and fifty miles each year thereafter, and the entire road was to be completed by July 1, 1876, under pain of forfeiture.” Also Episode 2 Review for the CPI conversion.↩

145. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “paper money, 102”); Episode 2 Script & Review (citing Minneapolis Fed CPI table: index 27 in 1861 to 47 in 1864).↩

146. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3): “That was by no means all of it, but it was enough for Judah to flash the first word to his Sacramento colleagues by the newly established telegraph: ‘We have drawn the elephant. Now let us see if we can harness him.’ … At the time, forty-four members of the House, seventeen senators, and the secretary of the Senate tried to sum it up. They gave Judah a signed written testimonial of appreciation…”↩

147. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#15, ch. 3 footnote): “He was the sponsor of the Morrill Act, which granted to each loyal state thirty thousand acres for each senator and representative for the purpose of endowing at least one state agriculture college. The ‘Morrill Land Grant’ act made possible the great state universities. It passed on July 2, 1862.” Also Episode 2 Script & Review for the 1860 census New York apportionment (31 representatives + 2 senators = 33 × 30,000 = 990,000 acres), Cornell University (1865), and the comparison to Stanford’s 8,353-acre campus.↩

148. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Sierra Nevada mountains: border determination on, 108-9, 149; Whitney, Josiah D., 108, 149; Sargent, Aaron A., 76, 77-78, 108”); Episode 2 Script & Review (citing Ambrose, p. 107, and Stanford’s Sept. 29, 1862 telegram).↩

149. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Ambrose, p. 107: ~15 miles × $16,000 difference = $240,000, roughly $6.5 million in modern prices).↩

150. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Ambrose, p. 107: Sargent’s quote, “Here you see, my pertinacity and Abraham’s faith moved mountains,” and the alternative report attributing “Here is a case in which Abraham’s faith has moved mountains” to Lincoln).↩

151. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Ambrose, p. 108: Judah telling Strong “the foothills do not begin here” and writing to Anna, “Rightly used it gives them unlimited credit throughout the world, and they would beggar it”).↩

152. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Judah, Theodore… Big Four conflicts with, 103-6, 108-9, 110-14, 115-16, 117, 124; death of, 115-17”); Episode 2 Script & Review (citing Ambrose, p. 108, and series DOC for death on Nov. 2, 1863, at age 37).↩

153. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Dodge, Grenville… Lincoln’s meetings with, 23-24, 26, 39, 86-88, 97, 270-71”); Episode 2 Script & Review (citing Ambrose, pp. 85–86).↩

154. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Pacific Railroad Act (1864), 94-96, 123, 143, 147, 254, 271; Washburne, E. B., 94-95; Stewart, Joseph P., 95-96; Sherman, Charles T., 96, 139”); Episode 2 Script & Review (citing Ambrose, pp. 94–95, 375).↩

155. Stephen E. Ambrose, *Nothing Like It in the World* (ambrose-nothing#0, Index: “Charles Crocker Contract and Finance Company, 105-6, 111, 165, 246-47, 321, 339, 375; Crédit Mobilier of America… creation of, 92-93; Contract and Finance Company vs., 105, 246; Pennsylvania Fiscal Agency, 92-93; Train, George Francis, 86, 88, 90, 92-93; Ames, Oakes… Crédit Mobilier and, 132, 190, 226, 227, 320, 336, 374-75; Ames, Oliver… CP loan made by, 103, 106, 113; Ames shovels, 103, 137”); Episode 2 Script & Review.↩

156. Episode 2 Script & Review (`podcast/ep02-the-toll.md`, citing Lincoln Log for March 18, 1865 patents to Bishop Joseph Sadoc Alemany for Mission San Juan Capistrano and Mission San Luis Rey; *Foster v. Mora*, 98 U.S. 425, for 46.74 acres vs. 44.40-acre state-lands patent note; San Gabriel patent Nov. 19, 1859; San Diego patent May 23, 1862; La Purísima 1874; and callback to Pío Pico’s $710 sale).↩

157. Stephen E. Ambrose, *Nothing Like It in the World: The Men Who Built the Transcontinental Railroad, 1863–1869* (`ambrose-nothing#0`, index p. 120; Episode 2 Script & Review, sec. 12): James Harvey Strobridge was Irish, over six feet tall, and thirty-seven when he took the work, stating that men generally earned their money when they worked for him.↩

158. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 119–120; Episode 2 Script & Review, sec. 12): at Bloomer Cut a black-powder blast cost Strobridge his right eye.↩

159. Episode 2 Script & Review, sec. 12 (citing Ambrose and Gordon H. Chang, *Ghosts of Gold Mountain*, `chang-ghosts-gold-mountain`): Chinese workers who learned English called Strobridge the “one-eyed bossy man,” and foremen clapped a hand over an eye to make crews think he was approaching.↩

160. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 149–150; Episode 2 Script & Review, sec. 12): Strobridge’s refusal, “I will not boss Chinese.”↩

161. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 107, 119, 148, 154; `ambrose-nothing#15`, ch. 3): white laborers repeatedly defected from the railroad grade to the Nevada silver mines.↩

162. Episode 2 Script & Review, sec. 12: flags the "Great Wall of China" retort by Charles Crocker as legend, contrasted with the payroll records.↩

163. Episode 2 Script & Review, sec. 12 (citing Chew / cprr.org research memo vs. Ambrose, *Nothing Like It in the World*, `ambrose-nothing#0`, pp. 149–150): DISPUTED date of first Chinese hiring—Chinese crews under contractors Hung Wah and Ah Toy appear on Central Pacific payrolls in January and February 1864, a year earlier than the traditional 1865 date in Ambrose.↩

164. Gordon H. Chang, *Ghosts of Gold Mountain: The Epic Story of the Chinese Who Built the Transcontinental Railroad* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): circulation of thousands of copies of a woodcut recruitment flyer in California and China, of which no copy survives.↩

165. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): Crocker admitted to Huntington that no one on the company side had translated the recruitment flyer back into English.↩

166. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Argon Audit sec. 2): historians cite 10,000–12,000 Chinese workers on the Central Pacific at peak, and 15,000–20,000 total across the entire build.↩

167. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): Chinese workers comprised about 90 percent of the Central Pacific workforce, "by far the largest single workforce in American industry to that date."↩

168. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index p. 164; Episode 2 Review, p. 163): Strobridge confronted Chinese strike or trouble leaders with an ax handle.↩

169. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 44–47, 164; Episode 2 Script & Review, secs. 04, 12): Crocker's exchange with Strobridge ("They are human creatures" / "you have got to do it… about as near brutes as they can get"), and callback to Crocker's June 1850 Green River pistol incident.↩

170. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 198–199, 314; Episode 2 Script & Review, sec. 12): Hanna Maria Strobridge lived in a fitted-out passenger car at the end of track with an awning and a caged canary.↩

171. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 198–199; Episode 2 Script & Review, sec. 12): called the "Heroine of the Central Pacific," she was the only white woman who saw the work through from beginning to end.↩

172. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#15`, ch. 3, n. 39): on November 27, 1861, Stanford, Crocker, Huntington, and Hopkins drew up articles of association for the Dutch Flat and Donner Lake Wagon Road Company (400 shares at $1,000 each; Crocker president, Hopkins secretary and treasurer; toll gates at Dutch Flat, Polley's Station, and Donner Lake).↩

173. Episode 2 Script, sec. 04; Open Issues for Part 2: Hopkins's February 1866 letter to Huntington on snow stopping work five or six months a year and paying teamsters $1.25 a ton is in the podcast script (and cf. `ambrose-nothing#0`, index pp. 195–196, 233), but was not in the review's verification table and is marked `[UNVERIFIED]`.↩

174. Episode 2 Script, sec. 04: Hopkins on teamsters preferring the old Pacific Road for its taverns, waiting girls, and barmaids.↩

175. Episode 2 Script, sec. 04: Hopkins's builder's arithmetic—six months if they reached the desert that season, eighteen months if they missed it.↩

176. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index p. 206; Episode 2 Script & Review, sec. 12): E. B. Crocker's letter to Huntington during the winter of 1866–1867 on buying up all the cheap goggles in the market for snow-blindness.↩

177. Episode 2 Script & Review, sec. 12 (citing Ambrose): E. B. Crocker's line, "Think of that… What an item in railroad building."↩

178. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#15`, ch. 3, n. 32; `ambrose-nothing#0`, index p. 244): Judah's report projected the longest tunnel above Donner Lake at 1,370 feet, and Summit Tunnel was engineered by Lewis M. Clement.↩

179. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 160–161, 199–201; Episode 2 Script & Review, sec. 12): three-man drilling crews at the granite face using fourteen-to-eighteen-pound sledges and black powder.↩

180. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 160–161; Episode 2 Script & Review, sec. 12): progress of six to twelve inches a day in the Summit Tunnel with shifts organized around the clock by Samuel Skerry Montague.↩

181. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 204–205; Episode 2 Script & Review, sec. 12): *Dutch Flat Enquirer*, Christmas Day, 1866, reporting four or five Chinese workers killed in a snow slide.↩

182. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 204–205; Episode 2 Script & Review, sec. 12): avalanche near the Summit Tunnel carrying away some twenty Chinese workers whose bodies were found after the spring thaw.↩

183. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index p. 204; Episode 2 Script & Review, sec. 12): historian Thomas W. Chinn's judgment that the loss of life was heavy.↩

184. Episode 2 Script & Review, sec. 12; Open Issues for Part 2: June 1870 train carrying boxed bones of perhaps 1,200 Chinese workers through Sacramento is hedged as “by one report”; the newspaper masthead was not in hand in the review and is marked `[UNVERIFIED]`.↩

185. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`) and Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 240–242; Episode 2 Script & Review, sec. 12): Monday, June 24, 1867 strike by 3,000 Chinese workers across nearly thirty miles from Cisco to Truckee.↩

186. Episode 2 Script & Review, sec. 12 (citing Chang and Ambrose, plus Minneapolis Fed CPI, 1867 index 42): strikers demanded $40 a month instead of $35 (roughly $840/month or $10,000/year today) and a ten-hour day instead of eleven.↩

187. Episode 2 Script & Review, sec. 12; Open Issues for Part 2: contemporary newspaper calling it the “greatest strike” is paraphrased because the specific newspaper masthead was not in hand; marked `[UNVERIFIED]`.↩

188. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 240–242; Episode 2 Script & Review, sec. 12): Mark Hopkins to Huntington on the strike (“if they succeed in this demand, then they control”).↩

189. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 240–242; Episode 2 Script & Review, sec. 12): E. B. Crocker’s letter (“The truth is they are getting smart”).↩

190. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 240–242; Episode 2 Script & Review, sec. 12): E. B. Crocker’s proposal to inundate California and Nevada with laborers—freedmen, Chinese, Japanese, all kinds.↩

191. Episode 2 Script & Review, sec. 12 (comparing Charles Crocker’s account in Ambrose with Chang, *Ghosts of Gold Mountain*, `chang-ghosts-gold-mountain`): DISPUTED strike demands (eight hours in Crocker’s telling vs. ten hours in company wires) and DISPUTED outcome (total defeat in Crocker’s telling vs. shaken partners and subsequent wage increases for skilled Chinese workers in Chang’s reading of the letters).↩

192. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): Charles Crocker’s rendering of the strike leaders’ words, “Eight hours a day is good for a white man, all the same good for a Chinaman.”↩

193. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): Charles Crocker’s pidgin reply to the strikers.↩

194. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 240–242; Episode 2 Script & Review, sec. 12): Crocker and the labor agents cutting off food and provisions to the striking camps.↩

195. Episode 2 Script & Review, sec. 12 (citing Chang and Ambrose): E. B. Crocker wrote that none of them went near the strikers for a week so as not to “exhibit anxiety.”↩

196. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 240–242; Episode 2 Script & Review, sec. 12): the strike ended after eight days.↩

197. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): Chang’s analysis of the partners’ letters showing they were shaken and that pay for skilled Chinese workers rose in the following months.↩

198. Episode 2 Script & Review, sec. 12 (citing E. B. Crocker’s business letter in Chang/Ambrose): “We want a hundred thousand chinamen here, so as to bring the price of labor down.”↩

199. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index p. 243; Episode 2 Script & Review, sec. 12): Huntington wrote that it would be all the better for them and the state if half a million came over in 1868.↩

200. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`, ch. 3; Episode 2 Script & Review, sec. 07): callback to Governor Leland Stanford’s January 10, 1862 inaugural address calling Chinese immigrants the “dregs of her population.”↩

201. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index p. 244; Episode 2 Script & Review, sec. 12): Chinese crews holed through the Summit Tunnel on August 3, 1867.↩

202. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 170–171, 201–202, 255, 290–292, 297–317, 327, 353, 371–372): the Central Pacific line along the Truckee and Humboldt rivers and across the Nevada desert into Utah, racing Grenville Dodge and Jack and Dan Casement’s Union Pacific crews and grading parallel roadbeds.↩

203. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 346–352; Episode 2 Script & Review, sec. 13, citing Ambrose ch. 16): Ten Mile Day began at 7:00 a.m. on April 28, 1869.↩

204. Episode 2 Script & Review, sec. 13 (citing Ambrose ch. 16, and Minneapolis Fed CPI, 1869 index 40): Crocker’s $10,000 bet with Thomas “Doc” Durant, roughly $250,000 (or $251,000) today.↩

205. Episode 2 Script & Review, sec. 13 (citing Ambrose ch. 16): DISPUTED / hedged whether Durant formally took the $10,000 bet (“Ambrose says Durant, reportedly, was sure they could not, and took the bet”).↩

206. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 346–352; Episode 2 Script & Review, sec. 13): by 7:00 p.m. on April 28, 1869, crews had laid ten miles and fifty-six feet of track up the slope toward Promontory.↩

207. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index p. 349; Episode 2 Script & Review, sec. 13; Argon Audit sec. 3, item 32): celebratory accounts named the eight Irish rail handlers (Michael Shay, Michael Kennedy, Michael Sullivan, Patrick Joyce, Thomas Daley, George Elliott, Edward Killeen, and Fred McNamara, with track boss George Coley) and no Chinese workers.↩

208. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 13): several thousand Chinese and Irish workers completed the ten-mile day, while official accounts recorded only the eight Irish names.↩

209. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 12): Strobridge seated his Chinese foreman, most likely labor contractor Hung Wah, at the head of a dinner table, reported under the heading “Chinese Laborers at Table.”↩

210. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 13; cf. `ambrose-nothing#0`, index p. 255 for Monument Point, Utah): Chinese workers, perhaps including Hung Wah, climbed a bluff near Monument Rock with a brown-glazed rice-wine jug from the Pearl River delta.↩

211. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 13): archaeologists found the jug a century and a half later; Chang wrote that the bottle, though mute, had something to say.↩

212. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 360–368; Episode 2 Script & Review, sec. 01, citing Ambrose ch. 17): Promontory Summit ceremony on May 10, 1869.↩

213. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 361–367; Episode 2 Script & Review, sec. 01): the Central Pacific’s *Jupiter* and the Union Pacific’s *No. 119* faced each other at Promontory.↩

214. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 359–360; Episode 2 Script & Review, sec. 13): unpaid Union Pacific workers detached and guarded Durant’s car until Oliver Ames and Durant wired the ransom/wages, delaying the ceremony two days.↩

215. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 360–367; Episode 2 Script & Review, sec. 01): Leland Stanford was the only one of the four Sacramento associates at Promontory, joined by Montague and Strobridge for the CP and Dodge, the Casement brothers, Dillon, and Reed for the UP.↩

216. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 360, 364–367; Episode 2 Script & Review, sec. 01): silver-headed maul, polished California laurel tie with pre-drilled hole, and the Golden Spike.↩

217. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 365–366; Episode 2 Script & Review, sec. 01): telegraph operator W. N. Shilling’s message: “To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer.”↩

218. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 363, 365; Episode 2 Script & Review, sec. 01): engineers George Booth and Sam Bradford pulled the locomotive whistles.↩

219. Episode 2 Script & Review, sec. 01 (citing Ambrose ch. 17): a veteran in the crowd recalled that “they all yelled like to bust.”↩

220. Episode 2 Script & Review, sec. 01 (citing Ambrose ch. 17): Stanford’s speech ran under three minutes; a Chinese squad in clean blue frocks and an Irish squad carried the last rails into the gap.↩

221. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 365–366; Episode 2 Script & Review, sec. 01, citing Ambrose ch. 17): Stanford swung the maul, missed the spike, and hit the rail.↩

222. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 365–366; Episode 2 Script & Review, sec. 01): Shilling closed the circuit and sent “done” across the continent.↩

223. Episode 2 Script & Review, sec. 01: officials announced the combined track laid by the two roads as 1,776 miles.↩

224. Episode 2 Script & Review, sec. 01 (citing Ambrose): after the dignitaries left, workers pulled the golden spike and laurel tie out and drove ordinary iron spikes into ordinary wood ties.↩

225. Episode 2 Script & Review, sec. 13: the California laurel tie burned in the 1906 San Francisco fire, while the golden spike is preserved in the art museum at Stanford.↩

226. Ambrose, *Nothing Like It in the World* (`ambrose-nothing#0`, index pp. 365, 366–367) and Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 13): Andrew J. Russell’s photograph *East and West Shaking Hands*.↩

227. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 13): Chang notes there is no evidence Russell staged the crowd to exclude Chinese workers.↩

228. Chang, *Ghosts of Gold Mountain* (`chang-ghosts-gold-mountain`; Episode 2 Script & Review, sec. 13): a blurred figure in the center of Russell’s frame, identifiable by clothes, hat, and stature as Chinese.↩

229. Episode 2 Script & Review, sec. 13: Transportation Secretary John Volpe at the 1969 Promontory centennial (“Who else but Americans could drill tunnels in mountains thirty feet deep in snow?”).↩

230. Richard Rayner, *The Associates: Four Capitalists Who Created California* (`winkler-corporations#54`, ch. 4, n. 6; Episode 2 Script & Review, sec. 11): in February 1873, during Representative Jeremiah Wilson’s congressional committee hearings, Mark Hopkins carried fifteen leather volumes of construction accounts to the basement furnace and burned them.↩

231. Rayner, *The Associates* (Episode 2 Script & Review, sec. 11): Huntington’s testimony to the Wilson committee, “My partner Hopkins is a peculiar man. He considered the paper no longer worth saving.”↩

232. Episode 2 Script & Review, sec. 11 (citing Rayner): Mark Hopkins died in March 1878, five years after burning the books in February 1873.↩

233. Episode 2 Script & Review, sec. 13 (citing Kilston research memo on Sheila Rothman and John E. Baur): 1852 statement by Dr. James Blake and post-1849 statement by San Francisco physician Dr. J. Praslow on California's climate.↩

234. Episode 2 Script & Review, sec. 13 (citing Kilston research memo on Rothman and Baur): illness motivated roughly a quarter of arrivals between 1840 and 1900, with the major rush after 1870 when a state report called California the "sanatorium of the world."↩

235. Episode 2 Script & Review, sec. 13: foreshadowing the post-1870 health migration, the "coughing rich," and Leland Stanford Jr.↩

236. Episode 2 Script & Review, sec. 13: the completed railroad as a private toll on every bushel of wheat moving out of California. ## Part 3↩

237. Collis P. Huntington to David D. Colton, November 30, 1874 (Letter No. 389), reprinted in *How Members of Congress Are Bribed* (Moore pamphlet), cited in Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (New York: W. W. Norton, 2011) (`[@white-railroaded]`), and Episode 3 Script & Review ("Friends"): "I think this coming session of Congress will be composed of the hungriest set of men that ever got together."↩

238. Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (New York: W. W. Norton, 2011) (`[@white-railroaded]`); Richard Rayner, *The Associates: Four Capitalists Who Created California* (2008/2009), cited in `[EXCERPT winkler-corporations#54]` (ch. 4, n. 6) and Episode 3 Script ("Friends").↩

239. Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (New York: W. W. Norton, 2011) (`[@white-railroaded]`); Richard Rayner, *The Associates: Four Capitalists Who Created California* (2008/2009), cited in `[EXCERPT winkler-corporations#54]` (ch. 4, n. 6) and Episode 3 Script ("Friends").↩

240. Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (New York: W. W. Norton, 2011) (`[@white-railroaded]`); Richard Rayner, *The Associates: Four Capitalists Who Created California* (2008/2009), cited in `[EXCERPT winkler-corporations#54]` (ch. 4, n. 6) and Episode 3 Script ("Friends").↩

241. *How Members of Congress Are Bribed* (Moore pamphlet, 1883 Colton trial record), cited in Richard White, *Railroaded* (`[@white-railroaded]`), Richard Rayner, *The Associates*, and Episode 3 Script & Review ("Friends").↩

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& Review (“Friends”).↩

244. *How Members of Congress Are Bribed* (Moore pamphlet, 1883 Colton trial record), cited in Richard White, *Railroaded* (`[@white-railroaded]`), Richard Rayner, *The Associates*, and Episode 3 Script & Review (“Friends”).↩

245. Collis P. Huntington to David D. Colton, November 30, 1874 (Letter No. 389), reprinted in *How Members of Congress Are Bribed* (Moore pamphlet), cited in Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (New York: W. W. Norton, 2011) (`[@white-railroaded]`), and Episode 3 Script & Review (“Friends”): “I think this coming session of Congress will be composed of the hungriest set of men that ever got together.”↩

246. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“The Bought Press”): Huntington to Hopkins on the AP agent in Washington; lobbyist Richard Franchot placing the Washington reporter on the Central Pacific payroll until AP editor James Simonton noticed the change in tone; and the 1875 bid to buy an exclusive AP membership for the *San Francisco Chronicle* at double the going rate to cut off the *Bulletin* and the *Call*.↩

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251. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“The Bought Press”): Huntington to Hopkins on the AP agent in Washington; lobbyist Richard Franchot placing the

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252. *San Francisco Chronicle* history; Richard T. Thieriot, “125 Years of the Chronicle,” *San Francisco Chronicle*, January 14, 1990, and Michael de Young, “History of the San Francisco Chronicle,” Bancroft Library, cited in Gray Brechin, *Imperial San Francisco: Urban Power, Earthly Ruin* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 1; `[EXCERPT brechin-imperial#69]`), and Episode 3 Script & Review (“The Chronicle”).↩

253. *San Francisco Chronicle* history; Richard T. Thieriot, “125 Years of the Chronicle,” *San Francisco Chronicle*, January 14, 1990, and Michael de Young, “History of the San Francisco Chronicle,” Bancroft Library, cited in Gray Brechin, *Imperial San Francisco: Urban Power, Earthly Ruin* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 1; `[EXCERPT brechin-imperial#69]`), and Episode 3 Script & Review (“The Chronicle”).↩

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257. Collis P. Huntington to David D. Colton, January 17, 1876 (Letter No. 107), in *How Members of Congress Are Bribed* (Moore pamphlet), and Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“Friends”): “It cost money to fix things so that I would know his bill would not pass. I believe that with $200,000 I can pass our bill.” (Minneapolis Fed CPI conversion: 1876 index 32 to 2026 index 1004.8, ×31.4 ≈ $6.3 million.)↩

258. Collis P. Huntington to David D. Colton, January 17, 1876 (Letter No. 107), in *How Members of Congress Are Bribed* (Moore pamphlet), and Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“Friends”): “It cost money to fix things so

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262. Collis P. Huntington to David D. Colton, January 29, 1876 (Letter No. 110) and March 23, 1876 (Letter No. 138), in *How Members of Congress Are Bribed* (Moore pamphlet), cited in Episode 3 Script & Review ("Friends"): "I keep on high ground."↩

263. Collis P. Huntington to David D. Colton, January 29, 1876 (Letter No. 110) and March 23, 1876 (Letter No. 138), in *How Members of Congress Are Bribed* (Moore pamphlet), cited in Episode 3 Script & Review ("Friends"): "I keep on high ground."↩

264. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review ("Friends" and "The Bought Press"): "Nineteenth-century Americans were not shocked by the corruption of the press."↩

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267. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review ("Friends" and "The Bought Press"): "Nineteenth-century Americans were not shocked by the corruption of the press."↩

268. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review ("Friends") on Tom Scott (Texas and Pacific), Jay Gould, Charles Perkins (Chicago, Burlington and Quincy), and John Murray Forbes.↩

269. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“Friends”) on Tom Scott (Texas and Pacific), Jay Gould, Charles Perkins (Chicago, Burlington and Quincy), and John Murray Forbes.↩

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271. Richard Rayner, *The Associates*, cited in Episode 3 Script & Review (“Nob Hill”); Diana Strazdes, “The Millionaire’s Palace: Leland Stanford’s Commission for Pottier & Stymus in San Francisco,” *Winterthur Portfolio* 36, no. 4 (Winter 2001): 213–16, 220, and Bertha Berner, *Mrs. Leland Stanford: An Intimate Account* (Stanford: Stanford University Press, 1935), 19–20, cited in Richard White, *Who Killed Jane Stanford?* (`[EXCERPT white-jane#1]`, ch. 1, n. 1–2). (Minneapolis Fed CPI conversion: $2 million in 1876 ×31.4 ≈ $63 million.)↩

272. Richard Rayner, *The Associates*, cited in Episode 3 Script & Review (“Nob Hill”); Diana Strazdes, “The Millionaire’s Palace: Leland Stanford’s Commission for Pottier & Stymus in San Francisco,” *Winterthur Portfolio* 36, no. 4 (Winter 2001): 213–16, 220, and Bertha Berner, *Mrs. Leland Stanford: An Intimate Account* (Stanford: Stanford University Press, 1935), 19–20, cited in Richard White, *Who Killed Jane Stanford?* (`[EXCERPT white-jane#1]`, ch. 1, n. 1–2). (Minneapolis Fed CPI conversion: $2 million in 1876 ×31.4 ≈ $63 million.)↩

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Jane Stanford? (`[EXCERPT white-jane#1]`, ch. 1, n. 1–2). (Minneapolis Fed CPI conversion: $2 million in 1876 ×31.4 ≈ $63 million.)↩

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279. Thorstein Veblen, *The Theory of the Leisure Class* (1899), cited in Episode 3 Script & Review (“Nob Hill”).↩

280. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (New York: Little, Brown, 2023) (`[@harris-palo-alto]`), cited in Episode 3 Script & Review (“The Sandlot”).↩

281. Episode 3 Script & Review (“The Story of Our Streets”) on the etymology of Nob Hill from *nabob* and *nawab*.↩

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283. Malcolm Harris, *Palo Alto* (`[@harris-palo-alto]`), cited in Episode 3 Script & Review (“The Sandlot”): “I will give the Central Pacific just three months to discharge their Chinamen, and if that is not done, Stanford and his crowd will have to take the consequences.”↩

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286. William Lilley III, “The Early Career of Francis G. Newlands, 1848–1897” (Ph.D. diss., Yale University, 1965), 67–69, 94, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 2, n. 10).↩

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292. Episode 3 Script & Review (“The Sandlot” and “Critique pass”) notes Leland Stanford’s 1862 gubernatorial inaugural phrases (“the dregs of her population,” “a degraded and distinct people”) and campaign phrase (“the dregs of Asia”) as carried over from Episode 2; flagged `[UNVERIFIED]` in the Part 3 open-issues log pending a primary text citation in this part’s source set.↩

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294. Richard Rayner, *The Associates*, ch. 12, and OpenSFHistory, cited in Episode 3 Script & Review (“Nob Hill” and “Yuma”): Mark Hopkins’s death on a Yuma siding on the night of March 28, 1878 (or March 29 in some notices), and Collis Huntington’s letter (“My old friend Mr. Hopkins has gone over the river…”).↩

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301. Online Archive of California finding aid and OpenSFHistory (July 4, 2021), cited in Episode 3 Script & Review ("Nob Hill") on Andrew Hallidie's Clay Street line (1873) and the opening of the California Street Cable Railroad on April 10, 1878.↩

302. Online Archive of California finding aid and OpenSFHistory (July 4, 2021), cited in Episode 3 Script & Review ("Nob Hill") on Andrew Hallidie's Clay Street line (1873) and the opening of the California Street Cable Railroad on April 10, 1878.↩

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304. Richard Rayner, *The Associates*, *San Francisco Examiner* (August 17, 1878), OpenSFHistory (1100 California Street), and *San Francisco Chronicle* / KQED retrospectives, cited in Episode 3 Script & Review ("Nob Hill") on Nicholas Yung's lot, Crocker's forty-foot spite fence (thirty feet in some later retellings), the ~$3,000 cost (×34.6 ≈ $100,000), Yung's threatened roof coffin, his move to Broderick Street, and his death in 1880.↩

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312. Richard Rayner, *The Associates*, cited in Episode 3 Script & Review (“Friends” and “Yuma”) on David Colton’s injury on October 8, 1878, his death two days later (horse fall vs. street rumor of a stabbing), and the partners pressing Ellen Colton to return her shares.↩

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316. Episode 3 Script & Review (“Nob Hill” and “Yuma”), citing Stanford Court Hotel history (905 California), OpenSFHistory, and Historic Hotels of America (Arabella Huntington’s 1915 gift of Huntington Park; William H. Crocker’s family land gift for Grace Cathedral after the April 18, 1906 earthquake and fire).↩

317. U.S. Department of State and National Archives records, cited in Episode 3 Script & Review (“Exclusion”) on President Rutherford B. Hayes’s March 1, 1879 veto of the Fifteen Passenger Bill under the Burlingame Treaty and subsequent treaty renegotiation (the Angell Treaty).↩

318. U.S. Department of State and National Archives records, cited in Episode 3 Script & Review (“Exclusion”) on President Rutherford B. Hayes’s March 1, 1879 veto of the Fifteen Passenger Bill under the Burlingame Treaty and subsequent treaty renegotiation (the Angell Treaty).↩

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320. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“The Bought Press” and “The Sandlot”) on the 1879 California Constitution, its anti-Chinese clauses, and the creation of the three-member elected Railroad Commission.↩

321. Richard White, *Railroaded* (`[@white-railroaded]`), cited in Episode 3 Script & Review (“The Bought Press” and “The Sandlot”) on the 1879 California Constitution, its anti-Chinese clauses, and the creation of the three-member elected Railroad Commission.↩

322. Henry George, *Progress and Poverty* (1879), cited verbatim in Episode 3 Script & Review (“The Wheat Field”): “A railroad company approaches a small town as a highwayman approaches his victim…”↩

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324. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

325. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

326. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

327. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

328. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

329. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

330. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

331. Episode 3 Script (“The Chronicle”) reports that one juror at Milton Kalloch’s trial said he would have done the same; marked `[UNVERIFIED]` per the Part 3 open-issues log as lacking a specific citation in the Episode 3 review table.↩

332. Irving McKee, “The Shooting of Charles de Young,” *Pacific Historical Review* 16, no. 3 (August 1947): 271–84, cited in Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 6; `[EXCERPT brechin-imperial#69]`); *San Francisco Chronicle* history and *New York Times*, April 24, 1880, cited in Episode 3 Script & Review (“The Chronicle”).↩

333. Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 4, n. 13, 24, 44; `[EXCERPT brechin-imperial#69]`), citing Arthur Chandler and Marvin Nathan, *The Fantastic Fair* (1993), Michael Corbett, *An Architectural History of the M.H. de Young Memorial Museum* (1994), and Upton Sinclair, *The Brass Check* (1920), 275; and Episode 3 Script & Review (“The Story of Our Streets”).↩

334. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

335. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

336. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

337. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

338. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

339. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

340. Pacific Railway Act of July 1, 1862, 12 Stat. 489, Sec. 3 (`[@pacific-railroad-act-1862]`), and *Curtner v. United States*, cited in Episode 3 Script & Review (“The Wheat Field”).↩

341. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

342. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

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351. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

352. *Visalia Times*, May 15, 1880, as reprinted in *History of Tulare County*, and Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”): “As has long been anticipated, the conflict.”↩

353. *Visalia Times*, May 15, 1880, as reprinted in *History of Tulare County*, and Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”): “As has long been anticipated, the conflict.”↩

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359. Richard Rayner, *The Associates*, ch. 13, cited in Episode 3 Script & Review (“The Wheat Field”) on the May 11, 1880 Mussel Slough shootout near Hanford.↩

360. Richard Rayner, *The Associates*, *San Francisco Examiner* (August 17, 1878), OpenSFHistory (1100 California Street), and *San Francisco Chronicle* / KQED retrospectives, cited in Episode 3 Script & Review (“Nob Hill”) on Nicholas Yung’s lot, Crocker’s forty-foot spite fence (thirty feet in some later retellings), the ~$3,000 cost (×34.6 ≈ $100,000), Yung’s threatened roof coffin, his move to Broderick Street, and his death in 1880.↩

361. Online Archive of California (George and Phoebe Apperson Hearst papers) and *San Francisco Chronicle* / *Examiner* history, cited in Episode 3 Script & Review (“The Story of Our Streets” and “Bierce and Hearst”); see also Gray Brechin, *Imperial San Francisco* (`[EXCERPT brechin-imperial#34]`, ch. 7, n. 11).↩

362. U.S. Bureau of the Census, Tenth Census (1880) compilations (75,132 Chinese residents in California out of a state population of 864,894, or 8.7%; 105,465 Chinese residents in the United States), cited in Episode 3 Script & Review (“The Sandlot” and “Exclusion”).↩

363. U.S. Bureau of the Census, Tenth Census (1880) compilations (75,132 Chinese residents in California out of a state population of 864,894, or 8.7%; 105,465 Chinese residents in the United States), cited in Episode 3 Script & Review (“The Sandlot” and “Exclusion”).↩

364. Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (`white-railroaded` in `references.bib`); Cardinal Podcast Ep. 3 script & review (§05 “The Bought Press”): *San Francisco Wasp*, February 19, 1881, “In the Hollow of His Hand,” showing Crocker and Huntington holding up Stanford with the three-member railroad commission created by the 1879 California constitution in his palm; 1886 memo to the president of the Union Pacific noting the New York *Star* did “as Huntington directs,” and White’s observation, “Nineteenth-century Americans were not shocked by the corruption of the press.”↩

365. Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (`white-railroaded` in `references.bib`); Cardinal Podcast Ep. 3 script & review (§05 “The Bought Press”): *San Francisco Wasp*, February 19, 1881, “In the Hollow of His Hand,” showing Crocker and Huntington holding up Stanford with the three-member railroad commission created by the 1879 California constitution in his palm; 1886 memo to the president of the Union Pacific noting the New York *Star* did “as Huntington directs,” and White’s observation, “Nineteenth-century Americans were not shocked by the corruption of the press.”↩

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368. Richard Rayner, *The Associates: Four Capitalists Who Created California* (2008/2009), cited in Cardinal Podcast Ep. 3 script & review (§11 “Bierce and Hearst”) and `winkler-corporations#54` (ch. 4, n. 6): Ambrose Bierce poem in the *San Francisco Wasp*, April 2, 1881 (“These mounds are green where lie the brave who fought and fell at Hanford. / O, point me out, I pray, the grave of Leland Stanford”), and Bierce’s moniker “Stealand Landford.”↩

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371. Cardinal Podcast Ep. 3 script & review (§10 “Exclusion”), citing U.S. State Department and National Archives milestone records, alongside Erika Lee, *At America’s Gates: Chinese Immigration during the Exclusion Era, 1882–1943* (`lee-americas-gates`) and Alexander Saxton, *The Indispensable Enemy* (`saxton-indispensable-enemy`): Rutherford B. Hayes’s March 1, 1879 veto of the fifteen-passenger bill under the Burlingame Treaty; the renegotiated Angell Treaty allowing “reasonable” restrictions; the Page Act of 1875; Chester A. Arthur’s veto of the 20-year ban and signing of the 10-year Chinese Exclusion Act on May 6, 1882; the Geary Act of 1892; the 1902 extension to Hawaii and the Philippines; and the 1943 repeal leaving a quota of 105 per year.↩

372. Cardinal Podcast Ep. 3 script & review (§10 “Exclusion”), citing U.S. State Department and National Archives milestone records, alongside Erika Lee, *At America’s Gates: Chinese Immigration during the Exclusion Era, 1882–1943* (`lee-americas-gates`) and Alexander Saxton, *The Indispensable Enemy* (`saxton-indispensable-enemy`): Rutherford B. Hayes’s March 1, 1879 veto of the fifteen-passenger bill

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380. Cardinal Podcast Ep. 3 script & review (§10 “Exclusion”), citing U.S. State Department and National Archives milestone records, alongside Erika Lee, *At America’s Gates: Chinese Immigration during the Exclusion Era, 1882–1943* (`lee-americas-gates`) and Alexander Saxton, *The Indispensable Enemy* (`saxton-indispensable-enemy`): Rutherford B. Hayes’s March 1, 1879 veto of the fifteen-passenger bill under the Burlingame Treaty; the renegotiated Angell Treaty allowing “reasonable” restrictions; the Page Act of 1875; Chester A. Arthur’s veto of the 20-year ban and signing of the 10-year Chinese Exclusion Act on May 6, 1882; the Geary Act of 1892; the 1902 extension to Hawaii and the Philippines; and the 1943 repeal leaving a quota of 105 per year.↩

381. Cardinal Podcast Ep. 3 script & review (§10 “Exclusion”), citing U.S. State Department and National Archives milestone records, alongside Erika Lee, *At America’s Gates: Chinese Immigration during the Exclusion Era, 1882–1943* (`lee-americas-gates`) and Alexander Saxton, *The Indispensable Enemy* (`saxton-indispensable-enemy`): Rutherford B. Hayes’s March 1, 1879 veto of the fifteen-passenger bill

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382. Cardinal Podcast Ep. 3 script & review (§10 "Exclusion"), citing U.S. State Department and National Archives milestone records, alongside Erika Lee, *At America's Gates: Chinese Immigration during the Exclusion Era, 1882–1943* (`lee-americas-gates`) and Alexander Saxton, *The Indispensable Enemy* (`saxton-indispensable-enemy`): Rutherford B. Hayes's March 1, 1879 veto of the fifteen-passenger bill under the Burlingame Treaty; the renegotiated Angell Treaty allowing "reasonable" restrictions; the Page Act of 1875; Chester A. Arthur's veto of the 20-year ban and signing of the 10-year Chinese Exclusion Act on May 6, 1882; the Geary Act of 1892; the 1902 extension to Hawaii and the Philippines; and the 1943 repeal leaving a quota of 105 per year.↩

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387. Cardinal Podcast Ep. 3 script & review (§06 “The Sandlot” and §10 “Exclusion”), citing 1880 U.S. Census compilations: 75,132 Chinese residents in California out of a state population of 864,894 (8.7 percent, or nearly one in twelve) and out of 105,465 Chinese residents nationally (about seven in ten).↩

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389. Cardinal Podcast Ep. 3 script & review (§06 “The Sandlot” and Critique Pass) and Open Issues notes: splits Stanford’s 1862 rhetoric between his inaugural address (“the dregs of her population,” “a degraded and distinct people”) and his campaign stump phrase (“the dregs of Asia”). Flagged `[UNVERIFIED]` in text per Open Issues because the Episode 3 review log cross-references Episode 2 without naming the primary print source in the Episode 3 table.↩

390. Cardinal Podcast Ep. 3 script & review (§06 “The Sandlot” and Critique Pass) and Open Issues notes: splits Stanford’s 1862 rhetoric between his inaugural address (“the dregs of her population,” “a degraded and distinct people”) and his campaign stump phrase (“the dregs of Asia”). Flagged `[UNVERIFIED]` in text per Open Issues because the Episode 3 review log cross-references Episode 2 without naming the primary print source in the Episode 3 table.↩

391. Cardinal Podcast Ep. 3 script & review (§10 “Exclusion”), citing U.S. State Department and National Archives milestone records, alongside Erika Lee, *At America’s Gates: Chinese Immigration during the Exclusion Era, 1882–1943* (`lee-americas-gates`) and Alexander Saxton, *The Indispensable Enemy* (`saxton-indispensable-enemy`): Rutherford B. Hayes’s March 1, 1879 veto of the fifteen-passenger bill under the Burlingame Treaty; the renegotiated Angell Treaty allowing “reasonable” restrictions; the Page Act of 1875; Chester A. Arthur’s veto of the 20-year ban and signing of the 10-year Chinese Exclusion Act on May 6, 1882; the Geary Act of 1892; the 1902 extension to Hawaii and the Philippines; and the 1943 repeal leaving a quota of 105 per year.↩

392. Richard Rayner, *The Associates*, cited in Cardinal Podcast Ep. 3 script & review (§09 “Birth of the Octopus”): *San Francisco Wasp* cartoon *The Curse of California* (August 19, 1882; Rayner credits “Edward Keller,” while other standard accounts credit G. Frederick Keller).↩

393. Richard White, *Railroaded* (`white-railroaded`), cited in Cardinal Podcast Ep. 3 script & review (§09 “Birth of the Octopus”): election of Democratic Governor George Stoneman in 1882; the 1884 special session of the California legislature (assembly 61–19 Democratic, senate 32–8 Democratic); nine of ten San Francisco Democratic senators following “Blind Boss” Christopher Buckley (citing Callow, “San Francisco’s Blind Boss,” *Pacific Historical Review*, 1956); Pacific Railway Commission estimate of ~$700,000 spent by the Southern Pacific on politics in 1884 (roughly $26 million in 2026 CPI dollars); and White’s verbatim line, “The political marriage of Leland Stanford and Christopher Buckley gave birth to the Octopus.”↩

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396. Cardinal Podcast Ep. 3 script & review (§04 “Friends” and §08 “Yuma”), citing Richard Rayner, *The Associates*, Richard White, *Railroaded* (`white-railroaded`), and the 1883 Ellen Colton lawsuit record / pamphlet *How Members of Congress Are Bribed* (Moore pamphlet, Letter nos. 389, 107, 110, and 138): Huntington to Colton on November 30, 1874 (“I think this coming session of Congress will be composed of the hungriest set of men that ever got together”), January 17, 1876 (“It cost money to fix things so that I would know his bill would not pass. I believe that with $200,000 I can pass our bill”), and January 29 and March 23, 1876 (“I keep on high ground”).↩

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418. Frank Norris, *The Octopus* (1901), cited in Cardinal Podcast Ep. 3 script & review (§09 “Birth of the Octopus”): “all the traffic would bear,” “The railroad could not be operated at a legitimate profit,” and “tentacles of steel.”↩

419. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (`harris-palo-alto`), and Cardinal Podcast Ep. 3 script & review (§13 “A Telegram from Florence”) and Open Issues notes: Leland Stanford Jr. born May 14, 1868, died March 13, 1884 in Florence, Italy, at age fifteen. Marked DISPUTED and `[UNVERIFIED]` regarding David Starr Jordan’s “Roman fever” vs. university histories’ “typhoid” per Open Issues note that the specific citation for Jordan vs. university histories is still needed.↩

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425. Richard White, *Railroaded* (`white-railroaded`), and Cardinal Podcast Ep. 3 script & review (§09 “Birth of the Octopus”): Stanford bought his way into the U.S. Senate in 1885 with Christopher Buckley’s help, with his term beginning March 4, 1885.↩

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427. Richard Rayner, *The Associates*, Gray Brechin, *Imperial San Francisco* (`brechin-imperial#34`, ch. 5, n. 46: “‘Hearst’s Boodle Wins,’ *New York Times*, 19 January 1887”), and Cardinal Podcast Ep. 3 script & review (§11 “Bierce and Hearst”, citing Biographical Directory of the U.S. Congress and *San Francisco Examiner* history): George Hearst’s mining fortune (Comstock, Homestake, Anaconda); Stanford’s support for Hearst in 1886 against Huntington ally Aaron Sargent; Hearst’s Senate appointment (March 23–August 4, 1886) and full term (from March 4, 1887 until his death on February 28, 1891); acquisition of the *Examiner* in 1880 and transfer to “W. R. Hearst, proprietor” on March 4, 1887; and Ambrose Bierce’s 1896 *Examiner* columns attacking Huntington (February 11, April 10, and April 12, 1896).↩

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436. Richard White, *Railroaded: The Transcontinentals and the Making of Modern America* (`white-railroaded` in `references.bib`); Cardinal Podcast Ep. 3 script & review (§05 "The Bought Press"): *San Francisco Wasp*, February 19, 1881, "In the Hollow of His Hand," showing Crocker and Huntington holding up Stanford with the three-member railroad commission created by the 1879 California constitution in his palm; 1886 memo to the president of the Union Pacific noting the New York *Star* did "as Huntington directs," and White's observation, "Nineteenth-century Americans were not shocked by the corruption of the press."↩

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439. Adam Winkler, *We the Corporations: How American Businesses Won Their Civil Rights* (`winkler-corporations#54`, ch. 4, nn. 1–8, 28–31, and `winkler-corporations#70`), and Cardinal Podcast Ep. 3 script & review (§12 "The Caption"): Southern Pacific's refusal to pay California property taxes where mortgage deductions were denied; the

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475. David Starr Jordan, *The Days of a Man*; university histories, via Podcast Ep. 4 Review Log (“Leland Jr. born May 14, 1868; died March 13, 1884, two months short of sixteen. Jordan’s memoir: ‘Roman fever.’ University histories: typhoid. DOC. Not treated as one diagnosis.”).↩

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477. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 7 nn. 5–6; Podcast Ep. 4 Script & Review Log): Jane Stanford describing going “into dear Leland’s room and kneel down and tell all my sorrows to the blessed Jesus. I feel my darling boy and my dear husband are pleading with me”; curators in 1893 moving the bedroom to the university museum “just as it was in your city home.”↩

478. David Starr Jordan, *The Days of a Man*, vol. 1, pp. 118, 354–55 (white-jane#1, ch. 9 n. 9); Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023; `harris-palo-alto`), via Podcast Ep. 4 Review Log (“‘The children of California shall be my children’ (Jordan). Harris: ‘our children,’ told to Jane”).↩

479. Herbert C. Nash, “Biographical Sketch,” in *In Memoriam: Leland Stanford, Jr.* (c. 1884), via Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 7 nn. 2–3, ch. 11 n. 9; Podcast Ep. 4 Script & Review Log).↩

480. David Starr Jordan, printed statement dictated by Leland and Jane Stanford in 1892, via Podcast Ep. 4 Script & Review Log (“Wills April 24, 1884, Paris, before Maud Lord Drake. Drake unknown until a

séance with the Grants in October 1884; ‘detected in fraud.’ Jane: ‘No spiritualistic influence… a babe unborn’”).↩

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484. Richard White, *Who Killed Jane Stanford?*, p. 45, and Bertha Berner, *Mrs. Leland Stanford: An Intimate Account* (Stanford University Press, 1935), pp. 42–44, 165–68 (white-jane#1, ch. 8 nn. 18–21, ch. 23 nn. 5–6; Podcast Ep. 4 Review Log).↩

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486. Ulysses S. Grant to Leland Stanford, June 2, 1884, and July 2, 1884, and Frederick Grant to Leland Stanford, 1889, Stanford Papers, via Richard White, *Who Killed Jane Stanford?*, and Podcast Ep. 4 Review Log (“He deposited with the firm on the 2nd of May $86,000”).↩

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488. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 8 nn. 6, 10–13; white-jane#38, p. 357; Podcast Ep. 4 Script & Review Log).↩

489. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 9 n. 9; white-jane#38, p. 362; Podcast Ep. 4 Script & Review Log): Jane Stanford on Harvard and Yale graduates being “of benefit to no one” and employing “six of them as car-conductors on the Market Street Line now.”↩

490. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 8 nn. 6, 10–13; white-jane#38, p. 357; Podcast Ep. 4 Script & Review Log).↩

491. Enabling Act of March 9, 1885 (“An act to advance learning, the arts and sciences, and to promote the public welfare”), in *The Leland Stanford Jr. University, Circular 1*, pp. 5–22, and *Stanford University: The Founding Grant with Amendments, Legislation, and Court Decrees* (white-jane#1, ch. 9 nn. 17–18; Podcast Ep. 4 Review Log).↩

492. Founding Grant of Stanford University, signed November 11, 1885 (berlin-troublemakers#39, “What Do We Do with These?” n. 24; white-jane#1, ch. 9 nn. 17–18; Podcast Ep. 4 Script & Review Log).↩

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495. Adam Winkler, *We the Corporations: How American Businesses Won Their Civil Rights* (Liveright, 2018; `winkler-we-the-corporations`, winkler-corporations#54, ch. 4 nn. 39, 44, 47, 54, 57); Founding Grant trustee list via Podcast Ep. 4 Script & Review Log.↩

496. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10; white-jane#38, p. 353; Podcast Ep. 4 Script & Review Log) on the promised $30 million endowment and the presidency first offered to Columbia’s Nicholas Murray Butler.↩

497. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 nn. 10–11; white-jane#38, p. 355; Podcast Ep. 4 Script & Review Log), citing George Crothers Memoranda (“Real Estate Owned by Estate of Leland Stanford, Deceased. Taxes were $6983; revenue was $6606”).↩

498. Gray Brechin, *Imperial San Francisco: Urban Power, Earthly Ruin* (University of California Press, 1999; `brechin-imperial-san-francisco`), and Richard White, *Who Killed Jane Stanford?* (white-jane#38, pp. 353, 358; Podcast Ep. 4 Script & Review Log).↩

499. Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`; brechin-imperial#34, ch. 7 n. 1), and Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 45 nn. 7–8; white-jane#38, p. 359; Podcast Ep. 4 Script & Review Log).↩

500. Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`; brechin-imperial#34, ch. 7 n. 1), and Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 45 nn. 7–8; white-jane#38, p. 359; Podcast Ep. 4 Script & Review Log).↩

501. Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`; brechin-imperial#34, ch. 7 n. 1), and Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 45 nn. 7–8; white-jane#38, p. 359; Podcast Ep. 4 Script & Review Log).↩

502. Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`; brechin-imperial#34, ch. 7 n. 1), and Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 45 nn. 7–8; white-jane#38, p. 359; Podcast Ep. 4 Script & Review Log).↩

503. Gordon H. Chang, *Ghosts of Gold Mountain: The Epic Story of the Chinese Who Built the Transcontinental Railroad* (Houghton Mifflin Harcourt, 2019; `chang-ghosts-gold-mountain`), and Podcast Ep. 4 Script & Review Log on Central Pacific labor and the Muwekma Ohlone baptismal records at Mission Dolores and Mission Santa Clara.↩

504. David Starr Jordan, *The Days of a Man*, via Podcast Ep. 4 Script & Review Log (“The Story of Our Streets”).↩

505. Mary Jo Ignoffo, via Winchester research file in Podcast Ep. 4 Script & Review Log; Richard White, *Who Killed Jane Stanford?* (Podcast Ep. 4 Review Log: “Cornerstone May 14, 1887, about fifteen miles away. No evidence the women met or corresponded”).↩

506. Mary Jo Ignoffo, via Winchester research file in Podcast Ep. 4 Script & Review Log; Richard White, *Who Killed Jane Stanford?* (Podcast Ep. 4 Review Log: “Cornerstone May 14, 1887, about fifteen miles away. No evidence the women met or corresponded”).↩

507. Winchester research file, Susy Smith’s *Prominent American Ghosts* (1967), History San Jose (Samuel Leib Papers, Sarah Winchester series, 1865–1922), and Richard White, *Who Killed Jane Stanford?*, p. 79, via Podcast Ep. 4 Script & Review Log.↩

508. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 7 nn. 5–6; Podcast Ep. 4 Script & Review Log): Jane Stanford describing going “into dear Leland’s room and kneel down and tell all my sorrows to the blessed Jesus. I feel my darling boy and my dear husband are pleading with me”; curators in 1893 moving the bedroom to the university museum “just as it was in your city home.”↩

509. Stanford University public history (555 students, about 130 women) and Richard White, *Who Killed Jane Stanford?* (559 students); David Starr Jordan opening-day address, October 1, 1891, via Podcast Ep. 4 Script & Review Log.↩

510. Stanford University public history (555 students, about 130 women) and Richard White, *Who Killed Jane Stanford?* (559 students); David Starr Jordan opening-day address, October 1, 1891, via Podcast Ep. 4 Script & Review Log.↩

511. Kenneth Whyte, *Hoover: An Extraordinary Life in Extraordinary Times*, via Podcast Ep. 4 Script & Review Log.↩

512. Kenneth Whyte, *Hoover: An Extraordinary Life in Extraordinary Times*, via Podcast Ep. 4 Script & Review Log.↩

513. Kenneth Whyte, *Hoover: An Extraordinary Life in Extraordinary Times*, via Podcast Ep. 4 Script & Review Log.↩

514. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 n. 13; white-jane#38, p. 358; Podcast Ep. 4 Script & Review Log).↩

515. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 n. 13; white-jane#38, p. 358; Podcast Ep. 4 Script & Review Log).↩

516. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 nn. 13–16; Podcast Ep. 4 Script & Review Log) and David C. Frederick, *Rugged Justice: The Ninth Circuit Court of Appeals and the American West, 1891–1941* (University of California Press, 1994), pp. 34–51.↩

517. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 nn. 13–16; Podcast Ep. 4 Script & Review Log) and David C. Frederick, *Rugged Justice: The Ninth Circuit Court of Appeals and the American West, 1891–1941* (University of California Press, 1994), pp. 34–51.↩

518. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 nn. 11–15; white-jane#38, p. 361; Podcast Ep. 4 Script & Review Log).↩

519. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 nn. 11–15; white-jane#38, p. 361; Podcast Ep. 4 Script & Review Log).↩

520. Richard White, *Who Killed Jane Stanford?* (white-jane#1, ch. 10 n. 17; white-jane#38, pp. 353, 362; Podcast Ep. 4 Script & Review Log), citing Joseph Choate to Jane Stanford, February 9, 1896, Stanford Papers.↩

521. Bureau of Navigation report, John D. Long diary, Admiral Hyman Rickover (1976 study), Naval History and Heritage Command, and Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`), via Podcast Ep. 4 Script & Review Log.↩

522. Bureau of Navigation report, John D. Long diary, Admiral Hyman Rickover (1976 study), Naval History and Heritage Command, and Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`), via Podcast Ep. 4 Script & Review Log.↩

523. Bureau of Navigation report, John D. Long diary, Admiral Hyman Rickover (1976 study), Naval History and Heritage Command, and Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`), via Podcast Ep. 4 Script & Review Log.↩

524. Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`), via Podcast Ep. 4 Script & Review Log.↩

525. National Archives (Newlands Resolution, July 7, 1898) and Gray Brechin, *Imperial San Francisco* (brechin-imperial#34, ch. 3 n. 38, quoting Hall McAllister Jr. to Frederick Sharon, October 6, 1898, Sharon Family papers, part 1, box 9), via Podcast Ep. 4 Review Log.↩

526. U.S. Department of State Office of the Historian and Senate ratification record (February 6, 1899, 57–27), via Podcast Ep. 4 Script & Review Log.↩

527. U.S. Department of State Office of the Historian and Gray Brechin, *Imperial San Francisco* (`brechin-imperial-san-francisco`), via Podcast Ep. 4 Script & Review Log.↩

528. Richard White, *Who Killed Jane Stanford?*, p. 104, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Podcast Ep. 4 Script & Review Log.↩

529. Smithsonian snapshot (March 27, 2019) and Kenneth Whyte, *Hoover*, p. 76, via Podcast Ep. 4 Script & Review Log.↩

530. Kenneth Whyte, *Hoover*, pp. 77–78, via Podcast Ep. 4 Script & Review Log.↩

531. Kenneth Whyte, *Hoover*, pp. 77–78, via Podcast Ep. 4 Script & Review Log.↩

532. Kenneth Whyte, *Hoover*, pp. xi, 75–80, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`, ch. 2.3, quoting Ian Phimister), via Podcast Ep. 4 Script & Review Log.↩

533. Kenneth Whyte, *Hoover*, pp. xi, 75–80, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`, ch. 2.3, quoting Ian Phimister), via Podcast Ep. 4 Script & Review Log.↩

534. Kenneth Whyte, *Hoover*, pp. xi, 75–80, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`, ch. 2.3, quoting Ian Phimister), via Podcast Ep. 4 Script & Review Log.↩

535. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 13–18, citing Crothers Papers, "Memoranda dictated immediately before the visit of Mrs. G. O. Wilson editor of the *Stanford Illustrated Review*, June 17, 1930," and p. 79 via Podcast Ep. 4 Script & Review, Sec. 06: in 1899 Jane deeded the $25 million endowment—roughly $1 billion in current grocery dollars—with no reservations or trust clauses to San Jose lawyer and trustee Samuel Franklin Leib alone; Crothers called it a "strange procedure").↩

536. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 13–18, citing Crothers Papers, "Memoranda dictated immediately before the visit of Mrs. G. O. Wilson editor of the *Stanford Illustrated Review*, June 17, 1930," and p. 79 via Podcast Ep. 4 Script & Review, Sec. 06: in 1899 Jane deeded the $25 million endowment—roughly $1 billion in current grocery dollars—with no reservations or trust clauses to San Jose lawyer and trustee Samuel Franklin Leib alone; Crothers called it a "strange procedure").↩

537. Podcast Ep. 4 Script & Review, Sec. 06 (citing Mary Jo Ignoffo and History San Jose, Samuel Leib Papers, Sarah Winchester series, 1865–1922; house address 525 South Winchester Boulevard).↩

538. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 9–11, citing Orrin Leslie Elliott, *Stanford University: The First 25 Years*, p. 282, "*Control of the Espee Passes to the Speyers*," *San Francisco Examiner*, Dec. 5, 1899, and Crothers to Considine, July 21, 1948: "Crothers's estimates of the loss increased over the years. By the 1940s he put it at $2 million a year in income and $100 million in value to the endowment"; white-jane#38, Index, pp. 355, 360, 362; Podcast Ep. 4 Script & Review, Sec. 10: May 31, 1899 sale for $11 million—roughly $440 million today—which Crothers initially put at $7 million, or roughly $280 million today, less than Speyer & Co. got from E. H. Harriman the next year).↩

539. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 9–11, citing Orrin Leslie Elliott, *Stanford University: The First 25 Years*, p. 282, "*Control of the Espee Passes to the Speyers*," *San Francisco Examiner*, Dec. 5, 1899, and Crothers to Considine, July 21, 1948: "Crothers's estimates of the loss increased over the years. By the 1940s he put it at $2 million a year in income and $100 million in value to the

endowment"; white-jane#38, Index, pp. 355, 360, 362; Podcast Ep. 4 Script & Review, Sec. 10: May 31, 1899 sale for $11 million—roughly $440 million today—which Crothers initially put at $7 million, or roughly $280 million today, less than Speyer & Co. got from E. H. Harriman the next year).↩

540. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 12–16, citing Crothers Papers, "Withdrawal of Illegal, Impossible, and Undesirable Provisions"; white-jane#38, Index, p. 355; Podcast Ep. 4 Script & Review, Sec. 10: May 31, 1899 Great Deed, "a great but defective grant," and Mountford Wilson's clause giving Jane the power "to grant, bargain, sell, convey, or lease").↩

541. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 12–16, citing Crothers Papers, "Withdrawal of Illegal, Impossible, and Undesirable Provisions"; white-jane#38, Index, p. 355; Podcast Ep. 4 Script & Review, Sec. 10: May 31, 1899 Great Deed, "a great but defective grant," and Mountford Wilson's clause giving Jane the power "to grant, bargain, sell, convey, or lease").↩

542. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 13 n. 12–16, citing Crothers Papers, "Withdrawal of Illegal, Impossible, and Undesirable Provisions"; white-jane#38, Index, p. 355; Podcast Ep. 4 Script & Review, Sec. 10: May 31, 1899 Great Deed, "a great but defective grant," and Mountford Wilson's clause giving Jane the power "to grant, bargain, sell, convey, or lease").↩

543. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 25, Ch. 13 n. 17, Ch. 17 n. 5, citing Article IX, California Constitution of 1879, Section 10, amended Nov. 6, 1900; Podcast Ep. 4 Script & Review, Sec. 10: vote of 137,667 to 67,737, tax exemption while used for education, and conditional tuition exemption for California students).↩

544. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 25, Ch. 13 n. 17, Ch. 17 n. 5, citing Article IX, California Constitution of 1879, Section 10, amended Nov. 6, 1900; Podcast Ep. 4 Script & Review, Sec. 10: vote of 137,667 to 67,737, tax exemption while used for education, and conditional tuition exemption for California students).↩

545. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 25, Ch. 13 n. 17, Ch. 17 n. 5, citing Article IX, California Constitution of 1879, Section 10, amended Nov. 6, 1900; Podcast Ep. 4 Script & Review, Sec. 10: vote of 137,667 to 67,737, tax exemption while used for education, and conditional tuition exemption for California students).↩

546. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 16 n. 14–16, Ch. 17 n. 1, 4–6, citing "Sudden Death of Barbara Jordan," *Call*, Sept. 14, 1900; Jordan to Ross, June 15, 1900, and Nov. 11, 1900; Mayor Phelan to Ross, Nov. 11, 1900; Podcast Ep. 4 Script & Review, Sec. 10).↩

547. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 16 n. 14–16, Ch. 17 n. 1, 4–6, citing "Sudden Death of Barbara Jordan," *Call*, Sept. 14, 1900; Jordan to Ross, June 15, 1900, and Nov. 11, 1900;

Mayor Phelan to Ross, Nov. 11, 1900; Podcast Ep. 4 Script & Review, Sec. 10).↩

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550. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 15 n. 7–13, Ch. 16 n. 1–6, citing "Warning Against Coolie 'Natives' and Japanese," *San Francisco Call*, May 8, 1900, p. 12, and Gordon H. Chang, "The Chinese and the Stanfords"; white-jane#38, Index, pp. 352, 355, 359, on bimetallism vs. gold standard, Bryan, *Honest Dollars*, street railways, and coolie labor; Podcast Ep. 4 Script & Review, Sec. 10).↩

551. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 15 n. 7–13, Ch. 16 n. 1–6, citing "Warning Against Coolie 'Natives' and Japanese," *San Francisco Call*, May 8, 1900, p. 12, and Gordon H. Chang, "The Chinese and the Stanfords"; white-jane#38, Index, pp. 352, 355, 359, on bimetallism vs. gold standard, Bryan, *Honest Dollars*, street railways, and coolie labor; Podcast Ep. 4 Script & Review, Sec. 10).↩

552. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 17 n. 6–8, citing Ross to *San Francisco Chronicle*, Nov. 14, 1900, and "Dr. Edward Ross Forced Out," *Chronicle*, Nov. 14, 1900, p. 1; white-jane#38, Index, p. 354: "Dr. Edward Ross Forced Out of Stanford University"; Podcast Ep. 4 Script & Review, Sec. 10, on Jordan's praise and Ross's "Judas kiss").↩

553. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 17 n. 6–8, citing Ross to *San Francisco Chronicle*, Nov. 14, 1900, and "Dr. Edward Ross Forced Out," *Chronicle*, Nov. 14, 1900, p. 1; white-jane#38, Index, p. 354: "Dr. Edward Ross Forced Out of Stanford University"; Podcast Ep. 4 Script & Review, Sec. 10, on Jordan's praise and Ross's "Judas kiss").↩

554. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 17 n. 6–8, citing Ross to *San Francisco Chronicle*, Nov. 14, 1900, and "Dr. Edward Ross Forced Out," *Chronicle*, Nov. 14, 1900, p. 1; white-jane#38, Index, p. 354: "Dr. Edward Ross Forced Out of Stanford University"; Podcast Ep. 4 Script & Review, Sec. 10, on Jordan's praise and Ross's "Judas kiss").↩

555. Richard White, *Who Killed Jane Stanford?*, p. 95 (via Podcast Ep. 4 Script & Review, Sec. 11: "Jane Stanford played the dowager queen").↩

556. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 17 n. 10–31, citing George Howard to Jordan, Nov. 17, 1900, and Jan. 12, 1901; letters of Morton Aldrich, Frank Fetter, Spencer, C. N. Little, and Edwin R. A. Seligman; and "Report of the Committee of Economists on the Dismissal of Edward A. Ross from Leland Stanford Jr. University," Feb. 20, 1901; Gray Brechin, *Imperial San Francisco*, brechin-imperial#34, Ch. 7 n. 48).↩

557. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 6–8, Ch. 21 n. 17–19, citing Crothers Papers, "Women Students," and Jane Stanford to Mrs. Mills, July 14, 1904; Podcast Ep. 4 Script & Review, Sec. 10, and Argon Audit #37: women about two in five by 1899, May 31, 1899 cap of 500 women negotiated by Crothers, and 1904 view of coeducation as unsettled).↩

558. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 6–8, Ch. 21 n. 17–19, citing Crothers Papers, "Women Students," and Jane Stanford to Mrs. Mills, July 14, 1904; Podcast Ep. 4 Script & Review, Sec. 10, and Argon Audit #37: women about two in five by 1899, May 31, 1899 cap of 500 women negotiated by Crothers, and 1904 view of coeducation as unsettled).↩

559. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 6–8, Ch. 21 n. 17–19, citing Crothers Papers, "Women Students," and Jane Stanford to Mrs. Mills, July 14, 1904; Podcast Ep. 4 Script & Review, Sec. 10, and Argon Audit #37: women about two in five by 1899, May 31, 1899 cap of 500 women negotiated by Crothers, and 1904 view of coeducation as unsettled).↩

560. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 6–8, Ch. 21 n. 17–19, citing Crothers Papers, "Women Students," and Jane Stanford to Mrs. Mills, July 14, 1904; Podcast Ep. 4 Script & Review, Sec. 10, and Argon Audit #37: women about two in five by 1899, May 31, 1899 cap of 500 women negotiated by Crothers, and 1904 view of coeducation as unsettled).↩

561. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 3, 6, Ch. 20 n. 7–19, citing Rev. D. Charles Gardner's Jan. 25, 1903 dedication address, Memorial Church Inscriptions, and R. Heber Newton correspondence; white-jane#38, Index, pp. 358, 361; Podcast Ep. 4 Script & Review, Sec. 12: "Koohinoor," facade inscription, immortality of the soul, and "spoke mainly to a sect of one").↩

562. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 3, 6, Ch. 20 n. 7–19, citing Rev. D. Charles Gardner's Jan. 25, 1903 dedication address, Memorial Church Inscriptions, and R. Heber Newton correspondence; white-jane#38, Index, pp. 358, 361; Podcast Ep. 4 Script & Review, Sec. 12: "Koohinoor," facade inscription, immortality of the soul, and "spoke mainly to a sect of one").↩

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566. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 8 n. 22–30, Ch. 23 n. 1–14, citing Thomas Welton Stanford Apports in Stanford Objects Collection, Berner's account of the 1903 Australia trip and medium Charles Bailey, and John Edgar Coover, *Experiments in Psychical Research at Leland Stanford Junior University*, 1917; Podcast Ep. 4 Script & Review, Sec. 12).↩

567. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 2, Ch. 21 n. 1–16, Ch. 26 n. 18, citing Jane Stanford to Archbishop Riordan, March 4, 1902; Crothers Papers, "Hazards of Volition Escaped"; and Feb. 22, 1905 clippings on "queening"; white-jane#38, Index, pp. 353–355; Podcast Ep. 4 Script & Review, Sec. 13).↩

568. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 2, Ch. 21 n. 1–16, Ch. 26 n. 18, citing Jane Stanford to Archbishop Riordan, March 4, 1902; Crothers Papers, "Hazards of Volition Escaped"; and Feb. 22, 1905 clippings on "queening"; white-jane#38, Index, pp. 353–355; Podcast Ep. 4 Script & Review, Sec. 13).↩

569. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 2, Ch. 21 n. 1–16, Ch. 26 n. 18, citing Jane Stanford to Archbishop Riordan, March 4, 1902; Crothers Papers, "Hazards of Volition Escaped"; and Feb. 22, 1905 clippings on "queening"; white-jane#38, Index, pp. 353–355; Podcast Ep. 4 Script & Review, Sec. 13).↩

570. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 2, Ch. 21 n. 1–16, Ch. 26 n. 18, citing Jane Stanford to Archbishop Riordan, March 4, 1902; Crothers Papers, "Hazards of Volition Escaped"; and Feb. 22, 1905 clippings on "queening"; white-jane#38, Index, pp. 353–355; Podcast Ep. 4 Script & Review, Sec. 13).↩

571. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 2, Ch. 21 n. 1–16, Ch. 26 n. 18, citing Jane Stanford to Archbishop Riordan, March 4, 1902; Crothers Papers, "Hazards of Volition Escaped"; and Feb. 22, 1905 clippings on "queening"; white-jane#38, Index, pp. 353–355; Podcast Ep. 4 Script & Review, Sec. 13).↩

572. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 2, Ch. 21 n. 1–16, Ch. 26 n. 18, citing Jane Stanford to Archbishop Riordan, March 4, 1902; Crothers Papers, "Hazards of Volition Escaped"; and Feb. 22, 1905 clippings on "queening"; white-jane#38, Index, pp. 353–355; Podcast Ep. 4 Script & Review, Sec. 13).↩

573. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 6, Ch. 21 n. 4–6, 20–21, citing "Resignation of Surviving Founder," Jane Stanford's July 6, 1903 inauguration address as President of the Board of Trustees, and the 1903 Santa Clara County Superior Court special proceeding).↩

574. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 24 n. 1–15, citing Horace Davis, Nicholas Murray Butler, and Julius Goebel correspondence, and Jane Stanford's July 6, 1904 letter and July 23, 1904 Address to the Board of Trustees, read Aug. 1, 1904; white-jane#38, Index, pp. 354, 360).↩

575. Gray Brechin, *Imperial San Francisco* (brechin-imperial#34, Ch. 3 n. 31–32, citing *Congressional Record*, 57th Cong., 1st sess., May 15, 1902, p. 5525, on Gen. Jacob Smith's statements; Podcast Ep. 4 Script & Review, Sec. 14, citing U.S. State Department Office of the Historian and court-martial reports on Maj. Edwin F. Glenn's Nov. 27, 1900 water cure of Tobeniano Ealdama at Igbaras, the Sept. 28, 1901 Balangiga attack, Smith's "Kill and burn" and "Everything over ten" orders, and Maj. Littleton Waller's countermand).↩

576. Gray Brechin, *Imperial San Francisco* (brechin-imperial#34, Ch. 3 n. 31–32, citing *Congressional Record*, 57th Cong., 1st sess., May 15, 1902, p. 5525, on Gen. Jacob Smith's statements; Podcast Ep. 4 Script & Review, Sec. 14, citing U.S. State Department Office of the Historian and court-martial reports on Maj. Edwin F. Glenn's Nov. 27, 1900 water cure of Tobeniano Ealdama at Igbaras, the Sept. 28, 1901 Balangiga attack, Smith's "Kill and burn" and "Everything over ten" orders, and Maj. Littleton Waller's countermand).↩

577. Gray Brechin, *Imperial San Francisco* (brechin-imperial#34, Ch. 3 n. 31–32, citing *Congressional Record*, 57th Cong., 1st sess., May 15, 1902, p. 5525, on Gen. Jacob Smith's statements; Podcast Ep. 4 Script & Review, Sec. 14, citing U.S. State Department Office of the Historian and court-martial reports on Maj. Edwin F. Glenn's Nov. 27, 1900 water cure of Tobeniano Ealdama at Igbaras, the Sept. 28, 1901 Balangiga attack, Smith's "Kill and burn" and "Everything over ten" orders, and Maj. Littleton Waller's countermand).↩

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579. Gray Brechin, *Imperial San Francisco* (brechin-imperial#34, Ch. 3 n. 31–32, citing *Congressional Record*, 57th Cong., 1st sess., May 15, 1902, p. 5525, on Gen. Jacob Smith's statements; Podcast Ep. 4 Script & Review, Sec. 14, citing U.S. State Department Office of the Historian and court-martial reports on Maj. Edwin F. Glenn's Nov. 27, 1900 water cure of Tobeniano Ealdama at Igbaras, the Sept. 28, 1901 Balangiga attack, Smith's "Kill and burn" and "Everything over ten" orders, and Maj. Littleton Waller's countermand).↩

580. Gray Brechin, *Imperial San Francisco* (brechin-imperial#69, Select Bibliography, and brechin-imperial#34, Ch. 3 n. 29, citing Samuel Clemens, “Mark Twain on American Imperialism”; Podcast Ep. 4 Script & Review, Sec. 14).↩

581. Gray Brechin, *Imperial San Francisco* (brechin-imperial#34, Ch. 3 n. 75, 77: “Eight hundred and fifty-seven recorded engagements followed Roosevelt’s announcement”); Podcast Ep. 4 Script & Review, Sec. 14 (citing State Department Office of the Historian casualty figures: more than 4,200 U.S. combatants, more than 20,000 Filipino combatants, and up to 200,000 Filipino civilians).↩

582. Gray Brechin, *Imperial San Francisco* (brechin-imperial#69, Select Bibliography, citing Carl E. Ackerman, “President Roosevelt in California,” *Sunset* 11, no. 2, June 1903, pp. 103–113; Podcast Ep. 4 Script & Review, Sec. 14: May 13, 1903 dedication of the Dewey Monument in Union Square, *Olympia* first, and Roosevelt’s quote).↩

583. Gray Brechin, *Imperial San Francisco* (brechin-imperial#69, Select Bibliography, citing Carl E. Ackerman, “President Roosevelt in California,” *Sunset* 11, no. 2, June 1903, pp. 103–113; Podcast Ep. 4 Script & Review, Sec. 14: May 13, 1903 dedication of the Dewey Monument in Union Square, *Olympia* first, and Roosevelt’s quote).↩

584. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 1 n. 4–30, Ch. 2 n. 1–13, Ch. 3 n. 1–11, and white-jane#38, Index, pp. 352–355, 357–359, 362; Podcast Ep. 4 Script & Review, Sec. 15: Jan. 14, 1905 Poland Spring poisoning at the Nob Hill mansion, Wakelee’s Pharmacy and Falkenau Assaying Co., Chief Wittman, Harry Morse, and Jules Callundan).↩

585. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 1 n. 4–30, Ch. 2 n. 1–13, Ch. 3 n. 1–11, and white-jane#38, Index, pp. 352–355, 357–359, 362; Podcast Ep. 4 Script & Review, Sec. 15: Jan. 14, 1905 Poland Spring poisoning at the Nob Hill mansion, Wakelee’s Pharmacy and Falkenau Assaying Co., Chief Wittman, Harry Morse, and Jules Callundan).↩

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587. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 5 n. 2–26, and white-jane#38, Index, pp. 355, 359, 360; Podcast Ep. 4 Script & Review, Secs. 06, 15: Jan. 23, 1905 visit to Hotel Vendome in San Jose, St. Francis Hotel, and Feb. 15 farewell letters to Crothers and May Hopkins).↩

588. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 5 n. 2–26, and white-jane#38, Index, pp. 355, 359, 360; Podcast Ep. 4 Script & Review, Secs. 06, 15: Jan. 23, 1905 visit to Hotel Vendome in San Jose, St. Francis Hotel, and Feb. 15 farewell letters to Crothers and May Hopkins).↩

589. Richard White, *Who Killed Jane Stanford?*, pp. 167–168 (and white-jane#1, Ch. 5 n. 21, Ch. 28 n. 1–5; white-jane#38, Index, pp. 354, 357, 360; Podcast Ep. 4 Script & Review, Sec. 15: Feb. 15, 1905 sailing of *S.S. Korea*, Berner spelled "Benner," Gen. Arthur MacArthur, $75,000 in jewels—roughly $2.8 million today—and Charles Sidney Dole).↩

590. Richard White, *Who Killed Jane Stanford?*, pp. 167–168 (and white-jane#1, Ch. 5 n. 21, Ch. 28 n. 1–5; white-jane#38, Index, pp. 354, 357, 360; Podcast Ep. 4 Script & Review, Sec. 15: Feb. 15, 1905 sailing of *S.S. Korea*, Berner spelled "Benner," Gen. Arthur MacArthur, $75,000 in jewels—roughly $2.8 million today—and Charles Sidney Dole).↩

591. Richard White, *Who Killed Jane Stanford?*, pp. 167–168 (and white-jane#1, Ch. 5 n. 21, Ch. 28 n. 1–5; white-jane#38, Index, pp. 354, 357, 360; Podcast Ep. 4 Script & Review, Sec. 15: Feb. 15, 1905 sailing of *S.S. Korea*, Berner spelled "Benner," Gen. Arthur MacArthur, $75,000 in jewels—roughly $2.8 million today—and Charles Sidney Dole).↩

592. Richard White, *Who Killed Jane Stanford?*, pp. 167–168 (and white-jane#1, Ch. 5 n. 21, Ch. 28 n. 1–5; white-jane#38, Index, pp. 354, 357, 360; Podcast Ep. 4 Script & Review, Sec. 15: Feb. 15, 1905 sailing of *S.S. Korea*, Berner spelled "Benner," Gen. Arthur MacArthur, $75,000 in jewels—roughly $2.8 million today—and Charles Sidney Dole).↩

593. Richard White, *Who Killed Jane Stanford?*, pp. 167–168 (and white-jane#1, Ch. 5 n. 21, Ch. 28 n. 1–5; white-jane#38, Index, pp. 354, 357, 360; Podcast Ep. 4 Script & Review, Sec. 15: Feb. 15, 1905 sailing of *S.S. Korea*, Berner spelled "Benner," Gen. Arthur MacArthur, $75,000 in jewels—roughly $2.8 million today—and Charles Sidney Dole).↩

594. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 2 n. 12: "'Attempt Made to Poison Mrs. Stanford,' *Mercury*, Feb. 19, 1905, mentions story broke Feb. 18").↩

595. Podcast Ep. 4 Script & Review, Sec. 15 (citing Theodore Roosevelt Center, Feb. 18, 1905, and Freer Gallery of Art provenance: Blue Room presentation by Minister Liang Cheng of Katharine Carl's portrait of Empress Dowager Cixi, painted for the 1904 St. Louis World's Fair).↩

596. Podcast Ep. 4 Script & Review, Sec. 15 (citing Theodore Roosevelt Center, Feb. 18, 1905, and Freer Gallery of Art provenance: Blue Room presentation by Minister Liang Cheng of Katharine Carl's portrait of Empress Dowager Cixi, painted for the 1904 St. Louis World's Fair).↩

597. Richard White, *Who Killed Jane Stanford?*, p. 168 (and white-jane#1, Ch. 28 n. 5–10, Ch. 31 n. 2; white-jane#38, Index, pp. 353, 357; Podcast Ep. 4 Script & Review, Secs. 01, 15: "well and hearty," Gov. George R. Carter appointed by Roosevelt in 1903 after consulting Jordan, and Feb. 21 check-in to Moana Hotel Room 120, with Berner and Hunt in Room 122).↩

598. Richard White, *Who Killed Jane Stanford?*, p. 168 (and white-jane#1, Ch. 28 n. 5–10, Ch. 31 n. 2; white-jane#38, Index, pp. 353, 357; Podcast Ep. 4 Script & Review, Secs. 01, 15: "well and hearty," Gov. George R. Carter appointed by Roosevelt in 1903 after consulting Jordan, and Feb. 21 check-in to Moana Hotel Room 120, with Berner and Hunt in Room 122).↩

599. Richard White, *Who Killed Jane Stanford?*, p. 168 (and white-jane#1, Ch. 28 n. 5–10, Ch. 31 n. 2; white-jane#38, Index, pp. 353, 357; Podcast Ep. 4 Script & Review, Secs. 01, 15: “well and hearty,” Gov. George R. Carter appointed by Roosevelt in 1903 after consulting Jordan, and Feb. 21 check-in to Moana Hotel Room 120, with Berner and Hunt in Room 122).↩

600. Richard White, *Who Killed Jane Stanford?*, p. 168 (and white-jane#1, Ch. 28 n. 5–10, Ch. 31 n. 2; white-jane#38, Index, pp. 353, 357; Podcast Ep. 4 Script & Review, Secs. 01, 15: “well and hearty,” Gov. George R. Carter appointed by Roosevelt in 1903 after consulting Jordan, and Feb. 21 check-in to Moana Hotel Room 120, with Berner and Hunt in Room 122).↩

601. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

602. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

603. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

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606. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in

the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

607. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 5 n. 13–17, Ch. 33 n. 7–16; white-jane#38, Index, pp. 352, 356, 359, 361: purchase of bicarbonate of soda at Stanford Pharmacy in Palo Alto from W. E. Jackson, and Berner’s relationship with pharmacist Paul J. Schwab).↩

608. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

609. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

610. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

611. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

612. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 28 n. 9–10, Ch. 29 n. 1–6; white-jane#38, Index, pp. 355, 356, 358; Podcast Ep. 4 Script & Review, Sec. 01: Feb. 28, 1905 ride along the windward coast past the Pali and Koolau Cliffs, picnic menu reported by Berner in the *Hawaiian Star*, Mrs. Grinbaum, cascara capsule, half teaspoon of bicarbonate of soda, “Give it time to get out of my stomach,” and May Hunt helping with the door lock).↩

613. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner’s Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, “Oh I am so ill. Get me a doctor,” “Oh God, forgive me my sins,” “Is my soul prepared to meet my dear ones?,” “This a horrible death to die,” and physical tetanic symptoms over ten minutes).↩

614. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner's Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, "Oh I am so ill. Get me a doctor," "Oh God, forgive me my sins," "Is my soul prepared to meet my dear ones?," "This a horrible death to die," and physical tetanic symptoms over ten minutes).↩

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618. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner's Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, "Oh I am so ill. Get me a doctor," "Oh God, forgive me my sins," "Is my soul prepared to meet my dear ones?," "This a horrible death to die," and physical tetanic symptoms over ten minutes).↩

619. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner's Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, "Oh I am so ill. Get me a doctor," "Oh God, forgive me my sins," "Is my soul prepared to meet my dear ones?," "This a horrible death to die," and physical tetanic symptoms over ten minutes).↩

620. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner's Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355,

358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, “Oh I am so ill. Get me a doctor,” “Oh God, forgive me my sins,” “Is my soul prepared to meet my dear ones?,” “This a horrible death to die,” and physical tetanic symptoms over ten minutes).↩

621. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner’s Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, “Oh I am so ill. Get me a doctor,” “Oh God, forgive me my sins,” “Is my soul prepared to meet my dear ones?,” “This a horrible death to die,” and physical tetanic symptoms over ten minutes).↩

622. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner’s Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, “Oh I am so ill. Get me a doctor,” “Oh God, forgive me my sins,” “Is my soul prepared to meet my dear ones?,” “This a horrible death to die,” and physical tetanic symptoms over ten minutes).↩

623. Richard White, *Who Killed Jane Stanford?*, pp. 171–174 (and white-jane#1, Ch. 29 n. 6–12, Ch. 39 n. 12, citing Coroner’s Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, Dr. H. V. Murray, and Dr. Francis Root Day; white-jane#38, Index, pp. 353, 355, 358; Podcast Ep. 4 Script & Review, Secs. 01, 15: Adam Heunisch, “Oh I am so ill. Get me a doctor,” “Oh God, forgive me my sins,” “Is my soul prepared to meet my dear ones?,” “This a horrible death to die,” and physical tetanic symptoms over ten minutes).↩

624. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 31 n. 6–9, Ch. 36 n. 1–26, citing Drs. Clifford B. Wood and Francis Root Day, *Autopsy on the Body of Mrs. Jane L. Stanford*, March 1, 1905, testimony of chemists R. A. Duncan and Edmund Shorey, and Coroner’s Inquest Verdict, p. 70; white-jane#38, Index, pp. 353, 354, 360, 362; Podcast Ep. 4 Script & Review, Sec. 15).↩

625. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 31 n. 6–9, Ch. 36 n. 1–26, citing Drs. Clifford B. Wood and Francis Root Day, *Autopsy on the Body of Mrs. Jane L. Stanford*, March 1, 1905, testimony of chemists R. A. Duncan and Edmund Shorey, and Coroner’s Inquest Verdict, p. 70; white-jane#38, Index, pp. 353, 354, 360, 362; Podcast Ep. 4 Script & Review, Sec. 15).↩

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testimony of chemists R. A. Duncan and Edmund Shorey, and Coroner's Inquest Verdict, p. 70; white-jane#38, Index, pp. 353, 354, 360, 362; Podcast Ep. 4 Script & Review, Sec. 15).↩

628. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 6 n. 5, Ch. 31 n. 5, 16–17, Ch. 34 n. 20, Ch. 35 n. 6, Ch. 36 n. 3–8; white-jane#38, Index, pp. 352, 355, 356, 361; Podcast Ep. 4 Script & Review, Sec. 15: Judge W. L. Stanley, Berner's $15,000 bequest—roughly $560,000 today—Ariel Lathrop calling her a medium, White's five adjectives, and Hawaiian officials declining to prosecute).↩

629. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 6 n. 5, Ch. 31 n. 5, 16–17, Ch. 34 n. 20, Ch. 35 n. 6, Ch. 36 n. 3–8; white-jane#38, Index, pp. 352, 355, 356, 361; Podcast Ep. 4 Script & Review, Sec. 15: Judge W. L. Stanley, Berner's $15,000 bequest—roughly $560,000 today—Ariel Lathrop calling her a medium, White's five adjectives, and Hawaiian officials declining to prosecute).↩

630. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 6 n. 5, Ch. 31 n. 5, 16–17, Ch. 34 n. 20, Ch. 35 n. 6, Ch. 36 n. 3–8; white-jane#38, Index, pp. 352, 355, 356, 361; Podcast Ep. 4 Script & Review, Sec. 15: Judge W. L. Stanley, Berner's $15,000 bequest—roughly $560,000 today—Ariel Lathrop calling her a medium, White's five adjectives, and Hawaiian officials declining to prosecute).↩

631. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 6 n. 5, Ch. 31 n. 5, 16–17, Ch. 34 n. 20, Ch. 35 n. 6, Ch. 36 n. 3–8; white-jane#38, Index, pp. 352, 355, 356, 361; Podcast Ep. 4 Script & Review, Sec. 15: Judge W. L. Stanley, Berner's $15,000 bequest—roughly $560,000 today—Ariel Lathrop calling her a medium, White's five adjectives, and Hawaiian officials declining to prosecute).↩

632. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 4 n. 2–8, Ch. 43 n. 1–6; white-jane#38, Index, pp. 351, 360, 362: Ah Wing, Wong Toy Wong at Mendocino State Hospital, and Tay Wang / Charles Eyre).↩

633. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 6 n. 5, Ch. 31 n. 5, 16–17, Ch. 34 n. 20, Ch. 35 n. 6, Ch. 36 n. 3–8; white-jane#38, Index, pp. 352, 355, 356, 361; Podcast Ep. 4 Script & Review, Sec. 15: Judge W. L. Stanley, Berner's $15,000 bequest—roughly $560,000 today—Ariel Lathrop calling her a medium, White's five adjectives, and Hawaiian officials declining to prosecute).↩

634. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 31 n. 4, Ch. 35 n. 1–13, Ch. 39 n. 1–9, Ch. 47 n. 6; white-jane#38, Index, pp. 353, 355, 359, 360; Podcast Ep. 4 Script & Review, Sec. 15: High Sheriff William Henry, Jordan, Hopkins, Jules Callundan, and Harry Reynolds sailing on the *S.S. Alameda*).↩

635. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 39 n. 19–22, Ch. 42 n. 1–2, 9–12, citing "Jordan and Hopkins United in a Statement," *Hawaiian Star*, March 15, 1905, and Central Union Church service; Podcast Ep. 4 Script & Review, Sec. 15).↩

636. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 41 n. 1–21, Ch. 44 n. 1, 12–21, Ch. 46 n. 14–31, citing Dr. Ernest C. Waterhouse's report and correspondence with Jordan, the March 21

return of the *Alameda*, and the report by Cooper Medical College's Dr. William Ophüls, Dr. Charles L. Morgan, and Dr. H. A. L. Ryfkogel; white-jane#38, Index, pp. 358, 359, 362).↩

637. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 7, citing Robert W. P. Cutler, *The Mysterious Death of Jane Stanford*, Stanford University Press, 2003; white-jane#38, Index, pp. 353, 362; Podcast Ep. 4 Script & Review, Sec. 15, on Dr. Wood calling the angina pectoris theory "simply rot").↩

638. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 14 n. 13, Ch. 24 n. 1–15, Ch. 45 n. 19, Ch. 47 n. 21, citing "Asa P. Stanford's Widow Sues Estate for a Million," *San Francisco Examiner*, June 5, 1906, p. 1; Podcast Ep. 4 Script & Review, Sec. 15).↩

639. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (references.bib: `harris-palo-alto`, 2023; Podcast Ep. 4 Script & Review, Sec. 15: "it is no longer particularly controversial to suggest Jordan was part of a conspiracy, most likely with Bertha Berner, and that no one has produced a definitive proof").↩

640. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 40 n. 6–17, citing Fremont Older, *My Own Story*, "Attempt to Conceal Stanford Murder by Natural Death Theory," *Bulletin*, March 6, 1905, "Unfounded Suspicions," *Post*, March 6, 1905, and *Oakland Tribune*, March 11, 1905; white-jane#38, Index, pp. 354, 357, 358, 359; Podcast Ep. 4 Script & Review, Sec. 15, on W. H. Mills paying the press for Harriman's Southern Pacific).↩

641. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 40 n. 6–17, citing Fremont Older, *My Own Story*, "Attempt to Conceal Stanford Murder by Natural Death Theory," *Bulletin*, March 6, 1905, "Unfounded Suspicions," *Post*, March 6, 1905, and *Oakland Tribune*, March 11, 1905; white-jane#38, Index, pp. 354, 357, 358, 359; Podcast Ep. 4 Script & Review, Sec. 15, on W. H. Mills paying the press for Harriman's Southern Pacific).↩

642. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 1, Ch. 45 n. 1–19, citing *Order of Exercises for the Funeral Service in the Memorial Church, Leland Stanford Jr. University, Friday the Twenty-fourth of March Nineteen and Five*; white-jane#38, Index, pp. 353, 354, 360; Podcast Ep. 4 Script & Review, Sec. 15: March 24 funeral, "Nearer, My God, to Thee," alma mater, steel chest sealed in sarcophagus on March 25, and workman cutting Feb. 28, 1905 into the marble on March 16, 1905).↩

643. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 1, Ch. 45 n. 1–19, citing *Order of Exercises for the Funeral Service in the Memorial Church, Leland Stanford Jr. University, Friday the Twenty-fourth of March Nineteen and Five*; white-jane#38, Index, pp. 353, 354, 360; Podcast Ep. 4 Script & Review, Sec. 15: March 24 funeral, "Nearer, My God, to Thee," alma mater, steel chest sealed in sarcophagus on March 25, and workman cutting Feb. 28, 1905 into the marble on March 16, 1905).↩

644. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 1, Ch. 45 n. 1–19, citing *Order of Exercises for the Funeral Service in the Memorial Church, Leland Stanford Jr. University, Friday the Twenty-*

fourth of March Nineteen and Five; white-jane#38, Index, pp. 353, 354, 360; Podcast Ep. 4 Script & Review, Sec. 15: March 24 funeral, “Nearer, My God, to Thee,” alma mater, steel chest sealed in sarcophagus on March 25, and workman cutting Feb. 28, 1905 into the marble on March 16, 1905).↩

645. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 47 n. 8–23, citing William James’s 1906 correspondence from Stanford and the April 18, 1906 earthquake damage to Memorial Church, the library, the gymnasium, the museum, and the Memorial Arch; Gray Brechin, *Imperial San Francisco*, brechin-imperial#34, Ch. 7 n. 1; Podcast Ep. 4 Script & Review, Sec. 16).↩

646. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 47 n. 8–23, citing William James’s 1906 correspondence from Stanford and the April 18, 1906 earthquake damage to Memorial Church, the library, the gymnasium, the museum, and the Memorial Arch; Gray Brechin, *Imperial San Francisco*, brechin-imperial#34, Ch. 7 n. 1; Podcast Ep. 4 Script & Review, Sec. 16).↩

647. Podcast Ep. 4 Script & Review, Sec. 16 (citing William Woodville Rockhill to Elihu Root, April 26, 1906: Cixi’s April 23–25, 1906 edict ordering 100,000 taels for San Francisco relief—returned by Rockhill under Roosevelt’s policy—and 40,000 taels sent to the Chinese minister in Washington for Chinese residents).↩

648. Podcast Ep. 4 Script & Review, Sec. 16 (citing Gray Brechin, *Imperial San Francisco*, National Park Service, and SFUSD records: Oct. 11, 1906 school board order sending Japanese students to the Oriental School, where Chinese students had been segregated since 1885 after *Tape v. Hurley*).↩

649. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Ch. 12 n. 13: David Starr Jordan, *The Human Harvest: A Study of the Decay of Races Through the Survival of the Unfit*, copyright 1907); Malcolm Harris, *Palo Alto* (`harris-palo-alto`, via Podcast Ep. 4 Script & Review, Sec. 07: Jordan’s 1906 appointment as inaugural chair of the Committee on Eugenics of the American Breeders’ Association).↩

650. Podcast Ep. 4 Script & Review, Sec. 06 (citing Winchester research file / Mary Jo Ignoffo: Sarah Winchester’s 1909 gift of $300,000 through Yale to endow the William Wirt Winchester Hospital for tuberculosis in New Haven, equal to thirty years of Jordan’s $10,000 salary).↩

651. Richard White, *Who Killed Jane Stanford?* (white-jane#38, Index, p. 361: Jane Stanford Way); Podcast Ep. 4 Script & Review, Sec. 07 (citing *Stanford Report*: Sept. 13, 2018 trustee vote renaming Serra Mall, Serra dorm, and Serra House while keeping Serra Street; Oct. 7, 2019 change from 450 Serra Mall to 450 Jane Stanford Way; and October 2020 removal of Jordan’s name from Jordan Hall, Building 420, and the Louis Agassiz statue). ## Part 5↩

652. Richard White, *Who Killed Jane Stanford?* (white-jane#38, Index, p. 361: Jane Stanford Way); Podcast Ep. 4 Script & Review, Sec. 07 (citing *Stanford Report*: Sept. 13, 2018 trustee vote renaming Serra Mall, Serra dorm, and Serra House while keeping Serra Street; Oct. 7,

2019 change from 450 Serra Mall to 450 Jane Stanford Way; and October 2020 removal of Jordan's name from Jordan Hall, Building 420, and the Louis Agassiz statue). ## Part 5↩

653. Kenneth Whyte, *Hoover: An Extraordinary Life in Extraordinary Times* (whyte-hoover#162, ch. 1 n. 1, ch. 2 nn. 9–19); Episode 5 script/review, §02.↩

654. Whyte, *Hoover* (whyte-hoover#162, ch. 2 nn. 20–24, citing Herbert Hoover to Nell May Hill, August 30, 1892, and Will Irwin, *Herbert Hoover: A Reminiscent Biography*, pp. 45, 48); Episode 5 script/review, §02.↩

655. Rebecca Solnit, *River of Shadows: Eadweard Muybridge and the Technological Wild West* (references.bib: `solnit-river-of-shadows`); Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (references.bib: `harris-palo-alto`); Episode 5 script/review, §06.↩

656. Episode 5 review, "This pass" table, citing Zócalo Public Square and the City of Pasadena history deck (about 150 Indianapolis neighbors led by Thomas Balch Elliott incorporated the Indiana Colony on January 31, 1874, on Rancho San Pascual).↩

657. Lyra Kilston, "Dr. Southern California," *Cabinet* 66 (2018–19), drawing on J. W. Wood, *Pasadena, California, Historical and Personal* (1917), pp. 120–121, and John E. Baur, *The Health Seekers of Southern California, 1870–1900* (Episode 5 review, "This pass" table).↩

658. Whyte, *Hoover*, p. 36 (whyte-hoover#162, ch. 2 nn. 25–26, 42; Episode 5 review, §02).↩

659. Whyte, *Hoover* (whyte-hoover#162, ch. 2; Episode 5 script/review, §02).↩

660. Whyte, *Hoover*, p. 43 (whyte-hoover#162, ch. 2 n. 40, ch. 4 nn. 2–4, citing Nancy Beck Young, *Lou Henry Hoover: Activist First Lady*, pp. 8–11; Episode 5 review, §02).↩

661. Whyte, *Hoover*, p. 43 (whyte-hoover#162, ch. 4 nn. 2–6; Episode 5 script/review, §02).↩

662. Whyte, *Hoover* (whyte-hoover#162, ch. 3 nn. 1–4, citing Hoover, *Memoirs: Years of Adventure*, p. 26; Episode 5 script/review, §02).↩

663. Whyte, *Hoover* (whyte-hoover#162, ch. 3 nn. 5–7, citing *Carson City Gold & Silver Min. Co. v. North Star Min. Co.*, 83 F. 341 (9th Cir. 1897), and Herbert Hoover, "Some Notes on 'Crossings,'" *Mining and Scientific Press* 72 (1896)).↩

664. Whyte, *Hoover* (whyte-hoover#162, ch. 3 nn. 7–15, citing George H. Nash, *The Life of Herbert Hoover: The Engineer, 1874–1914*, pp. 49–50, and Hoover to Lester Hinsdale, April 14, 1897; Episode 5 script/review, §02).↩

665. Whyte, *Hoover* (whyte-hoover#162, ch. 3 nn. 16–39, citing Herbert Hoover correspondence with Theodore Hoover and Lester Hinsdale from Coolgardie and Kalgoorlie, 1897–1898).↩

666. Whyte, *Hoover* (whyte-hoover#162, ch. 3 nn. 23–36; Episode 5 script/review, §02).↩

667. Whyte, *Hoover* (whyte-hoover#162, ch. 4 nn. 13–17, citing Herbert Hoover, “The Kaiping Coal Mines and Coal Field, Chihle Province, North China,” *Transactions of the Institution of Mining and Metallurgy* 10 (1901–2): 419–43).↩

668. Whyte, *Hoover*, p. 43 (whyte-hoover#162, ch. 4 nn. 7–13; Episode 5 script/review, §02).↩

669. Episode 5 script/review, §07, citing San Francisco Museum and UCSB American Presidency Project.↩

670. DISPUTED date: Episode 5 review opening note and script (§07) give May 13, 1903 (“Dewey Monument dedication is May 13, 1903, matching Episode 4”), while the Episode 5 review §07 table gives May 14, 1903, citing San Francisco Museum and UCSB Presidency Project; see also Carl E. Ackerman, “President Roosevelt in California,” *Sunset* 11, no. 2 (June 1903): 103–13 (brechin-imperial#69).↩

671. Episode 5 script/review, §07, paraphrasing the *Morning Oregonian*’s report of Roosevelt’s remarks.↩

672. *Morning Oregonian*, May 15, 1903, p. 2 (quoted in Episode 5 script/review, §07).↩

673. Richard White, *Who Killed Jane Stanford?* (white-jane#1, Preface n. 3: “Address delivered at the Dedication Service, Jan. 25, 1903, by Reverend D. Charles Gardner, Gardner Collection, b. 1”).↩

674. Episode 5 script/review, §01, citing Stanford quake06 timeline and repair pages and ORSL Memorial Church history.↩

675. Marquis James and Bessie Rowland James, *Biography of a Bank* (cited in Episode 5 script/review, §01); see also Felice A. Bonadio, *A.P. Giannini: Banker of America* (brechin-imperial#69).↩

676. Giannini Foundation account (Episode 5 script/review, §01).↩

677. James and James, *Biography of a Bank* (Episode 5 script/review, §01).↩

678. Whyte, *Hoover*, p. 100 (whyte-hoover#162, ch. 5 nn. 26–30, citing Richard Hartley, “Bewick, Moreing in Western Australian Gold Mining, 1897–1904,” *Labour History* 65 (November 1993): 8–12; Episode 5 script/review, §02).↩

679. Whyte, *Hoover*, p. 100 (Episode 5 script/review, §02: “Air at Sons of Gwalia ‘notably bad.’ Dust, fumes, falling rock”).↩

680. Whyte, *Hoover*, p. 100 (Episode 5 script/review, §02).↩

681. Whyte, *Hoover* (whyte-hoover#162, ch. 5 n. 25, citing T. A. Rickard, Walter Benton Ingalls, Herbert Hoover, and R. Gilman Brown, *The Economics of Mining* (New York: Engineering and Mining Journal, 1905), pp. 191–92).↩

682. *Stanford Magazine*, “The Vexing Legacy of Lewis Terman” (Episode 5 script/review, §04).↩

683. HP garage timeline and National Register of Historic Places nomination 07000307 (Episode 5 script/review, §09).↩

684. White, *Who Killed Jane Stanford?* (white-jane#1, ch. 47 nn. 1, 8, citing “Accepted Position at Stanford University,” *San Jose Mercury*, Feb. 23, 1905; Linda Simon, “William James at Stanford,” *California History* 69, no. 4 (Winter 1990/1991): 332–41; William James to Jordan, Sept. 1, 1904; and R. Laurence Moore, *In Search of White Crows: Spiritualism, Parapsychology, and American Culture*); Episode 5 script/review, §03.↩

685. White, *Who Killed Jane Stanford?* (white-jane#1, ch. 28–29, 31, citing Coroner’s Inquest testimony of Bertha Berner, May Hunt, Dr. Francis Howard Humphris, and Dr. Clifford B. Wood, and *Hawaiian Star*, March 1, 1905).↩

686. White, *Who Killed Jane Stanford?* (white-jane#1, ch. 31 nn. 6–9, ch. 36 nn. 12–25, ch. 41 nn. 1–21, ch. 42 nn. 2–12, citing Drs. Wood and Day autopsy report, March 1, 1905; Testimony of R. A. Duncan; Coroner’s Inquest Verdict, p. 70; and “Jordan Says Officially that Mrs. Stanford Died a Natural Death,” *San Francisco Examiner*, March 16, 1905).↩

687. White, *Who Killed Jane Stanford?*, pp. 281–82 (white-jane#1, ch. 47 nn. 8–16; Episode 5 script/review, §03).↩

688. William James to Alice James, Jan. 17, 1906, *Correspondence of William James*, vol. 11, p. 148, quoted in White, *Who Killed Jane Stanford?*, p. 281 (white-jane#1, ch. 47 n. 9; Episode 5 script/review, §03).↩

689. White, *Who Killed Jane Stanford?*, pp. 281–82 (white-jane#1, ch. 47 nn. 10–18; Episode 5 script/review, §03).↩

690. USGS 1906 earthquake page and White, *Who Killed Jane Stanford?* (Episode 5 script/review, §01).↩

691. William James to Miss Frances R. Morse, April 22, 1906, *The Letters of William James*, vol. 2, pp. 247–49, cited in White, *Who Killed Jane Stanford?* (white-jane#1, ch. 47 n. 19; Episode 5 script/review, §01).↩

692. White, *Who Killed Jane Stanford?* (white-jane#1, ch. 47 n. 19; Episode 5 script/review, §01).↩

693. Stanford quake06 timeline and repair pages; ORSL Memorial Church history (Episode 5 script/review, §01).↩

694. Gordon H. Chang, *Ghosts of Gold Mountain: The Epic Story of the Chinese Who Built the Transcontinental Railroad* (references.bib: `chang-ghosts-gold-mountain`; Episode 5 script/review, §01).↩

695. White, *Who Killed Jane Stanford?* (white-jane#1, ch. 8 n. 2, ch. 14 n. 14); Stanford earthquake timeline (Episode 5 script/review, §01).↩

696. “Ah Wing Account,” Stanford quake06 centennial gallery, cited in White, *Who Killed Jane Stanford?* (white-jane#1, ch. 4 n. 8, ch. 14 n. 14; Episode 5 script/review, §01).↩

697. DISPUTED details of Giannini’s 1906 wagon rescue: *Time* and UC Berkeley ARE Giannini Hall 75th anniversary page ($2 million under vegetables); Office of the Comptroller of the Currency bank history ($80,000 in coins under oranges); and Marquis James and Bessie Rowland James, *Biography of a Bank* (two of his stepfather’s produce wagons, plank on two barrels on the Washington Street wharf within two days), all documented in Episode 5 script/review, §01.↩

698. White, *Who Killed Jane Stanford?*, p. 282 (white-jane#1, ch. 47 n. 19; Episode 5 script/review, §03).↩

699. William James to Horace Davis, April 19, 1906, and David Starr Jordan to Horace Davis, May 20, 1906, Davis Papers, b. 1, f. 19, cited in White, *Who Killed Jane Stanford?*, p. 282 (white-jane#1, ch. 47 nn. 22–23; Episode 5 script/review, §03).↩

700. David Starr Jordan, *The Human Harvest: A Study of the Decay of Races Through the Survival of the Unfit* (1907; white-jane#1, ch. 12 nn. 13–14), and Jordan, “Foreclosing the Mortgage on War,” *World’s Work* 24, no. 2 (June 1912): 205–8, and “The Perennial Bogey of War,” *World’s Work* 25, no. 2 (December 1912): 191–96 (brechin-imperial#69).↩

701. Malcolm Harris, *Palo Alto* (references.bib: `harris-palo-alto`, bionomics chapter; Episode 5 script/review, §04); see also Lulu Miller, *Why Fish Don’t Exist* (white-jane#1, ch. 5 n. 1).↩

702. Alexandra Minna Stern, *Eugenic Nation: Faults and Frontiers of Better Breeding in Modern America*, 2nd ed. (references.bib: `stern-eugenic-nation`); Episode 5 script/review, §04.↩

703. Episode 5 script/review, §07.↩

704. Herbert Hoover, *The Principles of Mining* (New York: Hill Publishing, 1909), and Georgius Agricola, *De Re Metallica*, trans. Herbert Hoover and Lou Henry Hoover (1912), cited in Whyte, *Hoover* (whyte-hoover#162, ch. 6 nn. 33–39) and Brechin, *Imperial San Francisco* (brechin-imperial#69).↩

705. Malcolm Harris, *Palo Alto*, footnote in the bionomics chapter: “Lewis Terman’s young adulthood was marred by a long and debilitating course of tuberculosis, requiring his move to the mild California climate in the first place” (quoted in Episode 5 review, “This pass” table, and §04).↩

706. White, *Who Killed Jane Stanford?* (white-jane#1, ch. 23 nn. 4, 14, citing Orrin Leslie Elliott, *Stanford University: The First 25 Years*, p. 159, and John Edgar Coover, *Experiments in Psychical Research at Leland Stanford Junior University*, 1917); Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy* (references.bib: `lemann-big-test`); Episode 5 script/review, §04.↩

707. Episode 5 script/review, §06.↩

708. AFI Catalog and *Moving Picture World* (1911), quoted in Episode 5 script/review, §06; on the Motion Picture Patents Company, see Tim Wu, *The Master Switch: The Rise and Fall of Information Empires* (references.bib: `wu-master-switch`).↩

709. Whyte, *Hoover*, p. 124 (whyte-hoover#162, ch. 6 nn. 46–50; quoted in Episode 5 script/review, §03).↩

710. Whyte, *Hoover*, p. 124 (Episode 5 script/review, §03).↩

711. Whyte, *Hoover*, p. 124, and Memorandum, January 2, 1913, Stanford Academic and Financial Positions, Subject Files, Pre-Commerce Papers, HHPL (whyte-hoover#162, ch. 6 nn. 46–50; Episode 5 script/review, §03).↩

712. Paolo E. Coletta, "'The Most Thankless Task': Bryan and the California Alien Land Legislation," *Pacific Historical Review* 36, no. 2 (May 1967): 163–87, and Frank W. Van Nuys, "A Progressive Confronts the Race Question: Chester Rowell, the California Alien Land Act of 1913, and the Contradictions of Early Twentieth Century Racial Thought," *California History* 73, no. 1 (1994): 2–13 (brechin-imperial#69); Episode 5 script/review, §10.↩

713. DISPUTED Flagstaff departure: PBS SoCal *Lost LA* (noting the snow/cold story is widely regarded as apocryphal) and American Film Institute archive (recording the high mountains vs. plains account, Dustin Farnum's suggestion, December 20 arrival in Los Angeles, Christmas Eve barn lease at Selma and Vine, December 29 start of shooting, and DeMille playing a faro dealer to save $3), documented in Episode 5 script/review, §06; see also Scott Eyman, *Empire of Dreams: The Epic Life of Cecil B. DeMille*, and Simon Louvish, *Cecil B. DeMille: A Life in Art* (winkler-corporations#54, ch. 10 nn. 8–9).↩

714. PBS SoCal *Lost LA* / Robert Sklar, recording DeMille's note on 25 to 50 percent lower labor costs (Episode 5 script/review, §06).↩

715. Kenneth Whyte, *Hoover: An Extraordinary Life in Extraordinary Times* (2017), via Podcast Ep. 5 script & review, sec. 02 ("London Wall partner by 1914").↩

716. Whyte, *Hoover*, ch. 7 (whyte-hoover#162, citing Hoover, *Years of Adventure*, pp. 137–138), and Podcast Ep. 5 script & review, sec. 05 ("Sarajevo, June 28, 1914. Hoover dismissed it").↩

717. Whyte, *Hoover*, ch. 7 (whyte-hoover#162, citing Hoover to R. L. Wilbur, August 4, 1914, and George H. Nash, *The Life of Herbert Hoover: The Engineer, 1874–1914*, pp. 575–576), and Podcast Ep. 5 script & review, sec. 05 ("August 4, walked about a block to the U.S. consulate").↩

718. Whyte, *Hoover*, ch. 7 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 05 ("7.5 million people. About 20 percent of food grown at home. German harvest seizure and British blockade").↩

719. Whyte, *Hoover*, ch. 7 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 05 ("7.5 million people. About 20 percent of food grown at home. German harvest seizure and British blockade").↩

720. Whyte, *Hoover*, ch. 7 (whyte-hoover#162, citing Nash, *The Humanitarian, 1914–1917*, pp. 15–22, and Hoover, *Years of Adventure*, pp. 141–145), and Podcast Ep. 5 script & review, sec. 05 (“Shaler, 1,500 tons. CRB: Rotterdam, canal, monopoly”).↩

721. Whyte, *Hoover*, ch. 7 (whyte-hoover#162, citing Nash, *The Humanitarian, 1914–1917*, pp. 15–22, and Hoover, *Years of Adventure*, pp. 141–145), and Podcast Ep. 5 script & review, sec. 05 (“Shaler, 1,500 tons. CRB: Rotterdam, canal, monopoly”).↩

722. Whyte, *Hoover*, ch. 7 (whyte-hoover#162, citing Nash, *The Humanitarian, 1914–1917*, pp. 15–22, and Hoover, *Years of Adventure*, pp. 141–145), and Podcast Ep. 5 script & review, sec. 05 (“Shaler, 1,500 tons. CRB: Rotterdam, canal, monopoly”).↩

723. Whyte, *Hoover*, ch. 7 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 05 (“Cable: ‘Over one million people on bread line…’ and ‘upwards of twenty thousand tons foodstuffs monthly.’ First ship, 3,500 tons, November 21”).↩

724. Whyte, *Hoover*, about p. 146 and p. 206 (and whyte-hoover#162, ch. 7–8 notes citing George I. Gay, *Public Relations of the Commission for Relief in Belgium*, vol. 1, 1929), via Podcast Ep. 5 script & review, sec. 05 (“‘Enormous propaganda.’ ‘Detailed personal horror stuff.’ Shaw’s lapdogs from Canterbury to San Francisco”).↩

725. Whyte, *Hoover*, about p. 146 and p. 206 (and whyte-hoover#162, ch. 7–8 notes citing George I. Gay, *Public Relations of the Commission for Relief in Belgium*, vol. 1, 1929), via Podcast Ep. 5 script & review, sec. 05 (“‘Enormous propaganda.’ ‘Detailed personal horror stuff.’ Shaw’s lapdogs from Canterbury to San Francisco”).↩

726. Whyte, *Hoover*, about p. 146 and p. 206 (and whyte-hoover#162, ch. 7–8 notes citing George I. Gay, *Public Relations of the Commission for Relief in Belgium*, vol. 1, 1929), via Podcast Ep. 5 script & review, sec. 05 (“‘Enormous propaganda.’ ‘Detailed personal horror stuff.’ Shaw’s lapdogs from Canterbury to San Francisco”).↩

727. Whyte, *Hoover*, ch. 7 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 05 (“Cable: ‘Over one million people on bread line…’ and ‘upwards of twenty thousand tons foodstuffs monthly.’ First ship, 3,500 tons, November 21”).↩

728. Whyte, *Hoover*, ch. 7 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 05 (“Cable: ‘Over one million people on bread line…’ and ‘upwards of twenty thousand tons foodstuffs monthly.’ First ship, 3,500 tons, November 21”).↩

729. Lewis M. Terman, *The Measurement of Intelligence* (1916); *Stanford Magazine*, “The Vexing Legacy of Lewis Terman”; and Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy* (Farrar, Straus and Giroux, 1999; `lemann-big-test`), via Podcast Ep. 5 script & review, sec. 04.↩

730. Lewis M. Terman, *The Measurement of Intelligence* (1916); *Stanford Magazine*, “The Vexing Legacy of Lewis Terman”; and Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy*

(Farrar, Straus and Giroux, 1999; `lemann-big-test`), via Podcast Ep. 5 script & review, sec. 04.↩

731. Lewis M. Terman, *The Measurement of Intelligence* (1916); *Stanford Magazine*, "The Vexing Legacy of Lewis Terman"; and Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy* (Farrar, Straus and Giroux, 1999; `lemann-big-test`), via Podcast Ep. 5 script & review, sec. 04.↩

732. Lewis M. Terman, *The Measurement of Intelligence* (1916); *Stanford Magazine*, "The Vexing Legacy of Lewis Terman"; and Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy* (Farrar, Straus and Giroux, 1999; `lemann-big-test`), via Podcast Ep. 5 script & review, sec. 04.↩

733. Lewis M. Terman, *The Measurement of Intelligence* (1916); *Stanford Magazine*, "The Vexing Legacy of Lewis Terman"; and Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy* (Farrar, Straus and Giroux, 1999; `lemann-big-test`), via Podcast Ep. 5 script & review, sec. 04.↩

734. Lewis M. Terman, *The Measurement of Intelligence* (1916); *Stanford Magazine*, "The Vexing Legacy of Lewis Terman"; and Nicholas Lemann, *The Big Test: The Secret History of the American Meritocracy* (Farrar, Straus and Giroux, 1999; `lemann-big-test`), via Podcast Ep. 5 script & review, sec. 04.↩

735. *Stanford Magazine*, "The Vexing Legacy of Lewis Terman," via Podcast Ep. 5 script & review, sec. 04 ("Army tests, more than 1.7 million draftees").↩

736. *Stanford Magazine*, "The Vexing Legacy of Lewis Terman," via Podcast Ep. 5 script & review, sec. 04 ("Army tests, more than 1.7 million draftees").↩

737. Whyte, *Hoover*, ch. 9–10 (whyte-hoover#162, citing George H. Nash, *Master of Emergencies, 1917–1918*, and U.S. Food Administration Bulletin No. 2, 1917), and Podcast Ep. 5 script & review, sec. 05 ("Wilson, Food Administrator. Hooverize: clean plate, meatless, waste nothing").↩

738. Whyte, *Hoover*, ch. 9–10 (whyte-hoover#162, citing George H. Nash, *Master of Emergencies, 1917–1918*, and U.S. Food Administration Bulletin No. 2, 1917), and Podcast Ep. 5 script & review, sec. 05 ("Wilson, Food Administrator. Hooverize: clean plate, meatless, waste nothing").↩

739. Office of the Comptroller of the Currency (OCC) bank history and Felice A. Bonadio, *A.P. Giannini: Banker of America* (University of California Press, 1994, cited in brechin-imperial#69), via Podcast Ep. 5 script & review, sec. 01 ("Statewide branches by 1918. Renamed Bank of America in November 1930. By Giannini's death in 1949, largest bank, 500+ branches, about $6 billion in assets"; CPI 1949 = 71.5, 2026 = 1004.8, roughly $84 billion).↩

740. Hoover Institution timeline, and Whyte, *Hoover*, ch. 11 (whyte-hoover#162), via Podcast Ep. 5 script & review, sec. 05 ("June 1919 telegram to Ray Lyman Wilbur. $50,000. Send Ephraim D. Adams to

Paris. Lutz and Golder also collected"; CPI 1919 = 52.1, roughly $964,000).↩

741. Hoover Institution timeline, and Whyte, *Hoover*, ch. 11 (whyte-hoover#162), via Podcast Ep. 5 script & review, sec. 05 ("June 1919 telegram to Ray Lyman Wilbur. $50,000. Send Ephraim D. Adams to Paris. Lutz and Golder also collected"; CPI 1919 = 52.1, roughly $964,000).↩

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745. The Nobel Foundation, "John Steinbeck: Questions and Answers," NobelPrize.org (`nobel-steinbeck-qa`), confirming attendance at Stanford 1919–1925 without a degree; *The Stanford Daily*, "Steinbeck at Stanford," June 7, 2012 (`stanforddaily-steinbeck-2012`), for the 93-unit figure; and Gavin Jones, "Steinbeck at Stanford," Stanford Historical Society, March 2020 (`jones-steinbeck-stanford`).↩

746. Lewis Terman, *Genetic Studies of Genius*, and *Stanford Magazine* / Online Archive of California, via Podcast Ep. 5 script & review, sec. 04 ("Core of 643 children at IQ 135 or above. He enrolled his own children"). The exact start year 1921 is marked unverified as it does not appear in the Episode 5 review table or excerpts.↩

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748. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023; `harris-palo-alto`), and NobelPrize.org, via Podcast Ep. 5 script & review, sec. 04 ("Shockley scored 129, 'slightly subgenius.' Lewis taught until 1943 and died in 1956, the year of Shockley's Nobel").↩

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750. Hoover Institution timeline, via Podcast Ep. 5 script & review, sec. 05 (“1922, renamed Hoover War Library. 40,000 documents that year, in the Stanford library”).↩

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753. UC Berkeley Department of Agricultural and Resource Economics (ARE) Giannini Hall 75th anniversary history and Giannini Foundation, via Podcast Ep. 5 script & review, sec. 01 and 07 (“Giannini, $1.5 million in 1928, Giannini Foundation and Giannini Hall. About $29 million grocery”).↩

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755. Whyte, *Hoover*, ch. 17 and ch. 20 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 08 (“Elected November 1928. Crash, autumn 1929”).↩

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757. Bureau of Labor Statistics and Irving Bernstein / U.S. Department of Labor history, via Podcast Ep. 5 script & review, sec. 08 (“Modern CPS unemployment survey starts in 1940. 1933 estimate 24.9 percent, 12,830,000, about one in four. March the worst month. Farmers often uncounted. U.S. Steel, about 225,000 full-time in 1929, to zero full-time on April 1, 1933”).↩

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759. Office of the Comptroller of the Currency (OCC) bank history and Felice A. Bonadio, *A.P. Giannini: Banker of America* (University of California Press, 1994, cited in brechin-imperial#69), via Podcast Ep. 5 script & review, sec. 01 (“Statewide branches by 1918. Renamed Bank of America in November 1930. By Giannini’s death in 1949, largest bank, 500+ branches, about $6 billion in assets”; CPI 1949 = 71.5, 2026 = 1004.8, roughly $84 billion).↩

760. HistoryLink, file 23080, and Whyte, *Hoover*, ch. 20 n. 43 (whyte-hoover#162, citing Frank R. Kent, "Charley Michelson," *Scribner's Magazine*, September 1930), via Podcast Ep. 5 script & review, sec. 08 ("'Hooverville' first in print in 1930, credited to Charles Michelson of the DNC. Seattle: Jesse Jackson and about 20 men, October 1931, tidal flats south of Pioneer Square, former shipyard. Up to 1,200 people on about 9 acres, until 1941").↩

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768. Whyte, *Hoover*, ch. 24 n. 68 (whyte-hoover#162), and Podcast Ep. 5 script & review, sec. 08 (“Roosevelt beat Hoover in 1932. Warren Delano Jr., Russell and Company, a China fortune in opium”).↩

769. Whyte, *Hoover*, post-presidency chapters, via Podcast Ep. 5 script & review, sec. 05 and 08 (“Up to 1,500 letters a day. Anti-Hoover literature in the library. Relief at no longer making ten vital decisions a day”).↩

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775. *American Banker* and Hilbert Museum, via Podcast Ep. 5 review header and script, sec. 01 (“Bank of America, Giannini’s bank, lent Disney the money to finish *Snow White* and stayed through *Fantasia*”).↩

776. Podcast Ep. 5 script & review, sec. 10 (“Klystron: Russell and Sigurd Varian with William Hansen, late 1930s”).↩

777. Hewlett-Packard garage timeline and National Register of Historic Places nomination 07000307, via Podcast Ep. 5 script & review, sec. 09 (“367 Addison. Built about 1905 for Dr. John Spencer… September 1938: Packards downstairs, Hewlett in the shed, Ione Spencer upstairs… Both finished electrical engineering at Stanford in 1934. Packard at GE in Schenectady. Hewlett at Stanford and MIT. Terman fellowship, 1938, with Russell Varian. Lucile’s registrar paycheck and the books. Used Sears drill press in the car”; rent $45/month, roughly $1,100).↩

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783. Hewlett-Packard company history and National Register of Historic Places nomination 07000307, via Podcast Ep. 5 review header and sec. 09 (“$538 working capital: cash plus the drill press. Partnership January 1, 1939. Coin toss. Model 200A audio oscillator”; “HP’s own history: eight Model 200B oscillators, modified from the 200A, at $71.50 each, for *Fantasia*”; CPI 1938 = 42.3, $538 ≈ $12,800).↩

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789. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Podcast Ep. 5 script & review, sec. 10 (“About three weeks after Pearl Harbor, Terman was called to the Radio Research Laboratory at Harvard. Chaff / window. British collaboration. Terman in England. Normandy ghost fleets”).↩

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799. California Alien Land Law of 1913 (also cited in brechin-imperial#69: Paolo E. Coletta, "'The Most Thankless Task': Bryan and the California Alien Land Legislation," *Pacific Historical Review* 36, no. 2, 1967, and Frank W. Van Nuys, *California History* 73, no. 1, 1994), and Podcast Ep. 5 script & review, sec. 10 ("Property-loss estimate of about $400 million. Widely repeated. Not pinned to a single CWRIC page in this pass").↩

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811. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023; `harris-palo-alto`), and NobelPrize.org, via Podcast Ep. 5 script & review, sec. 04 (“Shockley scored 129, ‘slightly subgenius.’ Lewis taught until 1943 and died in 1956, the year of Shockley’s Nobel”).↩

812. Vannevar Bush, *Science, the Endless Frontier: A Report to the President* (United States Government Printing Office, July 1945; `bush-endless-frontier`), and Podcast Ep. 5 script & review, sec. 10.↩

813. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stuart W. Leslie, *The Cold War and American Science* (`leslie-cold-war-science`), and Rebecca S. Lowen, *Creating the Cold War University* (`lowen-cold-war-university`), via Podcast Ep. 5 script & review, sec. 11 (“Terman back in Palo Alto as the war ends, 1945 into 1946”).↩

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815. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stuart W. Leslie, *The Cold War and American Science* (`leslie-cold-war-science`), and Rebecca S. Lowen, *Creating the Cold War University* (`lowen-cold-war-university`), via Podcast Ep. 5 script & review, sec. 11 (“Terman back in Palo Alto as the war ends, 1945 into 1946”).↩

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817. Office of the Comptroller of the Currency (OCC) bank history and Felice A. Bonadio, *A.P. Giannini: Banker of America* (University of California Press, 1994, cited in brechin-imperial#69), via Podcast Ep. 5 script & review, sec. 01 (“Statewide branches by 1918. Renamed Bank of America in November 1930. By Giannini’s death in 1949, largest bank, 500+ branches, about $6 billion in assets”; CPI 1949 = 71.5, 2026 = 1004.8, roughly $84 billion).↩

818. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (Little, Brown, 2023; `harris-palo-alto`), and NobelPrize.org, via Podcast Ep. 5 script & review, sec. 04 (“Shockley scored 129, ‘slightly subgenius.’ Lewis taught until 1943 and died in 1956, the year of Shockley’s Nobel”).↩

819. Stanford University Founding Grant, November 11, 1885 (berlin-troublemakers#39, “What Do We Do with These?” n. 24; white-jane#1, ch. 15 n. 14), and Podcast Ep. 5 script & review, sec. 11 (“Founding Grant, 1885: no power to sell or convey. A lease is not a sale”). ## Part 6↩

820. Stanford University Founding Grant, November 11, 1885 (berlin-troublemakers#39, “What Do We Do with These?” n. 24; white-jane#1, ch. 15 n. 14), and Podcast Ep. 5 script & review, sec. 11 (“Founding Grant, 1885: no power to sell or convey. A lease is not a sale”). ## Part 6↩

821. Stanford University Founding Grant, November 11, 1885 (berlin-troublemakers#39, “What Do We Do with These?” n. 24; white-jane#1, ch. 15 n. 14), and Podcast Ep. 5 script & review, sec. 11 (“Founding Grant, 1885: no power to sell or convey. A lease is not a sale”). ## Part 6↩

822. Rana Mitter, *Forgotten Ally: China’s World War II, 1937–1945*, cited in Episode 6 Review (“In Asia the war begins July 7, 1937, at the Marco Polo Bridge. That is the Second Sino-Japanese War, not the 1894–95 war. Mitter’s range for Chinese dead is 14–20 million.”).↩

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war. Mitter's range for Chinese dead is 14–20 million.").↩

824. National Park Service, Richmond shipyards, and usmm.org on the SS *Robert E. Peary*, cited in Episode 6 Review ("Kaiser's four Richmond yards: 747 ships, 519 of them Liberty ships. Richmond from about 20,000 to more than 100,000. SS *Robert E. Peary*, keel to launch in 4 days, 15 hours, 29 minutes, November 8–12, 1942.").↩

825. National Park Service, Richmond shipyards, and usmm.org on the SS *Robert E. Peary*, cited in Episode 6 Review ("Kaiser's four Richmond yards: 747 ships, 519 of them Liberty ships. Richmond from about 20,000 to more than 100,000. SS *Robert E. Peary*, keel to launch in 4 days, 15 hours, 29 minutes, November 8–12, 1942.").↩

826. National Park Service, Richmond shipyards, and usmm.org on the SS *Robert E. Peary*, cited in Episode 6 Review ("Kaiser's four Richmond yards: 747 ships, 519 of them Liberty ships. Richmond from about 20,000 to more than 100,000. SS *Robert E. Peary*, keel to launch in 4 days, 15 hours, 29 minutes, November 8–12, 1942.").↩

827. Sally Denton, *The Profiteers: Bechtel and the Men Who Built the World*, cited in Episode 6 Script and Review ("Bechtel built Pacific bases from Alameda outward, and had partner yards. Not the same company as Kaiser.").↩

828. Carol Petillo, *Pacific Historical Review* (February 1979), and Philippine Executive Order No. 1 (January 3, 1942, via lawphil.net), cited in Episode 6 Review ("January 3, 1942, on Corregidor, Quezon's Executive Order No. 1 awarded MacArthur $500,000, with smaller sums to aides. Exposed by Carol Petillo, *Pacific Historical Review*, February 1979.").↩

829. Oakland City Council actions (March 19 and April 8, 1942; signs April 11, 1942) and 1968 I-580 freeway resolution, cited in Episode 6 Review ("Oakland's MacArthur Boulevard: council action March 19 and April 8, 1942; signs April 11, 1942, during the war. The I-580 freeway naming is a 1968 resolution.").↩

830. Densho, Tanforan detention facility article, and Miné Okubo, cited in Episode 6 Script and Review ("Tanforan, April 28–October 13, 1942. 8,033 people. 64 percent U.S. citizens. About 3,700 in horse stalls. Okubo: 'I was given a number and placed in a horse stall.' Korematsu held there. Stanford sent books. Food 37 cents a day; the Army spent 50 cents on a soldier. Conversion cost $1,188,000. … The track is named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, grantee of Rancho Buri Buri.").↩

831. Densho, Tanforan detention facility article, and Miné Okubo, cited in Episode 6 Script and Review ("Tanforan, April 28–October 13, 1942. 8,033 people. 64 percent U.S. citizens. About 3,700 in horse stalls. Okubo: 'I was given a number and placed in a horse stall.' Korematsu held there. Stanford sent books. Food 37 cents a day; the Army spent 50 cents on a soldier. Conversion cost $1,188,000. … The track is named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, grantee of Rancho Buri Buri.").↩

832. Densho, Tanforan detention facility article, and Miné Okubo, cited in Episode 6 Script and Review (“Tanforan, April 28–October 13, 1942. 8,033 people. 64 percent U.S. citizens. About 3,700 in horse stalls. Okubo: ‘I was given a number and placed in a horse stall.’ Korematsu held there. Stanford sent books. Food 37 cents a day; the Army spent 50 cents on a soldier. Conversion cost $1,188,000. … The track is named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, grantee of Rancho Buri Buri.”).↩

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835. Densho, Tanforan detention facility article, and Miné Okubo, cited in Episode 6 Script and Review (“Tanforan, April 28–October 13, 1942. 8,033 people. 64 percent U.S. citizens. About 3,700 in horse stalls. Okubo: ‘I was given a number and placed in a horse stall.’ Korematsu held there. Stanford sent books. Food 37 cents a day; the Army spent 50 cents on a soldier. Conversion cost $1,188,000. … The track is named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, grantee of Rancho Buri Buri.”).↩

836. Densho, Tanforan detention facility article, and Miné Okubo, cited in Episode 6 Script and Review (“Tanforan, April 28–October 13, 1942. 8,033 people. 64 percent U.S. citizens. About 3,700 in horse stalls. Okubo: ‘I was given a number and placed in a horse stall.’ Korematsu held there. Stanford sent books. Food 37 cents a day; the Army spent 50 cents on a soldier. Conversion cost $1,188,000. … The track is named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, grantee of Rancho Buri Buri.”).↩

837. Densho, Tanforan detention facility article, and Miné Okubo, cited in Episode 6 Script and Review (“Tanforan, April 28–October 13, 1942. 8,033 people. 64 percent U.S. citizens. About 3,700 in horse stalls. Okubo: ‘I was given a number and placed in a horse stall.’ Korematsu held there. Stanford sent books. Food 37 cents a day; the Army spent 50 cents on a soldier. Conversion cost $1,188,000. … The track is named for Toribio Tanforan, grandson-in-law of José Antonio Sánchez, grantee of Rancho Buri Buri.”).↩

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841. Hoover Institution tower and timeline pages, cited in Episode 6 Script and Review (“Hoover Tower, Arthur Brown Jr., 285 feet. $600,000 decided in 1938. Bells from the Belgian pavilion at the 1939 fair. ‘For Peace Alone Do I Ring.’ Dedicated June 20, 1941. Wartime listening post. 1959: trustees set the Institution apart inside the university; Hoover’s private-enterprise charge.”).↩

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845. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (`harris-palo-alto`), cited in Episode 6 Script and Review (“January 1945: Terman gets $25,000 from Donald Tresidder for the Microwave Laboratory, that name and no other. Office of Naval Research is the spigot.”).↩

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850. Teikoku Shoin statistics and Williamson (1951), cited in Episode 6 Script and Review ("Arthur MacArthur Jr. was military governor of the Philippines, May 5, 1900–July 4, 1901, and clashed with Taft. Douglas was a military adviser and field marshal, not the civil reformer… Japan: tenant-farmed land from 45.9 percent (November 1945) to 9.9 percent (August 1950). About 4.76 million acres by July 1950.").↩

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854. Kenneth Whyte, *Hoover: An Extraordinary Life in Extraordinary Times*, pp. 575–585, cited in Episode 6 Script and Review ("Famine survey: left LaGuardia March 17, 1946, in a C-54, age 71. 'Old family doctor.' Five weeks, about 35,000 miles, 25 countries.").↩

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859. Hoover Institution, “The Goebbels Diaries” and 1948 acquisition note, cited in Episode 6 Script and Review (“1946 offer of ‘the only original portions’ of the Goebbels diaries: handwritten 1925–26, typed 1942–43. The Nazi party archive at Hoover is microfilm; paper originals went to the Bundesarchiv.”).↩

860. Malcolm Harris, *Palo Alto* (`harris-palo-alto`); Rebecca S. Lowen, *Creating the Cold War University: The Transformation of Stanford* (`lowen-cold-war-university`); and Sally Denton, *The Profiteers*, ch. 10, cited in Episode 6 Script and Review (“SRI, 1946. Steve Bechtel Sr. a founding director. Tresidder fired the director rather than let one Navy contract turn the institute into a public charge, and died before naming a replacement. Sterling made Terman provost in 1954. Lowen, via Harris: Sterling was anticommunist and anti-Roosevelt enough to please Hoover.”).↩

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865. East Palo Alto general plan, cited in Episode 6 Script and Review ("San Francisquito Creek is the Santa Clara / San Mateo line. … Informal vote, December 12, 1925. Ravenswood and Runnymede lost to East Palo Alto. Annexation talk. The enabling law never came.").↩

866. East Palo Alto general plan, cited in Episode 6 Script and Review ("San Francisquito Creek is the Santa Clara / San Mateo line. … Informal vote, December 12, 1925. Ravenswood and Runnymede lost to East Palo Alto. Annexation talk. The enabling law never came.").↩

867. East Palo Alto general plan, cited in Episode 6 Script and Review ("San Francisquito Creek is the Santa Clara / San Mateo line. … Informal vote, December 12, 1925. Ravenswood and Runnymede lost to East Palo Alto. Annexation talk. The enabling law never came.").↩

868. Richard Rothstein, *The Color of Law: A Forgotten History of How Our Government Segregated America* (`rothstein-color-of-law`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), cited in Episode 6 Script and Review ("FHA as the postwar mortgage backer. HOLC maps. Redlining. Covenants. 1948: courts will not enforce racial covenants. Harris: the FHA still did not insure Black buyers.").↩

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871. Adam Winkler, *We the Corporations: How American Businesses Won Their Civil Rights* (winkler-corporations#54), ch. 8, n. 6 ("The Supreme Court decision ending racially restrictive covenants was Shelley v. Kraemer, 334 U.S. 1 (1948).").↩

872. Richard Rothstein, *The Color of Law: A Forgotten History of How Our Government Segregated America* (rothstein-color-of-law), and Malcolm Harris, *Palo Alto* (harris-palo-alto), cited in Episode 6 Script and Review ("FHA as the postwar mortgage backer. HOLC maps. Redlining. Covenants. 1948: courts will not enforce racial covenants. Harris: the FHA still did not insure Black buyers.").↩

873. Malcolm Harris, *Palo Alto* (harris-palo-alto), following Richard Rothstein, *The Color of Law* (rothstein-color-of-law), cited in Episode 6 Script and Review ("Peninsula Housing Association, late 1940s. Of the first 150 families, 2 percent Black. FHA would not insure. Bank of America would not lend what FHA refused. Sold to a whites-only developer. The subdivision is Ladera. About 260 acres.").↩

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877. Richard Rothstein, *The Color of Law* (rothstein-color-of-law), and Malcolm Harris, *Palo Alto* (harris-palo-alto), cited in Episode 6 Script and Review ("Blockbusting east of Highway 101. Within six years, exclusively white to 82 percent Black. Palo Alto zoned less than 10 percent of housing land for multifamily buildings. … Art Fong, a Chinese-American Hewlett-Packard recruit, could not find a rental in Palo Alto after the war.").↩

878. Malcolm Harris, *Palo Alto* (harris-palo-alto), and Vannevar Bush, *As We May Think* (*The Atlantic*, reprinted in *Life*, 1945), cited in Episode 6 Script and Review ("Navy radio. Read Bush's *Atlantic* piece, reprinted in *Life*, on the way to the Philippines on V-J Day. Ames, 1950. The idea was thought nuts.").↩

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880. *TIME*; *Oregon Statesman* (April 18, 1951); *Los Angeles Times* (April 19, 1951); and *New York Times*, cited in Episode 6 Script and Review ("Fired April 11, 1951. Landed in San Francisco April 17, first time in the continental United States in 14 years. Earl Warren met the plane. The next day, on the City Hall steps, he renounced political ambitions. July 7, 1952, he gave the Republican keynote and was put in nomination. Eisenhower won on the first ballot.").↩

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885. U.S. Department of State, *Foreign Relations of the United States* (FRUS), and Presidio history, cited in Episode 6 Script and Review ("September 8, 1951, Treaty of San Francisco at the War Memorial Opera House, Japan and 48 other nations. USSR, Poland, and Czechoslovakia refused. The same day the U.S.–Japan Security Treaty was signed at the Presidio. MacArthur did not sign.").↩

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888. U.S. Department of State, *Foreign Relations of the United States* (FRUS), and Presidio history, cited in Episode 6 Script and Review ("September 8, 1951, Treaty of San Francisco at the War Memorial Opera House, Japan and 48 other nations. USSR, Poland, and Czechoslovakia refused. The same day the U.S.–Japan Security Treaty was signed at the Presidio. MacArthur did not sign.").↩

889. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), ch. 3.1, citing Rebecca S. Lowen, *Creating the Cold War University* (`lowen-cold-war-university`), cited in Episode 6 Script and Review ("Defense budget $13 billion to $58 billion in 1951. While bombers leveled North Korean cities, military electronics money at Stanford tripled. By 1967 Stanford trailed only MIT and Caltech in defense contracts.").↩

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893. Richard Rothstein, *The Color of Law* (`rothstein-color-of-law`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), cited in Episode 6 Script and Review ("Blockbusting east of Highway 101. Within six years, exclusively white to 82 percent Black. Palo Alto zoned less than 10 percent of housing land for multifamily buildings. … Art Fong, a Chinese-American Hewlett-Packard recruit, could not find a rental in Palo Alto after the war.").↩

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10 percent of housing land for multifamily buildings. … Art Fong, a Chinese-American Hewlett-Packard recruit, could not find a rental in Palo Alto after the war.”).↩

896. Palo Alto local history, cited in Episode 6 Script and Review (“Page Mill Road, early 1860s, a lumber road. William Page, a New York timber man, mill near today’s California Avenue, in Mayfield.”); `event-ledger.csv` note on 1951 authorization vs. 1953 Varian move-in.↩

897. Stanford University Founding Grant (November 11, 1885), cited in Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 24), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), cited in Episode 6 Script and Review (“First Stanford Industrial Park lease: Varian’s research department, 1953. Terman was an investor and a director. Russell and Sigurd Varian, the klystron.”).↩

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901. Malcolm Harris, *Palo Alto* (`harris-palo-alto`); Rebecca S. Lowen, *Creating the Cold War University: The Transformation of Stanford* (`lowen-cold-war-university`); and Sally Denton, *The Profiteers*, ch. 10, cited in Episode 6 Script and Review (“SRI, 1946. Steve Bechtel Sr. a founding director. Tresidder fired the director rather than let one Navy contract turn the institute into a public charge, and died before naming a replacement. Sterling made Terman provost in 1954. Lowen, via Harris: Sterling was anticommunist and anti-Roosevelt enough to please Hoover.”).↩

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904. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), cited in Episode 6 Script and Review (“After Varian: Hewlett-Packard headquarters, General Electric, Kodak. Honors Cooperative Program, 1954. Companies paid double tuition to send employees ahead of the ordinary line. The loop: government sponsors microwave research, professors consult for Varian, Varian hires the students, Varian sells tubes to the government. … Carl Djerassi and Syntex, as the chemical cousin of the electronics park.”).↩

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907. Stanford University land record, cited in Episode 6 Script and Review (“1955: Stanford ceded 87.5 acres on Miranda Avenue for the Palo Alto VA.”).↩

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929. Arthur Rock, Computer History Museum oral history (2007), and Jay Last, Computer History Museum oral history (2007), cited in Episode 6 Script and Review (“Rock: 35 companies, all no. Last: about 20 or 30, all no, lawyers stopped the close calls.”).↩

930. Arthur Rock, Computer History Museum oral history (2007), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), cited in Episode 6 Script and Review (“Sherman Fairchild, largest individual IBM shareholder. Fairchild Camera and Instrument put in $1.5 million. Rock’s memory: the eight about 80 percent, Hayden Stone 20. … Fairchild Semiconductor in Mountain View. Noyce made eight. Shipping to IBM within a year.”).↩

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938. California Business and Professions Code § 16600 (recodified 1941 from California Civil Code § 1673 of 1872), and AnnaLee Saxenian, *Regional Advantage: Culture and Competition in Silicon Valley and Route 128* (`saxenian-regional-advantage`), cited in Episode 6 Script and Review.↩

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940. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), and Steve Blank, history of Lockheed Missiles Systems Division, cited in Episode 6 Script and Review (“1955: Lockheed’s missiles division planned to move from Los Angeles to Sunnyvale, near Ames at Moffett. Moffett’s large prewar contract was 1931, under Hoover. 1956: a research complex in the park, then a factory nearby. Employment at the missiles division: none in 1956, more than 28,000 by 1965. About 30,000 California engineers laid off in 1963–64.”).↩

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944. Hoover Institution tower and timeline pages, cited in Episode 6 Script and Review (“Hoover Tower, Arthur Brown Jr., 285 feet. $600,000 decided in 1938. Bells from the Belgian pavilion at the 1939 fair. ‘For Peace Alone Do I Ring.’ Dedicated June 20, 1941. Wartime listening post. 1959: trustees set the Institution apart inside the university; Hoover’s private-enterprise charge.”).↩

945. Dwight D. Eisenhower, Farewell Address (January 17, 1961), cited in Episode 6 Script and Review (“In the councils of government, we must guard against the acquisition of unwarranted influence, whether sought or unsought, by the military-industrial complex.”).↩

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948. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (`harris-palo-alto`), *Stanford Magazine*, and Stanford University land records, via Episode 6 script (Section 09) and review log: “Kesey, a Stanford writing fellow, in a government drug study at the Menlo Park VA on Willow Road. About $75 a session. Historians tie the study to CIA money. The careful accounts do not file it under the name MKUltra. The year is 1959 in some accounts and 1961 in others” and “1955: Stanford ceded 87.5 acres on Miranda Avenue for the Palo Alto VA. That is not the Menlo Park hospital.”↩

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951. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), and Smithsonian Institution oral history, via Episode 6 script (Section 09) and review log: Engelbart was a Navy radio technician who shipped for the Philippines

on V-J Day reading Vannevar Bush's *Atlantic* piece reprinted in *Life*, worked at Ames at Moffett Field in 1950, joined SRI with a small Air Force grant, and built the mouse prototype with Bill English in 1964.↩

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953. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Prometheus in the Pentagon — Bob Taylor," n. 46), citing John Markoff, *What the Dormouse Said: How the Sixties Counterculture Shaped the Personal Computer* (New York: Penguin Books, 2005), 51: "The NASA funding for Engelbart's NLS oNLine Systems was about $80,000. Taylor also told Licklider about Engelbart's work, which led to ARPA (where Licklider was then director) also funding the research."↩

954. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), and Smithsonian Institution oral history, via Episode 6 script (Section 09) and review log: Engelbart was a Navy radio technician who shipped for the Philippines on V-J Day reading Vannevar Bush's *Atlantic* piece reprinted in *Life*, worked at Ames at Moffett Field in 1950, joined SRI with a small Air Force grant, and built the mouse prototype with Bill English in 1964.↩

955. Episode 6 script (Section 09) and review log, citing jerrygarcia.com and Menlo Park city plaque: "May 5, 1965, the Warlocks at Magoo's Pizza, Menlo Park. December 4, 1965, San Jose Acid Test, first billing as the Grateful Dead. Brand co-produced the Trips Festival in January 1966."↩

956. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), and Martin A. Lee and Bruce Shlain, *Acid Dreams: The Complete Social History of LSD: The CIA, the Sixties, and Beyond* (cited in `levine-surveillance#72`, ch. 4, n. 14, and Episode 6 review log): Al Hubbard, ex-OSS, worked on SRI's Alternative Futures project in khakis, badge, and pistol, guiding trips and gathering opposition data on the New Left, though he was "not a CIA operative as such."↩

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958. SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log: campus learned of two chemical-warfare contracts in 1965 and Southeast Asia counterinsurgency contracts in 1967; SRI argued the chemical work was almost exclusively defensive; Harris documents SRI's work on the CIA's strategic-hamlet plan and counterinsurgency contracts reaching millions of dollars by 1968.↩

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960. Episode 6 review log (Hoover research note): "Friedman met Reagan in 1967."↩

961. SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log: campus learned of two chemical-warfare contracts in 1965 and Southeast Asia counterinsurgency contracts in 1967; SRI argued the chemical work was almost exclusively defensive; Harris documents SRI's work on the CIA's strategic-hamlet plan and counterinsurgency contracts reaching millions of dollars by 1968.↩

962. Yasha Levine, *Surveillance Valley* (`levine-surveillance#72`, ch. 1, nn. 1, 44, 59, 61), citing *Foreign Relations of the United States, 1961–1963*, vol. 1, *Vietnam, 1961*, Doc. 96, Richard J. Barber Associates, *The Advanced Research Projects Agency 1958–1974* (1975), and Sharon Weinberger, *The Imagineers of War* (2017), ch. 5.↩

963. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "What Do We Do with These? — Niels Reimers," nn. 13, 15, 21, 25), citing Agreement between the Board of Trustees of the Leland Stanford, Jr. University and Research Corporation (Jan. 1, 1956), Niels Reimers notes (June 27, Oct. 29, and Nov. 14, 1968), and Frank Newman to Kenneth M. Cuthbertson (July 19, 1968), Stanford University Office of Technology Licensing (SUOTL) archives.↩

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965. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "What Do We Do with These? — Niels Reimers," nn. 13, 15, 21, 25), citing Agreement between the Board of Trustees of the Leland Stanford, Jr. University and Research Corporation (Jan. 1, 1956), Niels Reimers notes (June 27, Oct. 29, and Nov. 14, 1968), and Frank Newman to Kenneth M. Cuthbertson (July 19, 1968), Stanford University Office of Technology Licensing (SUOTL) archives.↩

966. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, "Turn Your Backs on the Origins of Computing! — Bob Taylor," n. 24; "Prometheus in the Pentagon," n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

967. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

968. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

969. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

970. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

971. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

972. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

973. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

974. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

975. Adam Fisher oral history, Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Fred Turner, *From Counterculture to Cyberculture* (`turner-counterculture-cyberculture` in `references.bib`; `levine-surveillance#72`, ch. 4, nn. 25–27), and John Markoff, *What the Dormouse Said*, ch. 5, via Episode 6 script (Section 09) and review log.↩

976. Adam Fisher oral history, Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Fred Turner, *From Counterculture to Cyberculture* (`turner-counterculture-cyberculture` in `references.bib`; `levine-surveillance#72`, ch. 4, nn. 25–27), and John Markoff, *What the Dormouse Said*, ch. 5, via Episode 6 script (Section 09) and review log.↩

977. Adam Fisher oral history, Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Fred Turner, *From Counterculture to Cyberculture* (`turner-counterculture-cyberculture` in `references.bib`; `levine-surveillance#72`, ch. 4, nn. 25–27), and John Markoff, *What the Dormouse Said*, ch. 5, via Episode 6 script (Section 09) and review log.↩

978. Doug Engelbart, Smithsonian Oral History (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 24; “Prometheus in the Pentagon,” n. 51; `levine-surveillance#72`, ch. 4, n. 26, MouseSite), DARPA, and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 09) and review log.↩

979. Stanford University News Service (April 4, 1969), April 3rd Movement records (`a3m-records` in `references.bib`), and SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

980. Stanford University News Service (April 4, 1969), April 3rd Movement records (`a3m-records` in `references.bib`), and SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

981. Stanford University News Service (April 4, 1969), April 3rd Movement records (`a3m-records` in `references.bib`), and SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

982. Stanford University News Service (April 4, 1969), April 3rd Movement records (`a3m-records` in `references.bib`), and SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

983. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stanford University News Service, and April 3rd Movement records (`a3m-records` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

984. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stanford University News Service, and April 3rd Movement records (`a3m-records` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

985. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stanford University News Service, and April 3rd Movement records (`a3m-records` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

986. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stanford University News Service, and April 3rd Movement records (`a3m-records` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

987. Episode 6 script (Section 10) and review log, comparing Leslie Berlin's count ("nine days") and Malcolm Harris's phrasing ("more than a week") for the April 8–18, 1969 Applied Electronics Laboratory occupation.↩

988. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stanford University News Service, and April 3rd Movement records (`a3m-records` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

989. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), Stanford University News Service, and April 3rd Movement records (`a3m-records` in `references.bib`), via Episode 6 script (Section 10) and review log.↩

990. Stanford University News Service (April 14–17, 1969), via Episode 6 script (Section 10) and review log: "If we are successful here the next step will be to move to the Stanford Industrial Park."↩

991. Stanford University News Service (April 14–17, 1969), via Episode 6 script (Section 10) and review log: "If we are successful here the next step will be to move to the Stanford Industrial Park."↩

992. Stanford University News Service (April 14–17, 1969), via Episode 6 script (Section 10) and review log: "If we are successful here the next step will be to move to the Stanford Industrial Park."↩

993. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log, recording H. Bruce Franklin's remark to a provost and the four lines from the occupation newsletter song.↩

994. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log, recording H. Bruce Franklin's remark to a provost and the four lines from the occupation newsletter song.↩

995. Stanford University News Service (April 14–17, 1969), via Episode 6 script (Section 10) and review log: "If we are successful here the next step will be to move to the Stanford Industrial Park."↩

996. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "What Do We Do with These? — Niels Reimers," nn. 1–7, citing Ron Goben, "Police Break Up Palo Alto Jam-in by Stanford Mob," *Palo Alto Times*, May 16, 1969, and Stanford University News Service Chronology of Events, May 16, 1969), *Stanford Report*, "1969: A Turbulent Time" (`stanfordreport-1969` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log.↩

997. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Nerd Paradise — Al Alcorn," nn. 8–10), citing Seth Rosenfeld, *Subversives: The FBI's War on Student Radicals, and Reagan's Rise to Power* (2012), "Nation: Occupied Berkeley," *Time*, May 30, 1969 (482 arrested in a

single day), and Frederick Berry, Thomas Brooks, and Eugene Commons, “Terror in a Teapot,” *The Nation*, June 23, 1969 (“between May 15 and May 24, almost 1,000 arrests were made on the streets of Berkeley”).↩

998. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” nn. 1–7, citing Ron Goben, “Police Break Up Palo Alto Jam-in by Stanford Mob,” *Palo Alto Times*, May 16, 1969, and Stanford University News Service Chronology of Events, May 16, 1969), *Stanford Report*, “1969: A Turbulent Time” (`stanfordreport-1969` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log.↩

999. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” nn. 1–7, citing Ron Goben, “Police Break Up Palo Alto Jam-in by Stanford Mob,” *Palo Alto Times*, May 16, 1969, and Stanford University News Service Chronology of Events, May 16, 1969), *Stanford Report*, “1969: A Turbulent Time” (`stanfordreport-1969` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log.↩

1000. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” nn. 1–7, citing Ron Goben, “Police Break Up Palo Alto Jam-in by Stanford Mob,” *Palo Alto Times*, May 16, 1969, and Stanford University News Service Chronology of Events, May 16, 1969), *Stanford Report*, “1969: A Turbulent Time” (`stanfordreport-1969` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log.↩

1001. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” nn. 1–7, citing Ron Goben, “Police Break Up Palo Alto Jam-in by Stanford Mob,” *Palo Alto Times*, May 16, 1969, and Stanford University News Service Chronology of Events, May 16, 1969), *Stanford Report*, “1969: A Turbulent Time” (`stanfordreport-1969` in `references.bib`), and Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Section 10) and review log.↩

1002. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 12; “Make It Happen — Niels Reimers,” nn. 2–3), citing Richard Lyman to Department Chairmen and Principal Investigators (May 16, 1969), Niels Reimers to William S. Johnson, “Royalty Income Distribution” (May 21, 1969), and SUOTL financial records.↩

1003. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 12; “Make It Happen — Niels Reimers,” nn. 2–3), citing Richard Lyman to Department Chairmen and Principal Investigators (May 16, 1969), Niels Reimers to William S. Johnson, “Royalty Income Distribution” (May 21, 1969), and SUOTL financial records.↩

1004. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Prometheus in the Pentagon — Bob Taylor,” nn. 1–2, 37), and Yasha Levine, *Surveillance Valley* (`levine-surveillance#72`, ch. 2, nn. 62–

63; `levine-surveillance#73`, ch. 5, n. 18), citing Katie Hafner and Matthew Lyon, *Where Wizards Stay Up Late: The Origins of the Internet* (`hafner-lyon-wizards` in `references.bib`), 152–154, and Leonard Kleinrock, "Memoirs of the Sixties," in *The ARPANET Sourcebook* (2008), 96.↩

1005. Episode 6 script (Section 08) and review log: "December 12, 1969. Coppola and Lucas open American Zoetrope on Folsom Street."↩

1006. SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), and Stuart W. Leslie, *The Cold War and American Science: The Military-Industrial-Academic Complex at MIT and Stanford* (`leslie-cold-war-science` in `references.bib`, ch. 9), via Episode 6 script (Section 11) and review log.↩

1007. SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), and Stuart W. Leslie, *The Cold War and American Science: The Military-Industrial-Academic Complex at MIT and Stanford* (`leslie-cold-war-science` in `references.bib`, ch. 9), via Episode 6 script (Section 11) and review log.↩

1008. SRI International, *Appendix B* (`sri-appendix-b` in `references.bib`), and Stuart W. Leslie, *The Cold War and American Science: The Military-Industrial-Academic Complex at MIT and Stanford* (`leslie-cold-war-science` in `references.bib`, ch. 9), via Episode 6 script (Section 11) and review log.↩

1009. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "What Do We Do with These? — Niels Reimers," n. 10), citing "Towards an On-Campus ROTC Program at Stanford University: A Report and Recommendation by the Ad Hoc Committee [to the Faculty Senate]," Stanford University, April 2011.↩

1010. Malcolm Harris, *Palo Alto* (`harris-palo-alto`), via Episode 6 script (Sections 01, 02, 11) and review log.↩

1011. Episode 6 script (Section 08) and review log; Open Issues note identifies the quote as from Milton and Rose D. Friedman's memoir *Two Lucky People*, though the exact page citation is not present in the provided excerpts and is marked `[UNVERIFIED]`. ## Part 7↩

1012. Episode 6 script (Section 08) and review log; Open Issues note identifies the quote as from Milton and Rose D. Friedman's memoir *Two Lucky People*, though the exact page citation is not present in the provided excerpts and is marked `[UNVERIFIED]`. ## Part 7↩

1013. Episode 6 script (Section 08) and review log; Open Issues note identifies the quote as from Milton and Rose D. Friedman's memoir *Two Lucky People*, though the exact page citation is not present in the provided excerpts and is marked `[UNVERIFIED]`. ## Part 7↩

1014. Episode 7 script ("05. Myth Machines") and review log: American Zoetrope opened December 12, 1969, at 827 Folsom Street, San Francisco; Greek etymology *zoe* (life) and *tropos* (turn).↩

1015. Episode 7 script ("05. Myth Machines") and review log: American Zoetrope opened December 12, 1969, at 827 Folsom Street, San Francisco; Greek etymology *zoe* (life) and *tropos* (turn).↩

1016. Episode 7 script (“05. Myth Machines”) and Open Issues note: Walter Murch’s remark that none of the American Zoetrope crew “had family in the industry” is present in the podcast script but lacks an independent primary source in the review table; marked `[UNVERIFIED]`.↩

1017. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Hoover Institution / State Department biographical records and Laton McCartney, *Friends in High Places* (1988) for Shultz’s career sequence (MIT, University of Chicago dean, CASBS fellow at Stanford 1968–1969, Nixon Secretary of Labor, first OMB director, Secretary of the Treasury through 1974, Bechtel executive and president, part-time Stanford faculty, campus house and Oldsmobile, Secretary of State sworn July 16, 1982 through January 20, 1989, and Bohemian Grove attendance).↩

1018. Leslie Berlin, *Troublemakers: Silicon Valley’s Coming of Age* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” nn. 13–15): “Agreement between the Board of Trustees of the Leland Stanford, Jr. University and Research Corporation, Jan. 1, 1956, SUOTL… The inventor received $100 and 16.66 percent of the gross dollars received by Research Corporation; Stanford received 50 percent of the net.”↩

1019. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 25): “Frank Newman to Kenneth M. Cuthbertson, vice president for finance, July 19, 1968. In this letter, Newman promised that ‘we can operate on a low-risk yet effective basis,’ and after explaining that there would be ‘no added or special staff,’ proposed that $15,000 be set aside for outside attorneys’ fees for the pilot program. ‘The upside chance for gain may be considerable,’ he notes. He also mentioned ‘the ⅓ to the university, ⅓ to the department, and ⅓ to the inventor arrangement.’”↩

1020. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 2): “During the pilot period, Reimers and Newman implemented a flat 15 percent administrative charge, in addition to out-of-pocket expenses, against gross royalty income to cover costs. At the end of 1969, Newman ‘bowed out’ of the licensing program to begin work on the so-called Newman Report on educational policy. See ‘Technology Licensing Program Now Has Earned over $200,000,’ *Campus Report*, Nov. 15, 1972.”↩

1021. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 16): “In the Office of Technology Licensing’s first eighteen months, when only five inventions had been licensed, Reimers reported that ‘three petitions for title in inventions to sponsors (HEW twice and NSF) were first denied but were eventually granted after much effort including meetings in Washington, D.C. with agency officials’; Licensing Program Progress Report, July 20, 1970, SUOTL.”↩

1022. Episode 7 script (“02. The Ticket Booth”) and review log, citing Leslie Berlin, *Troublemakers*: Office of Technology Licensing made permanent by summer 1970 (a four-year-old official operation by summer 1974).↩

1023. Episode 7 script (“02. The Ticket Booth”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 19; “Make It Happen — Niels Reimers,” nn. 1, 6; “No Idea How You Start a Company,” n. 82): trailer behind Encina Hall, ferns and memorabilia, “MAKE IT HAPPEN” sign (Randy Block, “Wanted: Ideas to Manage,” *Stanford Daily*, March 29, 1976), Sally Hines, John Poitras (hired 1971), and Friday-afternoon popcorn and Henry Weinhard’s beer (“Management by Beer”).↩

1024. Episode 7 script (“02. The Ticket Booth”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 19; “Make It Happen — Niels Reimers,” nn. 1, 6; “No Idea How You Start a Company,” n. 82): trailer behind Encina Hall, ferns and memorabilia, “MAKE IT HAPPEN” sign (Randy Block, “Wanted: Ideas to Manage,” *Stanford Daily*, March 29, 1976), Sally Hines, John Poitras (hired 1971), and Friday-afternoon popcorn and Henry Weinhard’s beer (“Management by Beer”).↩

1025. Episode 7 script (“02. The Ticket Booth”) and review log, citing Don C. Hoefler, “Silicon Valley U.S.A.,” *Electronic News*, January 11, 1971 (Computer History Museum catalog 102770432); Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Building: 1972–1975,” nn. 1–2): “As late as December 1979, even local papers referred to ‘the Santa Clara Valley electronics industry’… The name ‘Silicon Valley’ first appeared in the *New York Times* in John H. Allan, ‘Whither Semiconductor Stocks?,’ *New York Times*, Nov. 11, 1975.” (Berlin’s note 1 gives Jan. 10, 1971; CHM catalog 102770432 verifies the Monday, Jan. 11, 1971 issue date.)↩

1026. Episode 7 script (“02. The Ticket Booth”) and review log, citing Don C. Hoefler, “Silicon Valley U.S.A.,” *Electronic News*, January 11, 1971 (Computer History Museum catalog 102770432); Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Building: 1972–1975,” nn. 1–2): “As late as December 1979, even local papers referred to ‘the Santa Clara Valley electronics industry’… The name ‘Silicon Valley’ first appeared in the *New York Times* in John H. Allan, ‘Whither Semiconductor Stocks?,’ *New York Times*, Nov. 11, 1975.” (Berlin’s note 1 gives Jan. 10, 1971; CHM catalog 102770432 verifies the Monday, Jan. 11, 1971 issue date.)↩

1027. Episode 7 script (“05. Myth Machines”), review log, and Open Issues note: American Zoetrope’s 1972 move to the Sentinel Building at 916 Kearny Street is verified in the review log; the callback to A. P. Giannini opening his bank in a remodeled North Beach saloon in 1904 has no specific citation in the Episode 7 review table (though Felice A. Bonadio, *A. P. Giannini: Banker of America*, is listed in `brechin-imperial#69`) and is marked `[UNVERIFIED]` per the Open Issues note.↩

1028. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 2): “During the pilot period, Reimers and Newman implemented a flat 15 percent administrative charge, in addition to out-of-pocket expenses, against gross royalty income to cover costs. At the end of 1969, Newman ‘bowed out’ of the licensing program to begin work on the so-called Newman Report on educational policy. See ‘Technology Licensing Program Now Has Earned over $200,000,’ *Campus Report*, Nov. 15, 1972.”↩

1029. Episode 7 script (“01. Sand Hill”) and review log, citing Leslie Berlin, *Troublemakers*: Kleiner Perkins began in 1972 at 3000 Sand Hill Road with an $8 million fund (about $64 million in 2026 CPI dollars), half from Pittsburgh industrialist Henry Hillman and half from individuals, trust funds, an insurance company, and Rockefeller University; subsequent move to the 29th floor of Two Embarcadero Center and Tom Perkins’s verbatim quote (“We didn’t want to be considered part of the flock. Eagles don’t flock, was our joke.”).↩

1030. Episode 7 script (“01. Sand Hill”) and review log, citing Leslie Berlin, *Troublemakers*: Kleiner Perkins began in 1972 at 3000 Sand Hill Road with an $8 million fund (about $64 million in 2026 CPI dollars), half from Pittsburgh industrialist Henry Hillman and half from individuals, trust funds, an insurance company, and Rockefeller University; subsequent move to the 29th floor of Two Embarcadero Center and Tom Perkins’s verbatim quote (“We didn’t want to be considered part of the flock. Eagles don’t flock, was our joke.”).↩

1031. Episode 7 script (“01. Sand Hill”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 19–22; “Looks Like $100 Million to Me!,” n. 6): Eugene Kleiner’s background in Vienna, WWII service, G.I. Bill, and Fairchild Semiconductor; Tom Perkins’s laser company acquired by Spectra-Physics, leadership of HP’s computer division, and study under Georges Doriot at Harvard Business School.↩

1032. Episode 7 script (“01. Sand Hill”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 19–22; “Looks Like $100 Million to Me!,” n. 6): Eugene Kleiner’s background in Vienna, WWII service, G.I. Bill, and Fairchild Semiconductor; Tom Perkins’s laser company acquired by Spectra-Physics, leadership of HP’s computer division, and study under Georges Doriot at Harvard Business School.↩

1033. Episode 7 script (“01. Sand Hill”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Hit in the Ass by Lightning — Al Alcorn,” n. 63): Don Valentine worked at Fairchild for eight years before leaving in 1967 with Charlie Sporck and Floyd Kvamme to revive National Semiconductor, then founded Sequoia Capital in 1972 on Sand Hill Road.↩

1034. Episode 7 script (“01. Sand Hill”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Hit in the Ass by Lightning — Al Alcorn,” n. 63): Don Valentine worked at Fairchild for eight years before leaving in 1967 with Charlie Sporck and Floyd Kvamme to revive National Semiconductor, then founded Sequoia Capital in 1972 on Sand Hill Road.↩

1035. Episode 7 script (“04. Pong to Apple”) and Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Nerd Paradise — Al Alcorn,” nn. 13–22; “Hit in the Ass by Lightning — Al Alcorn,” nn. 2, 5, 41): Al Alcorn’s prior work at Ampex in Redwood City and Nutting Associates (500 Logue Avenue), the 1972 launch of Atari with Nolan Bushnell and Ted Dabney, and Alcorn’s *Pong* sync-generator design using “less than 20, 50-cents chips.”↩

1036. Episode 7 script (“04. Pong to Apple”) and Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Nerd Paradise — Al Alcorn,” nn. 13–22; “Hit in the Ass by Lightning — Al Alcorn,” nn. 2, 5, 41): Al Alcorn’s prior work at Ampex in Redwood City and Nutting Associates (500 Logue Avenue), the 1972 launch of Atari with Nolan Bushnell and Ted Dabney, and Alcorn’s *Pong* sync-generator design using “less than 20, 50-cents chips.”↩

1037. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Hit in the Ass by Lightning — Al Alcorn,” nn. 48, 62): Spring 1974 Atari business plan (“Atari sold almost 7,000 Pongs. . . . The total volume for both Pong and Pong-type copies hit 45,000 units in 1973”) and Don Valentine’s receipt of 12,500 shares of Atari stock in exchange for organizational and marketing advice and help raising $2 million.↩

1038. Episode 7 script (“01. Sand Hill”) and review log, citing Leslie Berlin, *Troublemakers*: Kleiner Perkins began in 1972 at 3000 Sand Hill Road with an $8 million fund (about $64 million in 2026 CPI dollars), half from Pittsburgh industrialist Henry Hillman and half from individuals, trust funds, an insurance company, and Rockefeller University; subsequent move to the 29th floor of Two Embarcadero Center and Tom Perkins’s verbatim quote (“We didn’t want to be considered part of the flock. Eagles don’t flock, was our joke.”).↩

1039. Episode 7 script (“01. Sand Hill”) and review log, citing Leslie Berlin, *Troublemakers*: Kleiner Perkins began in 1972 at 3000 Sand Hill Road with an $8 million fund (about $64 million in 2026 CPI dollars), half from Pittsburgh industrialist Henry Hillman and half from individuals, trust funds, an insurance company, and Rockefeller University; subsequent move to the 29th floor of Two Embarcadero Center and Tom Perkins’s verbatim quote (“We didn’t want to be considered part of the flock. Eagles don’t flock, was our joke.”).↩

1040. Episode 7 script (“01. Sand Hill”) and review log, citing Leslie Berlin, *Troublemakers*, and the 1979 Department of Labor ERISA prudent-man clarification: 1974 ERISA enactment, 1978 capital-gains tax cut, 1979 Labor Department clarification judging prudence on the whole portfolio, and “more than half a billion” new dollars (about $2.3 billion in 2026 CPI dollars) entering venture capital within a year.↩

1041. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Hoover Institution / State Department biographical records and Laton McCartney, *Friends in High Places* (1988) for Shultz’s career sequence (MIT, University of Chicago dean, CASBS fellow at Stanford 1968–1969, Nixon Secretary of Labor, first OMB director, Secretary of the Treasury through 1974, Bechtel executive and president, part-time Stanford faculty, campus house and Oldsmobile, Secretary of State sworn July 16, 1982 through January 20, 1989, and Bohemian Grove attendance).↩

1042. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Hoover Institution / State Department biographical records and Laton McCartney, *Friends in High Places* (1988) for Shultz’s career sequence (MIT, University of Chicago dean, CASBS fellow at Stanford 1968–1969, Nixon Secretary of Labor, first OMB director, Secretary of the Treasury through 1974, Bechtel executive and president, part-time

Stanford faculty, campus house and Oldsmobile, Secretary of State sworn July 16, 1982 through January 20, 1989, and Bohemian Grove attendance).↩

1043. Episode 7 script (“02. The Ticket Booth”) and review log, citing Leslie Berlin, *Troublemakers*: Office of Technology Licensing made permanent by summer 1970 (a four-year-old official operation by summer 1974).↩

1044. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” nn. 8, 30): Victor K. McElheny, “Animal Gene Shifted to Bacteria; Aid Seen to Medicine and Farm,” *New York Times*, May 20, 1974; S. N. Cohen, A. C. Y. Chang, H. W. Boyer, and R. B. Helling, “Construction of Biologically Functional Bacterial Plasmids *in Vitro*,” *Proceedings of the National Academy of Sciences* 70 (1973): 3240–3244.↩

1045. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” nn. 8, 30): Victor K. McElheny, “Animal Gene Shifted to Bacteria; Aid Seen to Medicine and Farm,” *New York Times*, May 20, 1974; S. N. Cohen, A. C. Y. Chang, H. W. Boyer, and R. B. Helling, “Construction of Biologically Functional Bacterial Plasmids *in Vitro*,” *Proceedings of the National Academy of Sciences* 70 (1973): 3240–3244.↩

1046. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 25): Niels Reimers to File, Notes on a conversation with Stan Cohen, July 29, 1974, SUOTL (“Herb Boyer is willing to cooperate, also does not want personal gain”).↩

1047. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 25): Niels Reimers to File, Notes on a conversation with Stan Cohen, July 29, 1974, SUOTL (“Herb Boyer is willing to cooperate, also does not want personal gain”).↩

1048. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 12; “No Idea How You Start a Company,” nn. 11–13): criticism of patenting recombinant DNA as undermining the university’s mission, and the May 21, 1976 Stanford meeting with Paul Berg, Reimers, Cohen, and Joshua Lederberg.↩

1049. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 12; “No Idea How You Start a Company,” nn. 11–13): criticism of patenting recombinant DNA as undermining the university’s mission, and the May 21, 1976 Stanford meeting with Paul Berg, Reimers, Cohen, and Joshua Lederberg.↩

1050. Episode 7 script (“07. Bayh-Dole”) and review log, citing USPTO Patent 4,237,224 (“Process for producing biologically functional molecular chimeras,” filed Jan. 4, 1979, issued Dec. 2, 1980); Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” nn. 39, 44; “No Idea How You Start a Company,” n. 68): Bertram Rowland of Flehr, Hohbach, Test, Albritton & Herbert; parent application Ser. No. 520,591 filed Nov. 4, 1974, followed by

CIPs filed May 17, 1976, Nov. 9, 1978, and Jan. 4, 1979 (Ser. No. 1,000,021); Stanford–UC Agreement Concerning Rights in Invention, Aug. 29, 1975.↩

1051. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 56; “No Idea How You Start a Company,” nn. 5, 10, 14): 1975 Asilomar conference (150 scientists, organized in part by Maxine Singer), Cambridge Mayor Alfred E. Vellucci’s June 16, 1976 *Palo Alto Times* quote and May 16, 1977 letter to Philip Handler, the March 1977 NAS protest led by Jeremy Rifkin’s People’s Business Commission, and Reimers’s three-inch clipping file by late 1977.↩

1052. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 56; “No Idea How You Start a Company,” nn. 5, 10, 14): 1975 Asilomar conference (150 scientists, organized in part by Maxine Singer), Cambridge Mayor Alfred E. Vellucci’s June 16, 1976 *Palo Alto Times* quote and May 16, 1977 letter to Philip Handler, the March 1977 NAS protest led by Jeremy Rifkin’s People’s Business Commission, and Reimers’s three-inch clipping file by late 1977.↩

1053. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 23–30): Robert Swanson’s MIT degrees, Citicorp venture office, tenure at Kleiner & Perkins, Cetus board lunch, firing by Tom Perkins with payroll through the end of 1975, and Swanson’s verbatim recollection (“Maybe I’m not very good.”).↩

1054. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 23–30): Robert Swanson’s MIT degrees, Citicorp venture office, tenure at Kleiner & Perkins, Cetus board lunch, firing by Tom Perkins with payroll through the end of 1975, and Swanson’s verbatim recollection (“Maybe I’m not very good.”).↩

1055. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 23–30): Robert Swanson’s MIT degrees, Citicorp venture office, tenure at Kleiner & Perkins, Cetus board lunch, firing by Tom Perkins with payroll through the end of 1975, and Swanson’s verbatim recollection (“Maybe I’m not very good.”).↩

1056. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 31–48): Swanson’s cold calls, Herb Boyer’s cats Watson and Crick, Arthur Levinson walking to a pay phone to speak to a company, Brook Byers’s Bob Dylan comparison, and the naming of Genentech over “Herbob.”↩

1057. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 31–48): Swanson’s cold calls, Herb Boyer’s cats Watson and Crick, Arthur Levinson walking to a pay phone to speak to a company, Brook Byers’s Bob Dylan comparison, and the naming of Genentech over “Herbob.”↩

1058. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 31–48): Swanson’s cold calls, Herb Boyer’s cats Watson and Crick, Arthur Levinson walking to a pay phone to speak to a company, Brook Byers’s Bob Dylan comparison, and the naming of Genentech over “Herbob.”↩

1059. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 49, 56, 63): “Genentech Outline for Discussion, Kleiner & Perkins, 1 April 1976” and Swanson’s April 19, 1976 letter to Stanford and the University of California.↩

1060. Episode 7 script (“03. Patenting Life”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company,” nn. 31–48): Swanson’s cold calls, Herb Boyer’s cats Watson and Crick, Arthur Levinson walking to a pay phone to speak to a company, Brook Byers’s Bob Dylan comparison, and the naming of Genentech over “Herbob.”↩

1061. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “That Flips My Switch — Mike Markkula,” nn. 11–22; “There Are No Standards Yet — Mike Markkula,” nn. 5, 7, 22–24, 36–38; “Can You Imagine Your Grandmother Using One?,” n. 1): Steve Jobs at Atari (Sept. 1974–Dec. 1975), Steve Wozniak and the Homebrew Computer Club in Menlo Park, Mike Markkula’s Nov. 18, 1976 “Outline for Apple Computer Buisness [sic] Plan,” hiring of Mike Scott as president, and Regis McKenna’s agency (designer Rob Janoff) creating the Apple logo.↩

1062. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “That’s What I Did on Mondays — Mike Markkula,” n. 20): Markkula’s Intel stock options split twice after he joined in January 1971, with a $6.22 strike price against a market price of about $45 per share by 1975.↩

1063. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “That Flips My Switch — Mike Markkula,” nn. 11–22; “There Are No Standards Yet — Mike Markkula,” nn. 5, 7, 22–24, 36–38; “Can You Imagine Your Grandmother Using One?,” n. 1): Steve Jobs at Atari (Sept. 1974–Dec. 1975), Steve Wozniak and the Homebrew Computer Club in Menlo Park, Mike Markkula’s Nov. 18, 1976 “Outline for Apple Computer Buisness [sic] Plan,” hiring of Mike Scott as president, and Regis McKenna’s agency (designer Rob Janoff) creating the Apple logo.↩

1064. Episode 7 script (“04. Pong to Apple”) and review log, citing Sebastian Mallaby, *The Power Law* (`mallaby-power-law` in `references.bib`): Fall 1977 trip by Markkula and Jobs to Venrock on the 56th floor of 30 Rockefeller Plaza, Peter Crisp’s verbatim quote (“We went out in the hall, four or five of us, we looked at each other, and we shrugged our shoulders and said, What the hell?”), Venrock’s $300,000 investment for 10 percent of Apple at a $3 million valuation (about $1.7 million and $16 million in 2026 CPI dollars), and Arthur Rock’s $50,000 slice of Venrock’s allocation.↩

1065. Episode 7 script (“04. Pong to Apple”) and review log, citing Sebastian Mallaby, *The Power Law* (`mallaby-power-law` in `references.bib`): Fall 1977 trip by Markkula and Jobs to Venrock on the 56th floor of 30 Rockefeller Plaza, Peter Crisp’s verbatim quote (“We went out in the

hall, four or five of us, we looked at each other, and we shrugged our shoulders and said, What the hell?"), Venrock's $300,000 investment for 10 percent of Apple at a $3 million valuation (about $1.7 million and $16 million in 2026 CPI dollars), and Arthur Rock's $50,000 slice of Venrock's allocation.↩

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1067. Episode 7 script ("04. Pong to Apple") and review log, citing Sebastian Mallaby, *The Power Law* (`mallaby-power-law` in `references.bib`): Fall 1977 trip by Markkula and Jobs to Venrock on the 56th floor of 30 Rockefeller Plaza, Peter Crisp's verbatim quote ("We went out in the hall, four or five of us, we looked at each other, and we shrugged our shoulders and said, What the hell?"), Venrock's $300,000 investment for 10 percent of Apple at a $3 million valuation (about $1.7 million and $16 million in 2026 CPI dollars), and Arthur Rock's $50,000 slice of Venrock's allocation.↩

1068. Episode 7 script ("03. Patenting Life") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Make It Happen — Niels Reimers," n. 56; "No Idea How You Start a Company," nn. 5, 10, 14): 1975 Asilomar conference (150 scientists, organized in part by Maxine Singer), Cambridge Mayor Alfred E. Vellucci's June 16, 1976 *Palo Alto Times* quote and May 16, 1977 letter to Philip Handler, the March 1977 NAS protest led by Jeremy Rifkin's People's Business Commission, and Reimers's three-inch clipping file by late 1977.↩

1069. Episode 7 script ("03. Patenting Life") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Make It Happen — Niels Reimers," n. 56; "No Idea How You Start a Company," nn. 5, 10, 14): 1975 Asilomar conference (150 scientists, organized in part by Maxine Singer), Cambridge Mayor Alfred E. Vellucci's June 16, 1976 *Palo Alto Times* quote and May 16, 1977 letter to Philip Handler, the March 1977 NAS protest led by Jeremy Rifkin's People's Business Commission, and Reimers's three-inch clipping file by late 1977.↩

1070. Episode 7 script ("07. Bayh-Dole") and review log, citing USPTO Patent 4,237,224 ("Process for producing biologically functional molecular chimeras," filed Jan. 4, 1979, issued Dec. 2, 1980); Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Make It Happen — Niels Reimers," nn. 39, 44; "No Idea How You Start a Company," n. 68): Bertram Rowland of Flehr, Hohbach, Test, Albritton & Herbert; parent application Ser. No. 520,591 filed Nov. 4, 1974, followed by CIPs filed May 17, 1976, Nov. 9, 1978, and Jan. 4, 1979 (Ser. No. 1,000,021); Stanford–UC Agreement Concerning Rights in Invention, Aug. 29, 1975.↩

1071. Episode 7 script ("05. Myth Machines") and review log, citing Alvy Ray Smith's paperwork and the February 10, 1986 Pixar press release; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`,

"Postscript: The Troublemakers Today," nn. 11–16): Lucasfilm computer division founded 1979; Jobs pushed out of operational role at Apple in spring 1985 after clashing with John Sculley; Jobs's Jan. 30, 1986 $5 million certified check signed over to Lucasfilm by Smith and Ed Catmull plus $5 million capitalization ($10 million total, about $30 million in 2026 CPI dollars); Feb. 10, 1986 press release ("undisclosed sum in the millions," $125,000 Pixar Image Computer); ownership split (70% Jobs as chairman/investor, 30% ~40 employees including John Lasseter; Catmull president, Smith vice president).↩

1072. Episode 7 script ("01. Sand Hill") and review log, citing Leslie Berlin, *Troublemakers*, and the 1979 Department of Labor ERISA prudent-man clarification: 1974 ERISA enactment, 1978 capital-gains tax cut, 1979 Labor Department clarification judging prudence on the whole portfolio, and "more than half a billion" new dollars (about $2.3 billion in 2026 CPI dollars) entering venture capital within a year.↩

1073. Episode 7 script ("04. Pong to Apple") and review log, citing Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Young Maniacs — Mike Markkula," n. 42): Mike Markkula's December 1979 creed "The Apple Marketing Philosophy" (Empathy, Focus, Impute) and verbatim quote ("Our image is the combined result of everything the customer sees, hears, or feels from Apple.").↩

1074. Episode 7 script ("04. Pong to Apple") and review log, citing Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Young Maniacs — Mike Markkula," n. 42): Mike Markkula's December 1979 creed "The Apple Marketing Philosophy" (Empathy, Focus, Impute) and verbatim quote ("Our image is the combined result of everything the customer sees, hears, or feels from Apple.").↩

1075. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "This Is a Big Fucking Deal — Al Alcorn," nn. 7, 16–28; "Sitting in a Kiddie Seat — Al Alcorn," nn. 6, 9–26): 1976 Warner Communications acquisition of Atari, the Stella chip in the Atari VCS, Ray Kassar's background at Burlington Industries, Bushnell's late-1978 ouster, and Atari's >80% market share in home video games.↩

1076. Episode 7 script ("04. Pong to Apple") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Sitting in a Kiddie Seat — Al Alcorn," nn. 38–55): departure of the four programmers to found Activision, drop in 2600 cartridge releases (13 in 1978, 12 in 1979, 6 in 1980), Alcorn's verbatim quote ("We didn't need a business plan for Pong or the VCS."), the unreleased Cosmos holographic system at the January 1981 CES, and Alcorn's conclusion that "all creativity had ceased."↩

1077. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "This Is a Big Fucking Deal — Al Alcorn," nn. 7, 16–28; "Sitting in a Kiddie Seat — Al Alcorn," nn. 6, 9–26): 1976 Warner Communications acquisition of Atari, the Stella chip in the Atari VCS, Ray Kassar's background at Burlington Industries, Bushnell's late-1978 ouster, and Atari's >80% market share in home video games.↩

1078. Episode 7 script ("04. Pong to Apple") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Sitting in a Kiddie Seat — Al Alcorn," nn. 38–55): departure of the four programmers to found Activision, drop in 2600 cartridge releases (13 in 1978, 12 in 1979, 6 in

1980), Alcorn's verbatim quote ("We didn't need a business plan for Pong or the VCS."), the unreleased Cosmos holographic system at the January 1981 CES, and Alcorn's conclusion that "all creativity had ceased."↩

1079. Episode 7 script ("04. Pong to Apple") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Sitting in a Kiddie Seat — Al Alcorn," nn. 38–55): departure of the four programmers to found Activision, drop in 2600 cartridge releases (13 in 1978, 12 in 1979, 6 in 1980), Alcorn's verbatim quote ("We didn't need a business plan for Pong or the VCS."), the unreleased Cosmos holographic system at the January 1981 CES, and Alcorn's conclusion that "all creativity had ceased."↩

1080. Episode 7 script ("03. Patenting Life") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Introduction," n. 9; "Looks Like $100 Million to Me! — Niels Reimers and Bob Swanson," nn. 6–20): *Diamond v. Chakrabarty* (1980), Hambrecht & Quist underwriting, Genentech's Oct. 14, 1980 NASDAQ IPO (GENE) jumping from $35 to $89 in twenty minutes, closing valuation of $532 million (about $2.2 billion in 2026 CPI dollars), and Swanson's, Boyer's, and Kleiner Perkins's stakes each worth about $65 million at the close.↩

1081. Episode 7 script ("03. Patenting Life") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Introduction," n. 9; "Looks Like $100 Million to Me! — Niels Reimers and Bob Swanson," nn. 6–20): *Diamond v. Chakrabarty* (1980), Hambrecht & Quist underwriting, Genentech's Oct. 14, 1980 NASDAQ IPO (GENE) jumping from $35 to $89 in twenty minutes, closing valuation of $532 million (about $2.2 billion in 2026 CPI dollars), and Swanson's, Boyer's, and Kleiner Perkins's stakes each worth about $65 million at the close.↩

1082. Episode 7 script ("03. Patenting Life") and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Introduction," n. 9; "Looks Like $100 Million to Me! — Niels Reimers and Bob Swanson," nn. 6–20): *Diamond v. Chakrabarty* (1980), Hambrecht & Quist underwriting, Genentech's Oct. 14, 1980 NASDAQ IPO (GENE) jumping from $35 to $89 in twenty minutes, closing valuation of $532 million (about $2.2 billion in 2026 CPI dollars), and Swanson's, Boyer's, and Kleiner Perkins's stakes each worth about $65 million at the close.↩

1083. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Young Maniacs — Mike Markkula," nn. 3, 11): Apple Computer Prospectus, Dec. 12, 1980; Robert J. Cole, "An 'Orderly' Debut for Apple," *New York Times*, Dec. 13, 1980.↩

1084. Episode 7 script ("07. Bayh-Dole") and review log, citing USPTO Patent 4,237,224 ("Process for producing biologically functional molecular chimeras," filed Jan. 4, 1979, issued Dec. 2, 1980); Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, "Make It Happen — Niels Reimers," nn. 39, 44; "No Idea How You Start a Company," n. 68): Bertram Rowland of Flehr, Hohbach, Test, Albritton & Herbert; parent application Ser. No. 520,591 filed Nov. 4, 1974, followed by

CIPs filed May 17, 1976, Nov. 9, 1978, and Jan. 4, 1979 (Ser. No. 1,000,021); Stanford–UC Agreement Concerning Rights in Invention, Aug. 29, 1975.↩

1085. Episode 7 script (“07. Bayh-Dole”) and review log: Patent and Trademark Law Amendments Act (Bayh-Dole, P.L. 96-517) signed by President Jimmy Carter on December 12, 1980; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” nn. 14, 16, 49, 67–68; “No Idea How You Start a Company,” n. 8): Stanford’s Institutional Patent Agreement with NIH/HEW, Reimers’s collaboration with HEW Patent Branch Chief Norman J. Latker, and Doogab Yi, *The Recombinant University* (2008), p. 177.↩

1086. Episode 7 script (“07. Bayh-Dole”) and review log: Patent and Trademark Law Amendments Act (Bayh-Dole, P.L. 96-517) signed by President Jimmy Carter on December 12, 1980; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” nn. 14, 16, 49, 67–68; “No Idea How You Start a Company,” n. 8): Stanford’s Institutional Patent Agreement with NIH/HEW, Reimers’s collaboration with HEW Patent Branch Chief Norman J. Latker, and Doogab Yi, *The Recombinant University* (2008), p. 177.↩

1087. Episode 7 script (“07. Bayh-Dole”) and review log: Patent and Trademark Law Amendments Act (Bayh-Dole, P.L. 96-517) signed by President Jimmy Carter on December 12, 1980; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” nn. 14, 16, 49, 67–68; “No Idea How You Start a Company,” n. 8): Stanford’s Institutional Patent Agreement with NIH/HEW, Reimers’s collaboration with HEW Patent Branch Chief Norman J. Latker, and Doogab Yi, *The Recombinant University* (2008), p. 177.↩

1088. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Hoover Institution / State Department biographical records and Laton McCartney, *Friends in High Places* (1988) for Shultz’s career sequence (MIT, University of Chicago dean, CASBS fellow at Stanford 1968–1969, Nixon Secretary of Labor, first OMB director, Secretary of the Treasury through 1974, Bechtel executive and president, part-time Stanford faculty, campus house and Oldsmobile, Secretary of State sworn July 16, 1982 through January 20, 1989, and Bohemian Grove attendance).↩

1089. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Sally Denton as a single source for George Shultz’s 1982 confirmation hearing remark (“Private interests have no business with nuclear weapons.”) and Bechtel’s subsequent management of nuclear weapons laboratories.↩

1090. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Sally Denton as a single source for George Shultz’s 1982 confirmation hearing remark (“Private interests have no business with nuclear weapons.”) and Bechtel’s subsequent management of nuclear weapons laboratories.↩

1091. Episode 7 script (“08. The Bechtel Cabinet”) and review log, citing Hoover Institution / State Department biographical records and Laton McCartney, *Friends in High Places* (1988) for Shultz’s career sequence (MIT, University of Chicago dean, CASBS fellow at Stanford 1968–1969, Nixon Secretary of Labor, first OMB director, Secretary of the Treasury through 1974, Bechtel executive and president, part-time Stanford faculty, campus house and Oldsmobile, Secretary of State sworn July 16, 1982 through January 20, 1989, and Bohemian Grove attendance).↩

1092. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Looks Like $100 Million to Me! — Niels Reimers and Bob Swanson,” nn. 32–49): $10,000 signing fee and 0.5–1% end-product royalty structure, advertisements in *Science* (July 31, 1981), *Wall Street Journal* (Aug. 3, 1981), and *Nature* (Aug. 6, 1981), December 1, 1981 deadline, Finnegan, Henderson, Farabow & Garrett (“Bruno, Boris, and Brunhilde”), and Genentech’s eleventh-hour signing.↩

1093. Episode 7 script (“02. The Ticket Booth” and “07. Bayh-Dole”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Knew It Before They Did — Niels Reimers,” nn. 7–9, 13): Stanford’s 1983 licensing revenue of $2.5 million (about $8 million in 2026 CPI dollars), first in the nation, nearly 60% from Cohen-Boyer; Reimers’s estimate of >$50 million left on the table by refusing equity until 1989; and Stanford News Release, March 20, 1991 (“Niels Reimers, Long-Time Director of Stanford’s Patent Office Resigns; OTL to Gross $24 Million This Year”).↩

1094. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 2; “Conclusion: Wave After Wave,” nn. 4–5): PARC labs a mile from campus; Forest Baskett, Andreas Bechtolsheim, Bill Nowicki, and John Seamons, “The Sun Workstation: A Terminal System for the Stanford University Network,” March 30, 1980 invention disclosure; release of rights to Bechtolsheim after the design was deemed “marginally patentable.”↩

1095. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Turn Your Backs on the Origins of Computing! — Bob Taylor,” n. 2; “Conclusion: Wave After Wave,” nn. 4–5): PARC labs a mile from campus; Forest Baskett, Andreas Bechtolsheim, Bill Nowicki, and John Seamons, “The Sun Workstation: A Terminal System for the Stanford University Network,” March 30, 1980 invention disclosure; release of rights to Bechtolsheim after the design was deemed “marginally patentable.”↩

1096. Episode 7 script (“02. The Ticket Booth” and “07. Bayh-Dole”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Knew It Before They Did — Niels Reimers,” nn. 7–9, 13): Stanford’s 1983 licensing revenue of $2.5 million (about $8 million in 2026 CPI dollars), first in the nation, nearly 60% from Cohen-Boyer; Reimers’s estimate of >$50 million left on the table by refusing equity until 1989; and Stanford News Release, March 20, 1991 (“Niels Reimers, Long-Time Director of Stanford’s Patent Office Resigns; OTL to Gross $24 Million This Year”).↩

1097. Episode 7 script (“03. Patenting Life”) and review log, citing Sebastian Mallaby, *The Power Law*: as of 1984, the 14 investments in Kleiner Perkins’s first fund showed $208 million in combined profit (about $670 million in 2026 CPI dollars), 95% from Tandem and Genentech.↩

1098. Episode 7 script (“05. Myth Machines”) and review log, citing Jesse Hamlin, *SF State Magazine* (Fall 2010) and the *Los Angeles Times* obituary (Oct. 30, 2009): August Coppola’s tenure as dean of the College of Creative Arts at San Francisco State (1984–1992), $15 million legislative appropriation, 1991 “skybreaking” ceremony and verbatim quote (“The idea is to look up, rather than down. Look up to the sky, the clouds, teach young people to dream.”), and 1993 opening of the Fine Arts Building and 150-seat theater.↩

1099. Episode 7 script (“05. Myth Machines”) and review log, citing Alvy Ray Smith’s paperwork and the February 10, 1986 Pixar press release; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Postscript: The Troublemakers Today,” nn. 11–16): Lucasfilm computer division founded 1979; Jobs pushed out of operational role at Apple in spring 1985 after clashing with John Sculley; Jobs’s Jan. 30, 1986 $5 million certified check signed over to Lucasfilm by Smith and Ed Catmull plus $5 million capitalization ($10 million total, about $30 million in 2026 CPI dollars); Feb. 10, 1986 press release (“undisclosed sum in the millions,” $125,000 Pixar Image Computer); ownership split (70% Jobs as chairman/investor, 30% ~40 employees including John Lasseter; Catmull president, Smith vice president).↩

1100. Episode 7 script (“05. Myth Machines”) and review log, citing Alvy Ray Smith’s paperwork and the February 10, 1986 Pixar press release; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Postscript: The Troublemakers Today,” nn. 11–16): Lucasfilm computer division founded 1979; Jobs pushed out of operational role at Apple in spring 1985 after clashing with John Sculley; Jobs’s Jan. 30, 1986 $5 million certified check signed over to Lucasfilm by Smith and Ed Catmull plus $5 million capitalization ($10 million total, about $30 million in 2026 CPI dollars); Feb. 10, 1986 press release (“undisclosed sum in the millions,” $125,000 Pixar Image Computer); ownership split (70% Jobs as chairman/investor, 30% ~40 employees including John Lasseter; Catmull president, Smith vice president).↩

1101. Episode 7 script (“05. Myth Machines”) and review log, citing Alvy Ray Smith’s paperwork and the February 10, 1986 Pixar press release; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Postscript: The Troublemakers Today,” nn. 11–16): Lucasfilm computer division founded 1979; Jobs pushed out of operational role at Apple in spring 1985 after clashing with John Sculley; Jobs’s Jan. 30, 1986 $5 million certified check signed over to Lucasfilm by Smith and Ed Catmull plus $5 million capitalization ($10 million total, about $30 million in 2026 CPI dollars); Feb. 10, 1986 press release (“undisclosed sum in the millions,” $125,000 Pixar Image Computer); ownership split (70% Jobs as chairman/investor, 30% ~40 employees including John Lasseter; Catmull president, Smith vice president).↩

1102. Episode 7 script (“05. Myth Machines”) and review log, citing Alvy Ray Smith’s paperwork and the February 10, 1986 Pixar press release; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Postscript: The Troublemakers Today,” nn. 11–16): Lucasfilm computer division founded 1979; Jobs pushed out of operational role at

Apple in spring 1985 after clashing with John Sculley; Jobs's Jan. 30, 1986 $5 million certified check signed over to Lucasfilm by Smith and Ed Catmull plus $5 million capitalization ($10 million total, about $30 million in 2026 CPI dollars); Feb. 10, 1986 press release ("undisclosed sum in the millions," $125,000 Pixar Image Computer); ownership split (70% Jobs as chairman/investor, 30% ~40 employees including John Lasseter; Catmull president, Smith vice president).↩

1103. Episode 7 script ("06. The Story of Our Streets") and review log, citing Outside Lands and the Caltrans structure list: Presidio / Funston Avenue Tunnel renamed General Douglas MacArthur Tunnel on November 5, 1986 via Senate Concurrent Resolution 86 (introduced by Senator Milton Marks); distinction between General Arthur MacArthur (May 14, 1903 toast at the Dewey Monument dedication in Union Square) and General Douglas MacArthur (Corregidor 1942, occupation of Japan, fired by Truman in 1951).↩

1104. Episode 7 script ("06. The Story of Our Streets") and review log, citing Outside Lands and the Caltrans structure list: Presidio / Funston Avenue Tunnel renamed General Douglas MacArthur Tunnel on November 5, 1986 via Senate Concurrent Resolution 86 (introduced by Senator Milton Marks); distinction between General Arthur MacArthur (May 14, 1903 toast at the Dewey Monument dedication in Union Square) and General Douglas MacArthur (Corregidor 1942, occupation of Japan, fired by Truman in 1951).↩

1105. Episode 7 script ("09. The Grievance Paper") and review log, citing Max Chafkin, *The Contrarian* (2021): *The Stanford Review* founded in spring 1987 by Peter Thiel and Norman Book, first issue June 9, 1987, twelve-man masthead, and early staffers including Keith Rabois and David Sacks.↩

1106. Episode 7 script ("09. The Grievance Paper") and review log, citing Max Chafkin, *The Contrarian* (2021): *The Stanford Review* founded in spring 1987 by Peter Thiel and Norman Book, first issue June 9, 1987, twelve-man masthead, and early staffers including Keith Rabois and David Sacks.↩

1107. Episode 7 script ("09. The Grievance Paper") and review log, citing Max Chafkin, *The Contrarian* (2021): *The Stanford Review* founded in spring 1987 by Peter Thiel and Norman Book, first issue June 9, 1987, twelve-man masthead, and early staffers including Keith Rabois and David Sacks.↩

1108. Episode 7 script ("09. The Grievance Paper") and review log, citing Max Chafkin, *The Contrarian* (2021): *The Stanford Review* founded in spring 1987 by Peter Thiel and Norman Book, first issue June 9, 1987, twelve-man masthead, and early staffers including Keith Rabois and David Sacks.↩

1109. Episode 7 script ("08. The Bechtel Cabinet") and review log, citing Hoover Institution / State Department biographical records and Laton McCartney, *Friends in High Places* (1988) for Shultz's career sequence (MIT, University of Chicago dean, CASBS fellow at Stanford 1968–1969, Nixon Secretary of Labor, first OMB director, Secretary of the Treasury through 1974, Bechtel executive and president, part-time

Stanford faculty, campus house and Oldsmobile, Secretary of State sworn July 16, 1982 through January 20, 1989, and Bohemian Grove attendance).↩

1110. Episode 7 script (“02. The Ticket Booth” and “07. Bayh-Dole”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Knew It Before They Did — Niels Reimers,” nn. 7–9, 13): Stanford’s 1983 licensing revenue of $2.5 million (about $8 million in 2026 CPI dollars), first in the nation, nearly 60% from Cohen-Boyer; Reimers’s estimate of >$50 million left on the table by refusing equity until 1989; and Stanford News Release, March 20, 1991 (“Niels Reimers, Long-Time Director of Stanford’s Patent Office Resigns; OTL to Gross $24 Million This Year”).↩

1111. Episode 7 script (“02. The Ticket Booth” and “07. Bayh-Dole”) and review log; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Knew It Before They Did — Niels Reimers,” nn. 7–9, 13): Stanford’s 1983 licensing revenue of $2.5 million (about $8 million in 2026 CPI dollars), first in the nation, nearly 60% from Cohen-Boyer; Reimers’s estimate of >$50 million left on the table by refusing equity until 1989; and Stanford News Release, March 20, 1991 (“Niels Reimers, Long-Time Director of Stanford’s Patent Office Resigns; OTL to Gross $24 Million This Year”).↩

1112. Episode 7 script (“03. Patenting Life”) and review log, citing Leslie Berlin, *Troublemakers*: Robert Swanson stepped down as CEO of Genentech in 1990 after Roche bought a majority stake.↩

1113. Episode 7 script (“05. Myth Machines”) and review log, citing Jesse Hamlin, *SF State Magazine* (Fall 2010) and the *Los Angeles Times* obituary (Oct. 30, 2009): August Coppola’s tenure as dean of the College of Creative Arts at San Francisco State (1984–1992), $15 million legislative appropriation, 1991 “skybreaking” ceremony and verbatim quote (“The idea is to look up, rather than down. Look up to the sky, the clouds, teach young people to dream.”), and 1993 opening of the Fine Arts Building and 150-seat theater.↩

1114. Episode 7 script (“05. Myth Machines”) and review log, citing Jesse Hamlin, *SF State Magazine* (Fall 2010) and the *Los Angeles Times* obituary (Oct. 30, 2009): August Coppola’s tenure as dean of the College of Creative Arts at San Francisco State (1984–1992), $15 million legislative appropriation, 1991 “skybreaking” ceremony and verbatim quote (“The idea is to look up, rather than down. Look up to the sky, the clouds, teach young people to dream.”), and 1993 opening of the Fine Arts Building and 150-seat theater.↩

1115. Episode 7 script (“05. Myth Machines”) and review log, citing Jesse Hamlin, *SF State Magazine* (Fall 2010) and the *Los Angeles Times* obituary (Oct. 30, 2009): August Coppola’s tenure as dean of the College of Creative Arts at San Francisco State (1984–1992), $15 million legislative appropriation, 1991 “skybreaking” ceremony and verbatim quote (“The idea is to look up, rather than down. Look up to the sky, the clouds, teach young people to dream.”), and 1993 opening of the Fine Arts Building and 150-seat theater.↩

1116. Yasha Levine, *Surveillance Valley* (`levine-surveillance#72`, Ch. 4, nn. 41–69): NSFNET launched 1986 under Stephen Wolff, T-1 upgrade (1987) and T-3 upgrade (1989), IBM-MCI consortium Advanced

Network and Services (ANS) formed in 1990 ($9.3 million/year federal contract) and March 12, 1992 House hearing on NSFNET privatization, Al Gore's High Performance Computing Act of 1991 ($495 million, University of Illinois Mosaic browser spun off as Netscape), and 1993–1995 retirement of the public NSFNET backbone.↩

1117. Yasha Levine, *Surveillance Valley* (`levine-surveillance#72`, Ch. 4, nn. 41–69): NSFNET launched 1986 under Stephen Wolff, T-1 upgrade (1987) and T-3 upgrade (1989), IBM-MCI consortium Advanced Network and Services (ANS) formed in 1990 ($9.3 million/year federal contract) and March 12, 1992 House hearing on NSFNET privatization, Al Gore's High Performance Computing Act of 1991 ($495 million, University of Illinois Mosaic browser spun off as Netscape), and 1993–1995 retirement of the public NSFNET backbone.↩

1118. Yasha Levine, *Surveillance Valley* (`levine-surveillance#72`, Ch. 4, nn. 41–69): NSFNET launched 1986 under Stephen Wolff, T-1 upgrade (1987) and T-3 upgrade (1989), IBM-MCI consortium Advanced Network and Services (ANS) formed in 1990 ($9.3 million/year federal contract) and March 12, 1992 House hearing on NSFNET privatization, Al Gore's High Performance Computing Act of 1991 ($495 million, University of Illinois Mosaic browser spun off as Netscape), and 1993–1995 retirement of the public NSFNET backbone.↩

1119. Yasha Levine, *Surveillance Valley* (`levine-surveillance#72`, Ch. 4, nn. 1–2, 83–105) and Fred Turner, *From Counterculture to Cyberculture* (`turner-counterculture-cyberculture` in `references.bib`): January 1993 launch of *Wired* by Louis Rossetto Jr. and Kevin Kelly, backed by MIT Media Lab's Nicholas Negroponte and inspired by Stewart Brand's *Whole Earth* network.↩

1120. Episode 7 script ("05. Myth Machines") and review log, citing *Variety* and Alvy Ray Smith: November 1995 release of *Toy Story* and Pixar's IPO the following week ("largest IPO of the year").↩

1121. Ashlee Vance, *Elon Musk* (2015), cited in Podcast Episode 7 Script ("11. PayPal") and Podcast Episode 7 Review ("Claim log"): "'As the story goes,' Musk dropped out of a Stanford doctorate after two days. A later records request could not locate an admissions file. Physics professors there in 1995 did not remember him."↩

1122. Ashlee Vance, *Elon Musk* (2015), cited in Podcast Episode 7 Script ("11. PayPal") and Podcast Episode 7 Review ("Claim log"): "'As the story goes,' Musk dropped out of a Stanford doctorate after two days. A later records request could not locate an admissions file. Physics professors there in 1995 did not remember him."↩

1123. Ashlee Vance, *Elon Musk* (2015), cited in Podcast Episode 7 Script ("11. PayPal") and Podcast Episode 7 Review ("Claim log"): "'As the story goes,' Musk dropped out of a Stanford doctorate after two days. A later records request could not locate an admissions file. Physics professors there in 1995 did not remember him."↩

1124. Steven Levy, *In the Plex: How Google Thinks, Works, and Shapes Our Lives* (New York: Simon & Schuster, 2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review; Yasha Levine, *Surveillance Valley*

(levine-surveillance#73, ch. 5, nn. 20–21, 24), citing Terry Winograd's Stanford homepage and the NSF Digital Libraries Initiative.↩

1125. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review ("BackRub, PageRank as a name for Page, Lego rack").↩

1126. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "Hassan: 'We were using all the bandwidth of the network. And this was from a single machine.'"↩

1127. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "Hassan: 'We were using all the bandwidth of the network. And this was from a single machine.'"↩

1128. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "Hassan: 'We were using all the bandwidth of the network. And this was from a single machine.'"↩

1129. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "Hassan: 'We were using all the bandwidth of the network. And this was from a single machine.'"↩

1130. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "January 1997 emails to Vinod Khosla: $1.6 million total, $300,000 to Stanford, $200,000 salary, $400,000 bonus, $700,000 Excite stock; 10 percent traffic / $130,000 a day / $47 million a year… Hassan: 'He told us he wanted Excite's search engine to be eighty percent as good as the other search engines.'"↩

1131. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "January 1997 emails to Vinod Khosla: $1.6 million total, $300,000 to Stanford, $200,000 salary, $400,000 bonus, $700,000 Excite stock; 10 percent traffic / $130,000 a day / $47 million a year… Hassan: 'He told us he wanted Excite's search engine to be eighty percent as good as the other search engines.'"↩

1132. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "January 1997 emails to Vinod Khosla: $1.6 million total, $300,000 to Stanford, $200,000 salary, $400,000 bonus, $700,000 Excite stock; 10 percent traffic / $130,000 a day / $47 million a year… Hassan: 'He told us he wanted Excite's search engine to be eighty percent as good as the other search engines.'"↩

1133. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "January 1997 emails to Vinod Khosla: $1.6 million total, $300,000 to Stanford, $200,000 salary, $400,000 bonus, $700,000 Excite stock; 10 percent traffic / $130,000 a day / $47 million a year… Hassan: 'He told us he wanted Excite's search engine to be eighty percent as good as the other search engines.'"↩

1134. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script ("10. BackRub") and Review: "January 1997 emails to Vinod Khosla: $1.6 million total, $300,000 to Stanford, $200,000 salary, $400,000 bonus, $700,000 Excite stock; 10 percent traffic / $130,000 a day / $47 million a year… Hassan: 'He told us he wanted Excite's search engine to be eighty percent as good as the other search engines.'"↩

1135. Sergey Brin and Larry Page, “The PageRank Citation Ranking: Bringing Order to the Web,” Stanford University InfoLab, January 29, 1998, cited in Yasha Levine, *Surveillance Valley* (`levine-surveillance#73`, ch. 5, n. 35); Sergey Brin and Lawrence Page, “The Anatomy of a Large-Scale Hypertextual Web Search Engine,” *Computer Networks and ISDN Systems* 30, nos. 1–7 (1998): 107–117 (`references.bib`, `page-brin-anatomy`).↩

1136. Maryann Feldman, Alessandra Colaianni, and Kang Liu, cited in Podcast Episode 7 Script (“03. Patenting Life”) and Review (“468 licensees; about $255 million to Stanford and UC by the end of 2001; patents expired December 1997”).↩

1137. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Looks Like $100 Million to Me! — Niels Reimers and Bob Swanson,” n. 32): “Licensees were charged $10,000 per year until they had products, at which point they would switch to paying royalties that looked low (1 percent and as low as .5 percent for sales over $10 million)…”↩

1138. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “No Idea How You Start a Company — Niels Reimers and Bob Swanson,” n. 65: “Stan Cohen, in his interview with the author, says that he urged Reimers to license nonexclusively”), and Podcast Episode 7 Script (“03. Patenting Life”).↩

1139. Maryann Feldman, Alessandra Colaianni, and Kang Liu, cited in Podcast Episode 7 Script (“03. Patenting Life”) and Review (“468 licensees; about $255 million to Stanford and UC by the end of 2001; patents expired December 1997”).↩

1140. Podcast Episode 7 Script (“06. The Story of Our Streets”) and Review (“1998: NationsBank of Charlotte bought BankAmerica, kept the Bank of America name, headquarters in Charlotte”).↩

1141. Podcast Episode 7 Script (“06. The Story of Our Streets”) and Review (“1998: NationsBank of Charlotte bought BankAmerica, kept the Bank of America name, headquarters in Charlotte”).↩

1142. Podcast Episode 7 Script (“06. The Story of Our Streets”) and Part 7 Open Issues note (“Henry George reference on railroad land value capture (in script, no specific citation in review)”). See also John L. Thomas, *Alternative America: Henry George, Edward Bellamy, Henry Demarest Lloyd, and the Adversary Tradition* (1983), cited in Adam Winkler, *We the Corporations* (`winkler-corporations#54`, ch. 5, n. 19).↩

1143. Stanford University Founding Grant, November 11, 1885, cited in Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “What Do We Do with These? — Niels Reimers,” n. 24), and Podcast Episode 7 Script (“06. The Story of Our Streets”).↩

1144. Podcast Episode 7 Script (“06. The Story of Our Streets”) and Review (“Sand Hill as a gate, ‘four minutes from the campus’”).↩

1145. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Review (“Incorporated Sept. 4, 1998”); Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Conclusion: Wave After Wave,” nn. 5, 8, citing Forest Baskett, Andreas Bechtolsheim, Bill Nowicki, and John

Seamons, “The Sun Workstation: A Terminal System for the Stanford University Network,” March 30, 1980, and “Von Bechtolsheim: I Invested in Google to Solve My Own Problem”).↩

1146. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script (“10. BackRub”) and Review (“Incorporated Sept. 4, 1998. Wojcicki house, Santa Margarita Street, Menlo Park, bought for $615,000; garage and rooms at $1,700 a month. Craig Silverstein.”).↩

1147. Steven Levy, *In the Plex* (2011), cited in Podcast Episode 7 Script (“10. BackRub”) and Review (“Incorporated Sept. 4, 1998. Wojcicki house, Santa Margarita Street, Menlo Park, bought for $615,000; garage and rooms at $1,700 a month. Craig Silverstein.”).↩

1148. Stanford OTL FY2005 report, cited in Podcast Episode 7 Script (“10. BackRub”) and Review (“PageRank licensed for 1.8 million shares”).↩

1149. Stanford OTL FY2005 report, cited in Podcast Episode 7 Script (“10. BackRub”) and Review (“PageRank licensed for 1.8 million shares”).↩

1150. Adam Fisher, *Valley of Genius*, via Kevin Kelly, cited in Podcast Episode 7 Script (“10. BackRub”) and Review (“Page to Kevin Kelly: ‘We are not really interested in search. We are making an AI.’”).↩

1151. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Looks Like $100 Million to Me! — Niels Reimers and Bob Swanson,” n. 2): “In 1999, Genentech paid an additional $200 million to the university, both parties agreeing that ‘this settlement is not an admission that Genentech infringed UC’s patent or used the genetic material in question.’”↩

1152. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Postscript: The Troublemakers Today,” nn. 20, 22, citing Andrew Pollack, “Robert A. Swanson Dies at 52; Early Leader in Biotechnology,” *New York Times*, December 8, 1999, and David V. Goeddel and Arthur D. Levinson, “Obituary: Robert A. Swanson (1947–99),” *Nature* 403 [January 20, 2000]: 264), and Podcast Episode 7 Script (“03. Patenting Life”).↩

1153. Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“11. PayPal”) and Review (“Thanksgiving 1999, X.com card at an ATM. Soni: ‘Elon was very, very happy with that.’”).↩

1154. Ashlee Vance, *Elon Musk* (2015), cited in Podcast Episode 7 Script (“11. PayPal”) and Review (“Musk, Stanford talk, 2003: ‘Money is low bandwidth…’ and ‘It’s really just an entry in a database.’”).↩

1155. Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“11. PayPal”) and Review (“Levchin came from Kiev and, as a boy, wrote programs longhand because he had no machine… The fraud still ‘leaked through the system faster than they could patch it.’”).↩

1156. Max Chafkin, *The Contrarian* (2021), and Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“09. The Grievance Paper”, “11. PayPal”, “12. The Network”) and Review (“Rabois, Sacks, Book later at PayPal; Sacks as COO”).↩

1157. Max Chafkin, *The Contrarian* (2021), and Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“09. The Grievance Paper”, “11. PayPal”, “12. The Network”) and Review (“Rabois, Sacks, Book later at PayPal; Sacks as COO”).↩

1158. Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“11. PayPal”) and Review (“Levchin came from Kiev and, as a boy, wrote programs longhand because he had no machine… The fraud still ‘leaked through the system faster than they could patch it.’”).↩

1159. eBay Inc. press release, July 8, 2002 (filed with the SEC), cited in Podcast Episode 7 Script (“11. PayPal”) and Review (“July 8, 2002: eBay announces the purchase. Merger agreement dated July 7. 0.39 eBay shares per PayPal share. $1.5 billion on the July 5 price. About 60 percent of PayPal’s business on eBay. 46 million eBay users. Billpoint.”).↩

1160. Maryann Feldman, Alessandra Colaianni, and Kang Liu, cited in Podcast Episode 7 Script (“03. Patenting Life”) and Review; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 40, citing “Cohen-Boyer Royalties and Distribution FY 1979–80 Through FY 1998–1999,” SUOTL: “Precise amounts: total royalties: $254,763,248; total admin fee: $38,214,63[0]; total expenses: $1,801,761; total share for each school: $107,373,658. The $107 million to Stanford would be evenly split among inventor, department, and the university.”).↩

1161. Maryann Feldman, Alessandra Colaianni, and Kang Liu, cited in Podcast Episode 7 Script (“03. Patenting Life”) and Review; Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Make It Happen — Niels Reimers,” n. 40, citing “Cohen-Boyer Royalties and Distribution FY 1979–80 Through FY 1998–1999,” SUOTL: “Precise amounts: total royalties: $254,763,248; total admin fee: $38,214,63[0]; total expenses: $1,801,761; total share for each school: $107,373,658. The $107 million to Stanford would be evenly split among inventor, department, and the university.”).↩

1162. eBay Inc. press release, July 8, 2002 (filed with the SEC), cited in Podcast Episode 7 Script (“11. PayPal”) and Review (“July 8, 2002: eBay announces the purchase. Merger agreement dated July 7. 0.39 eBay shares per PayPal share. $1.5 billion on the July 5 price. About 60 percent of PayPal’s business on eBay. 46 million eBay users. Billpoint.”).↩

1163. SEC exhibit (October 3, 2002 closing disclosure) and Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“11. PayPal”, “12. The Network”) and Review (“Oct. 3, 2002 note, via Soni. Thiel: last day at the close of the markets. They hired friends, then friends’ friends, in concentric circles. ‘I have always known that people are the most valuable part of any business, and today I’m more certain than ever that this is the case.’”).↩

1164. SEC exhibit (October 3, 2002 closing disclosure) and Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“11. PayPal”, “12. The Network”) and Review (“Oct. 3, 2002 note, via Soni. Thiel: last day at the close of the markets. They hired friends, then friends’ friends,

in concentric circles. ‘I have always known that people are the most valuable part of any business, and today I’m more certain than ever that this is the case.’”).↩

1165. SEC exhibit (October 3, 2002 closing disclosure) and Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“11. PayPal”, “12. The Network”) and Review (“Oct. 3, 2002 note, via Soni. Thiel: last day at the close of the markets. They hired friends, then friends’ friends, in concentric circles. ‘I have always known that people are the most valuable part of any business, and today I’m more certain than ever that this is the case.’”).↩

1166. Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“12. The Network”) and Review (“Reid Hoffman, made COO against the board’s advice. A weekend at his grandparents’ house in Gualala; on the third day, a network of professional names, later LinkedIn. David Sacks left in charge while Thiel traveled, then not made chief executive. Jack Selby: trading again before the month was out. Ken Howery went too. Musk already out, putting money into a car and into rockets.”).↩

1167. Jimmy Soni, *The Founders* (2022), cited in Podcast Episode 7 Script (“12. The Network”) and Review (“Reid Hoffman, made COO against the board’s advice. A weekend at his grandparents’ house in Gualala; on the third day, a network of professional names, later LinkedIn. David Sacks left in charge while Thiel traveled, then not made chief executive. Jack Selby: trading again before the month was out. Ken Howery went too. Musk already out, putting money into a car and into rockets.”).↩

1168. Max Chafkin, *The Contrarian* (2021), cited in Podcast Episode 7 Script (“09. The Grievance Paper”) and Review (“Thiel later had an office on Sand Hill”).↩

1169. Stanford OTL FY2005 report; Lisa M. Krieger, “Stanford Earns $336 Million off Google Stock,” *San Jose Mercury News*, December 1, 2005; and *Stanford Office of Technology Licensing Annual Report 2014–2015*, cited in Podcast Episode 7 Script (“10. BackRub”), Review, and Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Knew It Before They Did — Niels Reimers,” nn. 10–11).↩

1170. Alvy Ray Smith, cited in Podcast Episode 7 Script (“05. Myth Machines”) and Review (“Disney purchase announced about $7.4 billion, January 2006. About $12 billion grocery”).↩

1171. Jimmy Soni, *The Founders* (2022), and Max Chafkin, *The Contrarian* (2021), cited in Podcast Episode 7 Script (“12. The Network”) and Review (“Five years later Fortune photographed the group as a crime family, and the joke got the name PayPal Mafia. Soni: the picture left people out and made them look more alike than the company had been.”).↩

1172. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Postscript: The Troublemakers Today,” nn. 23–24, citing Andrew Pollack, “Roche Agrees to Buy Genentech,” *New York Times*, March 12, 2009: “Genentech had a 3-for-2 stock split in March 1983 and then 2-for-1 splits in 1986, 1987, 1999, 2000, and 2004. Roche paid $95 per share at the acquisition”), and Podcast Episode 7 Script (“03. Patenting Life”).↩

1173. Leslie Berlin, *Troublemakers* (`berlin-troublemakers#39`, “Introduction: A Bit Like Love,” n. 4; “Knew It Before They Did — Niels Reimers,” nn. 11, 15, citing “OTL Financial Data 1970–2016”), and Stanford OTL overview, cited in Podcast Episode 7 Script (“02. The Ticket Booth”) and Review (“Cumulative licensing nearly $2 billion by about 2016; OTL overview about $1.91 billion through FY2017; three inventions (recombinant DNA, antibody patents, Google) out of roughly 12,000 evaluations”). ## Part 8↩

1174. Matt Sheehan, *The Transpacific Experiment*, with outside qualifier on China’s April 20, 1994 NCFC link vs. May 17, 1994 IHEP–SLAC link (Episode 8 review log).↩

1175. SUNY Buffalo and company biographies; U.S. Patent 5,920,859 (filed Feb. 5, 1997, granted 1999, assigned to IDD Information Services); U.S. Patent 6,285,999 (assigned to Leland Stanford Junior University, citing the IDD Hyperlink Search Engine as prior art) (Episode 8 review log).↩

1176. SUNY Buffalo and company biographies; U.S. Patent 5,920,859 (filed Feb. 5, 1997, granted 1999, assigned to IDD Information Services); U.S. Patent 6,285,999 (assigned to Leland Stanford Junior University, citing the IDD Hyperlink Search Engine as prior art) (Episode 8 review log).↩

1177. SUNY Buffalo and company biographies; U.S. Patent 5,920,859 (filed Feb. 5, 1997, granted 1999, assigned to IDD Information Services); U.S. Patent 6,285,999 (assigned to Leland Stanford Junior University, citing the IDD Hyperlink Search Engine as prior art) (Episode 8 review log).↩

1178. Matt Sheehan, *The Transpacific Experiment*, pp. 68–69; outside check on Eric Xu’s Texas A&M PhD and UC Berkeley postdoctoral fellowship (Episode 8 review log).↩

1179. Matt Sheehan, *The Transpacific Experiment*, pp. 68–69; outside check on Eric Xu’s Texas A&M PhD and UC Berkeley postdoctoral fellowship (Episode 8 review log).↩

1180. Matt Sheehan, *The Transpacific Experiment*, pp. 68–69; outside check on Eric Xu’s Texas A&M PhD and UC Berkeley postdoctoral fellowship (Episode 8 review log).↩

1181. Matt Sheehan, *The Transpacific Experiment*; Bloomberg and Stanford Graduate School of Business records for Greg Penner, Peninsula Capital, and Integrity Partners ($600,000 each; $1.2 million total seed round; Baidu founded January 2000) (Episode 8 review log).↩

1182. Matt Sheehan, *The Transpacific Experiment*; Bloomberg and Stanford Graduate School of Business records for Greg Penner, Peninsula Capital, and Integrity Partners ($600,000 each; $1.2 million total seed round; Baidu founded January 2000) (Episode 8 review log).↩

1183. Matt Sheehan, *The Transpacific Experiment*; Bloomberg and Stanford Graduate School of Business records for Greg Penner, Peninsula Capital, and Integrity Partners ($600,000 each; $1.2 million total seed round; Baidu founded January 2000) (Episode 8 review log).↩

1184. Matt Sheehan, *The Transpacific Experiment* (Google's Chinese-language launch in 2000, two-week block in September 2002, ~25% share by late 2002, Google.cn launch in 2006 under Kai-Fu Lee, Eric Schmidt's "evil scale," Baidu's poet advertisement, and late-2009 market shares of ~58% Baidu vs. ~36% Google) (Episode 8 review log).↩

1185. Episode 8 review log (Thefacebook launched at Harvard in February 2004 and moved to Palo Alto by summer 2004); Leslie Berlin, *Troublemakers*, Conclusion n. 1 (`berlin-troublemakers#39`, noting Facebook's later Menlo Park campus on the former Sun Microsystems site); Max Chafkin, *The Contrarian* (Episode 8 review log, noting Facebook's old University Avenue headquarters in Palo Alto).↩

1186. Episode 8 review log (Thefacebook launched at Harvard in February 2004 and moved to Palo Alto by summer 2004); Leslie Berlin, *Troublemakers*, Conclusion n. 1 (`berlin-troublemakers#39`, noting Facebook's later Menlo Park campus on the former Sun Microsystems site); Max Chafkin, *The Contrarian* (Episode 8 review log, noting Facebook's old University Avenue headquarters in Palo Alto).↩

1187. Shoshana Zuboff, *The Age of Surveillance Capitalism* (New York: PublicAffairs, 2019), in `references.bib` (`zuboff-surveillance-capitalism`) and Episode 8 review log.↩

1188. Telecommunications Act of 1996, 47 U.S.C. § 230; *Stratton Oakmont, Inc. v. Prodigy Services Co.* (N.Y. Sup. Ct. 1995); Congressional Research Service report, as cited in Episode 8 review log.↩

1189. Telecommunications Act of 1996, 47 U.S.C. § 230; *Stratton Oakmont, Inc. v. Prodigy Services Co.* (N.Y. Sup. Ct. 1995); Congressional Research Service report, as cited in Episode 8 review log.↩

1190. Telecommunications Act of 1996, 47 U.S.C. § 230; *Stratton Oakmont, Inc. v. Prodigy Services Co.* (N.Y. Sup. Ct. 1995); Congressional Research Service report, as cited in Episode 8 review log.↩

1191. Protection of Lawful Commerce in Arms Act (PLCAA), 2005; *Soto v. Bushmaster Firearms International, LLC* (Conn. Mar. 19, 2019, holding CUTPA is a valid predicate statute; U.S. Supreme Court cert. denied Nov. 12, 2019); Associated Press, *The Trace*, Justia, and SCOTUSblog, as cited in Episode 8 review log.↩

1192. Protection of Lawful Commerce in Arms Act (PLCAA), 2005; *Soto v. Bushmaster Firearms International, LLC* (Conn. Mar. 19, 2019, holding CUTPA is a valid predicate statute; U.S. Supreme Court cert. denied Nov. 12, 2019); Associated Press, *The Trace*, Justia, and SCOTUSblog, as cited in Episode 8 review log.↩

1193. Steve Jobs, Stanford Commencement Address, June 12, 2005 ("The focus of my entire adult life was gone, and it was devastating"), cited in Leslie Berlin, *Troublemakers*, Introduction n. 1 (`berlin-troublemakers#39`) and Episode 8 review log.↩

1194. Steve Jobs, Stanford Commencement Address, June 12, 2005 ("The focus of my entire adult life was gone, and it was devastating"), cited in Leslie Berlin, *Troublemakers*, Introduction n. 1 (`berlin-troublemakers#39`) and Episode 8 review log.↩

1195. Leslie Berlin, *Troublemakers*, "Young Maniacs — Mike Markkula," n. 42 (`berlin-troublemakers#39`, citing Walter Isaacson, *Steve Jobs*, p. 370, on Markkula's concept of "imputing"); Episode 8 script and Open Issues note (flagging the exact wording "a company's image is everything the customer sees, hears, or feels" as `[UNVERIFIED]` in the Episode 8 review log).↩

1196. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger, *New York*, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1197. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger, *New York*, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1198. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger, *New York*, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1199. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger, *New York*, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1200. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger, *New York*, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1201. Yasha Levine, *Surveillance Valley*, ch. 5, nn. 102–103, 106 (`levine-surveillance#73`, citing In-Q-Tel press release, June 25, 2003, on Keyhole's EarthViewer in Iraq, and In-Q-Tel's 2005 sale of 5,636 Google shares for $2.18 million).↩

1202. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger, *New York*, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1203. Max Chafkin, *The Contrarian: Peter Thiel and Silicon Valley's Pursuit of Power* (2021), ch. 8 notes (`chafkin-contrarian#75`, citing Andy Greenberg and Ryan Mac, *Forbes*, Sept. 2, 2013; Sharon Weinberger,

New York, Sept. 28, 2020; Shane Harris, *Washingtonian*, Jan. 31, 2012; Peter Waldman, Lizette Chapman, and Jordan Robertson, *Bloomberg Businessweek*, Apr. 19, 2018) and Episode 8 review log.↩

1204. Matt Sheehan, *The Transpacific Experiment* (Google's Chinese-language launch in 2000, two-week block in September 2002, ~25% share by late 2002, Google.cn launch in 2006 under Kai-Fu Lee, Eric Schmidt's "evil scale," Baidu's poet advertisement, and late-2009 market shares of ~58% Baidu vs. ~36% Google) (Episode 8 review log).↩

1205. Matt Sheehan, *The Transpacific Experiment* (Google's Chinese-language launch in 2000, two-week block in September 2002, ~25% share by late 2002, Google.cn launch in 2006 under Kai-Fu Lee, Eric Schmidt's "evil scale," Baidu's poet advertisement, and late-2009 market shares of ~58% Baidu vs. ~36% Google) (Episode 8 review log).↩

1206. Matt Sheehan, *The Transpacific Experiment* (Google's Chinese-language launch in 2000, two-week block in September 2002, ~25% share by late 2002, Google.cn launch in 2006 under Kai-Fu Lee, Eric Schmidt's "evil scale," Baidu's poet advertisement, and late-2009 market shares of ~58% Baidu vs. ~36% Google) (Episode 8 review log).↩

1207. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (`chafkin-contrarian#75`, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1208. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (`chafkin-contrarian#75`, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1209. Max Chafkin, *The Contrarian*, chs. 2–3 (`chafkin-contrarian#75`, citing Peter Thiel and David Sacks, *The Diversity Myth: Multiculturalism and the Politics of Intolerance at Stanford*, Independent Institute, 1995) and Episode 8 review log.↩

1210. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (`chafkin-contrarian#75`, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1211. Matt Sheehan, *The Transpacific Experiment*, pp. 162–163 (2008 Chinese deputy minister of culture question, "Who is the CEO of Hollywood?"; Wang Jianlin / Dalian Wanda's refused $4.9 billion bid

for 49% of Paramount; 2017 Chinese capital controls and ban on “irrational” overseas entertainment investment) (Episode 8 review log).↩

1212. Matt Sheehan, *The Transpacific Experiment*, pp. 162–163 (2008 Chinese deputy minister of culture question, “Who is the CEO of Hollywood?”; Wang Jianlin / Dalian Wanda’s refused $4.9 billion bid for 49% of Paramount; 2017 Chinese capital controls and ban on “irrational” overseas entertainment investment) (Episode 8 review log).↩

1213. Sebastian Mallaby, *The Power Law: Venture Capital and the Making of the New Future* (New York: Penguin Press, 2022), p. 334, in `references.bib` (`mallaby-power-law`) and Episode 8 review log.↩

1214. Sebastian Mallaby, *The Power Law: Venture Capital and the Making of the New Future* (New York: Penguin Press, 2022), p. 334, in `references.bib` (`mallaby-power-law`) and Episode 8 review log.↩

1215. *Wall Street Journal*, October 2023, and *GamesIndustry.biz* summary of the unsealed 2009 Remington / Freedom Group and Activision product-placement documents for the Adaptive Combat Rifle in *Call of Duty: Modern Warfare 2* and John Trull’s interview (Episode 8 review log).↩

1216. *Wall Street Journal*, October 2023, and *GamesIndustry.biz* summary of the unsealed 2009 Remington / Freedom Group and Activision product-placement documents for the Adaptive Combat Rifle in *Call of Duty: Modern Warfare 2* and John Trull’s interview (Episode 8 review log).↩

1217. *Wall Street Journal*, October 2023, and *GamesIndustry.biz* summary of the unsealed 2009 Remington / Freedom Group and Activision product-placement documents for the Adaptive Combat Rifle in *Call of Duty: Modern Warfare 2* and John Trull’s interview (Episode 8 review log).↩

1218. *Wall Street Journal*, October 2023, and *GamesIndustry.biz* summary of the unsealed 2009 Remington / Freedom Group and Activision product-placement documents for the Adaptive Combat Rifle in *Call of Duty: Modern Warfare 2* and John Trull’s interview (Episode 8 review log).↩

1219. *Wall Street Journal*, October 2023, and *GamesIndustry.biz* summary of the unsealed 2009 Remington / Freedom Group and Activision product-placement documents for the Adaptive Combat Rifle in *Call of Duty: Modern Warfare 2* and John Trull’s interview (Episode 8 review log).↩

1220. Matt Sheehan, *The Transpacific Experiment* (Google’s Chinese-language launch in 2000, two-week block in September 2002, ~25% share by late 2002, Google.cn launch in 2006 under Kai-Fu Lee, Eric Schmidt’s “evil scale,” Baidu’s poet advertisement, and late-2009 market shares of ~58% Baidu vs. ~36% Google) (Episode 8 review log).↩

1221. Matt Sheehan, *The Transpacific Experiment*, pp. 60, 80–83 (Operation Aurora hack targeting Gmail of a 20-year-old Tibetan activist at Stanford; Google’s Jan. 12, 2010 “A New Approach to China”; March 2010 Hong Kong redirect; 2014 block) (Episode 8 review log).↩

1222. Matt Sheehan, *The Transpacific Experiment*, pp. 60, 80–83 (Operation Aurora hack targeting Gmail of a 20-year-old Tibetan activist at Stanford; Google's Jan. 12, 2010 "A New Approach to China"; March 2010 Hong Kong redirect; 2014 block) (Episode 8 review log).↩

1223. Matt Sheehan, *The Transpacific Experiment*, pp. 60, 80–83 (Operation Aurora hack targeting Gmail of a 20-year-old Tibetan activist at Stanford; Google's Jan. 12, 2010 "A New Approach to China"; March 2010 Hong Kong redirect; 2014 block) (Episode 8 review log).↩

1224. Matt Sheehan, *The Transpacific Experiment*, pp. 60, 80–83 (Operation Aurora hack targeting Gmail of a 20-year-old Tibetan activist at Stanford; Google's Jan. 12, 2010 "A New Approach to China"; March 2010 Hong Kong redirect; 2014 block) (Episode 8 review log).↩

1225. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain's 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube's post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1226. Protection of Lawful Commerce in Arms Act (PLCAA), 2005; *Soto v. Bushmaster Firearms International, LLC* (Conn. Mar. 19, 2019, holding CUTPA is a valid predicate statute; U.S. Supreme Court cert. denied Nov. 12, 2019); Associated Press, *The Trace*, Justia, and SCOTUSblog, as cited in Episode 8 review log.↩

1227. Associated Press, NPR, Reuters, and PBS coverage of the Aug. 6, 2018 removals of Alex Jones / InfoWars by Apple, YouTube, Facebook, and Spotify, Twitter's Sept. 6, 2018 permanent ban, and Jones's Sandy Hook hoax claims and trial testimony (Episode 8 review log).↩

1228. Matt Sheehan, *The Transpacific Experiment*, pp. 232–234 (October 2013 DeLeon Realty 14-seat Mercedes bus tour on El Camino Real, Kim Heng, the Great Wall / Hoover Tower / Golden Gate brochure, the Menlo Park mansion stop, and the buyer's Bo Xilai remark) (Episode 8 review log).↩

1229. Matt Sheehan, *The Transpacific Experiment*, pp. 232–234 (October 2013 DeLeon Realty 14-seat Mercedes bus tour on El Camino Real, Kim Heng, the Great Wall / Hoover Tower / Golden Gate brochure, the Menlo Park mansion stop, and the buyer's Bo Xilai remark) (Episode 8 review log).↩

1230. Matt Sheehan, *The Transpacific Experiment*, p. 231 (China pouring more cement between 2011 and 2013 than the U.S. used in the entire twentieth century) (Episode 8 review log).↩

1231. Matt Sheehan, *The Transpacific Experiment*, pp. 232–234 (October 2013 DeLeon Realty 14-seat Mercedes bus tour on El Camino Real, Kim Heng, the Great Wall / Hoover Tower / Golden Gate brochure, the Menlo Park mansion stop, and the buyer's Bo Xilai remark) (Episode 8 review log).↩

1232. Matt Sheehan, *The Transpacific Experiment*, pp. 232–234 (October 2013 DeLeon Realty 14-seat Mercedes bus tour on El Camino Real, Kim Heng, the Great Wall / Hoover Tower / Golden Gate brochure, the

Menlo Park mansion stop, and the buyer's Bo Xilai remark) (Episode 8 review log).↩

1233. Matt Sheehan, *The Transpacific Experiment*, pp. 232–234 (October 2013 DeLeon Realty 14-seat Mercedes bus tour on El Camino Real, Kim Heng, the Great Wall / Hoover Tower / Golden Gate brochure, the Menlo Park mansion stop, and the buyer's Bo Xilai remark) (Episode 8 review log).↩

1234. Matt Sheehan, *The Transpacific Experiment*, pp. 232–234 (October 2013 DeLeon Realty 14-seat Mercedes bus tour on El Camino Real, Kim Heng, the Great Wall / Hoover Tower / Golden Gate brochure, the Menlo Park mansion stop, and the buyer's Bo Xilai remark) (Episode 8 review log).↩

1235. Matt Sheehan, *The Transpacific Experiment*, citing National Association of Realtors data (Chinese U.S. residential purchases quadrupling 2011–2015 to $28.6B; 53% in California in 2013; Palo Alto median home price rising 50% in 2012–2013 to nearly $2M; "the new Swiss bank account") (Episode 8 review log); Leslie Berlin, *Troublemakers*, Introduction n. 6 (`berlin-troublemakers#39`, citing Xin Jiang, *Palo Alto Weekly*, Oct. 27, 2016, that in 2015 the median home price in Palo Alto was $2.5 million).↩

1236. Matt Sheehan, *The Transpacific Experiment*, citing National Association of Realtors data (Chinese U.S. residential purchases quadrupling 2011–2015 to $28.6B; 53% in California in 2013; Palo Alto median home price rising 50% in 2012–2013 to nearly $2M; "the new Swiss bank account") (Episode 8 review log); Leslie Berlin, *Troublemakers*, Introduction n. 6 (`berlin-troublemakers#39`, citing Xin Jiang, *Palo Alto Weekly*, Oct. 27, 2016, that in 2015 the median home price in Palo Alto was $2.5 million).↩

1237. Matt Sheehan, *The Transpacific Experiment*, citing National Association of Realtors data (Chinese U.S. residential purchases quadrupling 2011–2015 to $28.6B; 53% in California in 2013; Palo Alto median home price rising 50% in 2012–2013 to nearly $2M; "the new Swiss bank account") (Episode 8 review log); Leslie Berlin, *Troublemakers*, Introduction n. 6 (`berlin-troublemakers#39`, citing Xin Jiang, *Palo Alto Weekly*, Oct. 27, 2016, that in 2015 the median home price in Palo Alto was $2.5 million).↩

1238. Matt Sheehan, *The Transpacific Experiment*, citing National Association of Realtors data (Chinese U.S. residential purchases quadrupling 2011–2015 to $28.6B; 53% in California in 2013; Palo Alto median home price rising 50% in 2012–2013 to nearly $2M; "the new Swiss bank account") (Episode 8 review log); Leslie Berlin, *Troublemakers*, Introduction n. 6 (`berlin-troublemakers#39`, citing Xin Jiang, *Palo Alto Weekly*, Oct. 27, 2016, that in 2015 the median home price in Palo Alto was $2.5 million).↩

1239. Matt Sheehan, *The Transpacific Experiment*, citing National Association of Realtors data (Chinese U.S. residential purchases quadrupling 2011–2015 to $28.6B; 53% in California in 2013; Palo Alto median home price rising 50% in 2012–2013 to nearly $2M; "the new Swiss bank account") (Episode 8 review log); Leslie Berlin,

Troublemakers, Introduction n. 6 (`berlin-troublemakers#39`, citing Xin Jiang, *Palo Alto Weekly*, Oct. 27, 2016, that in 2015 the median home price in Palo Alto was $2.5 million).↩

1240. Matt Sheehan, *The Transpacific Experiment* ($500,000 EB-5 visa threshold creating ten jobs; 9,128 visas issued to Chinese applicants in 2014, >85% of total; ~$700,000 in 2026 CPI dollars) (Episode 8 review log).↩

1241. Matt Sheehan, *The Transpacific Experiment*, pp. 4–5 (Tommy Sheehan driving Father Peter C. Yorke; Yorke's 1901 California Chinese Exclusion Convention keynote quote) (Episode 8 review log).↩

1242. Matt Sheehan, *The Transpacific Experiment*, pp. 267–268 (1869 Promontory Summit photograph vs. 2013 Bay Bridge opening with Mayors Ed Lee and Jean Quan: "This time the Chinese were in the middle") (Episode 8 review log).↩

1243. Matt Sheehan, *The Transpacific Experiment*, p. 106 (May 2014 hiring of Andrew Ng by Baidu to lead its Sunnyvale lab less than a mile from Google; Baidu valued at ~$60B, or ~$85B in 2026 CPI dollars) (Episode 8 review log).↩

1244. Matt Sheehan, *The Transpacific Experiment*, p. 106 (May 2014 hiring of Andrew Ng by Baidu to lead its Sunnyvale lab less than a mile from Google; Baidu valued at ~$60B, or ~$85B in 2026 CPI dollars) (Episode 8 review log).↩

1245. Christopher Wylie, *Mindf*ck: Cambridge Analytica and the Plot to Break America* (New York: Random House, 2019), and Reuters (August 2014 collection of 87M+ Facebook profiles via Aleksandr Kogan's app; Wylie's Harvard/Stanford legitimation quote) (Episode 8 review log; `chafkin-contrarian#75`, p. 357).↩

1246. Christopher Wylie, *Mindf*ck: Cambridge Analytica and the Plot to Break America* (New York: Random House, 2019), and Reuters (August 2014 collection of 87M+ Facebook profiles via Aleksandr Kogan's app; Wylie's Harvard/Stanford legitimation quote) (Episode 8 review log; `chafkin-contrarian#75`, p. 357).↩

1247. Christopher Wylie, *Mindf*ck* (2019), pp. 83, 112–114, as cited in `chafkin-contrarian#75` (p. 357) and Episode 8 review log (Wylie's account of summer 2014 meeting with Palantir staff at Soho Square, London).↩

1248. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain's 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube's post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1249. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain's 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube's post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1250. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain's 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube's post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1251. Matt Sheehan, *The Transpacific Experiment* (Li Zhifei's founding of Mobvoi, the backward *California* and *Silicon Valley* glass floor arrows, and Google's 2015 investment of ~$40–45 million) (Episode 8 review log).↩

1252. Matt Sheehan, *The Transpacific Experiment* (Li Zhifei's founding of Mobvoi, the backward *California* and *Silicon Valley* glass floor arrows, and Google's 2015 investment of ~$40–45 million) (Episode 8 review log).↩

1253. Sebastian Mallaby, *The Power Law* (2022), pp. 340–341 (Elizabeth Holmes at Stanford, engineering dean and Hoover Institution board members including George Shultz, $9B valuation / ~$12.7B in 2026 dollars, non-Sand Hill investors Walton $150M, Murdoch $121M, DeVos $100M, Cox $100M, Tim Draper angel investment, 100-vote shares, "faked it until they could make it") (Episode 8 review log).↩

1254. Sebastian Mallaby, *The Power Law* (2022), pp. 340–341 (Elizabeth Holmes at Stanford, engineering dean and Hoover Institution board members including George Shultz, $9B valuation / ~$12.7B in 2026 dollars, non-Sand Hill investors Walton $150M, Murdoch $121M, DeVos $100M, Cox $100M, Tim Draper angel investment, 100-vote shares, "faked it until they could make it") (Episode 8 review log).↩

1255. Sebastian Mallaby, *The Power Law* (2022), pp. 340–341 (Elizabeth Holmes at Stanford, engineering dean and Hoover Institution board members including George Shultz, $9B valuation / ~$12.7B in 2026 dollars, non-Sand Hill investors Walton $150M, Murdoch $121M, DeVos $100M, Cox $100M, Tim Draper angel investment, 100-vote shares, "faked it until they could make it") (Episode 8 review log).↩

1256. Sebastian Mallaby, *The Power Law* (2022), pp. 340–341 (Elizabeth Holmes at Stanford, engineering dean and Hoover Institution board members including George Shultz, $9B valuation / ~$12.7B in 2026 dollars, non-Sand Hill investors Walton $150M, Murdoch $121M, DeVos $100M, Cox $100M, Tim Draper angel investment, 100-vote shares, "faked it until they could make it") (Episode 8 review log).↩

1257. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (New York: Little, Brown, 2023), in `references.bib` (`harris-palo-alto`) and Episode 8 review log (William Draper III at Yale and in Skull and Bones with George H. W. Bush).↩

1258. Sebastian Mallaby, *The Power Law* (2022), pp. 340–341 (Elizabeth Holmes at Stanford, engineering dean and Hoover Institution board members including George Shultz, $9B valuation / ~$12.7B in 2026 dollars, non-Sand Hill investors Walton $150M, Murdoch $121M, DeVos $100M, Cox $100M, Tim Draper angel investment, 100-vote shares, "faked it until they could make it") (Episode 8 review log).↩

1259. Max Chafkin, *The Contrarian*, ch. 15 (chafkin-contrarian#75, p. 357, citing *Tampa Bay Times*, March 25, 2016, on the Pinellas County trial) and Episode 8 review log.↩

1260. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (chafkin-contrarian#75, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1261. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (chafkin-contrarian#75, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1262. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (chafkin-contrarian#75, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1263. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (chafkin-contrarian#75, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1264. Max Chafkin, *The Contrarian*, chs. 9, 13, 15 (chafkin-contrarian#75, citing Owen Thomas on *Valleywag*, Dec. 2007; Ryan Holiday, *Conspiracy*, 2018; Andrew Ross Sorkin, *New York Times*, May 25, 2016) and Episode 8 review log (March 2016 jury verdict of $115M compensatory and $25M punitive; Thiel's ~$10M legal funding and "specific deterrence" quote; Univision's $135M asset purchase; $31M Bollea settlement).↩

1265. Matt Sheehan, *The Transpacific Experiment*, pp. 162–163 (2008 Chinese deputy minister of culture question, "Who is the CEO of Hollywood?"; Wang Jianlin / Dalian Wanda's refused $4.9 billion bid for 49% of Paramount; 2017 Chinese capital controls and ban on "irrational" overseas entertainment investment) (Episode 8 review log).↩

1266. Matt Sheehan, *The Transpacific Experiment*, pp. 162–163 (2008 Chinese deputy minister of culture question, "Who is the CEO of Hollywood?"; Wang Jianlin / Dalian Wanda's refused $4.9 billion bid for 49% of Paramount; 2017 Chinese capital controls and ban on "irrational" overseas entertainment investment) (Episode 8 review log).↩

1267. *Wall Street Journal*, *Deadline* (January 2018), and NBC News, on Tencent acquiring just under 10% of David Ellison's Skydance at an ~$1.5 billion valuation (Episode 8 review log).↩

1268. Matt Sheehan, *The Transpacific Experiment*, pp. 123–124, 291 (*The Intercept*'s 2018 reporting on Google's Project Dragonfly, >740 employee signatures on protest letter, and Google effectively ending the project within a month) (Episode 8 review log).↩

1269. Matt Sheehan, *The Transpacific Experiment*, pp. 123–124, 291 (*The Intercept*'s 2018 reporting on Google's Project Dragonfly, >740 employee signatures on protest letter, and Google effectively ending the project within a month) (Episode 8 review log).↩

1270. Matt Sheehan, *The Transpacific Experiment*, pp. 54–55 (Donald Trump's 2018 private remark that "almost every student that comes over to this country is a spy") (Episode 8 review log).↩

1271. Associated Press, NPR, Reuters, and PBS coverage of the Aug. 6, 2018 removals of Alex Jones / InfoWars by Apple, YouTube, Facebook, and Spotify, Twitter's Sept. 6, 2018 permanent ban, and Jones's Sandy Hook hoax claims and trial testimony (Episode 8 review log).↩

1272. Associated Press, NPR, Reuters, and PBS coverage of the Aug. 6, 2018 removals of Alex Jones / InfoWars by Apple, YouTube, Facebook, and Spotify, Twitter's Sept. 6, 2018 permanent ban, and Jones's Sandy Hook hoax claims and trial testimony (Episode 8 review log).↩

1273. YouTube Official Blog (Jan. 25, 2019 announcement on reducing recommendations of "borderline" content affecting <1% of videos; Dec. 2019 claim of a 70% average U.S. drop in non-subscribed recommendation watch time of such content) (Episode 8 review log).↩

1274. YouTube Official Blog (Jan. 25, 2019 announcement on reducing recommendations of "borderline" content affecting <1% of videos; Dec. 2019 claim of a 70% average U.S. drop in non-subscribed recommendation watch time of such content) (Episode 8 review log).↩

1275. Protection of Lawful Commerce in Arms Act (PLCAA), 2005; *Soto v. Bushmaster Firearms International, LLC* (Conn. Mar. 19, 2019, holding CUTPA is a valid predicate statute; U.S. Supreme Court cert. denied Nov. 12, 2019); Associated Press, *The Trace*, Justia, and SCOTUSblog, as cited in Episode 8 review log.↩

1276. Protection of Lawful Commerce in Arms Act (PLCAA), 2005; *Soto v. Bushmaster Firearms International, LLC* (Conn. Mar. 19, 2019, holding CUTPA is a valid predicate statute; U.S. Supreme Court cert. denied Nov. 12, 2019); Associated Press, *The Trace*, Justia, and SCOTUSblog, as cited in Episode 8 review log.↩

1277. Protection of Lawful Commerce in Arms Act (PLCAA), 2005; *Soto v. Bushmaster Firearms International, LLC* (Conn. Mar. 19, 2019, holding CUTPA is a valid predicate statute; U.S. Supreme Court cert. denied Nov. 12, 2019); Associated Press, *The Trace*, Justia, and SCOTUSblog, as cited in Episode 8 review log.↩

1278. United States Federal Trade Commission, “FTC Imposes $5 Billion Penalty and Sweeping New Privacy Restrictions on Facebook,” July 2019, in `references.bib` (`ftc-facebook-2019`) and Episode 8 review log.↩

1279. Lisa M. Krieger, “Stanford Earns $336 Million off Google Stock,” *San Jose Mercury News*, Dec. 1, 2005, cited in Leslie Berlin, *Troublemakers*, “Knew It Before They Did — Niels Reimers,” n. 10 (`berlin-troublemakers#39`) and Episode 8 review log.↩

1280. CNN (July 22, 2019 and June 2, 2022), on the *Top Gun: Maverick* flight jacket Taiwan and Japan flag patches, Tencent’s withdrawal, and the 2022 release (Episode 8 review log).↩

1281. CNN (July 22, 2019 and June 2, 2022), on the *Top Gun: Maverick* flight jacket Taiwan and Japan flag patches, Tencent’s withdrawal, and the 2022 release (Episode 8 review log).↩

1282. *Platformer*, *Washington Post*, and NPR coverage of the Stanford Internet Observatory, founded in 2019 by Alex Stamos with Renée DiResta as research director (Episode 8 review log).↩

1283. *Stanford Report* and Stanford campus names records (Serra Mall renamed Jane Stanford Way effective Oct. 7, 2019, dedicated Nov. 14, 2019, with official address 450 Jane Stanford Way and Serra Street retained from Campus Drive East to El Camino Real; Jordan Hall renamed Building 420 in October 2020) (Episode 8 review log).↩

1284. *Stanford Report* and Stanford campus names records (Serra Mall renamed Jane Stanford Way effective Oct. 7, 2019, dedicated Nov. 14, 2019, with official address 450 Jane Stanford Way and Serra Street retained from Campus Drive East to El Camino Real; Jordan Hall renamed Building 420 in October 2020) (Episode 8 review log).↩

1285. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain’s 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube’s post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1286. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain’s 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube’s post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1287. Kevin Roose, *Rabbit Hole* (*New York Times* podcast, Episode 1, April 16, 2020) and *New York Times* reporting (June 8, 2019), on Caleb Cain’s 2014–2018 YouTube history (>12,000 videos, 2,500 searches), Stefan Molyneux, Natalie Wynn, and YouTube’s post-2012 watch-time optimization driving >70% of watch time (Episode 8 review log).↩

1288. Kevin Roose, *Rabbit Hole*, Episode 4 (*New York Times*, 2020), and Episode 7 / Episode 8 review logs (Susan Wojcicki as YouTube CEO in 2020 and 1998 Menlo Park garage landlord at $1,700/month).↩

1289. Hosseinmardi et al., *PNAS* (2021); Chen, Nyhan et al., *Science Advances* (2023); Hosseinmardi et al., *PNAS* (2024), as summarized in Episode 8 review log.↩

1290. Hosseinmardi et al., *PNAS* (2021); Chen, Nyhan et al., *Science Advances* (2023); Hosseinmardi et al., *PNAS* (2024), as summarized in Episode 8 review log.↩

1291. Hosseinmardi et al., *PNAS* (2021); Chen, Nyhan et al., *Science Advances* (2023); Hosseinmardi et al., *PNAS* (2024), as summarized in Episode 8 review log.↩

1292. Hosseinmardi et al., *PNAS* (2021); Chen, Nyhan et al., *Science Advances* (2023); Hosseinmardi et al., *PNAS* (2024), as summarized in Episode 8 review log.↩

1293. BBC, *The Verge*, CNBC, Associated Press, and CNN coverage of Alex Jones's Oct. 27, 2020 appearance with Tim Dillon on *The Joe Rogan Experience*, Rogan's ~$100M Spotify deal, and internal Spotify guidance (Episode 8 review log).↩

1294. BBC, *The Verge*, CNBC, Associated Press, and CNN coverage of Alex Jones's Oct. 27, 2020 appearance with Tim Dillon on *The Joe Rogan Experience*, Rogan's ~$100M Spotify deal, and internal Spotify guidance (Episode 8 review log).↩

1295. BBC, *The Verge*, CNBC, Associated Press, and CNN coverage of Alex Jones's Oct. 27, 2020 appearance with Tim Dillon on *The Joe Rogan Experience*, Rogan's ~$100M Spotify deal, and internal Spotify guidance (Episode 8 review log).↩

1296. Associated Press and PBS coverage of the 2021 Connecticut default liability ruling against Alex Jones for discovery violations (Episode 8 review log).↩

1297. OpenSecrets donor table drawn from FEC filings for Protect Ohio Values PAC ($10 million on March 12, 2021, followed by $3.5 million on April 18, 2022 and $1.5 million on April 25, 2022, totaling $15 million in support of J. D. Vance) (Episode 8 review log).↩

1298. Associated Press, February 15, 2022, and The Trace (Podcast Fact-Check Review, Episode 8): "Settlement Feb 15, 2022, $73 million, nine families, insurers, after Remington's second bankruptcy."↩

1299. OpenSecrets donor table, drawn from FEC filings (Podcast Fact-Check Review, Episode 8): "Thiel: $10 million (March 12, 2021), $3.5 million (April 18, 2022), $1.5 million (April 25, 2022), to Protect Ohio Values, for Vance."↩

1300. Bloomberg Law, *Texas Tribune*, and Reuters, May 24, 2024 (Podcast Fact-Check Review, Episode 8): "May 24, 2022, Robb Elementary: 19 children and 2 teachers killed. Salvador Ramos named once… Two suits filed May 24, 2024, by Josh Koskoff. *Mata v. Meta* in Los Angeles, about 45 family members, against Meta and Activision. *Rubio v. Daniel Defense* in Uvalde County, 31 family members."↩

1301. CNN, July 22, 2019, and June 2, 2022 (Podcast Fact-Check Review, Episode 8): “The 2022 film restores the flags and does not open in mainland China.”↩

1302. Associated Press and PBS (Podcast Fact-Check Review, Episode 8): “Texas: about $49–50 million (Aug 2022) cut on Aug 21, 2026, to about $6 million ($4.1 million compensatory plus about $1.5 million punitive)… Connecticut: default in 2021; jury $964 million plus $473 million; appeals to about $1.25 billion; cert denied Oct 14, 2025. Personal bankruptcy late 2022. Paid nothing as of August 2026.”↩

1303. Congressional Research Service Report R48642 (September 19, 2025), Nvidia Form 8-K (April 9, 2025), and Reuters (August 8, 2025; December 8, 2025; May 14, 2026) (Podcast Fact-Check Review, Episode 8): “Export controls. Oct 7, 2022: A100/H100-class chips and tools; Nvidia’s A800/H800. Oct 2023: loopholes closed; H20. Dec 2024: HBM, packaging, equipment. April 2025: H20 license, Nvidia 8-K, effectively a ban. Aug 2025: H20 licenses resume. Dec 2025: H200 allowed with a 25 percent cut to the U.S. government. May 2026: about ten Chinese firms cleared.”↩

1304. Faiz Siddiqui, as cited in Podcast Script and Fact-Check Review, Episode 8: “Siddiqui: ‘Any interest in participating in the Twitter deal?’ ‘Yes, of course.’ About $1 billion into the purchase.”↩

1305. Walter Isaacson and Reuters, as cited in Podcast Script and Fact-Check Review, Episode 8: “Twitter close, Oct 27, 2022, $44 billion. Sink. ‘The bird is freed.’ Executives fired the same day.”↩

1306. Walter Isaacson and Reuters, as cited in Podcast Script and Fact-Check Review, Episode 8: “Twitter close, Oct 27, 2022, $44 billion. Sink. ‘The bird is freed.’ Executives fired the same day.”↩

1307. Stanford University FY25 Annual Report and facts.stanford.edu (Podcast Fact-Check Review, Episode 8): “Endowment $40.8 billion at Aug 31, 2025. Payout $1.9 billion. Tuition $65,127. Nearly half of undergraduates receive need-based aid.”↩

1308. OpenSecrets donor table, drawn from FEC filings (Podcast Fact-Check Review, Episode 8): “Thiel: $10 million (March 12, 2021), $3.5 million (April 18, 2022), $1.5 million (April 25, 2022), to Protect Ohio Values, for Vance.”↩

1309. Theo Baker and *The Stanford Daily* (Podcast Script and Fact-Check Review, Episode 8): “Baker’s first Daily piece was Oct 24, 2022, on social life. First Tessier-Lavigne article: Nov 29, 2022. Tip toward PubPeer on Nov 1. Elisabeth Bik among the readers. ‘I called Sam, the Daily editor. “Holy shit,” he said. “We need to get going. How quickly can you write this?”’ Sam Catania. Board review announced the same day.”↩

1310. Theo Baker and *The Stanford Daily* (Podcast Script and Fact-Check Review, Episode 8): “Baker’s first Daily piece was Oct 24, 2022, on social life. First Tessier-Lavigne article: Nov 29, 2022. Tip toward PubPeer on Nov 1. Elisabeth Bik among the readers. ‘I called Sam, the Daily editor. “Holy shit,” he said. “We need to get going. How quickly can you write this?”’ Sam Catania. Board review announced the same day.”↩

1311. Theo Baker and *The Stanford Daily* (Podcast Script and Fact-Check Review, Episode 8): “Baker’s first Daily piece was Oct 24, 2022, on social life. First Tessier-Lavigne article: Nov 29, 2022. Tip toward PubPeer on Nov 1. Elisabeth Bik among the readers. ‘I called Sam, the Daily editor. “Holy shit,” he said. “We need to get going. How quickly can you write this?”’ Sam Catania. Board review announced the same day.”↩

1312. Theo Baker and *The Stanford Daily* (Podcast Script and Fact-Check Review, Episode 8): “Baker’s first Daily piece was Oct 24, 2022, on social life. First Tessier-Lavigne article: Nov 29, 2022. Tip toward PubPeer on Nov 1. Elisabeth Bik among the readers. ‘I called Sam, the Daily editor. “Holy shit,” he said. “We need to get going. How quickly can you write this?”’ Sam Catania. Board review announced the same day.”↩

1313. Special Committee of the Board of Trustees of Stanford University, *Report of the Special Committee of the Board of Trustees of Stanford University* (July 2023; `stanford-special-committee-2023` in `references.bib`), scientific panel report, and Mark Filip cover letter of July 17, 2023 (Podcast Fact-Check Review, Episode 8): “Panel, July 17, 2023, letter to the special committee, 365 Lasuen Street. Did not personally engage in research misconduct on the twelve papers… Manipulation in at least four of five primary papers, by others, in his lab. Each of the five has serious flaws in the presentation of research data. More than fifty thousand documents.”↩

1314. Special Committee of the Board of Trustees of Stanford University, *Report of the Special Committee of the Board of Trustees of Stanford University* (July 2023; `stanford-special-committee-2023` in `references.bib`), scientific panel report, and Mark Filip cover letter of July 17, 2023 (Podcast Fact-Check Review, Episode 8): “Panel, July 17, 2023, letter to the special committee, 365 Lasuen Street. Did not personally engage in research misconduct on the twelve papers… Manipulation in at least four of five primary papers, by others, in his lab. Each of the five has serious flaws in the presentation of research data. More than fifty thousand documents.”↩

1315. Stanford Board of Trustees statement and Marc Tessier-Lavigne letter, July 19, 2023 (`stanford-special-committee-2023` in `references.bib`; Podcast Fact-Check Review, Episode 8): “Board, July 19, 2023. Resignation effective Aug 31. Richard Saller interim Sept 1. Remained a tenured biology professor… Insufficient steps to correct the record.”↩

1316. Stanford Board of Trustees statement and Marc Tessier-Lavigne letter, July 19, 2023 (`stanford-special-committee-2023` in `references.bib`; Podcast Fact-Check Review, Episode 8): “Board, July 19, 2023. Resignation effective Aug 31. Richard Saller interim Sept 1. Remained a tenured biology professor… Insufficient steps to correct the record.”↩

1317. Theo Baker, p. 120 (Podcast Fact-Check Review, Episode 8): “Rosewood: ‘the modern clubhouse for Silicon Valley,’ on Sand Hill, on land owned by Stanford. Four-figure nights. Baker’s verdict: a row of dull buildings whose advantage is being close to the campus.”↩

1318. Theo Baker, p. 120 (Podcast Fact-Check Review, Episode 8): “Rosewood: ‘the modern clubhouse for Silicon Valley,’ on Sand Hill, on land owned by Stanford. Four-figure nights. Baker’s verdict: a row of dull buildings whose advantage is being close to the campus.”↩

1319. Callback to Episode 7 (Podcast Script and Fact-Check Review, Episode 8): “NationsBank / Charlotte, 1998.”↩

1320. Congressional Research Service Report R48642 (September 19, 2025), Nvidia Form 8-K (April 9, 2025), and Reuters (August 8, 2025; December 8, 2025; May 14, 2026) (Podcast Fact-Check Review, Episode 8): “Export controls. Oct 7, 2022: A100/H100-class chips and tools; Nvidia’s A800/H800. Oct 2023: loopholes closed; H20. Dec 2024: HBM, packaging, equipment. April 2025: H20 license, Nvidia 8-K, effectively a ban. Aug 2025: H20 licenses resume. Dec 2025: H200 allowed with a 25 percent cut to the U.S. government. May 2026: about ten Chinese firms cleared.”↩

1321. *The Wall Street Journal* via CNBC, Associated Press, and CNN (Podcast Fact-Check Review, Episode 8): “Feb 2024 renewal about $250 million, no longer exclusive (WSJ via CNBC). Oct 25, 2024, Trump in Austin. Nov 4, 2024, endorsement after a Musk interview: Musk made ‘the most compelling case.’”↩

1322. Bloomberg Law, *Texas Tribune*, and Reuters, May 24, 2024 (Podcast Fact-Check Review, Episode 8): “May 24, 2022, Robb Elementary: 19 children and 2 teachers killed. Salvador Ramos named once… Two suits filed May 24, 2024, by Josh Koskoff. *Mata v. Meta* in Los Angeles, about 45 family members, against Meta and Activision. *Rubio v. Daniel Defense* in Uvalde County, 31 family members.”↩

1323. *Texas Tribune* and complaint text (Podcast Script and Fact-Check Review, Episode 8): “Koskoff: ‘Just 23 minutes after midnight…’ through ‘trained him to use it.’… Complaint allegations: Modern Warfare title page, ‘totally murdered out,’ ‘Hunters Hunt,’ Instagram’s firearm-ad rule, ‘Menlo Park and Santa Monica.’”↩

1324. *Texas Tribune* and complaint text (Podcast Script and Fact-Check Review, Episode 8): “Koskoff: ‘Just 23 minutes after midnight…’ through ‘trained him to use it.’… Complaint allegations: Modern Warfare title page, ‘totally murdered out,’ ‘Hunters Hunt,’ Instagram’s firearm-ad rule, ‘Menlo Park and Santa Monica.’”↩

1325. *Platformer*, *The Washington Post*, and NPR (Podcast Fact-Check Review, Episode 8): “Stanford Internet Observatory, 2019, Alex Stamos, Renée DiResta. Lawsuits and House Judiciary subpoenas. June 2024: Platformer said it was being dismantled (Stamos gone, DiResta’s contract not renewed, staff told to look for jobs). Stanford said the work would continue under new leadership.”↩

1326. Paramount Form 8-K, CBS News, Associated Press, *Politico*, FCC-25-43 (July 24, 2025), and Reuters (August 7, 2025) (Podcast Fact-Check Review, Episode 8): “Paramount–Skydance agreement, July 7, 2024. July 2, 2025: Paramount pays $16 million to settle Trump’s *60 Minutes* suit. July 17, 2025: CBS cancels *The Late Show*, days after Colbert called the settlement, in his words, ‘a big fat bribe.’ Last season to end May 2026. July 24, 2025: FCC approves a deal of about $8 billion. Closes Aug. 7, 2025, as Paramount Skydance. Reuters: $8.4 billion.”↩

1327. *The Wall Street Journal* via CNBC, Associated Press, and CNN (Podcast Fact-Check Review, Episode 8): “Feb 2024 renewal about $250 million, no longer exclusive (WSJ via CNBC). Oct 25, 2024, Trump in Austin. Nov 4, 2024, endorsement after a Musk interview: Musk made ‘the most compelling case.’”↩

1328. Faiz Siddiqui, writing in December 2024 (Podcast Fact-Check Review, Episode 8): “Musk, more than $275 million via America PAC in the 2024 cycle.”↩

1329. CNN, Associated Press, NPR, and Law360 (Podcast Fact-Check Review, Episode 8): “Onion auction, Nov 2024, thrown out; Texas receiver order; April 2026 license-then-purchase; CEO said the families had not received a penny; stay; Texas Supreme Court refused to lift it on Sept 9, 2026. Jones left the InfoWars name.”↩

1330. Congressional Research Service Report R48642 (September 19, 2025), Nvidia Form 8-K (April 9, 2025), and Reuters (August 8, 2025; December 8, 2025; May 14, 2026) (Podcast Fact-Check Review, Episode 8): “Export controls. Oct 7, 2022: A100/H100-class chips and tools; Nvidia’s A800/H800. Oct 2023: loopholes closed; H20. Dec 2024: HBM, packaging, equipment. April 2025: H20 license, Nvidia 8-K, effectively a ban. Aug 2025: H20 licenses resume. Dec 2025: H200 allowed with a 25 percent cut to the U.S. government. May 2026: about ten Chinese firms cleared.”↩

1331. Associated Press and *The Guardian*, January 20, 2025 (Podcast Fact-Check Review, Episode 8): “Jan 20, 2025, moved indoors to the Capitol Rotunda because of the cold, which scrambled seating. Photographs and the AP and Guardian: Musk, Zuckerberg, Bezos, Pichai with the family, ahead of several cabinet nominees. Cook in that company of guests. TikTok’s chief in the room beside the intelligence nominee.”↩

1332. White House Executive Order (January 20, 2025), Congressional Research Service, and Reuters (May 28–29, 2025) (Podcast Fact-Check Review, Episode 8): “DOGE executive order, Jan 20, 2025: U.S. Digital Service renamed U.S. DOGE Service. Not a cabinet department. Temporary organization ends July 4, 2026. Access to unclassified records and software… Musk as a special government employee. About 130 days. Reuters, May 28, 2025: off-boarding begins that night.”↩

1333. Nvidia Form 10-K and earnings release for fiscal year ended January 26, 2025 (Podcast Fact-Check Review, Episode 8): “Nvidia, year ended Jan 26, 2025: revenue $130.5 billion, up 114 percent from $60.9 billion. Data center $115.2 billion.”↩

1334. Nvidia Form 10-K and earnings release for fiscal year ended January 26, 2025 (Podcast Fact-Check Review, Episode 8): “Nvidia, year ended Jan 26, 2025: revenue $130.5 billion, up 114 percent from $60.9 billion. Data center $115.2 billion.”↩

1335. *Financial Times* (January 2025), *The Verge*, House committee report, OpenAI memo (February 12, 2026), and Reuters (February 12, 2026) (Podcast Fact-Check Review, Episode 8): “OpenAI to the FT, Jan 2025: evidence DeepSeek used distillation, against OpenAI’s terms; evidence not published. House committee: ‘highly likely,’ relying on OpenAI. Feb 12, 2026 memo: continued distillation via obfuscated routers and resellers. No court finding.”↩

1336. Congressional Research Service Report R48642 (September 19, 2025), Nvidia Form 8-K (April 9, 2025), and Reuters (August 8, 2025; December 8, 2025; May 14, 2026) (Podcast Fact-Check Review, Episode 8): “Export controls. Oct 7, 2022: A100/H100-class chips and tools; Nvidia’s A800/H800. Oct 2023: loopholes closed; H20. Dec 2024: HBM, packaging, equipment. April 2025: H20 license, Nvidia 8-K, effectively a ban. Aug 2025: H20 licenses resume. Dec 2025: H200 allowed with a 25 percent cut to the U.S. government. May 2026: about ten Chinese firms cleared.”↩

1337. White House Executive Order (January 20, 2025), Congressional Research Service, and Reuters (May 28–29, 2025) (Podcast Fact-Check Review, Episode 8): “DOGE executive order, Jan 20, 2025: U.S. Digital Service renamed U.S. DOGE Service. Not a cabinet department. Temporary organization ends July 4, 2026. Access to unclassified records and software… Musk as a special government employee. About 130 days. Reuters, May 28, 2025: off-boarding begins that night.”↩

1338. Paramount Form 8-K, CBS News, Associated Press, *Politico*, FCC-25-43 (July 24, 2025), and Reuters (August 7, 2025) (Podcast Fact-Check Review, Episode 8): “Paramount–Skydance agreement, July 7, 2024. July 2, 2025: Paramount pays $16 million to settle Trump’s *60 Minutes* suit. July 17, 2025: CBS cancels *The Late Show*, days after Colbert called the settlement, in his words, ‘a big fat bribe.’ Last season to end May 2026. July 24, 2025: FCC approves a deal of about $8 billion. Closes Aug. 7, 2025, as Paramount Skydance. Reuters: $8.4 billion.”↩

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1340. Stanford University FY25 Annual Report and facts.stanford.edu (Podcast Fact-Check Review, Episode 8): “Endowment $40.8 billion at Aug 31, 2025. Payout $1.9 billion. Tuition $65,127. Nearly half of undergraduates receive need-based aid.”↩

1341. MyNewsLA (September 5, 2025) and Gibson Dunn (September 18, 2025) (Podcast Fact-Check Review, Episode 8): “Sept 5, 2025: Judge William F. Highberger dismissed Meta. ‘Limitless liability.’ Reaffirmed December 2025. Families appealed. Proximate cause; Meta’s conduct ‘so far attenuated… negligible or theoretical.’ Same day, Activision’s anti-SLAPP motion denied.”↩

1342. Paramount investor release and Reuters, October 6, 2025 (Podcast Fact-Check Review, Episode 8): “Oct. 6, 2025: Paramount buys The Free Press, reported at about $150 million, and names Bari Weiss editor in chief of CBS News, reporting directly to David Ellison.”↩

1343. Associated Press and PBS (Podcast Fact-Check Review, Episode 8): “Texas: about $49–50 million (Aug 2022) cut on Aug 21, 2026, to about $6 million ($4.1 million compensatory plus about $1.5 million

punitive)… Connecticut: default in 2021; jury $964 million plus $473 million; appeals to about $1.25 billion; cert denied Oct 14, 2025. Personal bankruptcy late 2022. Paid nothing as of August 2026.”↩

1344. U.S. Department of Justice, Central District of California, November 17, 2025 (Podcast Fact-Check Review, Episode 8): “Chenguang Gong, sentenced Nov 17, 2025, to 46 months, missile-detection infrared sensors, talent-program applications 2014–2022.”↩

1345. Netflix Form 10-K, Stanford Engineering Heroes, Netflix investor release (December 5, 2025), and Warner Bros. Discovery release (February 26, 2026) (Podcast Fact-Check Review, Episode 8): “Netflix founded Aug. 29, 1997, in Scotts Valley, by Reed Hastings and Marc Randolph. Headquarters later in Los Gatos. Hastings: Stanford MS in computer science, 1988, in artificial intelligence. Dec. 5, 2025: Netflix agrees to buy Warner Bros., HBO, and HBO Max… $82.7 billion enterprise value. Feb. 26, 2026: the Warner Bros. Discovery board calls Paramount’s bid superior. Netflix does not raise.”↩

1346. Congressional Research Service Report R48642 (September 19, 2025), Nvidia Form 8-K (April 9, 2025), and Reuters (August 8, 2025; December 8, 2025; May 14, 2026) (Podcast Fact-Check Review, Episode 8): “Export controls. Oct 7, 2022: A100/H100-class chips and tools; Nvidia’s A800/H800. Oct 2023: loopholes closed; H20. Dec 2024: HBM, packaging, equipment. April 2025: H20 license, Nvidia 8-K, effectively a ban. Aug 2025: H20 licenses resume. Dec 2025: H200 allowed with a 25 percent cut to the U.S. government. May 2026: about ten Chinese firms cleared.”↩

1347. U.S. Department of Justice release (January 29, 2026), Judge Vince Chhabria’s order (August 20, 2026), Reuters (August 20, 2026), and KQED (September 1, 2026) (Podcast Fact-Check Review, Episode 8): “Linwei Ding. Came for a master’s in 2010. More than 500 files; DOJ: more than two thousand pages. Convicted Jan 29, 2026, on seven espionage counts and seven trade-secret counts… Pitch: ‘We have experience with Google’s ten-thousand-card computational power platform; we just need to replicate and upgrade it.’… Aug 20, 2026: Judge Chhabria vacated all seven espionage counts; insufficient evidence he knew or intended a benefit to the PRC government. Trade-secret counts stand. Sentenced Sept 1, 2026: 12 months minus a day, $25,000 fine, more than $189,000 restitution.”↩

1348. *Financial Times* (January 2025), *The Verge*, House committee report, OpenAI memo (February 12, 2026), and Reuters (February 12, 2026) (Podcast Fact-Check Review, Episode 8): “OpenAI to the FT, Jan 2025: evidence DeepSeek used distillation, against OpenAI’s terms; evidence not published. House committee: ‘highly likely,’ relying on OpenAI. Feb 12, 2026 memo: continued distillation via obfuscated routers and resellers. No court finding.”↩

1349. Netflix Form 10-K, Stanford Engineering Heroes, Netflix investor release (December 5, 2025), and Warner Bros. Discovery release (February 26, 2026) (Podcast Fact-Check Review, Episode 8): “Netflix founded Aug. 29, 1997, in Scotts Valley, by Reed Hastings and Marc Randolph. Headquarters later in Los Gatos. Hastings: Stanford MS in computer science, 1988, in artificial intelligence. Dec. 5, 2025: Netflix

agrees to buy Warner Bros., HBO, and HBO Max… $82.7 billion enterprise value. Feb. 26, 2026: the Warner Bros. Discovery board calls Paramount's bid superior. Netflix does not raise."↩

1350. CNN, Associated Press, NPR, and Law360 (Podcast Fact-Check Review, Episode 8): "Onion auction, Nov 2024, thrown out; Texas receiver order; April 2026 license-then-purchase; CEO said the families had not received a penny; stay; Texas Supreme Court refused to lift it on Sept 9, 2026. Jones left the InfoWars name."↩

1351. Congressional Research Service Report R48642 (September 19, 2025), Nvidia Form 8-K (April 9, 2025), and Reuters (August 8, 2025; December 8, 2025; May 14, 2026) (Podcast Fact-Check Review, Episode 8): "Export controls. Oct 7, 2022: A100/H100-class chips and tools; Nvidia's A800/H800. Oct 2023: loopholes closed; H20. Dec 2024: HBM, packaging, equipment. April 2025: H20 license, Nvidia 8-K, effectively a ban. Aug 2025: H20 licenses resume. Dec 2025: H200 allowed with a 25 percent cut to the U.S. government. May 2026: about ten Chinese firms cleared."↩

1352. U.S. Department of Justice release (January 29, 2026), Judge Vince Chhabria's order (August 20, 2026), Reuters (August 20, 2026), and KQED (September 1, 2026) (Podcast Fact-Check Review, Episode 8): "Linwei Ding. Came for a master's in 2010. More than 500 files; DOJ: more than two thousand pages. Convicted Jan 29, 2026, on seven espionage counts and seven trade-secret counts… Pitch: 'We have experience with Google's ten-thousand-card computational power platform; we just need to replicate and upgrade it.'… Aug 20, 2026: Judge Chhabria vacated all seven espionage counts; insufficient evidence he knew or intended a benefit to the PRC government. Trade-secret counts stand. Sentenced Sept 1, 2026: 12 months minus a day, $25,000 fine, more than $189,000 restitution."↩

1353. Matt Sheehan, *The Transpacific Experiment*, pp. 54–55 (Podcast Script and Fact-Check Review, Episode 8): "Sheehan: Chinese American researchers charged with espionage, cases later unfounded. Trump, 2018, private: 'almost every student that comes over to this country is a spy.'"↩

1354. *Foreign Affairs*, RAND, Information Technology and Innovation Foundation (September 2024 and Hamilton Index 2026), and Matt Sheehan, *The Transpacific Experiment*, p. 214 (Podcast Fact-Check Review, Episode 8): "'China can manufacture but can't make California' as an argument. Jeffrey Ding: diffusion matters more than invention. RAND: weaker research-to-industry links. ITIF, Sept 2024… Hamilton Index 2026: leads 7 of 10. Dan Wang, *Breakneck*, via a Foreign Affairs review: engineering state versus a lawyerly society. Sheehan on 1980s Japan: investment kept 'generating jobs, income, and even innovation.'"↩

1355. Associated Press and PBS (Podcast Fact-Check Review, Episode 8): "Texas: about $49–50 million (Aug 2022) cut on Aug 21, 2026, to about $6 million ($4.1 million compensatory plus about $1.5 million punitive)… Connecticut: default in 2021; jury $964 million plus $473 million; appeals to about $1.25 billion; cert denied Oct 14, 2025. Personal bankruptcy late 2022. Paid nothing as of August 2026."↩

1356. CNN, Associated Press, NPR, and Law360 (Podcast Fact-Check Review, Episode 8): “Onion auction, Nov 2024, thrown out; Texas receiver order; April 2026 license-then-purchase; CEO said the families had not received a penny; stay; Texas Supreme Court refused to lift it on Sept 9, 2026. Jones left the InfoWars name.”↩

1357. Skydance release, SEC Form 8-K, Warner Bros. Discovery release, and Yahoo Finance, October 6, 2026 (Podcast Fact-Check Review, Episode 8): “Oct. 6, 2026: the purchase closes. Justice Kagan denies a last-minute petition. The company is renamed Skydance. David Ellison and Ynon Kreiz, co-chief executives. The Ellisons and RedBird hold all the voting shares. $47 billion in new equity from the Ellisons, RedBird, and the sovereign funds of Saudi Arabia, Abu Dhabi, and Qatar.”↩

1358. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (`harris-palo-alto` in `references.bib`), ch. 4 (Podcast Fact-Check Review, Episode 8): “Oracle: failed CIA contract Project Oracle at Ampex. Three employees leave. Third name, Oracle. First customer, the CIA. Founded 1977 with Bob Miner and Ed Oates. Reagan-era checks in plain white envelopes. Engineer, as Harris records it: ‘We bought this thing knowing damn well it wasn’t going to work. We’re buying an idea.’”↩

1359. Malcolm Harris, *Palo Alto: A History of California, Capitalism, and the World* (`harris-palo-alto` in `references.bib`), ch. 1.3 (Podcast Fact-Check Review, Episode 8): “Jan. 1880: Muybridge shows the zoopraxiscope at the Stanford mansion in San Francisco. Harris: ‘The age of movies began in California, not in Hollywood but in the Bay.’ Leland Stanford paid for the horse photographs.”↩

1360. Hollywood Historic Trust and California State Parks Office of Historic Preservation (Podcast Fact-Check Review, Episode 8): “1914: Cecil B. DeMille’s *The Squaw Man*, a western, shot in a barn at Selma and Vine. First feature made in Hollywood. Jesse L. Lasky Feature Play Company becomes Paramount.”↩

1361. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): “Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page’s lumber road and then the spine of the research park, Sand Hill, Rosewood.”↩

1362. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): “Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page’s lumber road and then the spine of the research park, Sand Hill, Rosewood.”↩

1363. Special Committee of the Board of Trustees of Stanford University, *Report of the Special Committee of the Board of Trustees of Stanford University* (July 2023; `stanford-special-committee-2023` in `references.bib`), scientific panel report, and Mark Filip cover letter of July 17, 2023 (Podcast Fact-Check Review, Episode 8): “Panel, July 17, 2023, letter to the special committee, 365 Lasuen Street. Did not personally engage in research misconduct on the twelve papers… Manipulation in at least four of five primary papers, by others, in his lab. Each of the five has serious flaws in the presentation of research data. More than fifty thousand documents.”↩

1364. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): “Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page’s lumber road and then the spine of the research park, Sand Hill, Rosewood.”↩

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1367. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): “Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page’s lumber road and then the spine of the research park, Sand Hill, Rosewood.”↩

1368. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): “Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and

now Building 420, Hoover, Terman, Page Mill as William Page's lumber road and then the spine of the research park, Sand Hill, Rosewood."↩

1369. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): "Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page's lumber road and then the spine of the research park, Sand Hill, Rosewood."↩

1370. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): "Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page's lumber road and then the spine of the research park, Sand Hill, Rosewood."↩

1371. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): "Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page's lumber road and then the spine of the research park, Sand Hill, Rosewood."↩

1372. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): "Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page's lumber road and then the spine of the research park, Sand Hill, Rosewood."↩

1373. Theo Baker, p. 120 (Podcast Fact-Check Review, Episode 8): "Rosewood: 'the modern clubhouse for Silicon Valley,' on Sand Hill, on land owned by Stanford. Four-figure nights. Baker's verdict: a row of dull buildings whose advantage is being close to the campus."↩

1374. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): "Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page's lumber road and then the spine of the research park, Sand Hill, Rosewood."↩

1375. Matt Sheehan, *The Transpacific Experiment*, pp. 267–268 (Podcast Fact-Check Review, Episode 8): “1869 golden-spike photograph: not a single Chinese face. 2013 Bay Bridge: mayors Ed Lee and Jean Quan. ‘This time the Chinese were in the middle.’”↩

1376. Stanford Campus Names page and *Stanford Report* (Podcast Script and Fact-Check Review, Episode 8): “Coda walk: Gálvez, Serra, Lasuen (La Purísima; 365 Lasuen), El Camino, Alvarado (1836; Jordan named the row), Stanford Avenue, Jane Stanford Way (effective Oct 7, 2019; dedicated Nov 14; address 450; Serra Street from Campus Drive East to El Camino kept the name), Jordan Hall renamed in October 2020 and now Building 420, Hoover, Terman, Page Mill as William Page’s lumber road and then the spine of the research park, Sand Hill, Rosewood.”↩